

XCPC Template Library

Morning Flower
and Evening Oath

Tongji University

双码风算法模板与数学速查

建设稿 · 2026-09-12

状态说明

本稿包含首批已进行本地验证的模块与完整来源覆盖表的入口。尚未完成 kuangbin、WIDA 全部条目的双码风迁移，尚未完成近三年中国赛站逐题覆盖审计，最大流、平衡树、双连通分量、可持久化等部分接口已有在线 AC 记录，具体题号、源码摘要及验证范围见 `verification/oj.json`。其余接口须分别核验，不能由同文件中某算法通过推断全部通过。本地对拍通过不代表对所有输入的绝对正确性证明；适用范围、数值上限与退化约定属于接口的一部分。

目录

使用与验证	x
第一部分 动态容器风格	1
第一章 Tarjan 与缩点	2
1.1 Tarjan 强连通分量与缩点	2
第二章 双连通分量与圆方树	4
2.1 Tarjan 点双、边双与圆方树	4
第三章 奇环顶点判定	7
3.1 属于简单奇环的全部顶点	7
第四章 删点连通性	9
4.1 由点双分解计算删点连通块数	9
第五章 边双缩点与边定向	10
5.1 边双缩点的大小与内部边数	10
5.2 无向边定向与强连通定向	11
第六章 桥树最少加边	12
6.1 连通无向图最少加边方案	12
第七章 数据结构	13
7.1 并查集	13
7.2 可撤销并查集	14
7.3 树状数组与第 k 小	15
7.4 区间加与区间和	15
7.5 64 位异或线性基	18
第八章 带位置线性基	21
8.1 带位置限制的异或线性基	21
第九章 线性空间求交	23
9.1 异或线性空间求交	23
9.2 Zassenhaus: 同时求和空间与交空间	24
第十章 线段树常用修改	25
10.1 线段树: 区间乘、加、赋值与模意义求和	25
第十一章 网络流	28
11.1 最大流与方案	28
11.2 势能最小费用最大流	30
11.3 上下界可行循环流	32

第十二章 最大权闭合子图	34
12.1 最大权闭合子图与选点方案	34
第十三章 图论	35
13.1 非负权最短路	35
13.2 强连通分量	36
13.3 2-SAT 与任意可行方案	37
13.4 Hopcroft-Karp 最大匹配与最小点覆盖	38
13.5 割点与桥	40
第十四章 字典序最小 2-SAT	42
14.1 递归染色与回滚求最小赋值	42
第十五章 有向欧拉路	44
15.1 有向图字典序最小欧拉路	44
第十六章 字典序单词链	46
16.1 字典序最小单词链	46
第十七章 无向欧拉路	47
17.1 无向图欧拉路与边方向	47
第十八章 混合图欧拉定向	49
18.1 欧拉回路与指定端点定向	49
18.2 任意起终点的混合欧拉路径	51
第十九章 图上异或行走	52
19.1 无向图最大异或行走	52
第二十章 树上算法	54
20.1 重链剖分与 LCA	54
第二十一章 离线最近公共祖先	56
21.1 离线 Tarjan LCA	56
第二十二章 在线最近公共祖先	58
22.1 欧拉序与 ST 表 LCA	58
第二十三章 倍增祖先与路径最值	60
23.1 倍增 LCA、祖先与最大边权	60
第二十四章 树上路径交	63
24.1 两条树路径的交集	63
第二十五章 点分治	64
25.1 点分治：树上距离点对计数	64
第二十六章 树上启发式合并	67
26.1 树上启发式合并：子树众数颜色和	67
第二十七章 虚树	70
27.1 虚树：关键点压缩与路径信息	70

第二十八章 字符串	73
28.1 前缀函数	73
28.2 KMP 匹配	73
28.3 Z 函数	74
28.4 Manacher 回文半径	74
28.5 最小循环表示	75
28.6 AC 自动机出现次数	75
28.7 倍增后缀数组与 LCP	78
28.8 后缀自动机	80
第二十九章 后缀查询与重复子串	82
29.1 后缀 LCP 与子串比较	82
29.2 两个前缀的最长公共后缀	83
29.3 重复子串 AA 的起止位置计数	84
第三十章 回文树	85
30.1 回文树 PAM: 回文后缀与出现次数	85
第三十一章 数论	87
31.1 模运算、素性、CRT 与类欧几里德	87
31.2 Pollard-Rho 因数分解	90
31.3 线性筛、欧拉函数与 Möbius 函数	91
31.4 静态模整数	92
31.5 阶乘组合数	93
第三十二章 批量模逆元	94
32.1 任意数组的批量逆元	94
第三十三章 混合进制中国剩余定理	95
33.1 Garner: 大周期 CRT 与混合进制	95
第三十四章 乘法阶与原根	97
34.1 乘法阶与原根	97
第三十五章 Carmichael 函数	100
35.1 Carmichael: 模乘法群的公共指数	100
第三十六章 素数模数组组合数	101
36.1 Lucas: 小素数模的大下标组合数	101
第三十七章 合数模数组组合数	103
37.1 合数模数组组合数 (扩展 Lucas)	103
第三十八章 二次剩余	106
38.1 素数模平方根 (Cipolla)	106
第三十九章 多项式插值	108
39.1 拉格朗日插值与连续点优化	108
第四十章 多项式	110
40.1 NTT、形式幂级数、FWT 与递推	110

第四十一章 代数与数论进阶	115
41.1 模高斯消元、行列式、矩阵树定理	115
41.2 杜教筛	118
41.3 扩展 BSGS	119
第四十二章 计算几何	121
42.1 整数精确二维几何	121
42.2 浮点点线圆构造	124
第四十三章 优化与可持久化	128
43.1 Max-plus 矩阵幂	128
43.2 离散李超树与最优直线编号	128
43.3 主席树：静态区间第 k 小	130
第四十四章 可持久化数组	133
44.1 可持久化数组：历史版本分叉与单点修改	133
第四十五章 树上路径第 k 小	136
45.1 树上路径点权第 k 小	136
第四十六章 动态区间顺序统计	139
46.1 动态区间第 k 小（树状数组套权值线段树）	139
第四十七章 区间不同数统计	142
47.1 静态区间不同数计数（前驱位置主席树）	142
第四十八章 图论进阶	144
48.1 矩形二分图最小权匹配	144
48.2 有向最小树形图	145
48.3 无向全局最小割	147
第四十九章 带权二分图匹配	149
49.1 二分图最大权匹配（缺边与可选匹配）	149
第五十章 精确几何进阶	151
50.1 整数最近点对与 Minkowski 和	151
50.2 整数三维共线、共面与方向	154
第五十一章 凸包支撑点查询	155
51.1 支撑点查询恢复下凸包	155
第五十二章 动态树	157
52.1 Link-cut tree：动态森林与路径异或	157
第五十三章 随机平衡树	161
53.1 FHQ Treap：可重集合与顺序统计	161
53.2 隐式 Treap：序列翻转、加值与区间和	163
第五十四章 伸展树	168
54.1 Splay：可重集合与顺序统计	168

第五十五章 状态序列维护	173
55.1 状态序列 Treap: 分组区间 GCD	173
第五十六章 一般图匹配	177
56.1 一般图最大基数匹配: 带花树	177
第五十七章 线性递推	180
57.1 Berlekamp–Massey: 最短线性递推	180
57.2 特征多项式求线性递推第 k 项	182
第二部分 传统接口风格	184
第五十八章 Tarjan 与缩点	185
58.1 Tarjan 强连通分量与缩点	185
第五十九章 双连通分量与圆方树	187
59.1 Tarjan 点双、边双与圆方树	187
第六十章 奇环顶点判定	190
60.1 属于简单奇环的全部顶点	190
第六十一章 删点连通性	192
61.1 由点双分解计算删点连通块数	192
第六十二章 边双缩点与边定向	193
62.1 边双缩点的大小与内部边数	193
62.2 无向边定向与强连通定向	194
第六十三章 桥树最少加边	195
63.1 连通无向图最少加边方案	195
第六十四章 数据结构	196
64.1 并查集	196
64.2 可撤销并查集	197
64.3 树状数组与第 k 小	198
64.4 区间加与区间和	199
64.5 64 位异或线性基	201
第六十五章 带位置线性基	203
65.1 带位置限制的异或线性基	203
第六十六章 线性空间求交	205
66.1 异或线性空间求交	205
66.2 Zassenhaus: 同时求和空间与交空间	206
第六十七章 线段树常用修改	207
67.1 线段树: 区间乘、加、赋值与模意义求和	207
第六十八章 网络流	210
68.1 最大流与方案	210
68.2 势能最小费用最大流	212
68.3 上下界可行循环流	215

第六十九章 最大权闭合子图	216
69.1 最大权闭合子图与选点方案	216
第七十章 图论	217
70.1 非负权最短路	217
70.2 强连通分量	218
70.3 2-SAT 与任意可行方案	219
70.4 Hopcroft-Karp 最大匹配与最小点覆盖	220
70.5 割点与桥	222
第七十一章 字典序最小 2-SAT	224
71.1 递归染色与回滚求最小赋值	224
第七十二章 有向欧拉路	226
72.1 有向图字典序最小欧拉路	226
第七十三章 字典序单词链	229
73.1 字典序最小单词链	229
第七十四章 无向欧拉路	230
74.1 无向图欧拉路与边方向	230
第七十五章 混合图欧拉定向	233
75.1 欧拉回路与指定端点定向	233
75.2 任意起终点的混合欧拉路径	235
第七十六章 图上异或行走	236
76.1 无向图最大异或行走	236
第七十七章 树上算法	238
77.1 重链剖分与 LCA	238
第七十八章 离线最近公共祖先	240
78.1 离线 Tarjan LCA	240
第七十九章 在线最近公共祖先	242
79.1 欧拉序与 ST 表 LCA	242
第八十章 倍增祖先与路径最值	244
80.1 倍增 LCA、祖先与最大边权	244
第八十一章 树上路径交	247
81.1 两条树路径的交集	247
第八十二章 点分治	248
82.1 点分治：树上距离点对计数	248
第八十三章 树上启发式合并	251
83.1 树上启发式合并：子树众数颜色和	251
第八十四章 虚树	254
84.1 虚树：关键点压缩与路径信息	254

第八十五章 字符串	257
85.1 前缀函数	257
85.2 KMP 匹配	257
85.3 Z 函数	258
85.4 Manacher 回文半径	258
85.5 最小循环表示	259
85.6 AC 自动机出现次数	259
85.7 倍增后缀数组与 LCP	262
85.8 后缀自动机	264
第八十六章 后缀查询与重复子串	266
86.1 后缀 LCP 与子串比较	266
86.2 两个前缀的最长公共后缀	267
86.3 重复子串 AA 的起止位置计数	268
第八十七章 回文树	269
87.1 回文树 PAM: 回文后缀与出现次数	269
第八十八章 数论	271
88.1 模运算、素性、CRT 与类欧几里德	271
88.2 Pollard-Rho 因数分解	274
88.3 线性筛、欧拉函数与 Möbius 函数	275
88.4 静态模整数	276
88.5 阶乘组合数	277
第八十九章 批量模逆元	278
89.1 任意数组的批量逆元	278
第九十章 混合进制中国剩余定理	279
90.1 Garner: 大周期 CRT 与混合进制	279
第九十一章 乘法阶与原根	280
91.1 乘法阶与原根	280
第九十二章 Carmichael 函数	283
92.1 Carmichael: 模乘法群的公共指数	283
第九十三章 素数模数组组合数	284
93.1 Lucas: 小素数模的大下标组合数	284
第九十四章 合数模数组组合数	285
94.1 合数模数组组合数 (扩展 Lucas)	285
第九十五章 二次剩余	288
95.1 素数模平方根 (Cipolla)	288
第九十六章 多项式插值	290
96.1 拉格朗日插值与连续点优化	290
第九十七章 多项式	292
97.1 NTT、形式幂级数、FWT 与递推	292

第九十八章 代数与数论进阶	297
98.1 模高斯消元、行列式、矩阵树定理	297
98.2 杜教筛	300
98.3 扩展 BSGS	301
第九十九章 计算几何	302
99.1 整数精确二维几何	302
99.2 浮点点线圆构造	305
第一百章 优化与可持久化	309
100.1 Max-plus 矩阵幂	309
100.2 离散李超树与最优直线编号	309
100.3 主席树：静态区间第 k 小	311
第一百零一章 可持久化数组	314
101.1 可持久化数组：历史版本分叉与单点修改	314
第一百零二章 树上路径第 k 小	317
102.1 树上路径点权第 k 小	317
第一百零三章 动态区间顺序统计	320
103.1 动态区间第 k 小（树状数组套权值线段树）	320
第一百零四章 区间不同数统计	323
104.1 静态区间不同数计数（前驱位置主席树）	323
第一百零五章 图论进阶	325
105.1 矩形二分图最小权匹配	325
105.2 有向最小树形图	326
105.3 无向全局最小割	328
第一百零六章 带权二分图匹配	329
106.1 二分图最大权匹配（缺边与可选匹配）	329
第一百零七章 精确几何进阶	331
107.1 整数最近点对与 Minkowski 和	331
107.2 整数三维共线、共面与方向	334
第一百零八章 凸包支撑点查询	335
108.1 支撑点查询恢复下凸包	335
第一百零九章 动态树	337
109.1 Link-cut tree：动态森林与路径异或	337
第一百一十章 随机平衡树	341
110.1 FHQ Treap：可重集合与顺序统计	341
110.2 隐式 Treap：序列翻转、加值与区间和	344
第一百一十一章 伸展树	348
111.1 Splay：可重集合与顺序统计	348

第一百一十二章 状态序列维护	353
112.1 状态序列 Treap: 分组区间 GCD	353
第一百一十三章 一般图匹配	357
113.1 一般图最大基数匹配: 带花树	357
第一百一十四章 线性递推	360
114.1 Berlekamp–Massey: 最短线性递推	360
114.2 特征多项式求线性递推第 k 项	361
第三部分 数学速查与维护	362
第一百一十五章 数学速查: 条件先于公式	363
115.1 整除、卷积与筛法	363
115.2 模运算的适用条件	363
115.3 任意数组的批量逆元	363
115.4 扩展 CRT: 逐式合并	364
115.5 Garner: 总周期可以超过整数范围	364
115.6 素数模平方根	364
115.7 乘法阶与原根	365
115.8 Carmichael: 最小公共指数	365
115.9 Lucas: 按素数进制逐位计算	365
115.10 合数模数组合数与插值	366
115.11 阶乘表: 组合数与排列数	366
115.12 组合计数	366
115.13 群作用计数	367
115.14 生成函数与递推	367
115.15 线性代数与生成树	367
115.16 模线性方程组: 工厂工期	367
115.17 概率、期望与博弈	367
115.18 数值算法与几何	368
115.19 线性递推: 系数方向与外推条件	368
第一百一十六章 来源与覆盖维护	369
116.1 参考基线	369
116.2 后续覆盖项	369

使用与验证

英文署名已于 QOJ 队伍账号 `ucup-team7502` 核对。两套风格依据训练提交 2932575、2930805、2931279 提取，不擅自将共享队号的提交归属到个人。

代码采用 C++20。以赛时快速抄写为先，一行一条语句，函数体展开。无状态的简单算法使用独立函数，有状态算法使用轻量 `struct`。动态版默认需要 `bits/stdc++.h` 和 `cassert`；传统版各头文件列明依赖。抄写单个函数或结构体时需同时抄写其依赖结构。同名的不同风格实现通常用于不同提交，不建议混在同一编译单元中。点号除特别注明外为 1-based，字符串和多项式为 0-based；所有区间约定应以条目说明为准。

测试源码位于 `tests/`，固定种子便于复现。验证包含独立暴力参考、边界案例、静态与动态实现、AddressSanitizer 和 UndefinedBehaviorSanitizer。`verification/` 保存实际运行记录与源码摘要；OJ 状态只在获得可查提交结果后登记。最大流记录为 <https://www.luogu.com.cn/record/297501480> 与 <https://www.luogu.com.cn/record/297502364>。

原始来源目录共登记 713 项，见 `docs/coverage.csv`。它包含不同版本、例题及公式，不应将其数目视为独立算法数，也不能将“已登记”视为“已完成”。

第一部分

动态容器风格

第一章 Tarjan 与缩点

1.1 Tarjan 强连通分量与缩点

TarjanSCC(n), 点号 1..n; add(u,v) 加有向边, run() 从头计算。bel[u] 为 1..cnt 的强连通分量编号, 跨分量边总从大编号指向小编号, 可按 cnt..1 做拓扑 DP。dag() 返回缩点邻接表, 保留重边; 若 DP 需要对不同邻点去重, 调用者应排序去重。支持自环、重边、不连通图和重复 run。时间及图存储 $O(n+m)$, 递归 DFS 另需 $O(n)$ 调用栈; 本地验证应配置足够栈内存。注意已有 SCC 模板的编号方向与本模板相反。

```
1 struct TarjanSCC
2 {
3     int n, timer = 0, cnt = 0;
4     vector<vector<int>> g;
5     vector<int> dfn, low, bel, st;
6
7     TarjanSCC(int n) : n(n), g(n + 1)
8     {
9     }
10
11     void add(int u, int v)
12     {
13         assert(1 <= u && u <= n && 1 <= v && v <= n);
14         g[u].push_back(v);
15     }
16
17     void dfs(int u)
18     {
19         dfn[u] = low[u] = ++timer;
20         st.push_back(u);
21         for (int v : g[u])
22         {
23             if (!dfn[v])
24             {
25                 dfs(v);
26                 low[u] = min(low[u], low[v]);
27             }
28             else if (!bel[v])
29                 low[u] = min(low[u], dfn[v]);
30         }
31         if (low[u] == dfn[u])
32         {
33             ++cnt;
34             while (true)
35             {
36                 int v = st.back();
37                 st.pop_back();
38                 bel[v] = cnt;
39                 if (v == u)
40                     break;
41             }
42         }
43     }
```

```
44
45 // IDs 1..cnt are in reverse topological order. Recursive DFS needs O(n) stack.
46 void run()
47 {
48     timer = 0;
49     cnt = 0;
50     dfn.assign(n + 1, 0);
51     low = dfn;
52     bel = dfn;
53     st.clear();
54     for (int u = 1; u <= n; u++)
55     {
56         if (!dfn[u])
57             dfs(u);
58     }
59 }
60
61 // Retains parallel edges. Use after run().
62 vector<vector<int>> dag() const
63 {
64     vector<vector<int>> result(cnt + 1);
65     for (int u = 1; u <= n; u++)
66     {
67         for (int v : g[u])
68         {
69             if (bel[u] != bel[v])
70                 result[bel[u]].push_back(bel[v]);
71         }
72     }
73     return result;
74 }
75 };
```

第二章 双连通分量与圆方树

2.1 Tarjan 点双、边双与圆方树

Biconnected(n), 点号 1..n; add(u,v) 返回原边的 0-based 编号, run() 从头计算。blocks 为点双的顶点列表: 桥的两端构成一个块, 孤立点独立成块; 同一个割点可出现在多个块。cut[u] 标记割点, bridge[id] 标记桥; bel[u] 是 1..cnt 的边双编号。block_forest() 返回圆方森林, 原点 1..n, blocks[i] 的方点编号 n+i+1; 孤立点也连接自己的单点方块。bridge_forest() 返回边双缩点森林, 邻接项为 (相邻分量, 原桥编号)。重边按边号处理; 自环保留原边编号但不参与 DFS, 不是桥, 不额外产生点双。全程 $O(n+m)$, 递归 DFS, 分量栈不替代递归调用。两种森林必须在 run 后获取。

```
1 struct Biconnected
2 {
3     int n, timer = 0, cnt = 0;
4     vector<vector<pair<int, int>>> g;
5     vector<pair<int, int>> edges;
6     vector<vector<int>> blocks;
7     vector<int> dfn, low, cut, bridge, bel, st;
8
9     Biconnected(int n) : n(n), g(n + 1)
10    {
11    }
12
13    int add(int u, int v)
14    {
15        assert(1 <= u && u <= n && 1 <= v && v <= n);
16        int id = edges.size();
17        edges.push_back({u, v});
18        if (u != v)
19        {
20            g[u].push_back({v, id});
21            g[v].push_back({u, id});
22        }
23        return id;
24    }
25
26    void dfs(int u, int parent_edge)
27    {
28        dfn[u] = low[u] = ++timer;
29        st.push_back(u);
30        int children = 0;
31        for (auto [v, id] : g[u])
32        {
33            if (id == parent_edge)
34                continue;
35            if (!dfn[v])
36            {
37                children++;
38                dfs(v, id);
39                low[u] = min(low[u], low[v]);
40                if (low[v] > dfn[u])
41                    bridge[id] = 1;
```

```

42         if (low[v] >= dfn[u])
43         {
44             if (parent_edge != -1 || children > 1)
45                 cut[u] = 1;
46             blocks.push_back({u});
47             while (true)
48             {
49                 int x = st.back();
50                 st.pop_back();
51                 blocks.back().push_back(x);
52                 if (x == v)
53                     break;
54             }
55         }
56     }
57     else
58         low[u] = min(low[u], dfn[v]);
59 }
60 if (parent_edge == -1 && children == 0)
61     blocks.push_back({u});
62 }
63
64 void paint(int u)
65 {
66     bel[u] = cnt;
67     for (auto [v, id] : g[u])
68     {
69         if (!bridge[id] && !bel[v])
70             paint(v);
71     }
72 }
73
74 void run()
75 {
76     timer = 0;
77     cnt = 0;
78     dfn.assign(n + 1, 0);
79     low = dfn;
80     cut = dfn;
81     bel = dfn;
82     bridge.assign(edges.size(), 0);
83     blocks.clear();
84     st.clear();
85     for (int u = 1; u <= n; u++)
86     {
87         if (!dfn[u])
88         {
89             dfs(u, -1);
90             st.pop_back();
91         }
92     }
93     for (int u = 1; u <= n; u++)
94     {
95         if (bel[u])
96             continue;
97         ++cnt;

```



```

98         paint(u);
99     }
100 }
101
102 // Original vertices 1..n; block i has node n+i+1. Includes isolated vertices.
103 vector<vector<int>> block_forest() const
104 {
105     vector<vector<int>> tree(n + blocks.size() + 1);
106     for (int i = 0; i < (int)blocks.size(); i++)
107     {
108         int b = n + i + 1;
109         for (int u : blocks[i])
110         {
111             tree[u].push_back(b);
112             tree[b].push_back(u);
113         }
114     }
115     return tree;
116 }
117
118 // Edge-connected component IDs 1..cnt; forest edges carry original edge IDs.
119 vector<vector<pair<int, int>>> bridge_forest() const
120 {
121     vector<vector<pair<int, int>>> tree(cnt + 1);
122     for (int id = 0; id < (int)edges.size(); id++)
123     {
124         if (!bridge[id])
125             continue;
126         auto [u, v] = edges[id];
127         u = bel[u];
128         v = bel[v];
129         tree[u].push_back({v, id});
130         tree[v].push_back({u, id});
131     }
132     return tree;
133 }
134 };

```

第三章 奇环顶点判定

3.1 属于简单奇环的全部顶点

先对 Biconnected 调用 run, 再调用 odd_cycle_vertices(graph), 返回 answer[1..n], 值为 1 表示该点属于某个长度至少为 3 的简单奇环, answer[0]=0。允许重边、断开图及空图; 自环不计入这里的奇环。某点双非二分当且仅当其中存在奇环, 此时该点双每个顶点都属于某个奇环; 答案为所有非二分点双的顶点并集。通过圆方森林把每条非自环边分到唯一点双: 两端中森林深度较大者的父方点即该块, 同深时二者父方点相同。随后只对块内边染色, 避免在每个块中重复扫描割点的全部邻边。两次 DFS 均递归, 分解后额外 $O(n+m)$ 时间和空间。可重排 blocks, 不能改变分解内容。六点以内所有简单图枚举奇环对拍, 另有重边/自环、长链、星形和大奇环测试; POJ2942 补图驱动本地验证, 在线待补。依赖 biconnected.hpp, 传统版函数为 Odd_Cycle_Vertices。

```
1 // After run(). Simple odd cycles have length >= 3; self-loops are ignored.
2 inline vector<int> odd_cycle_vertices(const Biconnected &graph)
3 {
4     int n = graph.n;
5     auto forest = graph.block_forest();
6     vector<int> parent(forest.size(), depth(forest.size()));
7     auto root_tree = [&](auto &&self, int u, int p) -> void
8     {
9         parent[u] = p;
10        for (int v : forest[u])
11            if (v != p)
12            {
13                depth[v] = depth[u] + 1;
14                self(self, v, u);
15            }
16    };
17    for (int u = 1; u <= n; u++)
18        if (!depth[u])
19        {
20            depth[u] = 1;
21            root_tree(root_tree, u, 0);
22        }
```

```

23     vector<vector<pair<int, int>>> edges(graph.blocks.size());
24     for (auto [u, v] : graph.edges)
25         if (u != v)
26         {
27             int block = parent[depth[u] >= depth[v] ? u : v] - n - 1;
28             assert(0 <= block && block < (int)edges.size());
29             edges[block].push_back({u, v});
30         }
31     vector<vector<int>> g(n + 1);
32     vector<int> color(n + 1), answer(n + 1);
33     for (int block = 0; block < (int)edges.size(); block++)
34     {
35         for (int u : graph.blocks[block])
36         {
37             g[u].clear();
38             color[u] = 0;
39         }
40         for (auto [u, v] : edges[block])
41         {
42             g[u].push_back(v);
43             g[v].push_back(u);
44         }
45         bool bipartite = true;
46         auto paint = [&](auto &&self, int u) -> void
47         {
48             for (int v : g[u])
49                 if (!color[v])
50                 {
51                     color[v] = -color[u];
52                     self(self, v);
53                 }
54             else if (color[v] == color[u])
55                 bipartite = false;
56         };
57         int root = graph.blocks[block][0];
58         color[root] = 1;
59         paint(paint, root);
60         if (!bipartite)
61             for (int u : graph.blocks[block])
62                 answer[u] = 1;
63     }
64     return answer;
65 }

```

第四章 删点连通性

4.1 由点双分解计算删点连通块数

在 `Biconnected.run()` 后调用 `removal_components(graph)`, 返回 (原连通块数 `before`, 删点结果 `after`)。 `after[u]` 为独立删除原顶点 `u` 后剩余连通块数, 点号 `1..n`; `after[0]=0` 不参与计算。传统版 `Removal_Components(graph)` 自动推导 `Biconnected_Graph<N>` 的 `N`。 `before` 等于边双数量减去桥数量; `after[u]` 等于 `before-1` 加包含 `u` 的非单点点双数量。单点块必须跳过, 因此孤立点删去后会少一个连通块。空图返回 `(0,{0})`。不修改原分解, $O(n+m)$ 时间、 $O(n)$ 额外空间。若只需要割点、桥和删点增量, 直接用更短的 `Lowlink/Low_Link`; 已经需要点双分解时可用本函数, 避免再次 DFS。

```
1 // Call after run(). Each vertex is deleted independently from the original graph.
2 inline pair<int, vector<int>> removal_components(const Biconnected &graph)
3 {
4     int before = graph.cnt - count(graph.bridge.begin(), graph.bridge.end(), 1);
5     vector<int> after(graph.n + 1, before - 1);
6     after[0] = 0;
7     for (const auto &block : graph.blocks)
8         if (block.size() > 1)
9             for (int u : block)
10                 after[u]++;
11     return {before, after};
12 }
```

第五章 边双缩点与边定向

5.1 边双缩点的大小与内部边数

在 `Biconnected.run()` 后构造 `EdgeCompression(graph)`, 传统版 `Edge_Compression(graph)` 自动推导 `Biconnected_Graph<N>` 的 `N`。 `n` 为边双数量; `size[c]` 为第 `c` 块的原顶点数, `inside[c]` 为块内逻辑无向边数, 分量号 `1..n`, `0` 号槽忽略。 `edges` 每项为 {块 `u`, 块 `v`, 原边 `id`} , 只含跨块的桥, 每条桥出现一次, 按原边号递增; 是无向森林, 不承诺端点为父子方向。 重边逐条计数, 自环在所属块内部计一次, 不能用邻接表度数直接计边而漏掉这些约定。 空图返回 `n=0`、长度 `1` 的零计数数组和空边表。 时间 $O(N+M)$, 输出空间 $O(N+M)$, N/M 指原图规模。

```
1 struct EdgeCompression
2 {
3     int n;
4     vector<int> size, inside;
5     vector<array<int, 3>> edges;
6
7     EdgeCompression(const Biconnected &graph) : n(graph.cnt), size(n + 1), inside(n + 1)
8     {
9         for (int u = 1; u <= graph.n; u++)
10             size[graph.bel[u]]++;
11         for (int id = 0; id < (int)graph.edges.size(); id++)
12         {
13             auto [u, v] = graph.edges[id];
14             u = graph.bel[u];
15             v = graph.bel[v];
16             if (u == v)
17                 inside[u]++;
18             else
19                 edges.push_back({u, v, id});
20         }
21     }
22 };
```

5.2 无向边定向与强连通定向

`orient_edges(graph)` 接受 Lowlink 对象，传统版 `Orient_Edges` 接受 `Low_Link`。返回长度等于逻辑边数的方向数组，第 `id` 项为该边定向后的 {起点, 终点}，顶点 1-based、边 `id` 为 0-based。每条无向边仅处理一次：DFS 树边向下，非树边指向祖先，自环保留 `{u,u}`，重边不合并。支持非连通图；每个边双块内定向后强连通，桥只给出某个方向，不保证跨桥互相可达。若要求整个非空图强连通，需另外 `run/Run` 检查 `components==1` 且无桥；不满足则不存在这样的方向。该函数只读取邻接表与边数，不依赖已有 DFS 状态，不修改原图，调用时无须先 `run`。复杂度 $O(N+M)$ 时间、 $O(N+M)$ 额外空间，DFS 保持递归。

```

1 // Tree edges go down; other edges go to ancestors. Handles each logical edge once.
2 inline vector<pair<int, int>> orient_edges(const Lowlink &graph)
3 {
4     vector<pair<int, int>> direction(graph.edges);
5     vector<int> seen(graph.n + 1);
6     auto dfs = [&](auto &&self, int u) -> void
7     {
8         seen[u] = 1;
9         for (auto [v, id] : graph.g[u])
10         {
11             if (direction[id].first)
12                 continue;
13             direction[id] = {u, v};
14             if (!seen[v])
15                 self(self, v);
16         }
17     };
18     for (int u = 1; u <= graph.n; u++)
19         if (!seen[u])
20             dfs(dfs, u);
21     return direction;
22 }
```

第六章 桥树最少加边

6.1 连通无向图最少加边方案

先对非空连通无向图调用 `Biconnected.run()`，再调用 `bridge_augmentation(graph)`。返回原图点号构成的新增边列表，不修改原图。允许新增重边，不产生自环；不适用于非连通图。已无桥时返回空列表。桥树有 L 个叶子时最少加 $(L+1)/2$ 条边（整数除法）。递归 DFS 顺序收集叶子，奇数时复制首叶，前后半段配对。每条桥一侧的叶子是循环连续区间，半周配对必跨过该割，故所有桥被消除；每个叶子至少需要一个新边端点，得到同样的下界。分解后本函数 $O(n+m)$ 时间与空间。依赖 `biconnected.hpp`。P2860 双风格完整程序本地验证；新增方案以逐边删除连通性独立验证，尚无本库在线 AC。

```
1 // Connected, nonempty graph after run(); new parallel edges are allowed.
2 inline vector<pair<int, int>> bridge_augmentation(const Biconnected &graph)
3 {
4     assert(graph.n > 0 && graph.cnt > 0);
5     int bridges = count(graph.bridge.begin(), graph.bridge.end(), 1);
6     assert(graph.cnt - bridges == 1);
7     if (graph.cnt == 1)
8         return {};
9     auto tree = graph.bridge_forest();
10    vector<int> representative(graph.cnt + 1);
11    for (int u = 1; u <= graph.n; u++)
12        representative[graph.bel[u]] = u;
13    vector<int> leaves;
14    auto dfs = [&](auto &&self, int u, int parent) -> void
15    {
16        if (tree[u].size() == 1)
17            leaves.push_back(u);
18        for (auto [v, id] : tree[u])
19            if (v != parent)
20                self(self, v, u);
21    };
22    dfs(dfs, 1, 0);
23    if (leaves.size() % 2)
24        leaves.push_back(leaves[0]);
25    int half = leaves.size() / 2;
26    vector<pair<int, int>> added;
27    for (int i = 0; i < half; i++)
28        added.push_back({representative[leaves[i]], representative[leaves[i + half]]});
29    return added;
30 }
```

第七章 数据结构

7.1 并查集

`init(n)`、`find(x)`、`merge(x,y)`，点号 $1..n$ ；`siz[find(x)]` 为集合大小。路径压缩与按大小合并，均摊 $O(\alpha(n))$ 。

```
1 struct DSU
2 {
3     vector<int> fa, siz;
4
5     DSU(int n = 0)
6     {
7         init(n);
8     }
9
10    void init(int n)
11    {
12        fa.resize(n + 1);
13        iota(fa.begin(), fa.end(), 0);
14        siz.assign(n + 1, 1);
15    }
16
17    int find(int x)
18    {
19        while (x != fa[x])
20        {
21            fa[x] = fa[fa[x]];
22            x = fa[x];
23        }
24        return x;
25    }
26
27    bool merge(int x, int y)
28    {
29        x = find(x);
30        y = find(y);
31        if (x == y)
32            return false;
33        if (siz[x] < siz[y])
34            swap(x, y);
35        fa[y] = x;
36        siz[x] += siz[y];
37        return true;
38    }
39 };
```


7.2 可撤销并查集

构造后 `snapshot()` 保存历史长度，`rollback(t)` 回退。禁止路径压缩；`find/merge` 为 $O(\log n)$ ，回退每次成功合并 $O(1)$ 。

```

1 struct RollbackDSU
2 {
3     vector<int> fa, siz;
4     vector<pair<int, int>> his;
5
6     RollbackDSU(int n) : fa(n + 1), siz(n + 1, 1)
7     {
8         iota(fa.begin(), fa.end(), 0);
9     }
10
11     int find(int x) const
12     {
13         while (x != fa[x])
14             x = fa[x];
15         return x;
16     }
17
18     int snapshot() const
19     {
20         return (int)his.size();
21     }
22
23     bool merge(int x, int y)
24     {
25         x = find(x);
26         y = find(y);
27         if (x == y)
28             return false;
29         if (siz[x] < siz[y])
30             swap(x, y);
31         his.push_back({y, siz[x]});
32         fa[y] = x;
33         siz[x] += siz[y];
34         return true;
35     }
36
37     void rollback(int t)
38     {
39         assert(0 <= t && t <= snapshot());
40         while (snapshot() > t)
41         {
42             auto [y, s] = his.back();
43             his.pop_back();
44             siz[fa[y]] = s;
45             fa[y] = y;
46         }
47     }
48 };

```

7.3 树状数组与第 k 小

$\text{add}(x,v)$ 、 $\text{sum}(x)$ 、 $\text{query}(l,r)$ ，下标 $1..n$ ，闭区间； $\text{kth}(k)$ 要求所有频率非负且 $k>0$ ，超过总和返回 $n+1$ 。单次 $O(\log n)$ 。

```

1 template <class T = long long> struct Fenwick
2 {
3     int n;
4     vector<T> a;
5
6     Fenwick(int n = 0) : n(n), a(n + 1)
7     {
8     }
9
10    void add(int x, T v)
11    {
12        assert(x >= 1);
13        for (; x <= n; x += x & -x)
14            a[x] += v;
15    }
16
17    T sum(int x) const
18    {
19        T ans = 0;
20        for (; x; x -= x & -x)
21            ans += a[x];
22        return ans;
23    }
24
25    T query(int l, int r) const
26    {
27        return sum(r) - sum(l - 1);
28    }
29
30    // Nonnegative frequencies; returns n+1 if k exceeds total.
31    int kth(T k) const
32    {
33        assert(k > 0);
34        int x = 0, b = 1;
35        while (b <= n / 2)
36            b *= 2;
37        for (; b; b >>= 1)
38            if (x + b <= n && a[x + b] < k)
39            {
40                k -= a[x + b];
41                x += b;
42            }
43        return x + 1;
44    }
45 };

```

7.4 区间加与区间和

初值全零； $\text{add}(l,r,v)$ 、 $\text{query}(l,r)$ ， $1 \leq l \leq r \leq n$ 。维护 sum 与加法 tag ；所有中间和与长度乘积须在 long long 范围内。单次 $O(\log n)$ 。

```

1 struct LazySeg
2 {
3     int n;
4     vector<long long> sum, tag;
5
6     LazySeg(int n) : n(n), sum(4 * n + 4), tag(4 * n + 4)
7     {
8         assert(n > 0);
9     }
10
11     void apply(int p, int l, int r, long long v)
12     {
13         sum[p] += (r - l + 1) * v;
14         tag[p] += v;
15     }
16
17     void push(int p, int l, int r)
18     {
19         int m = (l + r) / 2;
20         apply(p * 2, l, m, tag[p]);
21         apply(p * 2 + 1, m + 1, r, tag[p]);
22         tag[p] = 0;
23     }
24
25     void add(int l, int r, long long v)
26     {
27         add(1, 1, n, l, r, v);
28     }
29
30     void add(int p, int l, int r, int ql, int qr, long long v)
31     {
32         if (ql <= l && r <= qr)
33         {
34             apply(p, l, r, v);
35             return;
36         }
37         push(p, l, r);
38         int m = (l + r) / 2;
39         if (ql <= m)
40             add(p * 2, l, m, ql, qr, v);
41         if (qr > m)
42             add(p * 2 + 1, m + 1, r, ql, qr, v);
43         sum[p] = sum[p * 2] + sum[p * 2 + 1];
44     }
45
46     long long query(int l, int r)
47     {
48         return query(1, 1, n, l, r);
49     }
50
51     long long query(int p, int l, int r, int ql, int qr)
52     {
53         if (ql <= l && r <= qr)
54             return sum[p];
55         push(p, l, r);

```

```
56     int m = (l + r) / 2;
57     long long ans = 0;
58     if (ql <= m)
59         ans += query(p * 2, l, m, ql, qr);
60     if (qr > m)
61         ans += query(p * 2 + 1, m + 1, r, ql, qr);
62     return ans;
63 }
64 };
```

7.5 64 位异或线性基

值类型 U 为 `unsigned long long`，完整支持 64 位。默认空基；`init/Init` 清空。`insert/Insert` 返回是否增秩；`contains/Contains` 判断可表示性（允许空子集，所以 0 总可表示）。`query/Query(x=0)` 返回 x 与所张成空间异或后的最大无符号值。`kth/Kth(k,nonempty=true)` 返回第 k 小的不同异或值，1-based，参数 k 为 `__uint128_t`，非法序号返回 `nullopt`。默认只允许非空输入子集，0 仅在 `dependent` 为真时出现；传 `false` 则包括空子集所得的 0。`minimum/Minimum` 同此约定，空输入的非空最小值不存在。`merge/Merge` 将另一基所代表的输入多重集并入；重复元素和自合并可新增异或为 0 的非空方案。`rank` 是维数，`dependent` 表示存在非空零异或子集。插入、成员判断、最大值为 $O(64)$ ；合并 $O(64^2)$ 。`kth` 会在需要时 `rebuild/Rebuild` 规范化缓存，首次有效查询 $O(64^2)$ ，后续 $O(64)$ ；仅新增独立向量使缓存失效。所有成员仅供只读，不要直接改数组或标志。

```

1 struct XorBasis
2 {
3     using U = unsigned long long;
4     using Wide = __uint128_t;
5     U a[64]{};
6     vector<U> p;
7     int rank = 0;
8     bool dependent = false, dirty = false;
9
10    void init()
11    {
12        fill(a, a + 64, 0);
13        p.clear();
14        rank = 0;
15        dependent = false;
16        dirty = false;
17    }
18
19    bool insert(U x)
20    {
21        for (int i = 63; i >= 0; i--)
22            if (x >> i & 1)
23            {
24                if (!a[i])
25                {
26                    a[i] = x;
27                    rank++;
28                    dirty = true;
29                    return true;
30                }
31                x ^= a[i];
32            }
33        dependent = true;
34        return false;
35    }

```

```
37     bool contains(U x) const
38     {
39         for (int i = 63; i >= 0; i--)
40             if (x >> i & 1)
41                 x ^= a[i];
42         return x == 0;
43     }
44
45     U query(U x = 0) const
46     {
47         for (int i = 63; i >= 0; i--)
48             x = max(x, x ^ a[i]);
49         return x;
50     }
51
52     void rebuild()
53     {
54         vector<U> b(a, a + 64);
55         for (int i = 0; i < 64; i++)
56             if (b[i])
57                 for (int j = i + 1; j < 64; j++)
58                     if (b[j] >> i & 1)
59                         b[j] ^= b[i];
60         p.clear();
61         for (U x : b)
62             if (x)
63                 p.push_back(x);
64         dirty = false;
65     }
```

```

67 // k is 1-based; nonempty excludes the empty input subset, not value zero.
68 optional<U> kth(Wide k, bool nonempty = true)
69 {
70     if (!k)
71         return nullopt;
72     Wide index = k;
73     if (!nonempty || dependent)
74         index--;
75     if (index >= (Wide(1) << rank))
76         return nullopt;
77     if (dirty)
78         rebuild();
79     U answer = 0;
80     for (int i = 0; i < rank; i++)
81         if (index >> i & 1)
82             answer ^= p[i];
83     return answer;
84 }
85
86 optional<U> minimum(bool nonempty = true)
87 {
88     return kth(1, nonempty);
89 }
90
91 void merge(const XorBasis &other)
92 {
93     dependent |= other.dependent;
94     for (U x : other.a)
95         if (x)
96             insert(x);
97 }
98 };

```

第八章 带位置线性基

8.1 带位置限制的异或线性基

默认构造为空，init/Init 可清空。insert/Insert(x,at) 插入 unsigned long long 值及非负 int 标签，支持完整 64 位，标签可重复、乱序。contains/Contains(x,lower=0) 判断 x 能否由标签至少为 lower 的输入异或得到；query/Query(lower=0,x=0) 返回 x 与这些输入子集异或后的最大值；rank/Rank(lower=0) 返回对应空间维数。lower 为非负 long long，可大于所有 int 标签，空集仍可产生 0。全部操作 $O(64)$ ，空间 $O(64)$ 。消元时同一主元优先保留标签较大的向量，较旧向量继续消元；按阈值筛选后仍是正确线性基。标签是组合可用范围的界，不是选点方案；数组仅供只读。区间 [l,r] 应按右端点排序询问，扫描并仅插入到 r，再查询 lower=l；接口本身没有上界，插入超过 r 后不能只用 lower 恢复旧区间。若需保留历史，可复制每个前缀的对象，但空间为 $O(64n)$ 。不提供非空零异或判定或第 k 小；普通线性基的 dependent 不能直接移植到所有阈值。

```
1 struct PositionBasis
2 {
3     using U = unsigned long long;
4     U a[64];
5     int pos[64];
6
7     PositionBasis()
8     {
9         init();
10    }
11
12    void init()
13    {
14        fill(a, a + 64, 0);
15        fill(pos, pos + 64, -1);
16    }
```



```

18 void insert(U x, int at)
19 {
20     assert(at >= 0);
21     for (int i = 63; i >= 0; i--)
22         if (x >> i & 1)
23             {
24                 if (!a[i])
25                     {
26                         a[i] = x;
27                         pos[i] = at;
28                         return;
29                     }
30                 if (at > pos[i])
31                     {
32                         swap(a[i], x);
33                         swap(pos[i], at);
34                     }
35                 x ^= a[i];
36             }
37 }
38
39 bool contains(U x, long long lower = 0) const
40 {
41     assert(lower >= 0);
42     for (int i = 63; i >= 0; i--)
43         if ((x >> i & 1) && pos[i] >= lower)
44             x ^= a[i];
45     return x == 0;
46 }
47
48 U query(long long lower = 0, U x = 0) const
49 {
50     assert(lower >= 0);
51     for (int i = 63; i >= 0; i--)
52         if (pos[i] >= lower)
53             x = max(x, x ^ a[i]);
54     return x;
55 }
56
57 int rank(long long lower = 0) const
58 {
59     assert(lower >= 0);
60     int answer = 0;
61     for (int i = 0; i < 64; i++)
62         answer += pos[i] >= lower;
63     return answer;
64 }
65 };

```

第九章 线性空间求交

9.1 异或线性空间求交

`basis_intersection(a,b)` 返回 `span(a)` 与 `span(b)` 的交空间的一组独立生成元，结果类型 `XorBasis`；传统版 `Basis_Intersection` 接受并返回 `Xor_Basis`。支持完整 64 位无符号值，不修改输入，也不要求先 rebuild。使用第一个空间的主元，消去第二个空间的向量，同时记录来自第二个空间的部分；消元为零时，该部分属于交空间。时间 $O(64^2)$ ，额外空间 $O(64)$ 。输出按独立生成元解释：dependent 为 false，输入的重复次数及非空零异或关系不传播；表示的是子空间，不是两组原始输入元素的集合交。枚举整个交空间应调用 `result.kth(k,false)`，包含零；零空间返回空基，`rank=0`、`contains(0)=true`。

```
1 inline XorBasis basis_intersection(const XorBasis &a, const XorBasis &b)
2 {
3     using U = unsigned long long;
4     U pivot[64], from_b[64]{};
5     copy(a.a, a.a + 64, pivot);
6     XorBasis result;
7     for (U value : b.a)
8     {
9         U x = value, part = value;
10        for (int i = 63; i >= 0 && x; i--)
11            if (x >> i & 1)
12            {
13                if (!pivot[i])
14                {
15                    pivot[i] = x;
16                    from_b[i] = part;
17                    break;
18                }
19                x ^= pivot[i];
20                part ^= from_b[i];
21            }
22        if (!x && part)
23            result.insert(part);
24    }
25    return result;
26 }
```

9.2 Zassenhaus：同时求和空间与交空间

`basis_sum_intersection(a,b)` 返回和空间基、交空间基这一对结果。传统接口名为 `Basis_Sum_Intersection`。和空间由两组基共同张成，一般不是两个空间的集合并；两结果都是独立生成元，其 `dependent` 均为 `false`。将 `a` 的向量 `x` 编成 `(x,x)`，`b` 的向量 `y` 编成 `(y,0)`，用 `__uint128_t` 储存高低各 64 位并按高位到低位消元。高半部非零的行投影构成和空间的基，高半部为零的行取低半部构成交空间的基。必须先提升到 128 位再左移 64；不能用 64 位变量移位 64。时间 $O(128^2)$ ，额外空间 $O(128)$ 。结果满足 $\dim(A+B)+\dim(A \cap B)=\dim(A)+\dim(B)$ ，但维数恒等式只是检查条件，不能单独证明返回向量正确。包含空空间、相同空间和输入别名。

```

1 // Return independent generators of (sum space, intersection space).
2 inline pair<XorBasis, XorBasis> basis_sum_intersection(const XorBasis &a,
3                                                         const XorBasis &b)
4 {
5     using W = __uint128_t;
6     W pivot[128]{};
7     auto insert = [&](W x)
8     {
9         for (int i = 127; i >= 0 && x; i--)
10             if (x >> i & 1)
11             {
12                 if (!pivot[i])
13                 {
14                     pivot[i] = x;
15                     return;
16                 }
17                 x ^= pivot[i];
18             }
19     };
20     for (auto x : a.a)
21         insert((W(x) << 64) | x);
22     for (auto x : b.a)
23         insert(W(x) << 64);
24     XorBasis sum, intersection;
25     for (int i = 0; i < 128; i++)
26         if (pivot[i])
27         {
28             if (i < 64)
29                 intersection.insert((unsigned long long)pivot[i]);
30             else
31                 sum.insert((unsigned long long)(pivot[i] >> 64));
32         }
33     return {sum, intersection};
34 }
```

第十章 线段树常用修改

10.1 线段树：区间乘、加、赋值与模意义求和

`AffineSegTree(a,mod)` 传入非空 0-based 初值数组；查询与修改采用 1-based 非空闭区间。`update(l,r,mul,add)` 将区间元素 x 改为 $mul * x + add$ ；乘法用 $(mul, 0)$ ，加法用 $(1, add)$ ，赋值用 $(0, value)$ 。`query(l,r)` 返回区间和。 $1 \leq mod \leq INT_MAX$ ，模数无需为素数，初值和修改参数允许 long long 全范围，入口统一归一化。旧标记 (m, b) 后应用新标记 (a, c) ，复合为 $(m * a, b * a + c)$ ，乘法取模后再相加。建树 $O(n)$ ，单次 $O(\log n)$ ，空间 $O(n)$ 。内部 `build`、`apply` 等函数不作为公共接口；动态版重建新对象，传统版可重复 `Init`。

```
1 struct AffineSegTree
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         ll sum = 0, mul = 1, add = 0;
8     };
9
10    int n, mod;
11    vector<Node> t;
12
13    AffineSegTree(const vector<ll> &a, int mod) : n(a.size()), mod(mod), t(4 * n + 4)
14    {
15        assert(n > 0 && mod > 0);
16        build(1, 1, n, a);
17    }
18
19    ll norm(ll x) const
20    {
21        x %= mod;
22        if (x < 0)
23            x += mod;
24        return x;
25    }
26
27    void pull(int p)
28    {
29        t[p].sum = (t[p * 2].sum + t[p * 2 + 1].sum) % mod;
30    }
31
32    void build(int p, int l, int r, const vector<ll> &a)
33    {
34        t[p] = Node{};
35        if (l == r)
36        {
37            t[p].sum = norm(a[l - 1]);
38            return;
39        }
40        int m = (l + r) / 2;
41        build(p * 2, l, m, a);
42        build(p * 2 + 1, m + 1, r, a);
```

```

43     pull(p);
44 }
45
46 // New transform follows the old pending transform: x -> x*a+b.
47 void apply(int p, int len, ll a, ll b)
48 {
49     t[p].sum = (t[p].sum * a % mod + len * b % mod) % mod;
50     t[p].mul = t[p].mul * a % mod;
51     t[p].add = (t[p].add * a % mod + b) % mod;
52 }
53
54 void push(int p, int l, int r)
55 {
56     if (t[p].mul == 1 && t[p].add == 0)
57         return;
58     int m = (l + r) / 2;
59     apply(p * 2, m - l + 1, t[p].mul, t[p].add);
60     apply(p * 2 + 1, r - m, t[p].mul, t[p].add);
61     t[p].mul = 1;
62     t[p].add = 0;
63 }
64
65 void update(int l, int r, ll a, ll b)
66 {
67     assert(1 <= l && l <= r && r <= n);
68     update(1, 1, n, l, r, norm(a), norm(b));
69 }
70
71 void update(int p, int l, int r, int ql, int qr, ll a, ll b)
72 {
73     if (ql <= l && r <= qr)
74     {
75         apply(p, r - l + 1, a, b);
76         return;
77     }
78     push(p, l, r);
79     int m = (l + r) / 2;
80     if (ql <= m)
81         update(p * 2, l, m, ql, qr, a, b);
82     if (qr > m)
83         update(p * 2 + 1, m + 1, r, ql, qr, a, b);
84     pull(p);
85 }
86
87 ll query(int l, int r)
88 {
89     assert(1 <= l && l <= r && r <= n);
90     return query(1, 1, n, l, r);
91 }
92
93 ll query(int p, int l, int r, int ql, int qr)
94 {
95     if (ql <= l && r <= qr)
96         return t[p].sum;

```

```
97     push(p, l, r);
98     int m = (l + r) / 2;
99     ll answer = 0;
100    if (ql <= m)
101        answer += query(p * 2, l, m, ql, qr);
102    if (qr > m)
103        answer += query(p * 2 + 1, m + 1, r, ql, qr);
104    return answer % mod;
105 }
106 };
```

第十一章 网络流

11.1 最大流与方案

Dinic(n), add(u,v,c) 返回偶数原边编号; flow(s,t,limit) 返回本次新增流; used(id) 查询原边净流, cut(s) 返回残量可达点集。允许重边、自环, 容量非负, $s \neq t$; 重复求流须使用相同源汇。只有求尽最大流后 cut 才是最小割。一般 $O(V^2E)$, 递归增广深度 $O(V)$ 。总流不得超过 long long 上限。

```
1 struct Dinic
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int from, to;
8         ll cap, initial;
9     };
10
11     int n;
12     vector<Edge> e;
13     vector<vector<int>> g;
14     vector<int> dep, cur;
15
16     Dinic(int n) : n(n), g(n + 1), dep(n + 1), cur(n + 1)
17     {
18     }
19
20     int add(int u, int v, ll c)
21     {
22         assert(c >= 0);
23         int id = (int)e.size();
24         e.push_back({u, v, c, c});
25         e.push_back({v, u, 0, 0});
26         g[u].push_back(id);
27         g[v].push_back(id + 1);
28         return id;
29     }
30
31     bool bfs(int s, int t)
32     {
33         fill(dep.begin(), dep.end(), -1);
34         queue<int> q;
35         dep[s] = 0;
36         q.push(s);
37         while (!q.empty())
38         {
39             int u = q.front();
40             q.pop();
41             for (int id : g[u])
42                 if (e[id].cap && dep[e[id].to] < 0)
43                 {
44                     dep[e[id].to] = dep[u] + 1;
```

```

45         q.push(e[id].to);
46     }
47 }
48 return dep[t] >= 0;
49 }
50
51 ll dfs(int u, int t, ll f)
52 {
53     if (u == t)
54         return f;
55     for (int &i = cur[u]; i < (int)g[u].size(); i++)
56     {
57         int id = g[u][i], v = e[id].to;
58         if (!e[id].cap || dep[v] != dep[u] + 1)
59             continue;
60         ll d = dfs(v, t, min(f, e[id].cap));
61         if (d)
62         {
63             e[id].cap -= d;
64             e[id ^ 1].cap += d;
65             return d;
66         }
67     }
68     return 0;
69 }
70
71 // Returns additional flow; reuse only with the same source/sink.
72 ll flow(int s, int t, ll limit = LLONG_MAX)
73 {
74     assert(s != t && limit >= 0);
75     ll ans = 0;
76     while (ans < limit && bfs(s, t))
77     {
78         fill(cur.begin(), cur.end(), 0);
79         while (ans < limit)
80         {
81             ll d = dfs(s, t, limit - ans);
82             if (!d)
83                 break;
84             ans += d;
85         }
86     }
87     return ans;
88 }
89
90 ll used(int id) const
91 {
92     return e[id].initial - e[id].cap;
93 }
94
95 vector<int> cut(int s)

```



```

96     {
97         bfs(s, s);
98         vector<int> ans;
99         for (int u = 1; u <= n; u++)
100             if (dep[u] >= 0)
101                 ans.push_back(u);
102         return ans;
103     }
104 };

```

11.2 势能最小费用最大流

$\text{add}(u, v, \text{cap}, \text{cost})$ 返回边编号, $\text{flow}(s, t, \text{limit})$ 返回新增流量及 128 位费用, $\text{used}(id)$ 返回原边流量。允许负费用, 初始残量网络必须没有负环; 检测到负环抛异常。Bellman-Ford 初始化 $O(VE)$, 每次增广 $O(E \log V)$; 增广次数不能简单假设为 $O(V)$ 。另要求 $\sum(\text{cap} \times |\text{cost}|) < 2^{120}$, 防止总费用超过 int128 。

```

1 struct MinCostFlow
2 {
3     using ll = long long;
4     using i128 = __int128_t;
5
6     struct Edge
7     {
8         int from, to;
9         ll cap, cost, initial;
10    };
11
12    int n;
13    vector<Edge> e;
14    vector<vector<int>> g;
15
16    MinCostFlow(int n) : n(n), g(n + 1)
17    {
18    }
19
20    int add(int u, int v, ll cap, ll cost)
21    {
22        assert(cap >= 0 && cost != LLONG_MIN);
23        int id = (int)e.size();
24        e.push_back({u, v, cap, cost, cap});
25        e.push_back({v, u, 0, -cost, 0});
26        g[u].push_back(id);
27        g[v].push_back(id + 1);
28        return id;
29    }
30
31    ll used(int id) const
32    {
33        return e[id].initial - e[id].cap;
34    }
35
36    // Negative costs allowed; initial residual graph must have no negative cycle.
37    pair<ll, i128> flow(int s, int t, ll limit = LLONG_MAX)
38    {
39        assert(s != t && limit >= 0);

```

```

40  const i128 inf = i128(1) << 120;
41  vector<i128> h(n + 1, 0), d(n + 1);
42  vector<int> pre(n + 1);
43  // Super-source Bellman-Ford: feasible potentials for every vertex.
44  for (int it = 0; it < n; it++)
45  {
46      bool changed = false;
47      for (auto &a : e)
48          if (a.cap && h[a.to] > h[a.from] + a.cost)
49          {
50              h[a.to] = h[a.from] + a.cost;
51              changed = true;
52          }
53      if (!changed)
54          break;
55      if (it == n - 1)
56          throw invalid_argument("negative cycle");
57  }
58  ll f = 0;
59  i128 cost = 0;
60  while (f < limit)
61  {
62      fill(d.begin(), d.end(), inf);
63      d[s] = 0;
64      priority_queue<pair<i128, int>,
65                    vector<pair<i128, int>>,
66                    greater<pair<i128, int>>>
67          q;
68      q.push({0, s});
69      while (!q.empty())
70      {
71          auto [du, u] = q.top();
72          q.pop();
73          if (du != d[u])
74              continue;
75          for (int id : g[u])
76          {
77              auto &a = e[id];
78              i128 nd = du + a.cost + h[u] - h[a.to];
79              if (a.cap && nd < d[a.to])
80              {
81                  d[a.to] = nd;
82                  pre[a.to] = id;
83                  q.push({nd, a.to});
84              }
85          }
86      }
87      if (d[t] == inf)
88          break;
89      for (int u = 1; u <= n; u++)
90          if (d[u] != inf)
91              h[u] += d[u];
92      ll take = limit - f;
93      for (int u = t; u != s; u = e[pre[u]].from)
94          take = min(take, e[pre[u]].cap);
95      for (int u = t; u != s; u = e[pre[u]].from)

```

```

96         {
97             int id = pre[u];
98             e[id].cap -= take;
99             e[id ^ 1].cap += take;
100             cost += i128(take) * e[id].cost;
101         }
102         f += take;
103     }
104     return {f, cost};
105 }
106 };

```

11.3 上下界可行循环流

$\text{add}(u, v, lo, hi)$, $0 \leq lo \leq hi$; $\text{solve}()$ 仅调用一次, 返回可行性, 仅成功后读取 $\text{used}(id)$ 。约束为每点流量守恒; 不是带源汇最大流。所有上下界及其总和须在 long long 范围内。依赖 Dinic。

```

1 struct BoundedCirculation
2 {
3     using ll = long long;
4     int n;
5     Dinic g;
6     vector<ll> balance, lower;
7     vector<int> ids;
8     bool solved = false, feasible = false;
9
10    BoundedCirculation(int n) : n(n), g(n + 2), balance(n + 1)
11    {
12    }
13
14    int add(int u, int v, ll lo, ll hi)
15    {
16        assert(!solved && 0 <= lo && lo <= hi);
17        int id = ids.size();
18        ids.push_back(g.add(u, v, hi - lo));
19        lower.push_back(lo);
20        balance[u] -= lo;
21        balance[v] += lo;
22        return id;
23    }
24
25    bool solve()
26    {
27        assert(!solved);
28        solved = true;
29        ll need = 0;
30        for (int u = 1; u <= n; u++)
31            if (balance[u] > 0)
32            {
33                g.add(n + 1, u, balance[u]);
34                need += balance[u];
35            }
36        else if (balance[u] < 0)

```

```
37         g.add(u, n + 2, -balance[u]);
38     return feasible = (g.flow(n + 1, n + 2) == need);
39 }
40
41 ll used(int id) const
42 {
43     assert(feasible);
44     return lower[id] + g.used(ids[id]);
45 }
46 };
```

第十二章 最大权闭合子图

12.1 最大权闭合子图与选点方案

`maximum_closure(weight, dependencies)` 返回 (最大权值, 选中点列表), `weight` 为 0-based `vector<long long>`, 图中点号 `1..n`。依赖边 `(u,v)` 表示选 `u` 必须选 `v`, 允许环、自环与重边; 允许空选集, 因此答案不小于 0。依赖 `flow.hpp` 中的 `Dinic`。正权连源、负权连汇, 依赖边容量取正权总和加 1; 要求所有正权之和严格小于 `LLONG_MAX`。负权先提升到 `int128` 再取负, 并截断到依赖边容量, 因此 `LLONG_MIN` 也可处理。返回残量图源点可达的原顶点, 权值为正权总和减最小割; 超大负权边不会出现在最优割中, 截断不改变原问题最优值。每次调用重建网络。复杂度沿用 `Dinic`: 一般 $O(V^2E)$, 空间 $O(V+E)$, 增广保持递归。返回一种最优闭合集, 不承诺最大或最小选点个数。

```
1 inline pair<long long, vector<int>>
2 maximum_closure(const vector<long long> &weight,
3                 const vector<pair<int, int>> &dependencies)
4 {
5     using I = __int128_t;
6     int n = weight.size(), s = n + 1, t = n + 2;
7     I positive = 0;
8     for (long long w : weight)
9         if (w > 0)
10             positive += w;
11     assert(positive < LLONG_MAX);
12     long long inf = (long long)positive + 1;
13     Dinic graph(n + 2);
14     for (int u = 1; u <= n; u++)
15     {
16         long long w = weight[u - 1];
17         if (w > 0)
18             graph.add(s, u, w);
19         else if (w < 0)
20             graph.add(u, t, (long long)min(-I(w), I(inf)));
21     }
22     for (auto [u, v] : dependencies)
23     {
24         assert(1 <= u && u <= n && 1 <= v && v <= n);
25         graph.add(u, v, inf);
26     }
27     long long answer = (long long)positive - graph.flow(s, t);
28     vector<int> selected;
29     for (int u : graph.cut(s))
30         if (u <= n)
31             selected.push_back(u);
32     return {answer, selected};
33 }
```

第十三章 图论

13.1 非负权最短路

`add(u,v,w)`、`run(s)`，`dis` 为距离，`pre` 为前驱，`path(t)` 返回点序列。不可达用 `LLONG_MAX`；要求所有有限最短距离严格小于该哨兵， $w \geq 0$ 。 $O((V+E)\log V)$ 。

```
1 struct Dijkstra
2 {
3     using ll = long long;
4     static constexpr ll inf = LLONG_MAX;
5     int n;
6     vector<vector<pair<int, ll>>> g;
7     vector<ll> dis;
8     vector<int> pre;
9
10    Dijkstra(int n) : n(n), g(n + 1)
11    {
12    }
13
14    void add(int u, int v, ll w)
15    {
16        assert(w >= 0);
17        g[u].push_back({v, w});
18    }
19
20    void run(int s)
21    {
22        dis.assign(n + 1, inf);
23        pre.assign(n + 1, -1);
24        dis[s] = 0;
25        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<pair<ll, int>>> q;
26        q.push({0, s});
27        while (!q.empty())
28        {
29            auto [d, u] = q.top();
30            q.pop();
31            if (d != dis[u])
32                continue;
33            for (auto [v, w] : g[u])
34                if (w < inf - d && d + w < dis[v])
35                {
36                    dis[v] = d + w;
37                    pre[v] = u;
38                    q.push({dis[v], v});
39                }
40        }
41    }
42
43    vector<int> path(int t) const
44    {
```

```

45     if (dis[t] == inf)
46         return {};
47     vector<int> ans;
48     for (; t != -1; t = pre[t])
49         ans.push_back(t);
50     reverse(ans.begin(), ans.end());
51     return ans;
52 }
53 };

```

13.2 强连通分量

`add(u,v)`、`run()`、`bel[1..n]` 为分量编号，`cnt` 为数量；分量编号按缩点图拓扑序递增。递归 Kosaraju，按用户要求保留 DFS 写法； $O(V+E)$ ，同时存正反图。

```

1 struct SCC
2 {
3     int n, cnt = 0;
4     vector<vector<int>> g, rg;
5     vector<int> bel;
6
7     SCC(int n) : n(n), g(n + 1), rg(n + 1)
8     {
9     }
10
11     void add(int u, int v)
12     {
13         g[u].push_back(v);
14         rg[v].push_back(u);
15     }
16
17     void dfs1(int u, vector<int> &vis, vector<int> &order)
18     {
19         vis[u] = 1;
20         for (int v : g[u])
21         {
22             if (!vis[v])
23                 dfs1(v, vis, order);
24         }
25         order.push_back(u);
26     }
27
28     void dfs2(int u)
29     {
30         bel[u] = cnt;
31         for (int v : rg[u])
32         {
33             if (!bel[v])
34                 dfs2(v);
35         }
36     }
37
38     // Recursive Kosaraju; component IDs are in topological order, 1..cnt.
39     void run()
40     {

```

```

41     vector<int> vis(n + 1), order;
42     cnt = 0;
43     bel.assign(n + 1, 0);
44     for (int s = 1; s <= n; s++)
45     {
46         if (!vis[s])
47             dfs1(s, vis, order);
48     }
49     reverse(order.begin(), order.end());
50     for (int s : order)
51     {
52         if (!bel[s])
53         {
54             ++cnt;
55             dfs2(s);
56         }
57     }
58 }
59 };

```

13.3 2-SAT 与任意可行方案

add(x,a,y,b) 添加 (x=a) 或 (y=b), 变量 1..n; solve() 返回可满足性, 仅成功时读取 ans。依赖同文件 SCC; $O(V+E)$, 不保证字典序最小。

```

1 struct TwoSAT
2 {
3     int n;
4     SCC g;
5     vector<int> ans;
6
7     TwoSAT(int n) : n(n), g(2 * n)
8     {
9     }
10
11     int id(int x, bool value) const
12     {
13         return 2 * x - 1 + value;
14     }
15
16     void add(int x, bool a, int y, bool b)
17     {
18         g.add(id(x, !a), id(y, b));
19         g.add(id(y, !b), id(x, a));
20     }
21
22     bool solve()
23     {
24         g.run();
25         ans.assign(n + 1, 0);

```



```

26     for (int x = 1; x <= n; x++)
27     {
28         if (g.bel[id(x, 0)] == g.bel[id(x, 1)])
29             return false;
30         ans[x] = g.bel[id(x, 1)] > g.bel[id(x, 0)];
31     }
32     return true;
33 }
34 };

```

13.4 Hopcroft–Karp 最大匹配与最小点覆盖

左右两侧分别编号 $1..n$ 、 $1..m$ ，可有空侧及重边； $\text{add}(u,v)$ 加边， $\text{solve}()$ 返回最大匹配总数， $l[u]$ 与 $r[v]$ 为匹配端点，0 表示未匹配。保留已有合法匹配，可加边后再次 solve ；不支持删边或手动更改匹配数组。BFS 用 $\text{dep}[0]$ 记录最短终点层，递归 DFS 只沿最短层增广，并使用当前弧。 $V=n+m$ ，单次 solve 最坏 $O((V+E)\sqrt{V})$ ，空间 $O(V+E)$ 。 $\text{cover}()$ 在 solve 后、下一次改图前调用，返回 (左侧点列表, 右侧点列表)，为最小点覆盖；其补集是最大独立集。覆盖与匹配方案均可直接输出。

```

1 struct BipartiteMatching
2 {
3     int n, m;
4     vector<vector<int>> g;
5     vector<int> l, r, dep, cur;
6
7     BipartiteMatching(int n, int m)
8         : n(n), m(m), g(n + 1), l(n + 1), r(m + 1), dep(n + 1), cur(n + 1)
9     {
10    }
11
12    void add(int u, int v)
13    {
14        assert(1 <= u && u <= n && 1 <= v && v <= m);
15        g[u].push_back(v);
16    }
17
18    bool bfs()
19    {
20        queue<int> q;
21        fill(dep.begin(), dep.end(), n + 1);
22        fill(cur.begin(), cur.end(), 0);
23        for (int u = 1; u <= n; u++)
24        {
25            if (!l[u])
26            {
27                dep[u] = 0;
28                q.push(u);
29            }
30        }
31        while (!q.empty())
32        {
33            int u = q.front();
34            q.pop();
35            if (dep[u] >= dep[0])
36                continue;

```

```

37     for (int v : g[u])
38     {
39         int next = r[v];
40         if (dep[next] != n + 1)
41             continue;
42         dep[next] = dep[u] + 1;
43         if (next)
44             q.push(next);
45     }
46 }
47 return dep[0] != n + 1;
48 }
49
50 bool dfs(int u)
51 {
52     if (!u)
53         return true;
54     for (int &i = cur[u]; i < (int)g[u].size(); i++)
55     {
56         int v = g[u][i];
57         if (dep[r[v]] == dep[u] + 1 && dfs(r[v]))
58         {
59             l[u] = v;
60             r[v] = u;
61             return true;
62         }
63     }
64     dep[u] = n + 1;
65     return false;
66 }
67
68 // Retains an existing matching; edges may be added between calls.
69 int solve()
70 {
71     int answer = 0;
72     for (int u = 1; u <= n; u++)
73         answer += (l[u] != 0);
74     while (bfs())
75     {
76         for (int u = 1; u <= n; u++)
77         {
78             if (!l[u] && dfs(u))
79                 answer++;
80         }
81     }
82     return answer;
83 }
84
85 // Call after solve(), before adding more edges. Returns left/right vertex IDs.
86 pair<vector<int>, vector<int>> cover() const
87 {
88     vector<int> left, right;
89     for (int u = 1; u <= n; u++)
90     {
91         if (dep[u] == n + 1)
92             left.push_back(u);

```

```

93     }
94     for (int v = 1; v <= m; v++)
95     {
96         if (r[v] && dep[r[v]] != n + 1)
97             right.push_back(v);
98     }
99     return {left, right};
100 }
101 };

```

13.5 割点与桥

无向多重图，点号 1..n；add/Insert(u,v) 返回逻辑无向边的 0-based 编号，自环和重边均可输入。run/Run 后 cut[u] 为割点标记，bridge[id] 为桥标记，components 为原图连通块数。delta[u] 是独立删除顶点 u 及其关联边后增加的连通块数，剩余块数为 components+delta[u]；孤立点 delta=-1，普通非割点为 0，割点为正。根的增量为 DFS 子树数减 1；非根统计满足 low[child]>=dfn[u] 的子树数。n=1 的点删除后剩余 0 块。DFS 按父边号跳过，支持重边、非连通图和重复运行。复杂度 $O(n+m)$ ，递归 DFS；新增边后需重新 run。每个删点结果都针对原图，不表示逐次删除。

```

1 struct Lowlink
2 {
3     int n, timer = 0, components = 0;
4     vector<vector<pair<int, int>>> g;
5     vector<int> dfn, low, cut, bridge, delta;
6     int edges = 0;
7
8     Lowlink(int n) : n(n), g(n + 1)
9     {
10    }
11
12     int add(int u, int v)
13     {
14         g[u].push_back({v, edges});
15         g[v].push_back({u, edges});
16         return edges++;
17    }

```

```

19 void dfs(int u, int pe)
20 {
21     dfn[u] = low[u] = ++timer;
22     int children = 0;
23     for (auto [v, id] : g[u])
24     {
25         if (id == pe)
26             continue;
27         if (!dfn[v])
28         {
29             ++children;
30             dfs(v, id);
31             low[u] = min(low[u], low[v]);
32             if (low[v] > dfn[u])
33                 bridge[id] = 1;
34             if (pe != -1 && low[v] >= dfn[u])
35             {
36                 cut[u] = 1;
37                 delta[u]++;
38             }
39         }
40         else
41             low[u] = min(low[u], dfn[v]);
42     }
43     if (pe == -1 && children > 1)
44         cut[u] = 1;
45     if (pe == -1)
46         delta[u] = children - 1;
47 }
48
49 void run()
50 {
51     timer = 0;
52     components = 0;
53     dfn.assign(n + 1, 0);
54     low = dfn;
55     cut = dfn;
56     delta = dfn;
57     bridge.assign(edges, 0);
58     for (int u = 1; u <= n; u++)
59         if (!dfn[u])
60         {
61             components++;
62             dfs(u, -1);
63         }
64 }
65 };

```

第十四章 字典序最小 2-SAT

14.1 递归染色与回滚求最小赋值

构造 LexTwoSAT(n), 变量号 $1..n$ 。add(x,a,y,b) 加入子句 $(x=a) \text{ OR } (y=b)$, force(x,v) 强制变量值; solve() 成功后 answer[$1..n$] 是按变量编号、0 优先于 1 的字典序最小赋值。失败时 answer 为空, $n=0$ 时可满足, answer[0] 始终仅为占位。每个未确定变量先尝试 0, 递归传播所有蕴含; 冲突时仅撤销本次新增标记, 再试 1。history 是撤销记录, 不替代递归调用栈。可重复求解, 追加子句会清空旧答案; marked/history 仅为内部工作区, 失败时不可当作方案。内部 $\text{id}(x,v)=2(x-1)+v$, 从 0 开始, 可用异或 1 取反; 不要混用另一份 SCC 2-SAT 的 1-based 字面量编号。最坏 $O(n(n+m))$ 时间、 $O(n+m)$ 空间, 递归栈 $O(n)$; 100000 变量测试是特定蕴含链, 不代表一般大规模输入的性能保证。HDU1814 完整程序本地对拍, 在线待补。

```
1 struct LexTwoSAT
2 {
3     int n;
4     vector<vector<int>> g;
5     vector<int> marked, history, answer;
6
7     LexTwoSAT(int n) : n(n), g(2 * n), marked(2 * n)
8     {
9         assert(n >= 0);
10    }
11
12    int id(int x, bool value) const
13    {
14        assert(1 <= x && x <= n);
15        return 2 * (x - 1) + value;
16    }
17
18    // (x == a) OR (y == b). Internal literal IDs are 0-based.
19    void add(int x, bool a, int y, bool b)
20    {
21        int u = id(x, a), v = id(y, b);
22        g[u ^ 1].push_back(v);
23        g[v ^ 1].push_back(u);
24        answer.clear();
25    }
26
27    void force(int x, bool value)
28    {
29        add(x, value, x, value);
30    }
```

```
32 bool paint(int u)
33 {
34     if (marked[u ^ 1])
35         return false;
36     if (marked[u])
37         return true;
38     marked[u] = 1;
39     history.push_back(u);
40     for (int v : g[u])
41         if (!paint(v))
42             return false;
43     return true;
44 }
45
46 bool solve()
47 {
48     fill(marked.begin(), marked.end(), 0);
49     history.clear();
50     answer.clear();
51     for (int x = 1; x <= n; x++)
52     {
53         int u = id(x, false);
54         if (marked[u] || marked[u ^ 1])
55             continue;
56         int checkpoint = history.size();
57         if (!paint(u))
58         {
59             while ((int)history.size() > checkpoint)
60             {
61                 marked[history.back()] = 0;
62                 history.pop_back();
63             }
64             if (!paint(u ^ 1))
65                 return false;
66         }
67     }
68     answer.assign(n + 1, 0);
69     for (int x = 1; x <= n; x++)
70         answer[x] = marked[id(x, true)];
71     return true;
72 }
73 };
```

第十五章 有向欧拉路

15.1 有向图字典序最小欧拉路

构造 `DirectedEuler(n)`, $n \geq 1$, 点号 $1..n$; `add(u,v)` 返回从 0 开始的逻辑边编号, 支持自环与重边。`run()` 返回是否存在欧拉路; 参数 `start` 默认 0, `lexicographic` 默认 `true`。`start=0` 自动选择最小可行起点, 否则固定起点; 孤立点不影响有边部分的连通性。成功后 `vertices` 长度为 $m+1$, `edge_ids` 长度为 m , 第 i 条原边从 `vertices[i]` 指向 `vertices[i+1]`; 失败清空两数组。无边时返回单点路径, 自动起点为 1。可重复 `run`, `add` 会清空旧方案。默认求顶点序列字典序最小方案; 关闭排序后只保证任意方案, 邻接顺序可能保留上次排序。开启排序总时间 $O(n+m \log(m+1))$, 否则 $O(n+m)$, 空间 $O(n+m)$ 。DFS 递归深度可能达到 $m+1$, 不是 n 。度数检查后, 以消费边数判定是否遍历全部非孤立部分, 逆后序形成连续走法。小图全部走法对拍同时验证存在性、字典序及边编号, P7771 双风格完整程序本地验证, 在线待补。

```
1 struct DirectedEuler
2 {
3     int n;
4     vector<vector<pair<int, int>>> g;
5     vector<pair<int, int>> edges;
6     vector<int> in, out, next, vertices, edge_ids;
7
8     DirectedEuler(int n) : n(n), g(n + 1), in(n + 1), out(n + 1), next(n + 1)
9     {
10         assert(n > 0);
11     }
12
13     int add(int u, int v)
14     {
15         assert(1 <= u && u <= n && 1 <= v && v <= n);
16         int id = edges.size();
17         edges.push_back({u, v});
18         g[u].push_back({v, id});
19         out[u]++;
20         in[v]++;
21         vertices.clear();
22         edge_ids.clear();
23         return id;
24     }
25
26     void dfs(int u, int incoming)
27     {
28         while (next[u] < (int)g[u].size())
29         {
30             auto [v, id] = g[u][next[u]++];
31             dfs(v, id);
32         }
33         vertices.push_back(u);
34         if (incoming != -1)
35             edge_ids.push_back(incoming);
36     }
```

```

38 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
39 bool run(int start = 0, bool lexicographic = true)
40 {
41     assert(0 <= start && start <= n);
42     vertices.clear();
43     edge_ids.clear();
44     int source = 0, sink = 0;
45     for (int u = 1; u <= n; u++)
46     {
47         int delta = out[u] - in[u];
48         if (delta == 1 && !source)
49             source = u;
50         else if (delta == -1 && !sink)
51             sink = u;
52         else if (delta != 0)
53             return false;
54     }
55     if (bool(source) != bool(sink))
56         return false;
57     if (source)
58     {
59         if (start && start != source)
60             return false;
61         start = source;
62     }
63     if (!start)
64     {
65         start = 1;
66         for (int u = 1; u <= n; u++)
67             if (out[u])
68             {
69                 start = u;
70                 break;
71             }
72     }
73     if (lexicographic)
74         for (int u = 1; u <= n; u++)
75             sort(g[u].begin(), g[u].end());
76     fill(next.begin(), next.end(), 0);
77     dfs(start, -1);
78     if (edge_ids.size() != edges.size())
79     {
80         vertices.clear();
81         edge_ids.clear();
82         return false;
83     }
84     reverse(vertices.begin(), vertices.end());
85     reverse(edge_ids.begin(), edge_ids.end());
86     return true;
87 }
88 };

```


第十六章 字典序单词链

16.1 字典序最小单词链

`word_chain(words)` 返回可空的编号序列 (optional), 有解时为原输入的 0-based 单词编号顺序, 每个单词恰用一次; 相邻单词首尾字母相接, 单词序列字典序最小。无解返回 `nullopt`, 空输入返回有值的空序列。单词必须非空且仅含小写 a-z, 重复词视为不同编号。先对原编号按单词排序, 按此顺序加边, 再调用 `DirectedEuler.run(0,false)`; 不可让底层重新按终点排序。欧拉回路从最小出现首字母开始, 非回路起点由度数差确定。O(m log m) 次字符串比较及 O(L+m) 其他工作, L 为总字符数; 空间 O(m), 递归栈 O(m)。依赖 `directed_euler.hpp`; 传统版调用 `Word_Chain`, 内部使用 `Directed_Euler<26>`。全排列对拍、重复词/前缀词、20 万相同词及 POJ2337 完整输出本地验证, 在线待补。

```
1 // Lowercase nonempty words. Returns original 0-based IDs; duplicates stay distinct.
2 inline optional<vector<int>> word_chain(const vector<string> &words)
3 {
4     int n = words.size();
5     vector<int> order(n);
6     iota(order.begin(), order.end(), 0);
7     for (const auto &word : words)
8     {
9         assert(!word.empty());
10        for (char c : word)
11            assert('a' <= c && c <= 'z');
12    }
13    sort(order.begin(),
14         order.end(),
15         [&](int a, int b)
16         {
17             if (words[a] != words[b])
18                 return words[a] < words[b];
19             return a < b;
20         });
21    DirectedEuler graph(26);
22    for (int id : order)
23        graph.add(words[id].front() - 'a' + 1, words[id].back() - 'a' + 1);
24    // Preserve word order in each adjacency list; do not sort by destination.
25    if (!graph.run(0, false))
26        return nullopt;
27    vector<int> result;
28    for (int id : graph.edge_ids)
29        result.push_back(order[id]);
30    return result;
31 }
```

第十七章 无向欧拉路

17.1 无向图欧拉路与边方向

构造 `UndirectedEuler(n)`, $n \geq 1$, 点号 $1..n$; `add(u,v)` 返回从 0 开始的逻辑边号, 自环和重边均保留。`run()` 返回是否存在欧拉路, 参数 `start` 默认 0 (自动), `lexicographic` 默认 `true`。自动选最小可行起点; 指定起点须满足奇度条件且位于有边部分。成功得到 $m+1$ 个 `vertices` 和 m 个 `edge_ids`, 失败清空两数组。第 i 条边方向是 `vertices[i]` 到 `vertices[i+1]`, 与 `edges[edge_ids[i]]` 比较可判断是否反向, 自环任选方向。`used` 按逻辑边号标记, 使两条半边共同只消费一次; 自环贡献度数 2。孤立点不影响有边部分; 无边图返回指定点或点 1 的单点序列。默认求顶点序列字典序最小方案; 关闭排序只保证任意方案, 可能保留上次排序。可重复 `run`, 加边清空旧方案。排序开启时 $O(n+m \log(m+1))$, 否则 $O(n+m)$, 空间 $O(n+m)$, DFS 递归深度可能达到 $m+1$ 。全小图走法、原边号与方向独立验证, P2731 和 SGU101 完整程序本地验证; 在线待补。

```
1 struct UndirectedEuler
2 {
3     int n;
4     vector<vector<pair<int, int>>> g;
5     vector<pair<int, int>> edges;
6     vector<int> next, used, vertices, edge_ids;
7
8     UndirectedEuler(int n) : n(n), g(n + 1), next(n + 1)
9     {
10         assert(n > 0);
11     }
12
13     int add(int u, int v)
14     {
15         assert(1 <= u && u <= n && 1 <= v && v <= n);
16         int id = edges.size();
17         edges.push_back({u, v});
18         g[u].push_back({v, id});
19         g[v].push_back({u, id});
20         vertices.clear();
21         edge_ids.clear();
22         return id;
23     }
24
25     void dfs(int u, int incoming)
26     {
27         while (next[u] < (int)g[u].size())
28         {
29             auto [v, id] = g[u][next[u]++];
30             if (used[id])
31                 continue;
32             used[id] = 1;
33             dfs(v, id);
34         }
35         vertices.push_back(u);
36         if (incoming != -1)
37             edge_ids.push_back(incoming);
38     }
```

```

40 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
41 bool run(int start = 0, bool lexicographic = true)
42 {
43     assert(0 <= start && start <= n);
44     vertices.clear();
45     edge_ids.clear();
46     vector<int> odd;
47     for (int u = 1; u <= n; u++)
48         if (g[u].size() % 2)
49             odd.push_back(u);
50     if (!odd.empty() && odd.size() != 2)
51         return false;
52     if (!odd.empty())
53     {
54         if (start && start != odd[0] && start != odd[1])
55             return false;
56         if (!start)
57             start = odd[0];
58     }
59     if (!start)
60     {
61         start = 1;
62         for (int u = 1; u <= n; u++)
63             if (!g[u].empty())
64             {
65                 start = u;
66                 break;
67             }
68     }
69     if (lexicographic)
70         for (int u = 1; u <= n; u++)
71             sort(g[u].begin(), g[u].end());
72     fill(next.begin(), next.end(), 0);
73     used.assign(edges.size(), 0);
74     dfs(start, -1);
75     if (edge_ids.size() != edges.size())
76     {
77         vertices.clear();
78         edge_ids.clear();
79         return false;
80     }
81     reverse(vertices.begin(), vertices.end());
82     reverse(edge_ids.begin(), edge_ids.end());
83     return true;
84 }
85 };

```

第十八章 混合图欧拉定向

18.1 欧拉回路与指定端点定向

`mixed_euler_orientation(n, edges, start, finish)` 返回可空的定向边序列。每条输入为 $\{u, v, \text{type}\}$, $\text{type}=1$ 固定 u 到 v , $\text{type}=0$ 可反转; 点号 $1..n$, $n \geq 1$ 。省略两个端点或均传 0 时请求任意起点的欧拉回路; 两个正端点请求指定起终点的欧拉路径, 相等时为指定起点回路。无解返回 `nullopt`, 有解返回与原边顺序对应的端点对; 无边图可返回有值空序列。忽略无关孤立点, 但正端点须在有边部分。先任意定向可反转边, 扣除目标出入度差; 每条反转将两端差值分别改变 -2 和 $+2$, 奇数差无解, 否则用容量 1 的反转边流调整。检查底层无向连通性后, 满流即得到所需方向。最大流图 $n+2$ 点、至多 $m+n$ 条正向边, 总时间可按 Dinic 的 $O((n+2)^2(m+n))$ 上界计, 空间 $O(n+m)$ 。依赖 `flow.hpp`; 返回边交给 `DirectedEuler` 可生成连续走法, 指定端点时保持指定起点。定向不保证字典序最小。

```
1 // Edge = {u,v,type}: type 1 is fixed u->v; type 0 may be reversed.
2 // start=finish=0 requests a circuit; positive endpoints request a specified trail.
3 inline optional<vector<pair<int, int>>> mixed_euler_orientation(
4     int n, const vector<array<int, 3>> &edges, int start = 0, int finish = 0)
5 {
6     assert(n > 0 && 0 <= start && start <= n && 0 <= finish && finish <= n);
7     assert(bool(start) == bool(finish));
8     vector<int> parent(n + 1), id;
9     vector<long long> degree(n + 1), balance(n + 1);
10    iota(parent.begin(), parent.end(), 0);
11    auto find = [&](auto &&self, int u) -> int
12    {
13        if (parent[u] != u)
14            parent[u] = self(self, parent[u]);
15        return parent[u];
16    };
17    Dinic flow(n + 2);
18    vector<pair<int, int>> direction;
19    for (auto [u, v, type] : edges)
20    {
21        assert(1 <= u && u <= n && 1 <= v && v <= n);
22        assert(type == 0 || type == 1);
23        parent[find(find, u)] = find(find, v);
24        degree[u]++;
25        degree[v]++;
26        balance[u]++;
27        balance[v]--;
28        id.push_back(type == 0 ? flow.add(u, v, 1) : -1);
29        direction.push_back({u, v});
30    }
```

```

31     if (!edges.empty())
32     {
33         int root = find(find, edges[0][0]);
34         for (int u = 1; u <= n; u++)
35             if (degree[u] && find(find, u) != root)
36                 return nullopt;
37         if (start && (!degree[start] || !degree[finish]))
38             return nullopt;
39     }
40     if (start)
41     {
42         balance[start]--;
43         balance[finish]++;
44     }
45     int source = n + 1, sink = n + 2;
46     long long need = 0;
47     for (int u = 1; u <= n; u++)
48     {
49         if (balance[u] % 2)
50             return nullopt;
51         if (balance[u] > 0)
52         {
53             flow.add(source, u, balance[u] / 2);
54             need += balance[u] / 2;
55         }
56         else if (balance[u] < 0)
57             flow.add(u, sink, -balance[u] / 2);
58     }
59     if (flow.flow(source, sink) != need)
60         return nullopt;
61     for (int i = 0; i < (int)edges.size(); i++)
62         if (id[i] != -1 && flow.used(id[i]))
63             swap(direction[i].first, direction[i].second);
64     return direction;
65 }

```

18.2 任意起终点的混合欧拉路径

`mixed_euler_trail(n, edges)` 返回任意可行欧拉路径的定向边序列，格式同上一节。忽略边原方向统计度数，自环计 2；若奇度点为 0 个，请求回路；若恰有 2 个，尝试这两个端点的两种顺序；否则无解。最多两次定向求解。返回方向可交给 `DirectedEuler` 自动选择起点，恢复顶点与原边编号序列。枚举所有可反转边方向、全部固定端点及自由端点对拍；逐边检查固定边不变和连续走法，另有 10 万点长流量路径测试。POJ1637 双驱动本地验证，仅检查回路存在性；完整方向和开放路径依靠本地证据，在线待补。

```

1 // Any endpoints: zero odd degrees requests a circuit; two odd degrees tries both
2 // orders.
3 inline optional<vector<pair<int, int>>>
4 mixed_euler_trail(int n, const vector<array<int, 3>> &edges)
5 {
6     assert(n > 0);
7     vector<long long> degree(n + 1);
8     vector<int> odd;
9     for (auto [u, v, type] : edges)
10     {
11         assert(1 <= u && u <= n && 1 <= v && v <= n);
12         degree[u]++;
13         degree[v]++;
14     }
15     for (int u = 1; u <= n; u++)
16         if (degree[u] % 2)
17             odd.push_back(u);
18     if (odd.empty())
19         return mixed_euler_orientation(n, edges);
20     if (odd.size() != 2)
21         return nullopt;
22     auto answer = mixed_euler_orientation(n, edges, odd[0], odd[1]);
23     if (answer)
24         return answer;
25     return mixed_euler_orientation(n, edges, odd[1], odd[0]);
26 }
```

第十九章 图上异或行走

19.1 无向图最大异或行走

XorWalk(n) 建图，点号 1..n，边权为 unsigned long long，支持完整 64 位。add(u,v,w) 加无向边，允许重边、自环和零权；init(n) 清空图。build(root=1) 用递归 DFS 预处理 root 的连通块，query(u,v) 返回该块内 u 到 v 的最大边权异或和；任一点未被本次 build 覆盖则返回 nullopt，即使这两点在另一连通块相连。加入新边后必须重新 build；可更换根重复构建。空图可初始化，但不能 build。这里的行走允许重复点、重复边；不是最大异或简单路径。u=v 也可能通过绕环得到正值。DFS 距离 dist 给出一条树路径的异或值，每条已访问边形成 dist[u] xor dist[v] xor w 的闭合行走值，加入线性基；两点答案为 cycles.query(dist[u] xor dist[v])。build 后 vis、dist 可只读；cycles 表示环异或空间，不保留具体环或行走方案，不能将其非空子集标志解释为行走非空性。建图 O(n+m)，build 为 O(n+64m)，每次 query 为 O(64)，动态空间 O(n+m+64)。

```
1 struct XorWalk
2 {
3     using U = unsigned long long;
4     int n;
5     bool built = false;
6     vector<vector<pair<int, U>>> g;
7     vector<int> vis;
8     vector<U> dist;
9     XorBasis cycles;
10
11     XorWalk(int n)
12     {
13         init(n);
14     }
15
16     void init(int size)
17     {
18         assert(size >= 0);
19         n = size;
20         g.clear();
21         g.resize(size_t(n) + 1);
22         vis.assign(size_t(n) + 1, 0);
23         dist.assign(size_t(n) + 1, 0);
24         cycles.init();
25         built = false;
26     }
```

```

28 void add(int u, int v, U w)
29 {
30     assert(1 <= u && u <= n && 1 <= v && v <= n);
31     g[u].push_back({v, w});
32     g[v].push_back({u, w});
33     built = false;
34 }
35
36 void dfs(int u)
37 {
38     vis[u] = 1;
39     for (auto [v, w] : g[u])
40     {
41         if (!vis[v])
42         {
43             dist[v] = dist[u] ^ w;
44             dfs(v);
45         }
46         else
47             cycles.insert(dist[u] ^ dist[v] ^ w);
48     }
49 }
50
51 void build(int root = 1)
52 {
53     assert(1 <= root && root <= n);
54     fill(vis.begin(), vis.end(), 0);
55     fill(dist.begin(), dist.end(), 0);
56     cycles.init();
57     dfs(root);
58     built = true;
59 }
60
61 optional<U> query(int u, int v) const
62 {
63     assert(built && 1 <= u && u <= n && 1 <= v && v <= n);
64     if (!vis[u] || !vis[v])
65         return nullopt;
66     return cycles.query(dist[u] ^ dist[v]);
67 }
68 };

```


第二十章 树上算法

20.1 重链剖分与 LCA

`add(u,v)` 建无向树; `build(root)` 后读取 `fa`、`dep`、`siz`、`son`、`top`、`dfn`、`rk`。子树对应 $[dfn[u], dfn[u] + siz[u] - 1]$ 。
`lca(u,v)` 为 $O(\log n)$, `path(u,v,work,edge)` 分解路径; `work(l,r)` 仅适用于可交换聚合, 边权模式排除 LCA 位置。
输入必须是连通无环树, 建树 $O(n)$, 两遍递归 DFS; 支持改变根后重复 `build`, `dfn` 会重新编号。

```
1 struct HLD
2 {
3     int n, timer = 0;
4     vector<vector<int>> g;
5     vector<int> fa, dep, siz, son, top, dfn, rk;
6
7     HLD(int n)
8         : n(n),
9           g(n + 1),
10          fa(n + 1),
11          dep(n + 1),
12          siz(n + 1),
13          son(n + 1),
14          top(n + 1),
15          dfn(n + 1),
16          rk(n + 1)
17     {
18     }
19
20     void add(int u, int v)
21     {
22         g[u].push_back(v);
23         g[v].push_back(u);
24     }
25
26     void dfs1(int u, int p)
27     {
28         fa[u] = p;
29         siz[u] = 1;
30         son[u] = 0;
31         for (int v : g[u])
32         {
33             if (v == p)
34                 continue;
35             dep[v] = dep[u] + 1;
36             dfs1(v, u);
37             siz[u] += siz[v];
38             if (!son[u] || siz[v] > siz[son[u]])
39                 son[u] = v;
40         }
41     }
42
43     void dfs2(int u, int t)
44     {
```

```

45     top[u] = t;
46     dfn[u] = ++timer;
47     rk[timer] = u;
48     if (son[u])
49         dfs2(son[u], t);
50     for (int v : g[u])
51     {
52         if (v != fa[u] && v != son[u])
53             dfs2(v, v);
54     }
55 }
56
57 // Connected tree only. Recursive DFS; subtree interval is [dfn, dfn+siz-1].
58 void build(int root = 1)
59 {
60     assert(1 <= root && root <= n);
61     timer = 0;
62     dep[root] = 0;
63     dfs1(root, root);
64     dfs2(root, root);
65     assert(timer == n);
66 }
67
68 int lca(int u, int v) const
69 {
70     while (top[u] != top[v])
71     {
72         if (dep[top[u]] < dep[top[v]])
73             swap(u, v);
74         u = fa[top[u]];
75     }
76     return dep[u] < dep[v] ? u : v;
77 }
78
79 // Commutative operations only; edge=true excludes the LCA's position.
80 template <class F> void path(int u, int v, F work, bool edge = false) const
81 {
82     while (top[u] != top[v])
83     {
84         if (dep[top[u]] < dep[top[v]])
85             swap(u, v);
86         work(dfn[top[u]], dfn[u]);
87         u = fa[top[u]];
88     }
89     if (dep[u] > dep[v])
90         swap(u, v);
91     if (dfn[u] + edge <= dfn[v])
92         work(dfn[u] + edge, dfn[v]);
93 }
94 };

```

第二十一章 离线最近公共祖先

21.1 离线 Tarjan LCA

构造 `OfflineLCA(n)`, `add(u,v)` 加无向树边, `add_query(u,v)` 登记查询并返回从 0 开始的编号, `run(root=1)` 后读取 `answer[id]`。输入必须是非空树, 点号 $1..n$; 允许同点、重复查询及零个查询。`run` 可重复调用或换根, 不清空查询; 追加查询后需重新 `run`, 旧答案在重算前不可按新根使用。DFS 与并查集查找均递归。并查集按大小合并并压缩路径, `ancestor[find(u)]` 保存已合并部分对应的当前树祖先; 不能直接拿并查集代表当 LCA。每个子树完成后合并到父点并更新 `ancestor`; 查询另一端已完成时, 从其集合读取祖先。总时间 $O((n+q) \alpha(n))$, 空间 $O(n+q)$ 。全部点对和根的父链对拍、50 万点链及 P3379、POJ1470 完整驱动本地验证, 在线待补。

```
1 struct OfflineLCA
2 {
3     int n;
4     vector<vector<int>>> g;
5     vector<vector<pair<int, int>>>> queries;
6     vector<int> parent, size, ancestor, done, answer;
7
8     OfflineLCA(int n)
9         : n(n),
10          g(n + 1),
11          queries(n + 1),
12          parent(n + 1),
13          size(n + 1),
14          ancestor(n + 1),
15          done(n + 1)
16     {
17     }
18
19     void add(int u, int v)
20     {
21         assert(1 <= u && u <= n && 1 <= v && v <= n);
22         g[u].push_back(v);
23         g[v].push_back(u);
24     }
25
26     int add_query(int u, int v)
27     {
28         assert(1 <= u && u <= n && 1 <= v && v <= n);
29         int id = answer.size();
30         answer.push_back(0);
31         queries[u].push_back({v, id});
32         queries[v].push_back({u, id});
33         return id;
34     }
```

```

36     int find(int u)
37     {
38         if (parent[u] != u)
39             parent[u] = find(parent[u]);
40         return parent[u];
41     }
42
43     void merge(int u, int v)
44     {
45         u = find(u);
46         v = find(v);
47         if (size[u] < size[v])
48             swap(u, v);
49         parent[v] = u;
50         size[u] += size[v];
51     }
52
53     void dfs(int u, int p)
54     {
55         for (int v : g[u])
56             if (v != p)
57             {
58                 dfs(v, u);
59                 merge(u, v);
60                 ancestor[find(u)] = u;
61             }
62         done[u] = 1;
63         for (auto [v, id] : queries[u])
64             if (done[v])
65                 answer[id] = ancestor[find(v)];
66     }
67
68     // Input must be a nonempty tree. Query IDs are 0-based.
69     void run(int root = 1)
70     {
71         assert(1 <= root && root <= n);
72         iota(parent.begin(), parent.end(), 0);
73         iota(ancestor.begin(), ancestor.end(), 0);
74         fill(size.begin(), size.end(), 1);
75         fill(done.begin(), done.end(), 0);
76         fill(answer.begin(), answer.end(), 0);
77         dfs(root, 0);
78     }
79 };

```

第二十二章 在线最近公共祖先

22.1 欧拉序与 ST 表 LCA

构造 EulerLCA(n), add(u,v) 加无向树边, build(root=1) 预处理; lca(u,v) 返回最近公共祖先, distance(u,v) 返回无权路径边数。输入必须为非空树, 点号 1..n; 允许同点查询和重复 build 换根。只可在 build 后查询, 加边会清除 ready 标志。递归 DFS 在进入顶点及从每个儿子返回时记录顶点, euler 长度为 2n-1, first[u] 是从 0 开始的首次出现位置; depth[root]=0。两点首次出现位置间深度最小的顶点就是 LCA, 用 ST 表保存该顶点。预处理 $O(n \log n)$ 时间和空间, 每次查询 $O(1)$ 。ST 内部下标从 0 开始; first、depth、euler 可直接读取。50 万点链、全部小树根与点对以及 P3379、POJ1330 双风格完整程序本地验证; 在线待补。

```
1 struct EulerLCA
2 {
3     int n;
4     bool ready = false;
5     vector<vector<int>> g, st;
6     vector<int> first, depth, euler, lg;
7
8     EulerLCA(int n) : n(n), g(n + 1), first(n + 1), depth(n + 1)
9     {
10
11
12     void add(int u, int v)
13     {
14         assert(1 <= u && u <= n && 1 <= v && v <= n);
15         g[u].push_back(v);
16         g[v].push_back(u);
17         ready = false;
18     }
19
20     int shallower(int u, int v) const
21     {
22         return depth[u] <= depth[v] ? u : v;
23     }
```

```

25 void dfs(int u, int parent)
26 {
27     first[u] = euler.size();
28     euler.push_back(u);
29     for (int v : g[u])
30         if (v != parent)
31             {
32                 depth[v] = depth[u] + 1;
33                 dfs(v, u);
34                 euler.push_back(u);
35             }
36 }
37
38 // Input must be a nonempty tree. Rebuilding may change the root.
39 void build(int root = 1)
40 {
41     assert(1 <= root && root <= n);
42     euler.clear();
43     depth[root] = 0;
44     dfs(root, 0);
45     int m = euler.size();
46     lg.assign(m + 1, 0);
47     for (int i = 2; i <= m; i++)
48         lg[i] = lg[i / 2] + 1;
49     st.assign(lg[m] + 1, {});
50     st[0] = euler;
51     for (int k = 1; k <= lg[m]; k++)
52     {
53         st[k].resize(m - (1 << k) + 1);
54         for (int i = 0; i + (1 << k) <= m; i++)
55             st[k][i] = shallower(st[k - 1][i], st[k - 1][i + (1 << (k - 1))]);
56     }
57     ready = true;
58 }
59
60 int lca(int u, int v) const
61 {
62     assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
63     int l = first[u], r = first[v];
64     if (l > r)
65         swap(l, r);
66     int k = lg[r - l + 1];
67     return shallower(st[k][l], st[k][r - (1 << k) + 1]);
68 }
69
70 int distance(int u, int v) const
71 {
72     int z = lca(u, v);
73     return depth[u] + depth[v] - 2 * depth[z];
74 }
75 };

```

第二十三章 倍增祖先与路径最值

23.1 倍增 LCA、祖先与最大边权

构造 `LiftingLCA(n)`, `add(u,v,w=1)` 加边, `build(root=1)` 预处理非空无向树, 点号 `1..n`。 `jump(u,k)` 返回 `k` 级祖先, `k` 非负, 超过根返回 `0`; `lca(u,v)` 求最近公共祖先; `distance` 返回路径边数, 不是权值和; `max_edge` 返回路径最大边权, 同点返回 `nullopt`, 其他情况返回有值的 `optional<long long>`。支持负权和 `LLONG_MIN`, `LOW` 只是最大值合并的单位元, 不表示空路径。DFS 递归, 根深度为 `0`; `up[k][u]` 与 `mx[k][u]` 是 `2` 的 `k` 次方长度向上段的祖先与最大边权, `0` 号节点为哨兵。可重复 `build` 换根; 修改图后必须重建。预处理 $O(n \log n)$ 时间与空间, 各查询 $O(\log n)$ 。不需要边权最值时, 可删去 `mx` 和 `max_edge` 相关部分。全小树点对/根、极值权重及 50 万点负权链本地验证; P3379、POJ1330 驱动只验证 LCA, 最大边权仍是本地证据。

```
1 struct LiftingLCA
2 {
3     using ll = long long;
4     static constexpr ll LOW = numeric_limits<ll>::lowest();
5     int n, levels;
6     bool ready = false;
7     vector<vector<pair<int, ll>>> g;
8     vector<vector<int>> up;
9     vector<vector<ll>> mx;
10    vector<int> depth;
11
12    LiftingLCA(int n) : n(n), g(n + 1), depth(n + 1)
13    {
14        assert(n > 0);
15        levels = 1;
16        while ((1LL << levels) <= n)
17            levels++;
18        up.assign(levels, vector<int>(n + 1));
19        mx.assign(levels, vector<ll>(n + 1, LOW));
20    }
21
22    void add(int u, int v, ll w = 1)
23    {
24        assert(1 <= u && u <= n && 1 <= v && v <= n);
25        g[u].push_back({v, w});
26        g[v].push_back({u, w});
27        ready = false;
28    }
```

```

30 void dfs(int u, int parent, ll weight)
31 {
32     up[0][u] = parent;
33     mx[0][u] = weight;
34     for (int k = 1; k < levels; k++)
35     {
36         int p = up[k - 1][u];
37         up[k][u] = up[k - 1][p];
38         mx[k][u] = max(mx[k - 1][u], mx[k - 1][p]);
39     }
40     for (auto [v, w] : g[u])
41         if (v != parent)
42         {
43             depth[v] = depth[u] + 1;
44             dfs(v, u, w);
45         }
46     }
47
48     // Nonempty tree only. Node 0 means no ancestor.
49     void build(int root = 1)
50     {
51         assert(1 <= root && root <= n);
52         depth[root] = 0;
53         dfs(root, 0, LOW);
54         ready = true;
55     }
56
57     int jump(int u, long long steps) const
58     {
59         assert(ready && 1 <= u && u <= n && steps >= 0);
60         if (steps > depth[u])
61             return 0;
62         for (int k = 0; k < levels; k++)
63             if (steps >> k & 1)
64                 u = up[k][u];
65         return u;
66     }

```



```

68  int lca(int u, int v) const
69  {
70      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
71      if (depth[u] < depth[v])
72          swap(u, v);
73      u = jump(u, depth[u] - depth[v]);
74      if (u == v)
75          return u;
76      for (int k = levels - 1; k >= 0; k--)
77          if (up[k][u] != up[k][v])
78              {
79                  u = up[k][u];
80                  v = up[k][v];
81              }
82      return up[0][u];
83  }
84
85  int distance(int u, int v) const
86  {
87      return depth[u] + depth[v] - 2 * depth[lca(u, v)];
88  }
89
90  optional<ll> max_edge(int u, int v) const
91  {
92      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
93      if (u == v)
94          return nullopt;
95      if (depth[u] < depth[v])
96          swap(u, v);
97      ll answer = LOW;
98      int delta = depth[u] - depth[v];
99      for (int k = 0; k < levels; k++)
100          if (delta >> k & 1)
101              {
102                  answer = max(answer, mx[k][u]);
103                  u = up[k][u];
104              }
105      if (u == v)
106          return answer;
107      for (int k = levels - 1; k >= 0; k--)
108          if (up[k][u] != up[k][v])
109              {
110                  answer = max(answer, max(mx[k][u], mx[k][v]));
111                  u = up[k][u];
112                  v = up[k][v];
113              }
114      return max(answer, max(mx[0][u], mx[0][v]));
115  }
116  };

```

第二十四章 树上路径交

24.1 两条树路径的交集

`path_intersection(a,b,c,d,depth,lca)` 返回路径 `a-b` 与 `c-d` 的公共路径，字段 `u`、`v` 是交集端点，`vertices` 是公共顶点数；空交返回 `{0,0,0}`，单点交的 `u=v`。点号从 1 开始，`depth` 数组与 `lca` 回调必须来自同一棵非空树、同一个根，深度按边数计算，不能传带权根距离。支持数组或 `vector` 深度表，回调接收两个点并返回 LCA。将 `c`、`d` 分别投影到 `a-b` 得 `p`、`q`；投影是 `lca(a,b)`、`lca(a,x)`、`lca(b,x)` 中最深者。若 `p≠q`，`c-d` 必经过 `p-q`；若 `p=q`，检验 `p` 是否在 `c-d` 上即可区分单点交与空交。不依赖顶点编号大小。常数 LCA 调用、 $O(1)$ 额外空间；使用 EulerLCA 时为 $O(1)$ ，HLD/倍增时为 $O(\log n)$ 。公共边数为 $\max(0, \text{vertices}-1)$ 。全小树根与四端点、随机换号树用路径点集独立验证；P3398 完整程序只检查是否相交，端点及数量由本地验证覆盖。

```
1 struct PathIntersection
2 {
3     int u, v, vertices;
4 };
5
6 // Same nonempty tree and root for depth[] and lca(). Empty result is {0,0,0}.
7 template <class Depth, class Lca>
8 PathIntersection
9 path_intersection(int a, int b, int c, int d, const Depth &depth, Lca lca)
10 {
11     int z = lca(a, b);
12     auto project = [&](int x)
13     {
14         int p = z;
15         for (int v : {lca(a, x), lca(b, x)})
16             if (depth[v] > depth[p])
17                 p = v;
18         return p;
19     };
20     auto distance = [&](int u, int v)
21     {
22         return 1LL * depth[u] + depth[v] - 2LL * depth[lca(u, v)];
23     };
24     int p = project(c), q = project(d);
25     if (p == q && distance(c, p) + distance(p, d) != distance(c, d))
26         return {0, 0, 0};
27     return {p, q, int(distance(p, q) + 1)};
28 }
```

第二十五章 点分治

25.1 点分治：树上距离点对计数

CentroidPairs(n), 点号 $1..n$; add(u,v,w) 建树, 允许森林但禁止环、重边、自环, 边权非负; 所有简单路径长度须能用 long long 表示。build() 预处理, 可重复调用; 改变边后须重建。count_leq(k) 统计距离 $\leq k$ 的无序不同顶点对, count_exact(k) 统计距离恰为 k 的点对; 负 k 返回 0, 零距离只统计不同顶点, 跨连通块不配对。parent[c] 是点分树父节点, 根为 0。all[c] 为该层分治块到重心的距离, 含重心距离 0; part[c] 为去掉重心后各子块的距离。预处理 $O(n \log^2 n)$, 空间 $O(n \log n)$, 每次计数 $O(n \log n)$; 固定树上多次查询可以复用排序结果。DFS 递归, 距离和比较采用减法避免溢出。不是动态点分树更新模板。

```
1 struct CentroidPairs
2 {
3     using ll = long long;
4     int n;
5     vector<vector<pair<int, ll>>> g;
6     vector<int> siz, removed, parent;
7     vector<vector<ll>> all;
8     vector<vector<vector<ll>>> part;
9
10    CentroidPairs(int n)
11        : n(n),
12          g(n + 1),
13          siz(n + 1),
14          removed(n + 1),
15          parent(n + 1),
16          all(n + 1),
17          part(n + 1)
18    {
19    }
20
21    void add(int u, int v, ll w)
22    {
23        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
24        g[u].push_back({v, w});
25        g[v].push_back({u, w});
26    }
27
28    void size_dfs(int u, int p)
29    {
30        siz[u] = 1;
31        for (auto [v, w] : g[u])
32        {
33            if (v == p || removed[v])
34                continue;
35            size_dfs(v, u);
36            siz[u] += siz[v];
37        }
38    }
39
40    int centroid(int u, int p, int total)
41    {
```

```

42     for (auto [v, w] : g[u])
43     {
44         if (v != p && !removed[v] && siz[v] > total / 2)
45             return centroid(v, u, total);
46     }
47     return u;
48 }
49
50 void collect(int u, int p, ll d, vector<ll> &out)
51 {
52     out.push_back(d);
53     for (auto [v, w] : g[u])
54     {
55         if (v == p || removed[v])
56             continue;
57         assert(d <= numeric_limits<ll>::max() - w);
58         collect(v, u, d + w, out);
59     }
60 }
61
62 void decompose(int entry, int p)
63 {
64     size_dfs(entry, 0);
65     int c = centroid(entry, 0, siz[entry]);
66     removed[c] = 1;
67     parent[c] = p;
68     all[c] = {0};
69     for (auto [v, w] : g[c])
70     {
71         if (removed[v])
72             continue;
73         part[c].push_back({});
74         auto &d = part[c].back();
75         collect(v, c, w, d);
76         sort(d.begin(), d.end());
77         all[c].insert(all[c].end(), d.begin(), d.end());
78     }
79     sort(all[c].begin(), all[c].end());
80     for (auto [v, w] : g[c])
81     {
82         if (!removed[v])
83             decompose(v, c);
84     }
85 }
86
87 // Input must be a forest. Rebuild after changing edges.
88 void build()
89 {
90     fill(removed.begin(), removed.end(), 0);
91     for (int u = 1; u <= n; u++)
92     {
93         all[u].clear();
94         part[u].clear();
95     }
96     for (int u = 1; u <= n; u++)
97     {

```

```

98         if (!removed[u])
99             decompose(u, 0);
100     }
101 }
102
103 ll pairs(const vector<ll> &d, ll k) const
104 {
105     int l = 0, r = (int)d.size() - 1;
106     ll answer = 0;
107     while (l < r)
108     {
109         if (d[l] <= k - d[r])
110         {
111             answer += r - l;
112             l++;
113         }
114         else
115             r--;
116     }
117     return answer;
118 }
119
120 // Unordered pairs of distinct connected vertices. Nonnegative edge weights.
121 ll count_leq(ll k) const
122 {
123     if (k < 0)
124         return 0;
125     ll answer = 0;
126     for (int c = 1; c <= n; c++)
127     {
128         answer += pairs(all[c], k);
129         for (const auto &d : part[c])
130             answer -= pairs(d, k);
131     }
132     return answer;
133 }
134
135 ll count_exact(ll k) const
136 {
137     if (k < 0)
138         return 0;
139     return count_leq(k) - (k ? count_leq(k - 1) : 0);
140 }
141 };

```

第二十六章 树上启发式合并

26.1 树上启发式合并：子树众数颜色和

SubtreeColors(n), 点号 1..n, add(u,v) 建无向树; 要求图连通且无环, n=0 可空运行。run(colors,root=1) 传入长度 n+1 的 int 颜色数组, colors[0] 忽略。answer[u] 为子树内所有最高频颜色的原始编号之和, max_freq[u] 为该频次; 支持负颜色与稀疏编号, 内部离散化, 和用 long long。重复 run 可更换颜色或根。重儿子信息保留, 轻儿子通过 DFS 序区间加入; 清空整棵临时子树后再重置最大值, 不尝试逐次删除时维护众数。O(n log n) 时间, O(n) 空间, 全部 DFS 递归。insert 是内部增量统计, 可按题目修改统计量, 但相应清空规则也必须调整。

```
1 struct SubtreeColors
2 {
3     int n, best = 0;
4     long long sum = 0;
5     vector<vector<int>> g;
6     vector<int> siz, heavy, tin, tout, order, color, value, freq;
7     vector<int> max_freq;
8     vector<long long> answer;
9
10    SubtreeColors(int n) : n(n), g(n + 1)
11    {
12    }
13
14    void add(int u, int v)
15    {
16        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v);
17        g[u].push_back(v);
18        g[v].push_back(u);
19    }
20
21    void prepare(int u, int p)
22    {
23        siz[u] = 1;
24        tin[u] = order.size();
25        order.push_back(u);
26        for (int v : g[u])
27        {
28            if (v == p)
29                continue;
30            prepare(v, u);
31            siz[u] += siz[v];
32            if (siz[v] > siz[heavy[u]])
33                heavy[u] = v;
34        }
35        tout[u] = (int)order.size() - 1;
36    }
37
38    void insert(int u)
39    {
40        int c = color[u];
41        freq[c]++;
42        if (freq[c] > best)
```

```

43     {
44         best = freq[c];
45         sum = value[c];
46     }
47     else if (freq[c] == best)
48         sum += value[c];
49 }
50
51 void solve(int u, int p, bool keep)
52 {
53     for (int v : g[u])
54     {
55         if (v != p && v != heavy[u])
56             solve(v, u, false);
57     }
58     if (heavy[u])
59         solve(heavy[u], u, true);
60     for (int v : g[u])
61     {
62         if (v == p || v == heavy[u])
63             continue;
64         for (int i = tin[v]; i <= tout[v]; i++)
65             insert(order[i]);
66     }
67     insert(u);
68     max_freq[u] = best;
69     answer[u] = sum;
70     if (!keep)
71     {
72         for (int i = tin[u]; i <= tout[u]; i++)
73             freq[color[order[i]]]--;
74         best = 0;
75         sum = 0;
76     }
77 }
78
79 // A connected tree, colors[1..n]. Sum all colors tied for maximum frequency.
80 void run(const vector<int> &colors, int root = 1)
81 {
82     assert((int)colors.size() == n + 1);
83     siz.assign(n + 1, 0);
84     heavy = siz;
85     tin = siz;
86     tout = siz;
87     color = siz;
88     max_freq = siz;
89     answer.assign(n + 1, 0);
90     order.clear();
91     value.assign(colors.begin() + 1, colors.end());
92     sort(value.begin(), value.end());
93     value.erase(unique(value.begin(), value.end()), value.end());
94     freq.assign(value.size(), 0);
95     best = 0;
96     sum = 0;

```

```
97     if (!n)
98         return;
99     assert(1 <= root && root <= n);
100    for (int u = 1; u <= n; u++)
101        color[u] =
102            lower_bound(value.begin(), value.end(), colors[u]) - value.begin();
103    prepare(root, 0);
104    solve(root, 0, false);
105    }
106};
```


第二十七章 虚树

27.1 虚树：关键点压缩与路径信息

`VirtualTree(n)`, $n \geq 1$, 点号 $1..n$; `add(u,v,w=1)` 建非负权连通树, 无环和重边。 `prepare(root=1)` 递归预处理, 可换根重建; 根到任一点距离须在 `long long` 范围内。 `build(keys)` 自动去重并补齐必要 LCA, 返回 `vertices` 和 `edges`; 全部编号沿用原树, `vertices` 按 DFS 序排列, 非空时首项即虚树根。 `edges` 为 `(from,to,distance,minimum)`, 分别是祖先到后代的虚边、压缩路径长度和路径最小边权。空关键点返回空结果, 单点返回单节点; 原树根不会强制加入, 需要时由调用者加入 `keys`。边按后代 DFS 序排列, 可逆序做自底向上的 DP。不分配或清空 n 个查询邻接表; 每次仅返回 $O(k)$ 结果。预处理 $O(n \log n)$ 时间空间, 单次 $O(k \log k + k \log n)$ 。 `lca`、`ancestor` 也可直接使用, 要求有效点号且已经 `prepare`。构造用的祖先链栈属于虚树算法, 原树 DFS 保持递归。

```
1 struct VirtualTree
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int from, to;
8         ll distance, minimum;
9     };
10
11     struct Result
12     {
13         vector<int> vertices;
14         vector<Edge> edges;
15     };
16
17     int n, log = 1, timer = 0;
18     vector<vector<pair<int, ll>>> g;
19     vector<vector<int>> up;
20     vector<vector<ll>> mn;
21     vector<int> depth, tin, tout;
22     vector<ll> dist;
23
24     VirtualTree(int n) : n(n), g(n + 1)
25     {
26     }
27
28     void add(int u, int v, ll w = 1)
29     {
30         assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
31         g[u].push_back({v, w});
32         g[v].push_back({u, w});
33     }
34
35     void dfs(int u, int p)
36     {
37         tin[u] = ++timer;
38         up[0][u] = p;
39         for (int j = 1; j < log; j++)
```

```

40     {
41         up[j][u] = up[j - 1][up[j - 1][u]];
42         mn[j][u] = min(mn[j - 1][u], mn[j - 1][up[j - 1][u]]);
43     }
44     for (auto [v, w] : g[u])
45     {
46         if (v == p)
47             continue;
48         assert(dist[u] <= numeric_limits<ll>::max() - w);
49         depth[v] = depth[u] + 1;
50         dist[v] = dist[u] + w;
51         mn[0][v] = w;
52         dfs(v, u);
53     }
54     tout[u] = timer;
55 }
56
57 void prepare(int root = 1)
58 {
59     assert(1 <= root && root <= n);
60     log = 1;
61     while ((1LL << log) <= n)
62         log++;
63     up.assign(log, vector<int>(n + 1));
64     mn.assign(log, vector<ll>(n + 1, numeric_limits<ll>::max()));
65     depth.assign(n + 1, 0);
66     tin = depth;
67     tout = depth;
68     dist.assign(n + 1, 0);
69     timer = 0;
70     dfs(root, root);
71 }
72
73 bool ancestor(int u, int v) const
74 {
75     return tin[u] <= tin[v] && tout[v] <= tout[u];
76 }
77
78 int lca(int u, int v) const
79 {
80     if (ancestor(u, v))
81         return u;
82     if (ancestor(v, u))
83         return v;
84     for (int j = log - 1; j >= 0; j--)
85     {
86         if (!ancestor(up[j][u], v))
87             u = up[j][u];
88     }
89     return up[0][u];
90 }
91
92 Edge path(int u, int v) const
93 {
94     assert(u != v && ancestor(u, v));
95     ll minimum = numeric_limits<ll>::max();

```

```

96     int x = v, steps = depth[v] - depth[u];
97     for (int j = 0; j < log; j++)
98     {
99         if (steps >> j & 1)
100         {
101             minimum = min(minimum, mn[j][x]);
102             x = up[j][x];
103         }
104     }
105     return {u, v, dist[v] - dist[u], minimum};
106 }
107
108 // Uses original vertex IDs. Empty input gives an empty result.
109 Result build(vector<int> keys) const
110 {
111     if (keys.empty())
112         return {};
113     for (int u : keys)
114         assert(1 <= u && u <= n);
115     auto cmp = [&](int u, int v)
116     {
117         return tin[u] < tin[v];
118     };
119     sort(keys.begin(), keys.end(), cmp);
120     keys.erase(unique(keys.begin(), keys.end()), keys.end());
121     int k = keys.size();
122     for (int i = 1; i < k; i++)
123         keys.push_back(lca(keys[i - 1], keys[i]));
124     sort(keys.begin(), keys.end(), cmp);
125     keys.erase(unique(keys.begin(), keys.end()), keys.end());
126     Result result{keys, {}};
127     vector<int> st;
128     for (int u : keys)
129     {
130         while (!st.empty() && !ancestor(st.back(), u))
131             st.pop_back();
132         if (!st.empty())
133             result.edges.push_back(path(st.back(), u));
134         st.push_back(u);
135     }
136     return result;
137 }
138 };

```

第二十八章 字符串

28.1 前缀函数

`prefix_function(s)` 返回每个前缀的最长真 border 长度，0-based；空串返回空数组。O(n)，无状态，直接抄写函数。

```
1 inline vector<int> prefix_function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> p(n);
5     for (int i = 1; i < n; i++)
6     {
7         int j = p[i - 1];
8         while (j && s[i] != s[j])
9             j = p[j - 1];
10        p[i] = j + (s[i] == s[j]);
11    }
12    return p;
13 }
```

28.2 KMP 匹配

`kmp_match(s,t)` 返回非空模式串 `t` 在 `s` 中的所有起点 (0-based)，允许重叠。需要同章前缀函数。O(|s|+|t|)。

```
1 inline vector<int> kmp_match(const string &s, const string &t)
2 {
3     assert(!t.empty());
4     auto p = prefix_function(t);
5     vector<int> ans;
6     for (int i = 0, j = 0; i < (int)s.size(); i++)
7     {
8         while (j && s[i] != t[j])
9             j = p[j - 1];
10        j += (s[i] == t[j]);
11        if (j == (int)t.size())
12        {
13            ans.push_back(i - j + 1);
14            j = p[j - 1];
15        }
16    }
17    return ans;
18 }
```

28.3 Z 函数

`z_function(s)` 返回后缀 `s[i..]` 与原串的 LCP，约定 `z[0]=n`，空串返回空数组。O(n)。

```

1 inline vector<int> z_function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> a(n);
5     if (n)
6         a[0] = n;
7     for (int i = 1, l = 0, r = 0; i < n; i++)
8     {
9         if (i < r)
10            a[i] = min(r - i, a[i - l]);
11        while (i + a[i] < n && s[a[i]] == s[i + a[i]])
12            ++a[i];
13        if (i + a[i] > r)
14        {
15            l = i;
16            r = i + a[i];
17        }
18    }
19    return a;
20 }
```

28.4 Manacher 回文半径

`manacher(s)` 返回 `odd` 与 `even`；`odd[i]` 的半径包含中心，`even[i]` 的中心在 `i` 前。字符串及下标为 0-based，空串合法，O(n)。

```

1 // odd[i] includes the center; even[i] is centered before i.
2 inline pair<vector<int>, vector<int>> manacher(const string &s)
3 {
4     int n = (int)s.size();
5     vector<int> odd(n), even(n);
6     for (int i = 0, l = 0, r = -1; i < n; i++)
7     {
8         int k = i > r ? 1 : min(odd[l + r - i], r - i + 1);
9         while (i - k >= 0 && i + k < n && s[i - k] == s[i + k])
10            ++k;
11        odd[i] = k--;
12        if (i + k > r)
13        {
14            l = i - k;
15            r = i + k;
16        }
17    }
18    for (int i = 0, l = 0, r = -1; i < n; i++)
19    {
20        int k = i > r ? 0 : min(even[l + r - i + 1], r - i + 1);
21        while (i - k - 1 >= 0 && i + k < n && s[i - k - 1] == s[i + k])
22            ++k;

```

```

23     even[i] = k--;
24     if (i + k > r)
25     {
26         l = i - k - 1;
27         r = i + k;
28     }
29 }
30 return {odd, even};
31 }

```

28.5 最小循环表示

`minimum_rotation(s)` 返回字典序最小循环表示的起点 (0-based), 空串返回 0; 按无符号字节比较, 与 `string` 字典序一致。O(n)。

```

1 inline int minimum_rotation(const string &s)
2 {
3     int n = (int)s.size(), i = 0, j = 1, k = 0;
4     if (!n)
5         return 0;
6     while (i < n && j < n && k < n)
7     {
8         unsigned char a = s[(i + k) % n], b = s[(j + k) % n];
9         if (a == b)
10            {
11                ++k;
12                continue;
13            }
14         if (a > b)
15            {
16                i += k + 1;
17                if (i == j)
18                    ++i;
19            }
20         else
21            {
22                j += k + 1;
23                if (i == j)
24                    ++j;
25            }
26         k = 0;
27     }
28     return min(i, j);
29 }

```

28.6 AC 自动机出现次数

字母表 `a..z`, `add(s)` 返回终止状态, 要求模式非空; 全部插入后 `build()`, 只允许一次构建。`count(text)` 返回每个状态对应模式的出现次数, 可重复查询; 重复模式共享状态。构建 $O(26N)$, 查询 $O(|text|+N)$ 。

```

1 struct AhoCorasick
2 {
3     struct Node
4     {

```

```

5      array<int, 26> go{};
6      int fail = 0;
7  };
8
9  vector<Node> a{Node{}};
10 vector<int> order;
11 bool built = false;
12
13 int add(const string &s)
14 {
15     assert(!built && !s.empty());
16     int u = 0;
17     for (char ch : s)
18     {
19         int c = ch - 'a';
20         assert(0 <= c && c < 26);
21         if (!a[u].go[c])
22         {
23             int v = (int)a.size();
24             a[u].go[c] = v;
25             a.push_back(Node{});
26         }
27         u = a[u].go[c];
28     }
29     return u;
30 }
31
32 void build()
33 {
34     assert(!built);
35     built = true;
36     queue<int> q;
37     for (int v : a[0].go)
38         if (v)
39             q.push(v);
40     while (!q.empty())
41     {
42         int u = q.front();
43         q.pop();
44         order.push_back(u);
45         for (int c = 0; c < 26; c++)
46         {
47             int v = a[u].go[c];
48             if (v)
49             {
50                 a[v].fail = a[a[u].fail].go[c];
51                 q.push(v);
52             }
53             else
54                 a[u].go[c] = a[a[u].fail].go[c];
55         }
56     }
57 }
58
59 vector<long long> count(const string &s) const
60 {

```

```
61     assert(built);
62     vector<long long> ans(a.size());
63     int u = 0;
64     for (char ch : s)
65     {
66         u = a[u].go[ch - 'a'];
67         ++ans[u];
68     }
69     for (int i = (int)order.size() - 1; i >= 0; i--)
70         ans[a[order[i]].fail] += ans[order[i]];
71     return ans;
72 }
73 };
```


28.7 倍增后缀数组与 LCP

构造 `SuffixArray(s)` 支持任意字节串，按 `unsigned char` 的 0..255 数值比较，包含内嵌零字节；也可构造 `SuffixArray(a,alphabet)`，整数序列要求 $0 \leq a[i] < \text{alphabet}$ 、 $1 \leq \text{alphabet} < \text{INT_MAX}$ 。`init(a,alphabet)` 可重复重建，传统版为 `Init`。长度须小于 `INT_MAX`。`sa[k]` 为第 k 个非空后缀起点，`rk[i]` 为排名，全部 0-based；`lcp[k]` 为 `sa[k-1]` 与 `sa[k]` 的最长公共前缀，`lcp[0]=0`。空序列合法，三数组均为空。无需手动追加哨兵，也不输出空后缀。每轮按第二关键字已有顺序、第一关键字稳定计数排序，倍增 $O(n \log n + \text{alphabet})$ ，空间 $O(n + \text{alphabet})$ ；字节串 `alphabet=256`。大整数先自行保序离散化，避免按最大原值开桶。Kasai 求 `lcp` 为 $O(n)$ 。不同起点 x,y 的 LCP 为 `lcp[min(rk[x],rk[y])+1..max(rk[x],rk[y])]` 的最小值；本结构不内建 RMQ，需要频繁查询时另接静态 RMQ。同一起点的 LCP 为 $n-x$ 。覆盖倍增构造，不替代 DC3/SA-IS。

```

1 struct SuffixArray
2 {
3     vector<int> sa, rk, lcp;
4
5     SuffixArray(const string &s)
6     {
7         vector<int> a;
8         a.reserve(s.size());
9         for (unsigned char c : s)
10             a.push_back(c);
11         init(a, 256);
12     }
13
14     SuffixArray(const vector<int> &s, int alphabet)
15     {
16         init(s, alphabet);
17     }
18
19     void init(const vector<int> &s, int alphabet)
20     {
21         assert(s.size() < INT_MAX);
22         assert(1 <= alphabet && alphabet < INT_MAX);
23         int n = (int)s.size();
24         sa.resize(n);
25         rk.resize(n);
26         lcp.assign(n, 0);
27         if (!n)
28             return;
29         vector<int> order(n), old(n), cnt(max(n + 1, alphabet + 1));
30         int m = alphabet;
31         for (int i = 0; i < n; i++)
32         {
33             assert(0 <= s[i] && s[i] < alphabet);
34             rk[i] = s[i] + 1;
35             cnt[rk[i]]++;
36         }
37         for (int i = 1; i <= m; i++)
38             cnt[i] += cnt[i - 1];
39         for (int i = n - 1; i >= 0; i--)
40             sa[--cnt[rk[i]]] = i;

```

```

41     for (int k = 1; k < n; k *= 2)
42     {
43         int p = 0;
44         for (int i = n - k; i < n; i++)
45             order[p++] = i;
46         for (int i : sa)
47             if (i >= k)
48                 order[p++] = i - k;
49         fill(cnt.begin(), cnt.begin() + m + 1, 0);
50         for (int x : rk)
51             cnt[x]++;
52         for (int i = 1; i <= m; i++)
53             cnt[i] += cnt[i - 1];
54         for (int i = n - 1; i >= 0; i--)
55             sa[--cnt[rk[order[i]]]] = order[i];
56         old.swap(rk);
57         auto second = [&](int i)
58         {
59             return i < n - k ? old[i + k] : 0;
60         };
61         m = 1;
62         rk[sa[0]] = m;
63         for (int i = 1; i < n; i++)
64         {
65             int a = sa[i - 1], b = sa[i];
66             if (old[a] != old[b] || second(a) != second(b))
67                 m++;
68             rk[b] = m;
69         }
70         if (m == n)
71             break;
72     }
73     for (int i = 0; i < n; i++)
74         rk[sa[i]] = i;
75     for (int i = 0, k = 0; i < n; i++)
76     {
77         if (!rk[i])
78         {
79             k = 0;
80             continue;
81         }
82         int j = sa[rk[i] - 1];
83         while (k < n - i && k < n - j && s[i + k] == s[j + k])
84             ++k;
85         lcp[rk[i]] = k;
86         if (k)
87             --k;
88     }
89 }
90 };

```

28.8 后缀自动机

`extend(c)` 接收整数 0..25; `distinct()` 返回不同非空子串数, `counts()` 返回 `endpos` 次数且不修改结构, 可重复调用。状态 `a`, 初始状态 0, `link` 根为 -1; 空间 $O(26n)$, 构建 $O(26n)$ 。

```

1 struct SuffixAutomaton
2 {
3     struct Node
4     {
5         array<int, 26> go{};
6         int link = -1, len = 0;
7         long long occ = 0;
8     };
9
10    vector<Node> a{Node{}};
11    int last = 0;
12
13    void extend(int c)
14    {
15        assert(0 <= c && c < 26);
16        int cur = (int)a.size();
17        a.push_back(Node{});
18        a[cur].len = a[last].len + 1;
19        a[cur].occ = 1;
20        int p = last;
21        while (p != -1 && !a[p].go[c])
22        {
23            a[p].go[c] = cur;
24            p = a[p].link;
25        }
26        if (p == -1)
27            a[cur].link = 0;
28        else
29        {
30            int q = a[p].go[c];
31            if (a[p].len + 1 == a[q].len)
32                a[cur].link = q;
33            else
34            {
35                int clone = (int)a.size();
36                Node copy = a[q];
37                a.push_back(copy);
38                a[clone].len = a[p].len + 1;
39                a[clone].occ = 0;
40                while (p != -1 && a[p].go[c] == q)
41                {
42                    a[p].go[c] = clone;
43                    p = a[p].link;
44                }
45                a[q].link = a[cur].link = clone;
46            }
47        }
48        last = cur;
49    }
50
51    vector<long long> counts() const

```

```
52     {
53         vector<int> bucket(a[last].len + 1), order(a.size());
54         vector<long long> ans(a.size());
55         for (auto &v : a)
56             ++bucket[v.len];
57         partial_sum(bucket.begin(), bucket.end(), bucket.begin());
58         for (int i = 0; i < (int)a.size(); i++)
59             {
60                 order[--bucket[a[i].len]] = i;
61                 ans[i] = a[i].occ;
62             }
63         for (int i = (int)order.size() - 1; i > 0; i--)
64             ans[a[order[i]].link] += ans[order[i]];
65         return ans;
66     }
67
68     long long distinct() const
69     {
70         long long ans = 0;
71         for (int i = 1; i < (int)a.size(); i++)
72             ans += a[i].len - a[a[i].link].len;
73         return ans;
74     }
75 };
```

第二十九章 后缀查询与重复子串

29.1 后缀 LCP 与子串比较

先构造 SuffixArray, 再构造 SuffixLCP(sa); 传统版对应 Suffix_Array 和 Suffix_LCP。init(sa) / Init(sa) 可重新建表。复制所需排名与 LCP 数据, 不保存原字符串或源对象引用, 源对象可销毁或重建。query(x,y) 返回 s[x..n) 与 s[y..n) 的最长公共前缀, $0 \leq x, y \leq n$, 允许空后缀; 同起点返回 n-x。compare(l1,r1,l2,r2) 比较两个 0-based 半开子串, 允许空串, 返回 -1/0/1; 传统接口为 Query/Compare。若公共前缀覆盖较短子串, 按长度比较, 否则原后缀排名给出首个不同字符的顺序。字节排序及整数字母表约束继承 SuffixArray。预处理稀疏表 $O(n \log n)$ 时间、空间, 每次查询 $O(1)$, 无需哈希和原串。n、rk、lg、st 可读, 禁止调用方破坏。

```
1 struct SuffixLCP
2 {
3     int n;
4     vector<int> rk, lg;
5     vector<vector<int>>> st;
6
7     SuffixLCP(const SuffixArray &suffix)
8     {
9         init(suffix);
10    }
11
12    void init(const SuffixArray &suffix)
13    {
14        n = suffix.sa.size();
15        rk = suffix.rk;
16        lg.assign(n + 1, 0);
17        for (int i = 2; i <= n; i++)
18            lg[i] = lg[i / 2] + 1;
19        st.assign(lg[n] + 1, {});
20        st[0] = suffix.lcp;
21        for (int k = 1; k <= lg[n]; k++)
22        {
23            int len = 1 << k;
24            st[k].resize(n - len + 1);
25            for (int i = 0; i <= n - len; i++)
26                st[k][i] = min(st[k - 1][i], st[k - 1][i + len / 2]);
27        }
28    }
```

```

30  int query(int x, int y) const
31  {
32      assert(0 <= x && x <= n && 0 <= y && y <= n);
33      if (x == y || x == n || y == n)
34          return n - max(x, y);
35      int a = rk[x], b = rk[y];
36      if (a > b)
37          swap(a, b);
38      int k = lg[b - a];
39      return min(st[k][a + 1], st[k][b - (1 << k) + 1]);
40  }
41
42  // Compare [l1,r1) and [l2,r2); return -1, 0 or 1.
43  int compare(int l1, int r1, int l2, int r2) const
44  {
45      assert(0 <= l1 && l1 <= r1 && r1 <= n);
46      assert(0 <= l2 && l2 <= r2 && r2 <= n);
47      int a = r1 - l1, b = r2 - l2;
48      if (query(l1, l2) >= min(a, b))
49          return (a > b) - (a < b);
50      return rk[l1] < rk[l2] ? -1 : 1;
51  }
52 };

```

29.2 两个前缀的最长公共后缀

`prefix_lcs(reversed,x,y)` 返回原序列 `s[0..x)` 与 `s[0..y)` 的最长公共后缀，端点满足 $0 \leq x, y \leq n$ 。reversed 必须在原序列的反转序列上构造 SuffixLCP，不能传正向对象；转换为 `reversed.query(n-x,n-y)`，单次 $O(1)$ 。传统版 Prefix_Lcs 接受 Suffix_LCP。只需前向 LCP 时不必构造反向对象；该接口中的 LCS 是最长公共后缀，不是最长公共子序列。对于两个任意子串，以右端点查询后还需截断到两个子串长度的最小值。

```

1  // reversed must be built on the reversed original sequence.
2  inline int prefix_lcs(const SuffixLCP &reversed, int x, int y)
3  {
4      assert(0 <= x && x <= reversed.n && 0 <= y && y <= reversed.n);
5      return reversed.query(reversed.n - x, reversed.n - y);
6  }

```

29.3 重复子串 AA 的起止位置计数

`square_counts(forward,reversed)` 返回 `(start,finish)` 两个长度 `n` 的 `long long` 数组, `start[i]/finish[i]` 为开始/结束于位置 `i` 的非空 AA 型子串个数; 结束位置是最后一个字符下标, 均 0-based。两对象必须分别对应同一原序列及其反序列, 长度相同, 允许空序列。传统接口为 `Square_Counts`。枚举半长 `len`, 并以其整数倍 `i` 为锚点; 向左匹配截断为 `len-1`, 向右截断为 `len`, 由 LCP/LCS 得出合法起点连续区间, 差分累计。左侧不能放宽到 `len`, 否则相邻锚点会重复计数。预处理查询器后, 此函数 $O(n \log n)$ 时间、 $O(n)$ 额外空间。不同出现位置、不同半长分别计数; 不要求 `A` 为最小周期。P1117 的 AABB 拆分数为所有分界点 `finish[i]*start[i+1]` 之和, 累加类型需按题目范围选择, 函数本身只返回起止计数。

```

1 // Count nonempty squares AA by their starting and ending positions.
2 inline pair<vector<long long>, vector<long long>>
3 square_counts(const SuffixLCP &forward, const SuffixLCP &reversed)
4 {
5     int n = forward.n;
6     assert(reversed.n == n);
7     vector<long long> start(n + 1), finish(n + 1);
8     for (int len = 1; len <= n / 2; len++)
9         for (int i = 0; i < n - len; i += len)
10            {
11                int j = i + len;
12                int left = min(len - 1, prefix_lcs(reversed, i, j));
13                int right = min(len, forward.query(i, j));
14                if (left + right < len)
15                    continue;
16                int l = i - left, r = i + right - len;
17                start[l]++;
18                start[r + 1]--;
19                finish[l + 2 * len - 1]++;
20                finish[r + 2 * len]--;
21            }
22     for (int i = 1; i < n; i++)
23     {
24         start[i] += start[i - 1];
25         finish[i] += finish[i - 1];
26     }
27     start.resize(n);
28     finish.resize(n);
29     return {start, finish};
30 }
```

第三十章 回文树

30.1 回文树 PAM: 回文后缀与出现次数

`add(ch)` 追加小写英文字母, 返回最长回文后缀节点, 亦保存于 `last`。节点 0 是长度 0 根, 1 是长度 -1 根, 实际回文节点从 2 开始; `distinct()` 为不同非空回文串数。`t[u].len` 为长度, `fail` 为最长真回文后缀链接, `suffixes` 为该节点的非空回文后缀数量; 追加后的 `t[last].suffixes` 就是本位置结尾的回文串数。`total` 为当前全部回文子串的出现次数之和。`end` 为首次出现时的 0-based 右端点, 可恢复原串子段; `hits` 仅统计作为最长达后缀的次数, 不能当出现次数。`occurrences()` 返回按节点编号的完整出现次数副本, 根置零, $O(n)$; 重复调用以及查询后继续追加均安全。`init()` 清空重建。固定 26 字符集下建树 $O(n)$, 空间 $O(26n)$ 。

```
1 struct PalindromicTree
2 {
3     struct Node
4     {
5         int next[26]{};
6         int len = 0, fail = 0, suffixes = 0, end = -1;
7         long long hits = 0;
8     };
9
10    vector<Node> t;
11    vector<int> s;
12    int last = 0;
13    long long total = 0;
14
15    PalindromicTree()
16    {
17        init();
18    }
19
20    void init()
21    {
22        t.assign(2, Node{});
23        t[0].fail = 1;
24        t[1].len = -1;
25        s.assign(1, -1);
26        last = 0;
27        total = 0;
28    }
29
30    int get_fail(int u) const
31    {
32        int pos = s.size() - 1;
33        while (s[pos - t[u].len - 1] != s[pos])
34            u = t[u].fail;
35        return u;
36    }
37
38    // Append a lowercase letter; return the longest palindromic suffix node.
39    int add(char ch)
40    {
41        int c = ch - 'a';
```



```

42     assert(0 <= c && c < 26);
43     s.push_back(c);
44     int u = get_fail(last);
45     if (!t[u].next[c])
46     {
47         int v = t.size();
48         t.push_back(Node{});
49         t[v].len = t[u].len + 2;
50         t[v].fail = t[get_fail(t[u].fail)].next[c];
51         t[v].suffixes = t[t[v].fail].suffixes + 1;
52         t[v].end = (int)s.size() - 2;
53         t[u].next[c] = v;
54     }
55     last = t[u].next[c];
56     t[last].hits++;
57     total += t[last].suffixes;
58     return last;
59 }
60
61 int distinct() const
62 {
63     return (int)t.size() - 2;
64 }
65
66 // Copy-and-propagate: repeated calls and later appends remain valid.
67 vector<long long> occurrences() const
68 {
69     vector<long long> count(t.size());
70     for (int u = 2; u < (int)t.size(); u++)
71         count[u] = t[u].hits;
72     for (int u = (int)t.size() - 1; u >= 2; u--)
73         count[t[u].fail] += count[u];
74     count[0] = 0;
75     count[1] = 0;
76     return count;
77 }
78 };

```

第三十一章 数论

31.1 模运算、素性、CRT 与类欧几里德

mul/power 支持无符号 64 位正模数，乘积用 uint128。prime 使用确定性 64 位底数。inverse(a,m) 无逆元返回 -1。crt(r,m,b,n) 合并 $x \equiv r \pmod{m}$ 与 $x \equiv b \pmod{n}$ ，要求 $m,n > 0$ 。从 $r=0$ 、 $m=1$ 开始逐项合并；成功后 r 为最小非负解、 m 为最小正周期。余数允许完整 long long 范围。无解返回 false；有解但新周期超过 LLONG_MAX 抛 overflow_error。两种失败都会先将 r 归一化， m 不变；失败后不能继续当作完整解使用。递归 exgcd 和乘法中间量用 int128，单次 $O(\log \min(m,n))$ ，抄写 CRT 时只需类型别名、exgcd 和 crt。floor_sum 为 $i=0..n-1$ 的 $\text{floor}((ai+b)/m)$ 之和，支持负 a 、 b ，返回 int128；约定 $n \leq 10^9$ 、 $m \leq 10^9$ ， a 、 b 为有符号 64 位。

```
1 struct NumberTheory
2 {
3     using ll = long long;
4     using ull = unsigned long long;
5     using i128 = __int128_t;
6     using u128 = __uint128_t;
7
8     static ull mul(ull a, ull b, ull m)
9     {
10         return u128(a) * b % m;
11     }
12
13     static ull power(ull a, ull b, ull m)
14     {
15         assert(m);
16         ull r = 1 % m;
17         a %= m;
18         for (; b >>= 1, a = mul(a, a, m))
19             if (b & 1)
20                 r = mul(r, a, m);
21         return r;
22     }
23
24     static bool prime(ull n)
25     {
26         if (n < 2)
27             return false;
28         for (ull p : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37})
29             if (n % p == 0)
30                 return n == p;
31         ull d = n - 1;
32         int s = 0;
33         while (!(d & 1))
34         {
35             d >>= 1;
36             ++s;
37         }
38         for (ull a :
39             {2ULL, 325ULL, 9375ULL, 28178ULL, 450775ULL, 9780504ULL, 1795265022ULL})
40         {
41             if (a % n == 0)
```

```

42         continue;
43         ull x = power(a, d, n);
44         if (x == 1 || x == n - 1)
45             continue;
46         bool ok = false;
47         for (int j = 1; j < s; j++)
48         {
49             x = mul(x, x, n);
50             if (x == n - 1)
51             {
52                 ok = true;
53                 break;
54             }
55         }
56         if (!ok)
57             return false;
58     }
59     return true;
60 }
61
62 static i128 exgcd(i128 a, i128 b, i128 &x, i128 &y)
63 {
64     if (!b)
65     {
66         x = 1;
67         y = 0;
68         return a;
69     }
70     i128 d = exgcd(b, a % b, y, x);
71     y -= a / b * x;
72     return d;
73 }
74
75 static ll inverse(ll a, ll m)
76 {
77     assert(m > 0);
78     a %= m;
79     if (a < 0)
80         a += m;
81     i128 x, y;
82     if (exgcd(a, m, x, y) != 1)
83         return -1;
84     return (x % m + m) % m;
85 }
86
87 // x = r (mod m); false means inconsistent. Throws if lcm exceeds int64.
88 static bool crt(ll &r, ll &m, ll b, ll n)
89 {
90     assert(m > 0 && n > 0);
91     r %= m;
92     if (r < 0)
93         r += m;
94     b %= n;
95     if (b < 0)
96         b += n;
97     i128 x, y;

```

```

98     ll g = (ll)exgcd(m, n, x, y);
99     i128 diff = i128(b) - r;
100     if (diff % g)
101         return false;
102     i128 q = n / g, k = (diff / g * x % q + q) % q, mod = i128(m) * q;
103     if (mod > LLONG_MAX)
104         throw overflow_error("CRT modulus");
105     r = (r + i128(m) * k) % mod;
106     m = (ll)mod;
107     return true;
108 }
109
110 // sum_{i=0}^{n-1} floor((a*i+b)/m), signed a,b; 0<=n<=1e9, 1<=m<=1e9.
111 static i128 floor_sum(ll n, ll m, ll a, ll b)
112 {
113     assert(n >= 0 && m > 0);
114     i128 ans = 0;
115     auto norm = [&](ll &v)
116     {
117         ll q = v / m;
118         if (v % m < 0)
119             --q;
120         v = (ll)(i128(v) - i128(q) * m);
121         return q;
122     };
123     ans += i128(norm(a)) * n * (n - 1) / 2;
124     ans += i128(norm(b)) * n;
125     while (true)
126     {
127         ans += i128(n) * (n - 1) / 2 * (a / m);
128         a %= m;
129         ans += i128(n) * (b / m);
130         b %= m;
131         i128 y = i128(a) * n + b;
132         if (y < m)
133             break;
134         n = (ll)(y / m);
135         b = (ll)(y % m);
136         swap(a, m);
137     }
138     return ans;
139 }
140 };

```

31.2 Pollard-Rho 因数分解

`factor(n)` 返回升序质因子（含重数）， $n \geq 1$ ；依赖 `NumberTheory`。返回因子经过整除和素性检查；Las Vegas 随机运行时间，固定默认种子方便复现，可由调用者更换。无确定性最坏时间保证。

```

1 struct PollardRho
2 {
3     using ull = unsigned long long;
4     using u128 = __uint128_t;
5     mt19937_64 rng;
6
7     PollardRho(ull seed = 712367821) : rng(seed)
8     {
9     }
10
11     ull rho(ull n)
12     {
13         if (n % 2 == 0)
14             return 2;
15         for (;;)
16         {
17             ull c = rng() % (n - 1) + 1, x = rng() % n, y = x, d = 1;
18             auto f = [&](ull v)
19             {
20                 return (NumberTheory::mul(v, v, n) + u128(c)) % n;
21             };
22             // Retry bounded attempts; Las Vegas: only exact divisors returned.
23             for (int i = 0; i < 200000 && d == 1; i++)
24             {
25                 x = f(x);
26                 y = f(f(y));
27                 d = gcd(x > y ? x - y : y - x, n);
28             }
29             if (1 < d && d < n)
30                 return d;
31         }
32     }
33
34     void split(ull n, vector<ull> &a)
35     {
36         if (n == 1)
37             return;
38         if (NumberTheory::prime(n))
39         {
40             a.push_back(n);
41             return;
42         }
43         ull d = rho(n);
44         split(d, a);
45         split(n / d, a);
46     }

```

```

47
48     vector<ull> factor(ull n)
49     {
50         assert(n >= 1);
51         vector<ull> a;
52         split(n, a);
53         sort(a.begin(), a.end());
54         return a;
55     }
56 };

```

31.3 线性筛、欧拉函数与 Möbius 函数

构造参数 $n \geq 0$, prime 为质数表, lp/phi/mu 为 $0..n$ 数组。phi[1]=mu[1]=1, $O(n)$ 时间和空间。

```

1 struct LinearSieve
2 {
3     vector<int> prime, lp, phi, mu;
4
5     LinearSieve(int n) : lp(n + 1), phi(n + 1), mu(n + 1)
6     {
7         if (n)
8             phi[1] = mu[1] = 1;
9         for (int i = 2; i <= n; i++)
10         {
11             if (!lp[i])
12             {
13                 lp[i] = i;
14                 prime.push_back(i);
15                 phi[i] = i - 1;
16                 mu[i] = -1;
17             }
18             for (int p : prime)
19             {
20                 if (p > n / i)
21                     break;
22                 int j = i * p;
23                 lp[j] = p;
24                 if (i % p == 0)
25                 {
26                     phi[j] = phi[i] * p;
27                     mu[j] = 0;
28                     break;
29                 }
30                 phi[j] = phi[i] * (p - 1);
31                 mu[j] = -mu[i];
32             }
33         }
34     }
35 };

```

31.4 静态模整数

模板参数必须为不超过 `INT_MAX` 的素数且 ≥ 2 。加减乘、`pow`、`inv`、除法；求逆要求非零，`pow` 指数非负。
复杂度：加减乘 $O(1)$ ，幂和逆 $O(\log \text{mod})$ 。

```

1 template <int mod> struct ModInt
2 {
3     int v;
4
5     ModInt(long long x = 0) : v((x % mod + mod) % mod)
6     {
7     }
8
9     ModInt operator+(ModInt b) const
10    {
11        return ModInt((long long)v + b.v);
12    }
13
14    ModInt operator-(ModInt b) const
15    {
16        return ModInt((long long)v - b.v);
17    }
18
19    ModInt operator*(ModInt b) const
20    {
21        return ModInt(1LL * v * b.v);
22    }
23
24    ModInt pow(long long e) const
25    {
26        assert(e >= 0);
27        ModInt a = *this, r = 1;
28        for (; e >>= 1, a = a * a)
29            if (e & 1)
30                r = r * a;
31        return r;
32    }
33
34    ModInt inv() const
35    {
36        assert(v);
37        return pow(mod - 2);
38    } // prime modulus
39
40    ModInt operator/(ModInt b) const
41    {
42        return *this * b.inv();
43    }
44 };

```

31.5 阶乘组合数

`Binomial<prime>(n=0)` 预处理 $0..n$, 要求 `prime` 为素数且 $0 \leq n < \text{prime}$ 。`init(n)` 只扩大当前上限, 较小或相同的 n 不改变表; `choose(n,k)` 返回 $C(n,k)$, `permute(n,k)` 返回 $n!/(n-k)!$ 。查询前显式预处理到 n , 查询本身不扩表。 $n < 0$, $k < 0$ 或 $k > n$ 返回 0, $C(0,0)=P(0,0)=1$ 。`fac`、`ifac` 为可直接读取的阶乘与逆阶乘表, 增长可能使旧引用失效。单次查询 $O(1)$; 从 `old` 扩到 n 新增 $O(n-\text{old}+\log \text{prime})$ 次域运算, 重分配还可能复制旧表。首次预处理 $O(n+\log \text{prime})$, 空间 $O(n)$ 。超过模数不能使用阶乘除法: 小素数用 `Lucas`, 小合数用 `exLucas`。

```

1  template <int mod> struct Binomial
2  {
3      using Z = ModInt<mod>;
4      vector<Z> fac{Z(1)}, ifac{Z(1)};
5
6      Binomial(int n = 0)
7      {
8          init(n);
9      }
10
11     void init(int n)
12     {
13         assert(0 <= n && n < mod);
14         int old = (int)fac.size() - 1;
15         if (n <= old)
16             return;
17         fac.resize(n + 1);
18         ifac.resize(n + 1);
19         for (int i = old + 1; i <= n; i++)
20             fac[i] = fac[i - 1] * i;
21         ifac[n] = fac[n].inv();
22         for (int i = n; i > old; i--)
23             ifac[i - 1] = ifac[i] * i;
24     }
25
26     Z choose(int n, int k) const
27     {
28         if (k < 0 || k > n)
29             return 0;
30         assert(n < (int)fac.size());
31         return fac[n] * ifac[k] * ifac[n - k];
32     }
33
34     Z permute(int n, int k) const
35     {
36         if (k < 0 || k > n)
37             return 0;
38         assert(n < (int)fac.size());
39         return fac[n] * ifac[n - k];
40     }
41 };

```


第三十二章 批量模逆元

32.1 任意数组的批量逆元

`batch_inverse(a,mod)` 返回与 `a` 同序的逆元向量，传统接口为 `Batch_Inverse`。要求 `mod` 为正 `long long`，每个输入可为完整 `long long`；只要某一项与 `mod` 不互素，整个结果返回 `nullopt`，不输出部分逆元。空输入返回有效的空向量；模数 1 约定返回全零，与单个逆元接口一致。模数可以是合数。函数按值接收数组并归一化；不再需要原数组时传 `move(a)` 可避免输入副本。依赖 `NumberTheory` 的 `inverse` 和 `exgcd`，传统版对应 `Inverse`、`Exgcd`。仅求一次逆元，时间 $O(n+\log \text{mod})$ ，空间 $O(n)$ ；128 位中间乘法防止溢出。

```
1 inline optional<vector<long long>> batch_inverse(vector<long long> a, long long mod)
2 {
3     assert(mod > 0);
4     int n = a.size();
5     vector<long long> prefix(n + 1, 1 % mod);
6     for (int i = 0; i < n; i++)
7     {
8         a[i] %= mod;
9         if (a[i] < 0)
10             a[i] += mod;
11         prefix[i + 1] = __int128_t(prefix[i]) * a[i] % mod;
12     }
13     long long inverse = NumberTheory::inverse(prefix[n], mod);
14     if (inverse == -1)
15         return nullopt;
16     for (int i = n - 1; i >= 0; i--)
17     {
18         long long value = a[i];
19         a[i] = __int128_t(inverse) * prefix[i] % mod;
20         inverse = __int128_t(inverse) * value % mod;
21     }
22     return a;
23 }
```

第三十三章 混合进制中国剩余定理

33.1 Garner：大周期 CRT 与混合进制

`garner_digits(b,moduli)` 返回按输入顺序排列的混合进制各位；`garner(b,moduli,target)` 返回最小非负 CRT 解对 `target` 的余数。传统接口为 `Garner_Digits`、`Garner`。两个输入向量等长，长度可为 0；模数为正且两两互素，允许模数 1，余数支持完整 `long long` 范围。`target` 为正 `long long`，与输入模数无需互素。输入不互素时抛 `invalid_argument`，即使方程有解也不适用；此时用扩展 CRT。空输入的规范解取 0。算法从不构造模数乘积，中间运算使用 `int128`。 k 个模数耗时 $O(k^2+k \log M)$ ，空间 $O(k)$ ； M 为最大输入模数。依赖 `NumberTheory` 的 `inverse` 和 `exgcd`，传统版对应 `Inverse`、`Exgcd`。

```
1 // Pairwise coprime positive moduli. Digits use the input order.
2 inline vector<long long> garner_digits(const vector<long long> &b,
3                                       const vector<long long> &moduli)
4 {
5     using ll = long long;
6     using i128 = __int128_t;
7     int n = b.size();
8     assert(moduli.size() == b.size());
9     vector<ll> value(n), coefficient(n, 1), digit(n);
10    for (ll m : moduli)
11        assert(m > 0);
12    for (int i = 0; i < n; i++)
13    {
14        ll m = moduli[i];
15        ll inverse = NumberTheory::inverse(coefficient[i], m);
16        if (inverse == -1)
17            throw invalid_argument("Garner moduli are not coprime");
18        i128 delta = (i128(b[i]) - value[i]) % m;
19        if (delta < 0)
20            delta += m;
21        digit[i] = delta * inverse % m;
22        for (int j = i + 1; j < n; j++)
23        {
24            value[j] = (value[j] + i128(coefficient[j]) * digit[i]) % moduli[j];
25            coefficient[j] = i128(coefficient[j]) * m % moduli[j];
26        }
27    }
28    return digit;
29 }
```

规范解对目标数取模

```
31 // Return the least nonnegative CRT solution modulo target, not all solutions.
32 inline long long
33 garner(const vector<long long> &b, const vector<long long> &moduli, long long target)
34 {
35     assert(target > 0);
36     auto digit = garner_digits(b, moduli);
37     long long answer = 0;
38     for (int i = (int)digit.size() - 1; i >= 0; i--)
39         answer = (__int128_t(answer) * moduli[i] + digit[i]) % target;
40     return answer;
41 }
```

第三十四章 乘法阶与原根

34.1 乘法阶与原根

构造 `PrimitiveRoot(n)`, 要求 $2 \leq n \leq \text{INT_MAX}$; `init(n)` 可重建。`order(a)` 返回 a 模 n 的乘法阶, a 不可逆时返回 0; `is_root(a)` 判断该剩余类是否为原根, a 可为完整 `long long`。`minimum()` 返回 $1..n-1$ 内最小原根, 无原根返回 -1; `all()` 返回升序的全部原根, 无原根返回空向量。`mod`、`phi`、`exists` 只读; `factors` 为 `phi` 的不同素因子, 辅助 `factor` 为试除分解。依赖 `NumberTheory` 的 `power`。初始化 $O(\sqrt{n})$, 求阶 $O(\log^2 n)$, 原根判定 $O((1+\omega(\phi))\log n)$ 。若最小原根为 G , 顺序枚举最坏检查 G 个候选, 不能假定 G 总是常数。枚举全部另需 $O(\phi \log \phi + c \log c)$ 时间、 $O(c)$ 空间, c 为原根数; 大模数只求最小原根或阶时, 不应调用 `all`。

```
1 struct PrimitiveRoot
2 {
3     int mod, phi;
4     bool exists;
5     vector<int> factors;
6
7     static vector<int> factor(int n)
8     {
9         vector<int> result;
10        for (int p = 2; p <= n / p; p++)
11            if (n % p == 0)
12            {
13                result.push_back(p);
14                while (n % p == 0)
15                    n /= p;
16            }
17        if (n > 1)
18            result.push_back(n);
19        return result;
20    }
21
22    PrimitiveRoot(int n)
23    {
24        init(n);
25    }
```

```
27 void init(int n)
28 {
29     assert(n >= 2);
30     mod = n;
31     phi = n;
32     for (int p : factor(n))
33         phi -= phi / p;
34     int odd = n;
35     if (odd % 2 == 0)
36         odd /= 2;
37     exists = n == 2 || n == 4 || (odd % 2 && factor(odd).size() == 1);
38     factors = factor(phi);
39 }
40
41 int order(long long a) const
42 {
43     a %= mod;
44     if (a < 0)
45         a += mod;
46     if (gcd(a, (long long)mod) != 1)
47         return 0;
48     int result = phi;
49     for (int p : factors)
50         while (result % p == 0 && NumberTheory::power(a, result / p, mod) == 1)
51             result /= p;
52     return result;
53 }
```

```

55     bool is_root(long long a) const
56     {
57         if (!exists)
58             return false;
59         a %= mod;
60         if (a < 0)
61             a += mod;
62         if (gcd(a, (long long)mod) != 1)
63             return false;
64         for (int p : factors)
65             if (NumberTheory::power(a, phi / p, mod) == 1)
66                 return false;
67         return true;
68     }
69
70     int minimum() const
71     {
72         if (!exists)
73             return -1;
74         for (int g = 1; g < mod; g++)
75             if (is_root(g))
76                 return g;
77         return -1;
78     }
79
80     vector<int> all() const
81     {
82         int g = minimum();
83         vector<int> roots;
84         if (g == -1)
85             return roots;
86         long long value = 1;
87         for (int k = 1; k <= phi; k++)
88         {
89             value = value * g % mod;
90             if (gcd(k, phi) == 1)
91                 roots.push_back(value);
92         }
93         sort(roots.begin(), roots.end());
94         return roots;
95     }
96 };

```

第三十五章 Carmichael 函数

35.1 Carmichael: 模乘法群的公共指数

`carmichael(n)` 返回 $\lambda(n)$, 即使所有与 n 互素的 a 都满足 a 的 $\lambda(n)$ 次幂模 n 为 1 的最小正指数; 约定 $\lambda(1)=1$ 。要求 $1 \leq n \leq \text{INT_MAX}$ 。传统接口为 `Carmichael`。试除分解后将各素数幂的指数取 lcm, 特别处理 2 的三次及以上幂。时间 $O(\sqrt{n})$, 额外空间 $O(1)$; 所有中间值不超过有符号 64 位范围, 结果不超过 n 。该指数只适用于可逆底数, 不能据此直接对非互素底数的幂降指数。函数不依赖原根模块。

```
1 inline int carmichael(int n)
2 {
3     assert(n >= 1);
4     long long answer = 1;
5     for (int p = 2; p <= n / p; p++)
6         if (n % p == 0)
7         {
8             int power = 1;
9             while (n % p == 0)
10             {
11                 n /= p;
12                 power *= p;
13             }
14             int value = power / p * (p - 1);
15             if (p == 2 && power >= 8)
16                 value /= 2;
17             answer = answer / gcd(answer, (long long)value) * value;
18         }
19     if (n > 1)
20         answer = answer / gcd(answer, (long long)(n - 1)) * (n - 1);
21     return answer;
22 }
```

第三十六章 素数模数组组合数

36.1 Lucas：小素数模的大下标组合数

Lucas(p) 预处理素数模数 p, choose(n,k) 返回 $C(n,k)$ 模 p; n、k 为完整 unsigned long long 范围, $k > n$ 返回 0, $C(0,0)=1$ 。init(p) 可更换模数并重新预处理; 不检测素性, 要求调用者保证 p 为素数且 $p \geq 2$ 。预处理时间 $O(p + \log p)$, 动态版空间 $O(p)$; 单次查询 $O(1 + \log_p(n+1))$ 。适合小素数模数, 通常 $p \leq 10^6$; 大素数不能因查询下标小就忽略这里整段 $O(p)$ 的预处理。fac、ifac 下标为 $0..p-1$, 模数和表只读; 中间乘法用 long long。合数模数请使用 exLucas。

```
1 struct Lucas
2 {
3     int mod;
4     vector<int> fac, ifac;
5
6     Lucas(int p)
7     {
8         init(p);
9     }
10
11     void init(int p)
12     {
13         assert(p >= 2);
14         mod = p;
15         fac.resize(p);
16         ifac.resize(p);
17         fac[0] = 1;
18         for (int i = 1; i < p; i++)
19             fac[i] = 1LL * fac[i - 1] * i % p;
20         long long a = fac[p - 1], r = 1;
21         for (int e = p - 2; e; e >>= 1)
22         {
23             if (e & 1)
24                 r = r * a % p;
25             a = a * a % p;
26         }
27         ifac[p - 1] = r;
28         for (int i = p - 1; i; i--)
29             ifac[i - 1] = 1LL * ifac[i] * i % p;
30     }
```



```
32 int choose(unsigned long long n, unsigned long long k) const
33 {
34     if (k > n)
35         return 0;
36     long long answer = 1;
37     while (n || k)
38     {
39         int a = n % mod, b = k % mod;
40         if (b > a)
41             return 0;
42         answer = answer * fac[a] % mod;
43         answer = answer * ifac[b] % mod;
44         answer = answer * ifac[a - b] % mod;
45         n /= mod;
46         k /= mod;
47     }
48     return answer;
49 }
50 };
```

第三十七章 合数模数组合数

37.1 合数模数组合数 (扩展 Lucas)

ExLucas(mod) 预处理固定模数, choose(n,k) 返回 $C(n,k)$ 模 mod; n、k 为 unsigned long long, $k > n$ 返回 0, mod=1 返回 0。适合 $\text{mod} \leq 10^6$ 的合数或素数模数, 不能用于 10^9 级素数的整段预处理。分解 mod 为互素素数幂 $q = p^e$, 预处理每段内不含 p 因子的前缀积; unit(n,f) 去除 $n!$ 中全部 p 因子, valuation 统计指数。分子分母指数之差不少于 e 时该项为零, 否则只对互素单位部分求逆, 最后 CRT 合并。预处理 $O(\sqrt{\text{mod}} + \sum q)$, 空间 $O(\sum(q+1))$; 每次查询 $O(\sum(\log_p(n+1) \cdot \log(n+1) + \log q))$, 求和遍历不同素因子。所有乘法用 long long; 同一对象可重复查询, factors、fac 与辅助函数视为内部数据。

```
1 struct ExLucas
2 {
3     using ull = unsigned long long;
4     using ll = long long;
5
6     struct Factor
7     {
8         int p, q, exponent, start, crt;
9     };
10
11     int mod;
12     vector<int> fac;
13     vector<Factor> factors;
14
15     static ll power(ll a, ull b, int m)
16     {
17         ll result = 1;
18         while (b)
19         {
20             if (b & 1)
21                 result = result * a % m;
22             a = a * a % m;
23             b >>= 1;
24         }
25         return result;
26     }
27
28     ExLucas(int m) : mod(m)
29     {
30         assert(m >= 1);
31         int x = m;
32         for (int p = 2; p <= x / p; p++)
33         {
34             if (x % p)
35                 continue;
36             int q = 1, e = 0;
37             do
38             {
39                 x /= p;
40                 q *= p;
41                 e++;
```

```

42         } while (x % p == 0);
43         add_factor(p, q, e);
44     }
45     if (x > 1)
46         add_factor(x, x, 1);
47 }
48
49 void add_factor(int p, int q, int e)
50 {
51     int start = fac.size();
52     fac.push_back(1);
53     for (int i = 1; i < q; i++)
54         fac.push_back(1ll(fac.back()) * (i % p ? i : 1) % q);
55     fac.push_back(fac.back());
56     int phi = q - q / p;
57     int crt = 1ll(mod / q) * power(mod / q % q, phi - 1, q) % mod;
58     factors.push_back({p, q, e, start, crt});
59 }
60
61 static ull valuation(ull n, int p)
62 {
63     ull result = 0;
64     while (n)
65     {
66         n /= p;
67         result += n;
68     }
69     return result;
70 }
71
72 ll unit(ull n, const Factor &f) const
73 {
74     ll result = 1;
75     while (n)
76     {
77         result = result * power(fac[f.start + f.q], n / f.q, f.q) % f.q;
78         result = result * fac[f.start + n % f.q] % f.q;
79         n /= f.p;
80     }
81     return result;
82 }

```

```
84     int choose(ull n, ull k) const
85     {
86         if (k > n || mod == 1)
87             return 0;
88         ll answer = 0;
89         for (const auto &f : factors)
90         {
91             ull e = valuation(n, f.p) - valuation(k, f.p) - valuation(n - k, f.p);
92             if (e >= ull(f.exponent))
93                 continue;
94             ll denominator = unit(k, f) * unit(n - k, f) % f.q;
95             ll residue =
96                 unit(n, f) * power(denominator, f.q - f.q / f.p - 1, f.q) % f.q;
97             residue = residue * power(f.p, e, f.q) % f.q;
98             answer = (answer + residue * f.crt) % mod;
99         }
100         return answer;
101     }
102 };
```

第三十八章 二次剩余

38.1 素数模平方根 (Cipolla)

`mod_sqrt(n,p)` 返回 `vector<long long>`, 按升序列出全部不同根; 无解返回空向量, 零或模数 2 返回单根。要求 p 为素数, $2 \leq p \leq \text{INT_MAX}$, n 支持完整 `long long` 范围并在入口归一化。先用 Euler 判别; $p \equiv 3 \pmod{4}$ 时直接快速幂, 其余使用 Cipolla 二次扩域。函数内保存固定种子的伪随机生成器; 随机性影响寻找非剩余的尝试次数, 不影响结果, 通常期望 $O(\log p)$ 时间、 $O(1)$ 辅助空间。乘法用 `long long`, 扩域实部的三因子乘积中间先取模; 指数中的 $p+1$ 先转 `long long`, 避免 `INT_MAX` 溢出。此版本不处理合数模数及 64 位大素数, 也不用于浮点开方。

```
1 inline vector<long long> mod_sqrt(long long n, int p)
2 {
3     using ll = long long;
4     assert(p >= 2);
5     n = (n % p + p) % p;
6     if (n == 0 || p == 2)
7         return {n};
8     auto power = [p](ll a, ll b)
9     {
10         ll result = 1;
11         while (b)
12         {
13             if (b & 1)
14                 result = result * a % p;
15             a = a * a % p;
16             b >>= 1;
17         }
18         return result;
19     };
20     if (power(n, (p - 1) / 2) != 1)
21         return {};
22     ll answer;
23     if (p % 4 == 3)
24         answer = power(n, (ll(p) + 1) / 4);
25     else
26     {
27         static mt19937_64 rng(712367);
28         ll a, w;
29         do
30         {
31             a = rng() % p;
32             w = (a * a % p - n + p) % p;
33         } while (power(w, (p - 1) / 2) != p - 1);
34         using P = pair<ll, ll>;
35         auto mul = [p, w](P x, P y)
36         {
37             ll real = (x.first * y.first + x.second * y.second % p * w) % p;
38             ll imag = (x.first * y.second + x.second * y.first) % p;
39             return P{real, imag};
40         };
41         P result{1, 0}, base{a, 1};
```

```
42     ll exponent = (ll(p) + 1) / 2;
43     while (exponent)
44     {
45         if (exponent & 1)
46             result = mul(result, base);
47         base = mul(base, base);
48         exponent >>= 1;
49     }
50     answer = result.first;
51 }
52 ll other = p - answer;
53 if (answer > other)
54     swap(answer, other);
55 return {answer, other};
56 }
```

第三十九章 多项式插值

39.1 拉格朗日插值与连续点优化

先声明 `Lagrange.init(xs,ys,p)` 使用任意横坐标, `xs`、`ys` 为等长 0-based `vector<long long>`; `consecutive(ys,p)` 专用于横坐标 $0..n-1$ 。两者均覆盖旧数据。要求 p 为素数, $2 \leq p \leq \text{INT_MAX}$, 点数 $n \leq p$, 任意横坐标取模后两两不同; 不是仅要求原整数不同。`query(k)` 返回次数小于 n 的插值多项式在 k 处的值, 允许 k 、`xs`、`ys` 为完整 `long long` 范围, 内部先归一化; 空数据约定零多项式。通用预处理 $O(n^2+n \log p)$, 连续点预处理 $O(n+\log p)$, 每次查询 $O(n)$, 空间 $O(n)$ 。查询利用前缀积与滚动后缀积, 无需求逆, 因此 k 与样本点模 p 相等时也正确。若求 $1^d + \dots + k^d$, 要求 $0 \leq d \leq p-2$, 先计算 $0..d+1$ 的前缀和作为 `ys`, 再用 `consecutive`; 用快速幂生成这些样本另花 $O(d \log(d+1))$ 。仅处理素数域中的精确插值, 不适用于浮点曲线拟合或重复节点的导数约束。

```
1 struct Lagrange
2 {
3     using ll = long long;
4     int n, mod;
5     vector<ll> x, w;
6
7     ll norm(ll a) const
8     {
9         a %= mod;
10        if (a < 0)
11            a += mod;
12        return a;
13    }
14
15    ll power(ll a, int b) const
16    {
17        ll result = 1;
18        while (b)
19        {
20            if (b & 1)
21                result = result * a % mod;
22            a = a * a % mod;
23            b >>= 1;
24        }
25        return result;
26    }
27
28    void init(const vector<ll> &xs, const vector<ll> &ys, int p)
29    {
30        n = ys.size();
31        mod = p;
32        assert(p >= 2 && n <= p && xs.size() == ys.size());
33        x.resize(n);
34        w.resize(n);
35        for (int i = 0; i < n; i++)
36            x[i] = norm(xs[i]);
37        for (int i = 0; i < n; i++)
38        {
39            ll denominator = 1;
```

```

40     for (int j = 0; j < n; j++)
41         if (i != j)
42             denominator = denominator * norm(x[i] - x[j]) % mod;
43     assert(denominator != 0);
44     w[i] = norm(ys[i]) * power(denominator, mod - 2) % mod;
45 }
46 }
47
48 void consecutive(const vector<ll> &ys, int p)
49 {
50     n = ys.size();
51     mod = p;
52     assert(p >= 2 && n <= p);
53     x.resize(n);
54     w.resize(n);
55     if (!n)
56         return;
57     vector<ll> invfac(n, 1);
58     ll factorial = 1;
59     for (int i = 1; i < n; i++)
60         factorial = factorial * i % mod;
61     invfac[n - 1] = power(factorial, mod - 2);
62     for (int i = n - 1; i > 0; i--)
63         invfac[i - 1] = invfac[i] * i % mod;
64     for (int i = 0; i < n; i++)
65     {
66         x[i] = i;
67         w[i] = norm(ys[i]) * invfac[i] % mod * invfac[n - 1 - i] % mod;
68         if ((n - 1 - i) & 1)
69             w[i] = norm(-w[i]);
70     }
71 }
72
73 ll query(ll k) const
74 {
75     k = norm(k);
76     vector<ll> pre(n + 1, 1);
77     for (int i = 0; i < n; i++)
78         pre[i + 1] = pre[i] * norm(k - x[i]) % mod;
79     ll answer = 0, suffix = 1;
80     for (int i = n - 1; i >= 0; i--)
81     {
82         answer = (answer + w[i] * pre[i] % mod * suffix) % mod;
83         suffix = suffix * norm(k - x[i]) % mod;
84     }
85     return answer;
86 }
87 };

```


第四十章 多项式

40.1 NTT、形式幂级数、FWT 与递推

依赖 number_theory.hpp。模数固定 998244353，乘法结果长度 $\leq 2^{23}$ ；inverse 要求 $a[0] \neq 0$ ，log 要求 $a[0]=1$ ，exp 要求 $a[0]=0$ ，截断长度 $n>0$ 。朴素逐层 NTT 实现的 FPS 为 $O(n \log n)$ 。FWT 的 op 为 $\wedge$ 、 $|$ 、 $\&$ ；BM 返回 $s[n]=\sum c[i-1]s[n-i]$ 的系数；recurrence 用 $O(k^2 \log n)$ 求第 n 项。BM 对有限样本给出的递推不自动保证未知后续项成立。

```
1 struct Polynomial
2 {
3     static constexpr int mod = 998244353;
4     using Z = ModInt<mod>;
5     using Poly = vector<Z>;
6
7     static void ntt(Poly &a, bool invert = false)
8     {
9         int n = (int)a.size();
10        assert(n > 0 && (n & (n - 1)) == 0 && n <= (1 << 23));
11        for (int i = 1, j = 0; i < n; i++)
12        {
13            int bit = n >> 1;
14            for (; j & bit; bit >>= 1)
15                j ^= bit;
16            j ^= bit;
17            if (i < j)
18                swap(a[i], a[j]);
19        }
20        for (int len = 2; len <= n; len *= 2)
21        {
22            Z step = Z(3).pow((mod - 1) / len);
23            if (invert)
24                step = step.inv();
25            for (int i = 0; i < n; i += len)
26            {
27                Z w = 1;
28                for (int j = 0; j < len / 2; j++, w = w * step)
29                {
30                    Z u = a[i + j], v = a[i + j + len / 2] * w;
31                    a[i + j] = u + v;
32                    a[i + j + len / 2] = u - v;
33                }
34            }
35        }
36        if (invert)
37        {
38            Z inv = Z(n).inv();
39            for (auto &x : a)
40                x = x * inv;
41        }
42    }
43 }
```

```

44 static Poly multiply(Poly a, Poly b)
45 {
46     if (a.empty() || b.empty())
47         return {};
48     int size = (int)a.size() + (int)b.size() - 1, n = 1;
49     assert(size <= (1 << 23));
50     while (n < size)
51         n *= 2;
52     a.resize(n);
53     b.resize(n);
54     ntt(a);
55     ntt(b);
56     for (int i = 0; i < n; i++)
57         a[i] = a[i] * b[i];
58     ntt(a, true);
59     a.resize(size);
60     return a;
61 }
62
63 static Poly inverse(const Poly &a, int n)
64 {
65     assert(n > 0 && !a.empty() && a[0].v);
66     Poly b{a[0].inv()};
67     while ((int)b.size() < n)
68     {
69         int m = min(n, 2 * (int)b.size());
70         Poly f(a.begin(), a.begin() + min(m, (int)a.size()));
71         auto t = multiply(f, b);
72         t.resize(m);
73         for (auto &x : t)
74             x = Z(0) - x;
75         t[0] = t[0] + 2;
76         b = multiply(b, t);
77         b.resize(m);
78     }
79     return b;
80 }
81
82 static Poly derivative(const Poly &a)
83 {
84     Poly b;
85     for (int i = 1; i < (int)a.size(); i++)
86         b.push_back(a[i] * i);
87     return b;
88 }
89
90 static Poly integral(const Poly &a)
91 {
92     Poly b(a.size() + 1);
93     vector<Z> inv(a.size() + 1);
94     if (!a.empty())
95         inv[1] = 1;
96     for (int i = 2; i <= (int)a.size(); i++)
97         inv[i] = Z(mod - mod / i) * inv[mod % i];
98     for (int i = 0; i < (int)a.size(); i++)
99         b[i + 1] = a[i] * inv[i + 1];

```

```

100     return b;
101 }
102
103 static Poly log(const Poly &a, int n)
104 {
105     assert(n > 0 && !a.empty() && a[0].v == 1);
106     auto b = multiply(derivative(a), inverse(a, n));
107     b.resize(n - 1);
108     return integral(b);
109 }
110
111 static Poly exp(const Poly &a, int n)
112 {
113     assert(n > 0 && (a.empty() || a[0].v == 0));
114     Poly b{1};
115     while ((int)b.size() < n)
116     {
117         int m = min(n, 2 * (int)b.size());
118         auto t = log(b, m);
119         for (int i = 0; i < m; i++)
120             t[i] = (i < (int)a.size() ? a[i] : Z(0)) - t[i];
121         t[0] = t[0] + 1;
122         b = multiply(b, t);
123         b.resize(m);
124     }
125     return b;
126 }
127
128 static void fwt(Poly &a, char op, bool invert = false)
129 {
130     int n = (int)a.size();
131     assert(n > 0 && (n & (n - 1)) == 0);
132     const Z inv2 = (mod + 1) / 2;
133     for (int len = 1; len < n; len *= 2)
134         for (int i = 0; i < n; i += len * 2)
135             for (int j = 0; j < len; j++)
136             {
137                 Z &u = a[i + j], &v = a[i + j + len], x = u, y = v;
138                 if (op == '^')
139                 {
140                     u = x + y;
141                     v = x - y;
142                     if (invert)
143                     {
144                         u = u * inv2;
145                         v = v * inv2;
146                     }
147                 }
148                 else if (op == '|')
149                     v = invert ? y - x : y + x;
150                 else
151                 {
152                     assert(op == '&');
153                     u = invert ? x - y : x + y;
154                 }
155             }

```

```

156     }
157
158     // c: s[n]=sum c[i-1]*s[n-i], i=1..L.
159     static Poly berlekamp_massey(const Poly &s)
160     {
161         Poly c{1}, b{1};
162         int l = 0, m = 1;
163         Z last = 1;
164         for (int n = 0; n < (int)s.size(); n++)
165         {
166             Z d = s[n];
167             for (int i = 1; i <= l; i++)
168                 d = d + c[i] * s[n - i];
169             if (!d.v)
170             {
171                 ++m;
172                 continue;
173             }
174             auto old = c;
175             Z coef = d / last;
176             if (c.size() < b.size() + m)
177                 c.resize(b.size() + m);
178             for (int j = 0; j < (int)b.size(); j++)
179                 c[j + m] = c[j + m] - coef * b[j];
180             if (2 * l <= n)
181             {
182                 l = n + 1 - l;
183                 b = old;
184                 last = d;
185                 m = 1;
186             }
187             else
188                 ++m;
189         }
190         c.erase(c.begin());
191         for (auto &x : c)
192             x = Z(0) - x;
193         return c;
194     }
195
196     static Z recurrence(Poly init, const Poly &c, unsigned long long n)
197     {
198         int k = (int)c.size();
199         if (n < init.size())
200             return init[n];
201         if (!k)
202             return 0;
203         assert((int)init.size() >= k);
204         auto combine = [&](const Poly &a, const Poly &b)
205         {
206             Poly t(2 * k - 1);
207             for (int i = 0; i < k; i++)
208                 for (int j = 0; j < k; j++)
209                     t[i + j] = t[i + j] + a[i] * b[j];
210             for (int i = 2 * k - 2; i >= k; i--)
211                 for (int j = 1; j <= k; j++)

```

```
212         t[i - j] = t[i - j] + t[i] * c[j - 1];
213     t.resize(k);
214     return t;
215 };
216 Poly a(k), r(k);
217 r[0] = 1;
218 if (k == 1)
219     a[0] = c[0];
220 else
221     a[1] = 1;
222 for (; n; n >= 1, a = combine(a, a))
223     if (n & 1)
224         r = combine(r, a);
225 Z ans = 0;
226 for (int i = 0; i < k; i++)
227     ans = ans + r[i] * init[i];
228 return ans;
229 }
230 };
```

第四十一章 代数与数论进阶

41.1 模高斯消元、行列式、矩阵树定理

素数模数。solve(a,n) 输入增广矩阵，返回 consistent、rank、特解 particular 和零空间基 kernel。determinant(empty)=1。spanning_trees(n,edges) 使用 0-based 无向图，允许重边，忽略自环；返回生成树数模 p。高斯与行列式 $O(n^3)$ ，矩阵幂 $O(n^3 \log e)$ 。

```
1 template <int mod> struct LinearAlgebra
2 {
3     using Z = ModInt<mod>;
4     using Matrix = vector<vector<Z>>>;
5
6     struct Solution
7     {
8         bool consistent;
9         int rank;
10        vector<Z> particular;
11        Matrix kernel;
12    };
13
14    // Augmented m x (n+1), prime modulus. Empty system needs explicit n.
15    static Solution solve(Matrix a, int n)
16    {
17        int m = (int)a.size(), row = 0;
18        vector<int> where(n, -1);
19        for (const auto &v : a)
20            assert((int)v.size() == n + 1);
21        for (int col = 0; col < n && row < m; col++)
22        {
23            int p = row;
24            while (p < m && !a[p][col].v)
25                ++p;
26            if (p == m)
27                continue;
28            swap(a[p], a[row]);
29            Z inv = a[row][col].inv();
30            for (int j = col; j <= n; j++)
31                a[row][j] = a[row][j] * inv;
32            for (int i = 0; i < m; i++)
33                if (i != row && a[i][col].v)
34                {
35                    Z f = a[i][col];
36                    for (int j = col; j <= n; j++)
37                        a[i][j] = a[i][j] - f * a[row][j];
38                }
39            where[col] = row++;
40        }
41        for (int i = row; i < m; i++)
42            if (a[i][n].v)
43                return {false, row, {}, {}};
44        Solution ans{true, row, vector<Z>(n), {}};
```

```

45     for (int j = 0; j < n; j++)
46         if (where[j] != -1)
47             ans.particular[j] = a[where[j]][n];
48     for (int j = 0; j < n; j++)
49         if (where[j] == -1)
50         {
51             vector<Z> v(n);
52             v[j] = 1;
53             for (int k = 0; k < n; k++)
54                 if (where[k] != -1)
55                     v[k] = Z(0) - a[where[k]][j];
56             ans.kernel.push_back(v);
57         }
58     return ans;
59 }
60
61 static Z determinant(Matrix a)
62 {
63     int n = (int)a.size();
64     Z ans = 1;
65     for (int i = 0; i < n; i++)
66     {
67         assert((int)a[i].size() == n);
68         int p = i;
69         while (p < n && !a[p][i].v)
70             ++p;
71         if (p == n)
72             return 0;
73         if (p != i)
74         {
75             swap(a[p], a[i]);
76             ans = Z(0) - ans;
77         }
78         ans = ans * a[i][i];
79         Z inv = a[i][i].inv();
80         for (int j = i + 1; j < n; j++)
81         {
82             Z f = a[j][i] * inv;
83             for (int k = i; k < n; k++)
84                 a[j][k] = a[j][k] - f * a[i][k];
85         }
86     }
87     return ans;
88 }
89
90 static Matrix multiply(const Matrix &a, const Matrix &b)
91 {
92     assert(!a.empty() && !b.empty());
93     int n = a.size(), m = b[0].size(), k = b.size();
94     assert((int)a[0].size() == k);
95     Matrix c(n, vector<Z>(m));
96     for (int i = 0; i < n; i++)
97         for (int t = 0; t < k; t++)
98             for (int j = 0; j < m; j++)
99                 c[i][j] = c[i][j] + a[i][t] * b[t][j];
100     return c;

```

```

101     }
102
103     static Matrix power(Matrix a, unsigned long long e)
104     {
105         int n = a.size();
106         assert(n > 0 && (int)a[0].size() == n);
107         Matrix r(n, vector<Z>(n));
108         for (int i = 0; i < n; i++)
109             r[i][i] = 1;
110         for (; e >>= 1, a = multiply(a, a))
111             if (e & 1)
112                 r = multiply(r, a);
113         return r;
114     }
115
116     // Undirected multigraph, vertices 0..n-1; loops ignored.
117     static Z spanning_trees(int n, const vector<pair<int, int>> &edges)
118     {
119         assert(n > 0);
120         Matrix lap(n, vector<Z>(n));
121         for (auto [u, v] : edges)
122             if (u != v)
123             {
124                 lap[u][u] = lap[u][u] + 1;
125                 lap[v][v] = lap[v][v] + 1;
126                 lap[u][v] = lap[u][v] - 1;
127                 lap[v][u] = lap[v][u] - 1;
128             }
129         lap.pop_back();
130         for (auto &row : lap)
131             row.pop_back();
132         return determinant(lap);
133     }
134 };

```


41.2 杜教筛

构造预筛上限 $B \geq 1$, $\text{mertens}(n)$ 返回 μ 的前缀和, $\text{totient_sum}(n)$ 返回 φ 的前缀和 (int128)。约定 $0 \leq n \leq 10^{11}$, 按整除分块递归并缓存; 大输入需要合理 B , 不能把常数 B 当作性能保证。

```

1 struct DuJiao
2 {
3     using ll = long long;
4     using I = __int128_t;
5     int limit;
6     vector<ll> pmu, pphi;
7     unordered_map<ll, ll> mmu;
8     unordered_map<ll, I> mphi;
9
10    DuJiao(int limit) : limit(limit), pmu(limit + 1), pphi(limit + 1)
11    {
12        assert(limit >= 1);
13        LinearSieve s(limit);
14        for (int i = 1; i <= limit; i++)
15        {
16            pmu[i] = pmu[i - 1] + s.mu[i];
17            pphi[i] = pphi[i - 1] + s.phi[i];
18        }
19    }
20
21    ll mertens(ll n)
22    {
23        if (n <= limit)
24            return pmu[n];
25        if (mmu.count(n))
26            return mmu[n];
27        ll ans = 1;
28        for (ll l = 2, r; l <= n; l = r + 1)
29        {
30            r = n / (n / l);
31            ans -= (r - l + 1) * mertens(n / l);
32        }
33        return mmu[n] = ans;
34    }
35
36    I totient_sum(ll n)
37    {
38        if (n <= limit)
39            return pphi[n];
40        if (mphi.count(n))
41            return mphi[n];
42        I ans = I(n) * (n + 1) / 2;
43        for (ll l = 2, r; l <= n; l = r + 1)
44        {
45            r = n / (n / l);
46            ans -= I(r - l + 1) * totient_sum(n / l);
47        }
48        return mphi[n] = ans;
49    }
50 };

```

41.3 扩展 BSGS

依赖 number_theory.hpp 的求逆与快速幂。solve(a,b,m) 返回最小 $x \geq 0$ 使 $a^x \equiv b \pmod{m}$ ，无解返回 -1； $1 \leq m \leq 10^{12}$ ，a、b 可为完整 long long 范围，入口归一化，约定 $0^0=1$ 。先剥离 gcd 并逐次检查较小指数，再在互质剩余问题上 BSGS。baby 表对同余幂保留最小指数 j，按递增大步 i 检查区间，因此首个命中即最小解。单次期望时间 $O(\sqrt{m} + \log m)$ ，空间 $O(\sqrt{m})$ ，哈希表先 reserve；基于 unordered_map 的界为期望界。每次调用独立建表；底数 1 及零指数解直接返回。

```

1 struct DiscreteLog
2 {
3     using ll = long long;
4
5     // 1<=m<=1e12; Expected O(sqrt(m)) time, O(sqrt(m)) storage. Smallest x>=0, or -1.
6     static ll solve(ll a, ll b, ll m)
7     {
8         assert(m >= 1 && m <= 1000000000000LL);
9         a = (a % m + m) % m;
10        b = (b % m + m) % m;
11        if (m == 1 || b == 1)
12            return 0;
13        if (a == 1)
14            return -1;
15        ll offset = 0, k = 1;
16        for (ll g; (g = gcd(a, m)) > 1;)
17        {
18            if (b == k)
19                return offset;
20            if (b % g)
21                return -1;
22            b /= g;
23            m /= g;
24            k = (__int128)k * (a / g) % m;
25            ++offset;
26        }
27        ll target = (__int128)b * NumberTheory::inverse(k, m) % m;
28        ll step = sqrtl(m) + 1;
29        unordered_map<ll, ll> baby;
30        baby.reserve(step);
31        ll cur = 1 % m;
32        for (ll j = 0; j < step; j++)
33        {
34            if (!baby.count(cur))
35                baby[cur] = j;
36            cur = (__int128)cur * a % m;
37        }
38        ll inv = NumberTheory::inverse(NumberTheory::power(a, step, m), m);
39        cur = target;

```

```
40     for (ll i = 0; i <= step; i++)
41     {
42         auto it = baby.find(cur);
43         if (it != baby.end())
44             return offset + i * step + it->second;
45         cur = (__int128)cur * inv % m;
46     }
47     return -1;
48 }
49 };
```

第四十二章 计算几何

42.1 整数精确二维几何

坐标绝对值 $\leq 10^{12}$ ，点数 $\leq 10^6$ 。叉积、点积、线段相交、凸包、点在多边形、点在凸包、旋转卡壳直径。凸包逆时针，不保留共线内点，不重复首点。contains 返回 0 外部、1 边界、2 内部。convex_contains/diameter2 输入必须为本模板 strict hull。PolarLess 要求排除零向量。

```
1 struct IntegerGeometry
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     // Contract: |coordinate|<=1e12. Differences/products widened before arithmetic.
7     struct Point
8     {
9         ll x = 0, y = 0;
10
11         bool operator<(Point b) const
12         {
13             return tie(x, y) < tie(b.x, b.y);
14         }
15
16         bool operator==(Point b) const
17         {
18             return x == b.x && y == b.y;
19         }
20     };
21
22     static I cross(Point a, Point b, Point c)
23     {
24         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
25     }
26
27     static I dot(Point a, Point b, Point c)
28     {
29         return (I(b.x) - a.x) * (I(c.x) - a.x) + (I(b.y) - a.y) * (I(c.y) - a.y);
30     }
31
32     static I dist2(Point a, Point b)
33     {
34         return dot(a, b, b);
35     }
36
37     static int sign(I x)
38     {
39         return (x > 0) - (x < 0);
40     }
41
42     static bool on_segment(Point p, Point a, Point b)
43     {
44         return cross(a, b, p) == 0 && dot(p, a, b) <= 0;
```

```

45     }
46
47     static bool intersect(Point a, Point b, Point c, Point d)
48     {
49         int x = sign(cross(a, b, c)), y = sign(cross(a, b, d));
50         int z = sign(cross(c, d, a)), w = sign(cross(c, d, b));
51         if (!x && on_segment(c, a, b))
52             return true;
53         if (!y && on_segment(d, a, b))
54             return true;
55         if (!z && on_segment(a, c, d))
56             return true;
57         if (!w && on_segment(b, c, d))
58             return true;
59         return x * y < 0 && z * w < 0;
60     }
61
62     static vector<Point> hull(vector<Point> p)
63     {
64         sort(p.begin(), p.end());
65         p.erase(unique(p.begin(), p.end()), p.end());
66         int n = (int)p.size();
67         if (n < 3)
68             return p;
69         vector<Point> h;
70         for (Point v : p)
71         {
72             while (h.size() > 1 && cross(h[h.size() - 2], h.back(), v) <= 0)
73                 h.pop_back();
74             h.push_back(v);
75         }
76         int low = (int)h.size();
77         for (int i = n - 2; i >= 0; i--)
78         {
79             while ((int)h.size() > low && cross(h[h.size() - 2], h.back(), p[i]) <= 0)
80                 h.pop_back();
81             h.push_back(p[i]);
82         }
83         h.pop_back();
84         return h; // CCW, no repeated endpoint, no collinear interior points.
85     }
86
87     static I area2(const vector<Point> &p)
88     {
89         I ans = 0;
90         int n = (int)p.size();
91         for (int i = 0; i < n; i++)
92             ans += I(p[i].x) * p[(i + 1) % n].y - I(p[i].y) * p[(i + 1) % n].x;
93         return ans;
94     }
95
96     // 0 outside, 1 boundary, 2 inside. Simple polygon; either orientation.
97     static int contains(const vector<Point> &p, Point q)
98     {
99         int winding = 0, n = (int)p.size();
100        for (int i = 0; i < n; i++)

```

```

101     {
102         Point a = p[i], b = p[(i + 1) % n];
103         if (on_segment(q, a, b))
104             return 1;
105         I c = cross(a, b, q);
106         if (a.y <= q.y && b.y > q.y && c > 0)
107             ++winding;
108         if (a.y > q.y && b.y <= q.y && c < 0)
109             --winding;
110     }
111     return winding ? 2 : 0;
112 }
113
114 // Input must be the strict CCW hull above. O(log n).
115 static int convex_contains(const vector<Point> &p, Point q)
116 {
117     int n = (int)p.size();
118     if (n < 3)
119         return contains(p, q);
120     I a = cross(p[0], p[1], q), b = cross(p[0], p[n - 1], q);
121     if (a < 0 || b > 0)
122         return 0;
123     if (!a)
124         return on_segment(q, p[0], p[1]) ? 1 : 0;
125     if (!b)
126         return on_segment(q, p[0], p[n - 1]) ? 1 : 0;
127     int l = 1, r = n - 1;
128     while (l + 1 < r)
129     {
130         int m = (l + r) / 2;
131         if (cross(p[0], p[m], q) >= 0)
132             l = m;
133         else
134             r = m;
135     }
136     I c = cross(p[l], p[r], q);
137     return c < 0 ? 0 : c == 0 ? 1 : 2;
138 }
139
140 static I diameter2(const vector<Point> &p)
141 {
142     int n = (int)p.size();
143     if (n < 2)
144         return 0;
145     if (n == 2)
146         return dist2(p[0], p[1]);
147     I ans = 0;
148     int j = 1;
149     for (int i = 0; i < n; i++)
150     {
151         int k = (i + 1) % n;
152         while (cross(p[i], p[k], p[(j + 1) % n]) > cross(p[i], p[k], p[j]))
153             j = (j + 1) % n;
154         ans = max({ans, dist2(p[i], p[j]), dist2(p[k], p[j])});
155     }
156     return ans;

```

```

157 }
158
159 struct PolarLess
160 {
161     static int half(Point p)
162     {
163         return p.y < 0 || (p.y == 0 && p.x < 0);
164     }
165
166     bool operator()(Point a, Point b) const
167     {
168         if (half(a) != half(b))
169             return half(a) < half(b);
170         I c = cross({0, 0}, a, b);
171         if (c)
172             return c > 0;
173         return dist2({0, 0}, a) < dist2({0, 0}, b);
174     }
175 };
176 };

```

42.2 浮点点线圆构造

long double, 显式 eps 默认 10^{-12} 。要求有限输入及适当尺度；接近相切时按容差分类，不能提供任意实数输入的精确拓扑保证。Result 区分 none/one/two/infinite/degenerate。圆圆交点仅在圆心和正半径完全相同时返回 infinite；零半径圆按点处理，不把近同心圆当作重合。圆相关交点容差按局部距离和半径缩放，不使用固定的单位长度下限。零长度线返回 degenerate；投影与点到线段距离将零线段视为点。overlap 返回交叠面积，小圆弓用级数避免三角函数相减消失。坐标缩放和误差预算仍须按题面复核。

```

1 struct RealGeometry
2 {
3     using R = long double;
4     // Tolerance is explicit, not used in sorting. Requires finite, well-scaled input.
5     R eps;
6
7     RealGeometry(R eps = 1e-12L) : eps(eps)
8     {
9         assert(eps > 0);
10    }
11
12    struct Point
13    {
14        R x = 0, y = 0;
15
16        Point operator+(Point b) const
17        {
18            return {x + b.x, y + b.y};
19        }
20
21        Point operator-(Point b) const
22        {
23            return {x - b.x, y - b.y};
24        }
25    }

```

```

26     Point operator*(R k) const
27     {
28         return {x * k, y * k};
29     }
30
31     Point operator/(R k) const
32     {
33         return {x / k, y / k};
34     }
35 };
36
37 struct Circle
38 {
39     Point o;
40     R r;
41 };
42 enum class Kind
43 {
44     none,
45     one,
46     two,
47     infinite,
48     degenerate
49 };
50
51 struct Result
52 {
53     Kind kind;
54     vector<Point> p;
55 };
56
57 static R dot(Point a, Point b)
58 {
59     return a.x * b.x + a.y * b.y;
60 }
61
62 static R cross(Point a, Point b)
63 {
64     return a.x * b.y - a.y * b.x;
65 }
66
67 static R norm(Point a)
68 {
69     return hypotl(a.x, a.y);
70 }
71
72 static Point perp(Point a)
73 {
74     return {-a.y, a.x};
75 }
76
77 Point projection(Point p, Point a, Point b) const
78 {
79     Point v = b - a;
80     R d = dot(v, v);
81     if (d == 0)

```



```

82     return a;
83     return a + v * (dot(p - a, v) / d);
84 }
85
86 R segment_distance(Point p, Point a, Point b) const
87 {
88     Point v = b - a;
89     R d = dot(v, v);
90     if (d == 0)
91         return norm(p - a);
92     R t = clamp(dot(p - a, v) / d, R(0), R(1));
93     return norm(p - (a + v * t));
94 }
95
96 Result line_intersection(Point a, Point b, Point c, Point d) const
97 {
98     Point u = b - a, v = d - c;
99     R nu = norm(u), nv = norm(v);
100    if (nu == 0 || nv == 0)
101        return {Kind::degenerate, {}};
102    R det = cross(u, v);
103    if (fabsl(det) <= eps * nu * nv)
104        return {fabsl(cross(c - a, u)) <= eps * nu * max(R(1), norm(c - a))
105                ? Kind::infinite
106                : Kind::none,
107                {}};
108    return {Kind::one, {a + u * (cross(c - a, v) / det)}};
109 }
110
111 Result line_circle(Point a, Point b, Circle c) const
112 {
113     assert(c.r >= 0);
114     Point v = b - a;
115     R len = norm(v);
116     if (len == 0)
117         return {Kind::degenerate, {}};
118     Point h = projection(c.o, a, b);
119     R d = norm(h - c.o), tol = eps * max(d, c.r);
120     if (d > c.r + tol)
121         return {Kind::none, {}};
122     if (fabsl(d - c.r) <= tol)
123         return {Kind::one, {h}};
124     R t = sqrtl(max(R(0), (c.r - d) * (c.r + d)));
125     Point w = v * (t / len);
126     return {Kind::two, {h - w, h + w}};
127 }
128
129 Result circles(Circle a, Circle b) const
130 {
131     assert(a.r >= 0 && b.r >= 0);
132     Point v = b.o - a.o;
133     R d = norm(v);
134     R tol = eps * max({d, a.r, b.r});
135     // Infinite intersections require identical positive-radius circles.
136     if (d == 0)
137         {

```

```

138     if (a.r != b.r)
139         return {Kind::none, {}};
140     if (a.r == 0)
141         return {Kind::one, {a.o}};
142     return {Kind::infinite, {}};
143 }
144 if (a.r == 0 && b.r == 0)
145     return {Kind::none, {}};
146 if (a.r == 0 || b.r == 0)
147 {
148     R radius = max(a.r, b.r);
149     if (fabsl(d - radius) > tol)
150         return {Kind::none, {}};
151     return {Kind::one, {a.r == 0 ? a.o : b.o}};
152 }
153 if (d > a.r + b.r + tol || d < fabsl(a.r - b.r) - tol)
154     return {Kind::none, {}};
155 R x = (d * d + (a.r - b.r) * (a.r + b.r)) / (2 * d);
156 Point h = a.o + v * (x / d);
157 // Near-concentric circles must not be mistaken for internal tangency.
158 if (fabsl(x) > a.r + tol)
159     return {Kind::none, {}};
160 if (fabsl(fabsl(x) - a.r) <= tol)
161     return {Kind::one, {h}};
162 R y = sqrtl(max(R(0), (a.r - x) * (a.r + x)));
163 Point w = perp(v) * (y / d);
164 return {Kind::two, {h - w, h + w}};
165 }
166
167 R overlap(Circle a, Circle b) const
168 {
169     assert(a.r >= 0 && b.r >= 0);
170     R d = norm(a.o - b.o), pi = acosl(-1.L);
171     if (d >= a.r + b.r)
172         return 0;
173     if (d <= fabsl(a.r - b.r))
174         return pi * min(a.r, b.r) * min(a.r, b.r);
175     R difference = (a.r - b.r) * (a.r + b.r);
176     R x = acosl(clamp((d * d + difference) / (2 * d * a.r), R(-1), R(1)));
177     R y = acosl(clamp((d * d - difference) / (2 * d * b.r), R(-1), R(1)));
178     auto segment = [](R angle)
179     {
180         if (angle < 1e-3L)
181         {
182             R z = angle * angle;
183             return angle * z * (R(2) / 3 + z * (-R(2) / 15 + z * R(4) / 315));
184         }
185         return angle - sinl(2 * angle) / 2;
186     };
187     return a.r * a.r * segment(x) + b.r * b.r * segment(y);
188 }
189 };

```

第四十三章 优化与可持久化

43.1 Max-plus 矩阵幂

$a[i][j]$ 为转移收益，不存在的边为 neg ； $power(k)$ 求恰好 k 步的最优转移，单位矩阵对角为 0。所有有限中间值绝对值小于 2^{119} 。 $O(n^3 \log k)$ ；2025 成都 B 的转移类型，具体题目的建模仍需调用者完成。

```
1 struct MaxPlusMatrix
2 {
3     using I = __int128_t;
4     static constexpr I neg = -(I(1) << 120);
5     int n;
6     vector<vector<I>> a;
7
8     MaxPlusMatrix(int n, bool identity = false) : n(n), a(n, vector<I>(n, neg))
9     {
10         if (identity)
11             for (int i = 0; i < n; i++)
12                 a[i][i] = 0;
13     }
14
15     MaxPlusMatrix operator*(const MaxPlusMatrix &b) const
16     {
17         assert(n == b.n);
18         MaxPlusMatrix c(n);
19         for (int i = 0; i < n; i++)
20             for (int k = 0; k < n; k++)
21                 if (a[i][k] != neg)
22                     for (int j = 0; j < n; j++)
23                         if (b.a[k][j] != neg)
24                             c.a[i][j] = max(c.a[i][j], a[i][k] + b.a[k][j]);
25         return c;
26     }
27
28     MaxPlusMatrix power(unsigned long long e) const
29     {
30         MaxPlusMatrix b = *this, r(n, true);
31         for (; e >>= 1, b = b * b)
32             if (e & 1)
33                 r = r * b;
34         return r;
35     }
36 };
```

43.2 离散李超树与最优直线编号

构造传入所有可能查询的横坐标， $add(\{k,b,id\})$ 插入直线， $query(x)$ 返回最小值与最小编号，要求 x 在预登记坐标内且至少有一条线。整数求值用 $int128$ ， k 、 b 、 x 为有符号 64 位且中间值可表示。单次 $O(\log n)$ 。

```
1 struct LiChao
2 {
```

```

3   using ll = long long;
4   using I = __int128_t;
5
6   struct Line
7   {
8       ll k = 0, b = 0;
9       int id = -1;
10
11       I value(ll x) const
12       {
13           return I(k) * x + b;
14       }
15   };
16
17   struct Node
18   {
19       Line line;
20       int left = -1, right = -1;
21   };
22
23   vector<ll> xs;
24   vector<Node> a{Node{}};
25
26   LiChao(vector<ll> x) : xs(x)
27   {
28       sort(xs.begin(), xs.end());
29       xs.erase(unique(xs.begin(), xs.end()), xs.end());
30       assert(!xs.empty());
31   }
32
33   bool better(Line u, Line v, ll x) const
34   {
35       return v.id == -1 ||
36              (u.id != -1 && pair{u.value(x), u.id} < pair{v.value(x), v.id});
37   }
38
39   void add(Line line)
40   {
41       assert(line.id >= 0);
42       add(0, 0, (int)xs.size() - 1, line);
43   }
44
45   void add(int p, int l, int r, Line line)
46   {
47       int m = (l + r) / 2;
48       if (better(line, a[p].line, xs[m]))
49           swap(line, a[p].line);
50       if (l == r || line.id == -1)
51           return;
52       bool left = better(line, a[p].line, xs[l]);
53       bool right = better(line, a[p].line, xs[r]);
54       if (!left && !right)
55           return;
56       int next = left ? a[p].left : a[p].right;
57       if (next == -1)
58           {

```

```

59     next = (int)a.size();
60     a.push_back(Node{});
61     if (left)
62         a[p].left = next;
63     else
64         a[p].right = next;
65 }
66 if (left)
67     add(next, l, m, line);
68 else
69     add(next, m + 1, r, line);
70 }
71
72 pair<I, int> query(ll x) const
73 {
74     int idx = lower_bound(xs.begin(), xs.end(), x) - xs.begin();
75     assert(idx < (int)xs.size() && xs[idx] == x);
76     int p = 0, l = 0, r = (int)xs.size() - 1;
77     Line best;
78     while (p != -1)
79     {
80         if (better(a[p].line, best, x))
81             best = a[p].line;
82         if (l == r)
83             break;
84         int m = (l + r) / 2;
85         if (idx <= m)
86         {
87             p = a[p].left;
88             r = m;
89         }
90         else
91         {
92             p = a[p].right;
93             l = m + 1;
94         }
95     }
96     assert(best.id != -1);
97     return {best.value(x), best.id};
98 }
99 };

```

43.3 主席树：静态区间第 k 小

构造传入 0-based 数组， $\text{kth}(l, r, k)$ 的区间使用 1-based 闭区间；要求 $1 \leq k \leq r - l + 1$ 。离散化后每个前缀一个版本，单次 $O(\log n)$ ，总空间 $O(n \log n)$ 。

```

1 struct PersistentKth
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     vector<Node> t{Node{}};

```

```

9     vector<int> root{0};
10    vector<long long> vals;
11
12    PersistentKth(const vector<long long> &a) : vals(a)
13    {
14        sort(vals.begin(), vals.end());
15        vals.erase(unique(vals.begin(), vals.end()), vals.end());
16        for (long long x : a)
17        {
18            int p = lower_bound(vals.begin(), vals.end(), x) - vals.begin();
19            root.push_back(add(root.back(), 0, (int)vals.size() - 1, p));
20        }
21    }
22
23    int add(int old, int l, int r, int x)
24    {
25        int p = t.size();
26        Node copy = t[old];
27        t.push_back(copy);
28        ++t[p].sum;
29        if (l < r)
30        {
31            int m = (l + r) / 2;
32            if (x <= m)
33            {
34                int v = add(t[old].l, l, m, x);
35                t[p].l = v;
36            }
37            else
38            {
39                int v = add(t[old].r, m + 1, r, x);
40                t[p].r = v;
41            }
42        }
43        return p;
44    }
45
46    long long kth(int l, int r, int k) const
47    {
48        assert(1 <= l && l <= r && r < (int)root.size() && 1 <= k && k <= r - l + 1);
49        int u = root[l - 1], v = root[r], L = 0, R = (int)vals.size() - 1;
50        while (L < R)
51        {
52            int m = (L + R) / 2, c = t[t[v].l].sum - t[t[u].l].sum;
53            if (k <= c)
54            {
55                u = t[u].l;
56                v = t[v].l;
57                R = m;
58            }
59            else

```

```
60         {
61             k -= c;
62             u = t[u].r;
63             v = t[v].r;
64             L = m + 1;
65         }
66     }
67     return vals[L];
68 }
69 };
```

第四十四章 可持久化数组

44.1 可持久化数组：历史版本分叉与单点修改

PersistentArray(a,changes=0), a 是非空 0-based 初值数组; 第 0 版为初始状态。set(version,pos,value) 基于任意已有版本修改 1-based 位置, 创建并返回新版本编号; copy(version) 以 $O(1)$ 时间复制版本; query(version,pos) 只读且不自动创建版本。values 支持 long long 全范围。每次修改复制根到叶的节点, 旧版本不变; build 为内部函数, 不可覆盖历史节点。建树 $O(n)$, 修改及查询 $O(\log n)$, 初始 $2n-1$ 个节点, 每次修改至多 $\text{ceil}(\log_2(n))+1$ 个; 总空间 $O(n + \text{修改数} * \log n + \text{版本数})$ 。changes 仅用于预留节点池, 低估后可扩容, 所有外部引用应使用下标, 不保存 vector 元素引用。节点下标用 int, 节点总量须可表示。

```
1 struct PersistentArray
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0;
8         ll value = 0;
9     };
10
11     int n;
12     vector<Node> t{Node{}};
13     vector<int> root;
14
15     PersistentArray(const vector<ll> &a, int changes = 0) : n(a.size())
16     {
17         assert(n > 0 && changes >= 0);
18         int levels = 1;
19         for (int x = n - 1; x; x >>= 1)
20             levels++;
21         t.reserve(2ULL * n + 1ULL * changes * levels + 1);
22         root.push_back(build(1, n, a));
23     }
24
25     int build(int l, int r, const vector<ll> &a)
26     {
27         int p = t.size();
28         t.push_back(Node{});
29         if (l == r)
30             t[p].value = a[l - 1];
31         else
32         {
33             int m = (l + r) / 2;
34             int left = build(l, m, a);
35             int right = build(m + 1, r, a);
36             t[p].l = left;
37             t[p].r = right;
38         }
39         return p;
40     }
41 }
```



```

42 int modify(int old, int l, int r, int pos, ll value)
43 {
44     int p = t.size();
45     Node copy = t[old];
46     t.push_back(copy);
47     if (l == r)
48         t[p].value = value;
49     else
50     {
51         int m = (l + r) / 2;
52         if (pos <= m)
53         {
54             int child = modify(t[old].l, l, m, pos, value);
55             t[p].l = child;
56         }
57         else
58         {
59             int child = modify(t[old].r, m + 1, r, pos, value);
60             t[p].r = child;
61         }
62     }
63     return p;
64 }
65
66 // Create a new version from any existing version. Positions are 1-based.
67 int set(int version, int pos, ll value)
68 {
69     assert(0 <= version && version < (int)root.size());
70     assert(1 <= pos && pos <= n);
71     int p = modify(root[version], 1, n, pos, value);
72     root.push_back(p);
73     return (int)root.size() - 1;
74 }
75
76 int copy(int version)
77 {
78     assert(0 <= version && version < (int)root.size());
79     root.push_back(root[version]);
80     return (int)root.size() - 1;
81 }
82
83 ll query(int version, int pos) const
84 {
85     assert(0 <= version && version < (int)root.size());
86     assert(1 <= pos && pos <= n);
87     int p = root[version], l = 1, r = n;
88     while (l < r)
89     {
90         int m = (l + r) / 2;
91         if (pos <= m)
92         {
93             p = t[p].l;
94             r = m;

```

```
95         }
96         else
97         {
98             p = t[p].r;
99             l = m + 1;
100        }
101    }
102    return t[p].value;
103 }
104 };
```

第四十五章 树上路径第 k 小

45.1 树上路径点权第 k 小

TreePathKth(a) 传入非空 0-based 权值数组，节点编号 1..n；add(u,v) 加无向边，要求恰好构成一棵树。build(root=1) 递归 DFS 建立根到各节点的频数版本与倍增表，可换根重建，旧版本自动清空。kth(u,v,k) 返回简单路径上第 k 小的点权，包含两个端点，每个节点计一次； $1 \leq k \leq$ 路径点数。lca(u,v) 返回当前预处理根下的最近公共祖先。使用 $\text{root}[u] + \text{root}[v] - \text{root}[\text{lca}] - \text{root}[\text{parent}(\text{lca})]$ ，编号 0 为全零版本，避免少算或重复算 LCA。支持重复值与完整 long long 权值，内部只统计频数；权值和树结构在查询阶段保持不变。预处理 $O(n \log n)$ ，每次查询 $O(\log n)$ ，空间 $O(n \log n)$ 。递归深度最坏 $O(n)$ ，保持递归写法。

```
1 struct TreePathKth
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, sum = 0;
8     };
9
10    int n, lg;
11    vector<vector<int>> g, up;
12    vector<int> root, dep;
13    vector<ll> value, vals;
14    vector<Node> t;
15
16    TreePathKth(const vector<ll> &a)
17        : n(a.size()), g(n + 1), root(n + 1), dep(n + 1), value(a)
18    {
19        assert(n > 0);
20        lg = 1;
21        while ((1LL << lg) <= n)
22            lg++;
23        up.assign(lg, vector<int>(n + 1));
24    }
25
26    void add(int u, int v)
27    {
28        g[u].push_back(v);
29        g[v].push_back(u);
30    }
31
32    int insert(int old, int l, int r, int x)
33    {
34        int p = t.size();
35        Node copy = t[old];
36        t.push_back(copy);
37        t[p].sum++;
38        if (l == r)
39            return p;
40        int m = (l + r) / 2;
41        if (x <= m)
```

```

42     {
43         int child = insert(t[old].l, l, m, x);
44         t[p].l = child;
45     }
46     else
47     {
48         int child = insert(t[old].r, m + 1, r, x);
49         t[p].r = child;
50     }
51     return p;
52 }
53
54 void dfs(int u, int p)
55 {
56     up[0][u] = p;
57     for (int j = 1; j < lg; j++)
58         up[j][u] = up[j - 1][up[j - 1][u]];
59     int x = lower_bound(vals.begin(), vals.end(), value[u - 1]) - vals.begin();
60     root[u] = insert(root[p], 0, (int)vals.size() - 1, x);
61     for (int v : g[u])
62     {
63         if (v == p)
64             continue;
65         dep[v] = dep[u] + 1;
66         dfs(v, u);
67     }
68 }
69
70 void build(int r = 1)
71 {
72     assert(1 <= r && r <= n);
73     vals = value;
74     sort(vals.begin(), vals.end());
75     vals.erase(unique(vals.begin(), vals.end()), vals.end());
76     t.assign(1, Node{});
77     dep[r] = 0;
78     dfs(r, 0);
79 }
80
81 int lca(int u, int v) const
82 {
83     if (dep[u] < dep[v])
84         swap(u, v);
85     int diff = dep[u] - dep[v];
86     for (int j = 0; j < lg; j++)
87         if (diff >> j & 1)
88             u = up[j][u];
89     if (u == v)
90         return u;
91     for (int j = lg - 1; j >= 0; j--)
92         if (up[j][u] != up[j][v])
93         {
94             u = up[j][u];
95             v = up[j][v];
96         }
97     return up[0][u];

```

```

98     }
99
100     ll kth(int u, int v, int k) const
101     {
102         int z = lca(u, v);
103         assert(1 <= k && k <= dep[u] + dep[v] - 2 * dep[z] + 1);
104         int a = root[u], b = root[v], c = root[z], d = root[up[0][z]];
105         int l = 0, r = (int)vals.size() - 1;
106         while (l < r)
107         {
108             int count = t[t[a].l].sum + t[t[b].l].sum - t[t[c].l].sum - t[t[d].l].sum;
109             int m = (l + r) / 2;
110             if (k <= count)
111             {
112                 a = t[a].l;
113                 b = t[b].l;
114                 c = t[c].l;
115                 d = t[d].l;
116                 r = m;
117             }
118             else
119             {
120                 k -= count;
121                 a = t[a].r;
122                 b = t[b].r;
123                 c = t[c].r;
124                 d = t[d].r;
125                 l = m + 1;
126             }
127         }
128         return vals[l];
129     }
130 };

```

第四十六章 动态区间顺序统计

46.1 动态区间第 k 小 (树状数组套权值线段树)

DynamicKth(a,future) 传入非空 0-based 初值, future 包含所有将来可能赋入的新值; 内部自动合并初值并离散化。set(pos,value) 单点赋值, kth(l,r,k) 查询闭区间第 k 小, 下标均为 1-based; $1 \leq k \leq r-l+1$ 。允许重复值与完整 long long 权值; 新值必须已在离散化集合内, 操作可按原顺序执行, 不保留历史版本。每个树状数组节点对应一棵可变权值线段树; 删除旧值后插入新值。空节点回收到 free_head 链, 重复修改不会按操作次数累计泄漏节点。设 D 为不同值数, $S=n+\text{future.size()}$, 预处理 $O(S \log S+n \log n \log(D+1))$, 单次修改及查询 $O(\log n \log(D+1))$ 。节点数上界 $B=(\text{ceil}(\log_2(D))+1) \cdot \sum \text{lowbit}(i)$, 求和 $i=1..n$; 空间 $O(S+n+B)$ 。update、modify、id 和节点池均为内部接口; 不能手动减去不存在的值。

```
1 struct DynamicKth
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, sum = 0;
8     };
9
10    int n, free_head = 0;
11    vector<int> bit, rank;
12    vector<ll> vals;
13    vector<Node> t{Node{}};
14
15    DynamicKth(const vector<ll> &a, vector<ll> future = {})
16        : n(a.size()), bit(n + 1), rank(n + 1)
17    {
18        assert(n > 0);
19        vals = future;
20        vals.insert(vals.end(), a.begin(), a.end());
21        sort(vals.begin(), vals.end());
22        vals.erase(unique(vals.begin(), vals.end()), vals.end());
23        for (int i = 1; i <= n; i++)
24        {
25            rank[i] = id(a[i - 1]);
26            modify(i, rank[i], 1);
27        }
28    }
29
30    int id(ll value) const
31    {
32        int p = lower_bound(vals.begin(), vals.end(), value) - vals.begin();
33        assert(p < (int)vals.size() && vals[p] == value);
34        return p;
35    }
36
37    int new_node()
38    {
39        if (!free_head)
```

```

40     {
41         t.push_back(Node{});
42         return (int)t.size() - 1;
43     }
44     int p = free_head;
45     free_head = t[p].l;
46     t[p] = Node{};
47     return p;
48 }
49
50 int update(int p, int l, int r, int x, int delta)
51 {
52     if (!p)
53         p = new_node();
54     t[p].sum += delta;
55     if (l < r)
56     {
57         int m = (l + r) / 2;
58         if (x <= m)
59         {
60             int child = update(t[p].l, l, m, x, delta);
61             t[p].l = child;
62         }
63         else
64         {
65             int child = update(t[p].r, m + 1, r, x, delta);
66             t[p].r = child;
67         }
68     }
69     if (!t[p].sum)
70     {
71         t[p].l = free_head;
72         free_head = p;
73         return 0;
74     }
75     return p;
76 }
77
78 void modify(int pos, int x, int delta)
79 {
80     for (int i = pos; i <= n; i += i & -i)
81         bit[i] = update(bit[i], 0, (int)vals.size() - 1, x, delta);
82 }

```

```

84 void set(int pos, ll value)
85 {
86     assert(1 <= pos && pos <= n);
87     int x = id(value);
88     if (rank[pos] == x)
89         return;
90     modify(pos, rank[pos], -1);
91     modify(pos, x, 1);
92     rank[pos] = x;
93 }
94
95 ll kth(int l, int r, int k) const
96 {
97     assert(1 <= l && l <= r && r <= n && 1 <= k && k <= r - l + 1);
98     vector<int> a, b;
99     for (int i = r; i; i -= i & -i)
100         a.push_back(bit[i]);
101     for (int i = l - 1; i; i -= i & -i)
102         b.push_back(bit[i]);
103     int L = 0, R = (int)vals.size() - 1;
104     while (L < R)
105     {
106         int count = 0;
107         for (int p : a)
108             count += t[t[p].l].sum;
109         for (int p : b)
110             count -= t[t[p].l].sum;
111         bool left = k <= count;
112         if (!left)
113             k -= count;
114         for (int &p : a)
115             p = left ? t[p].l : t[p].r;
116         for (int &p : b)
117             p = left ? t[p].l : t[p].r;
118         int m = (L + R) / 2;
119         if (left)
120             R = m;
121         else
122             L = m + 1;
123     }
124     return vals[L];
125 }
126 };

```


第四十七章 区间不同数统计

47.1 静态区间不同数计数（前驱位置主席树）

PersistentDistinct(a) 传入非空 0-based 数组，query(l,r) 返回闭区间内不同值的个数， $1 \leq l \leq r \leq n$ 。支持重复值与完整 long long 值域；建成后不支持改值，查询顺序任意。令 prev[i] 为同值上次出现的位置，未出现为 0；root[i] 在值域 0..n-1 中统计前 i 个 prev。区间内每种值恰有一次 prev[i]<l，因此答案为 root[r]-root[l-1] 中位置 0..l-1 的频数和。每个元素只复制一条节点路径，预处理 $O(n \log n)$ ，查询 $O(\log n)$ ，空间 $O(n \log n)$ 。动态版预留 $n \cdot (\text{ceil}(\log_2(n))+1)+1$ 个节点，避免扩容峰值；root 和节点池为内部数据。

```
1 struct PersistentDistinct
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     int n;
9     vector<Node> t{Node{}};
10    vector<int> root;
11
12    PersistentDistinct(const vector<long long> &a) : n(a.size()), root(n + 1)
13    {
14        assert(n > 0);
15        int height = 1;
16        while ((1LL << (height - 1)) < n)
17            height++;
18        t.reserve(size_t(n) * height + 1);
19        vector<long long> vals = a;
20        sort(vals.begin(), vals.end());
21        vals.erase(unique(vals.begin(), vals.end()), vals.end());
22        vector<int> last(vals.size());
23        for (int i = 1; i <= n; i++)
24        {
25            int x = lower_bound(vals.begin(), vals.end(), a[i - 1]) - vals.begin();
26            root[i] = insert(root[i - 1], 0, n - 1, last[x]);
27            last[x] = i;
28        }
29    }
30
31    int insert(int old, int l, int r, int x)
32    {
33        int p = t.size();
34        Node copy = t[old];
35        t.push_back(copy);
36        t[p].sum++;
37        if (l < r)
38        {
39            int m = (l + r) / 2;
40            if (x <= m)
41            {
42                int child = insert(t[old].l, l, m, x);
```

```

43         t[p].l = child;
44     }
45     else
46     {
47         int child = insert(t[old].r, m + 1, r, x);
48         t[p].r = child;
49     }
50 }
51 return p;
52 }
53
54 int query(int l, int r) const
55 {
56     assert(1 <= l && l <= r && r <= n);
57     int u = root[l - 1], v = root[r];
58     int L = 0, R = n - 1, answer = 0;
59     while (L < R)
60     {
61         int m = (L + R) / 2;
62         if (l - 1 <= m)
63         {
64             u = t[u].l;
65             v = t[v].l;
66             R = m;
67         }
68         else
69         {
70             answer += t[t[v].l].sum - t[t[u].l].sum;
71             u = t[u].r;
72             v = t[v].r;
73             L = m + 1;
74         }
75     }
76     return answer + t[v].sum - t[u].sum;
77 }
78 };

```

第四十八章 图论进阶

48.1 矩形二分图最小权匹配

`solve(cost)` 输入 $n \times m$ 完整权值矩阵, $0 \leq n \leq m$; 每行匹配不同列, 返回最小费用及列号 (0-based)。允许负费用, 使用 int128 势能; $O(n^2m)$ 。禁止边不能用未经证明的有限大常数随意替代, 当前接口仅表示完整二分图。

```
1 struct Hungarian
2 {
3     using I = __int128_t;
4
5     // Minimum cost injection from n rows to m columns, n<=m. Zero-based matrix.
6     static pair<I, vector<int>> solve(const vector<vector<long long>> &cost)
7     {
8         int n = cost.size();
9         if (!n)
10             return {0, {}};
11         int m = cost[0].size();
12         assert(n <= m);
13         vector<I> u(n + 1), v(m + 1);
14         vector<int> p(m + 1), way(m + 1);
15         const I inf = I(1) << 120;
16         for (int i = 1; i <= n; i++)
17         {
18             assert((int)cost[i - 1].size() == m);
19             p[0] = i;
20             int j0 = 0;
21             vector<I> dis(m + 1, inf);
22             vector<bool> used(m + 1);
23             do
24             {
25                 used[j0] = true;
26                 int i0 = p[j0], j1 = 0;
27                 I delta = inf;
28                 for (int j = 1; j <= m; j++)
29                     if (!used[j])
30                     {
31                         I cur = I(cost[i0 - 1][j - 1]) - u[i0] - v[j];
32                         if (cur < dis[j])
33                         {
34                             dis[j] = cur;
35                             way[j] = j0;
36                         }
37                         if (dis[j] < delta)
38                         {
39                             delta = dis[j];
40                             j1 = j;
41                         }
42                     }
43                 for (int j = 0; j <= m; j++)
44                     if (used[j])
45                     {
```

```

46         u[p[j]] += delta;
47         v[j] -= delta;
48     }
49     else
50         dis[j] -= delta;
51     j0 = j1;
52 } while (p[j0]);
53 do
54 {
55     int j1 = way[j0];
56     p[j0] = p[j1];
57     j0 = j1;
58 } while (j0);
59 }
60 vector<int> ans(n);
61 for (int j = 1; j <= m; j++)
62     if (p[j])
63         ans[p[j] - 1] = j - 1;
64 return {-v[0], ans};
65 }
66 };

```

48.2 有向最小树形图

`solve(n, root, edges)`, 点号 $0..n-1$, $n \geq 1$; 返回以 `root` 为根向外可达的最小树形图费用, 无解 `nullopt`。允许负权、重边、自环 (自环不选); $O(VE)$ 。当前仅返回费用, 不恢复边方案。所有中间值须远小于 2^{120} 。

```

1 struct Arborescence
2 {
3     using I = __int128_t;
4
5     struct Edge
6     {
7         int u, v;
8         I w;
9     };
10
11     // Directed minimum rooted OUT-arborescence cost, 0-based, or nullopt.
12     static optional<I> solve(int n, int root, vector<Edge> e)
13     {
14         const I inf = I(1) << 120;
15         I ans = 0;
16         while (true)
17         {
18             vector<I> in(n, inf);
19             vector<int> pre(n, -1), id(n, -1), vis(n, -1);
20             for (auto a : e)
21                 if (a.u != a.v && a.w < in[a.v])
22                 {
23                     in[a.v] = a.w;
24                     pre[a.v] = a.u;
25                 }
26             in[root] = 0;
27             for (I x : in)
28                 {

```

```

29         if (x == inf)
30             return nullopt;
31         ans += x;
32     }
33     int cnt = 0;
34     for (int i = 0; i < n; i++)
35     {
36         int u = i;
37         while (vis[u] != i && id[u] == -1 && u != root)
38         {
39             vis[u] = i;
40             u = pre[u];
41         }
42         if (u != root && id[u] == -1)
43         {
44             id[u] = cnt;
45             for (int v = pre[u]; v != u; v = pre[v])
46                 id[v] = cnt;
47             ++cnt;
48         }
49     }
50     if (!cnt)
51         return ans;
52     for (int i = 0; i < n; i++)
53         if (id[i] == -1)
54             id[i] = cnt++;
55     for (auto &a : e)
56     {
57         int v = a.v;
58         a.u = id[a.u];
59         a.v = id[a.v];
60         if (a.u != a.v)
61             a.w -= in[v];
62     }
63     root = id[root];
64     n = cnt;
65 }
66 }
67 };

```

48.3 无向全局最小割

`solve(w)`, $n \geq 2$; w 是对称非负权矩阵, 重边需先累加, 主对角置零。返回割权与一侧顶点编号 (0-based), 允许不连通图。 $O(n^3)$, 总权值小于 2^{120} 。

```

1 struct StoerWagner
2 {
3     using I = __int128_t;
4
5     // Symmetric nonnegative adjacency matrix; returns weight and one cut side.
6     static pair<I, vector<int>> solve(vector<vector<I>> w)
7     {
8         int n = w.size();
9         assert(n >= 2);
10        vector<int> active(n);
11        iota(active.begin(), active.end(), 0);
12        vector<vector<int>> group(n);
13        for (int i = 0; i < n; i++)
14            group[i] = {i};
15        I best = I(1) << 120;
16        vector<int> side;
17        while (active.size() > 1)
18        {
19            vector<I> dis(n);
20            vector<bool> used(n);
21            int prev = -1;
22            for (int step = 0; step < (int)active.size(); step++)
23            {
24                int sel = -1;
25                for (int v : active)
26                    if (!used[v] && (sel == -1 || dis[v] > dis[sel]))
27                        sel = v;
28                if (step + 1 == (int)active.size())
29                {
30                    if (dis[sel] < best)
31                    {
32                        best = dis[sel];
33                        side = group[sel];
34                    }
35                    for (int v : active)
36                        if (v != prev && v != sel)
37                            w[prev][v] = w[v][prev] = w[prev][v] + w[sel][v];
38                    group[prev].insert(
39                        group[prev].end(), group[sel].begin(), group[sel].end());
40                    active.erase(find(active.begin(), active.end(), sel));
41                    break;
42                }

```

```
43         used[sel] = true;
44         prev = sel;
45         for (int v : active)
46             if (!used[v])
47                 dis[v] += w[sel][v];
48     }
49 }
50 return {best, side};
51 }
52 };
```

第四十九章 带权二分图匹配

49.1 二分图最大权匹配（缺边与可选匹配）

WeightedMatching(n,m), 两侧编号 1..n、1..m。add(x,y,w) 加边，重边保留最大权；边权支持完整 long long 范围，内部势能与答案为 int128。solve() 要求全部左点匹配不同右点，无解返回 nullopt；n=m 时即最大权完美匹配。solve(true) 允许点不匹配，只最大化总权，不优先最大化边数。两种模式均从空匹配重算；l[x]、r[y] 为方案，0 表示未匹配，无解时全部清零。缺边用独立于合法边权的 -2^{120} 哨兵，仅用 add 修改；不把缺边当零权边。允许空侧。全匹配 $O(n^2m)$ ，可选模式增加 n 个虚拟零权右点，为 $O(n^2(m+n))$ ；空间 $O(nm+n+m)$ 。按最短松弛量扩展交错树：每次至少访问一个新右点，最多 n 次后增广；无有限松弛量即无解。

```
1 struct WeightedMatching
2 {
3     using I = __int128_t;
4     static constexpr I inf = I(1) << 120;
5     int n, m;
6     vector<vector<I>> w;
7     vector<int> l, r;
8
9     WeightedMatching(int n, int m)
10         : n(n), m(m), w(n + 1, vector<I>(m + 1, -inf)), l(n + 1), r(m + 1)
11     {
12     }
13
14     void add(int x, int y, long long weight)
15     {
16         assert(1 <= x && x <= n && 1 <= y && y <= m);
17         w[x][y] = max(w[x][y], I(weight));
18     }
19
20     optional<I> solve(bool allow_unmatched = false)
21     {
22         fill(l.begin(), l.end(), 0);
23         fill(r.begin(), r.end(), 0);
24         if (!allow_unmatched && n > m)
25             return nullopt;
26         int cols = m + (allow_unmatched ? n : 0);
27         vector<I> u(n + 1), v(cols + 1);
28         vector<int> p(cols + 1), way(cols + 1);
29         for (int i = 1; i <= n; i++)
30         {
31             p[0] = i;
32             int j0 = 0;
33             vector<I> dis(cols + 1, inf);
34             vector<bool> used(cols + 1);
35             do
36             {
37                 used[j0] = true;
38                 int x = p[j0], j1 = 0;
39                 I delta = inf;
40                 for (int j = 1; j <= cols; j++)
41                 {
```



```

42         if (used[j])
43             continue;
44         I weight = j <= m ? w[x][j] : 0;
45         if (weight != -inf)
46         {
47             I cur = -weight - u[x] - v[j];
48             if (cur < dis[j])
49             {
50                 dis[j] = cur;
51                 way[j] = j0;
52             }
53         }
54         if (dis[j] < delta)
55         {
56             delta = dis[j];
57             j1 = j;
58         }
59     }
60     if (delta == inf)
61         return nullopt;
62     for (int j = 0; j <= cols; j++)
63     {
64         if (used[j])
65         {
66             u[p[j]] += delta;
67             v[j] -= delta;
68         }
69         else if (dis[j] != inf)
70             dis[j] -= delta;
71     }
72     j0 = j1;
73 } while (p[j0]);
74 do
75 {
76     int j1 = way[j0];
77     p[j0] = p[j1];
78     j0 = j1;
79 } while (j0);
80 }
81 I answer = 0;
82 for (int j = 1; j <= m; j++)
83 {
84     if (!p[j])
85         continue;
86     l[p[j]] = j;
87     r[j] = p[j];
88     answer += w[p[j]][j];
89 }
90 return answer;
91 }
92 };

```

第五十章 精确几何进阶

50.1 整数最近点对与 Minkowski 和

`closest_pair(points)` 返回最短距离平方，少于两个点为 `nullopt`，重复点返回 0， $O(n \log n)$ ，全程精确整数。
`minkowski` 输入 strict CCW 凸包，返回不重复首点的凸包；非退化 $O(n+m)$ ，退化时 $O((n+m)\log(n+m))$ 。坐标及坐标和须满足整数几何范围。

```
1 struct GeometryExtra
2 {
3     using G = IntegerGeometry;
4     using P = G::Point;
5     using I = G::I;
6
7     // No pair exists for n<2. Exact squared distance,  $O(n \log n)$ .
8     static optional<I> closest_pair(vector<P> p)
9     {
10         int n = p.size();
11         if (n < 2)
12             return nullopt;
13         sort(p.begin(), p.end());
14         for (int i = 1; i < n; i++)
15             if (p[i] == p[i - 1])
16                 return I(0);
17         vector<P> tmp(n);
18         auto by_y = [](P a, P b)
19         {
20             return tie(a.y, a.x) < tie(b.y, b.x);
21         };
22         function<I(int, int)> solve = [&](int l, int r) -> I
23         {
24             if (r - l <= 3)
25             {
26                 I best = I(1) << 120;
27                 for (int i = l; i < r; i++)
28                     for (int j = i + 1; j < r; j++)
29                         best = min(best, G::dist2(p[i], p[j]));
30                 sort(p.begin() + l, p.begin() + r, by_y);
31                 return best;
32             }
33             int m = (l + r) / 2;
34             long long x = p[m].x;
35             I best = min(solve(l, m), solve(m, r));
36             merge(p.begin() + l,
37                 p.begin() + m,
38                 p.begin() + m,
39                 p.begin() + r,
40                 tmp.begin() + l,
41                 by_y);
42             copy(tmp.begin() + l, tmp.begin() + r, p.begin() + l);
43             vector<P> strip;
44             for (int i = l; i < r; i++)
```

```

45         if ((I(p[i].x) - x) * (I(p[i].x) - x) < best)
46         {
47             for (int j = (int)strip.size() - 1; j >= 0; j--)
48             {
49                 I dy = I(p[i].y) - strip[j].y;
50                 if (dy * dy >= best)
51                     break;
52                 best = min(best, G::dist2(p[i], strip[j]));
53             }
54             strip.push_back(p[i]);
55         }
56         return best;
57     };
58     return solve(0, n);
59 }
60
61 // Inputs strict CCW hulls. Coordinate sums must obey IntegerGeometry's bound.
62 static vector<P> minkowski(vector<P> a, vector<P> b)
63 {
64     if (a.empty() || b.empty())
65         return {};
66     auto plus = [](P u, P v)
67     {
68         return P{u.x + v.x, u.y + v.y};
69     };
70     if (a.size() < 3 || b.size() < 3)
71     {
72         vector<P> p;
73         for (P u : a)
74             for (P v : b)
75                 p.push_back(plus(u, v));
76         return G::hull(p);
77     }
78     auto start = [](vector<P> &p)
79     {
80         auto it = min_element(p.begin(),
81                               p.end(),
82                               [](P u, P v)
83                               {
84                                   return tie(u.y, u.x) < tie(v.y, v.x);
85                               });
86         rotate(p.begin(), it, p.end());
87     };
88     start(a);
89     start(b);
90     int n = a.size(), m = b.size(), i = 0, j = 0;
91     vector<P> result;
92     while (i < n || j < m)
93     {
94         result.push_back(plus(a[i % n], b[j % m]));
95         if (i == n)
96         {
97             ++j;
98             continue;
99         }
100         if (j == m)

```

```
101         {
102             ++i;
103             continue;
104         }
105         P u{a[(i + 1) % n].x - a[i].x, a[(i + 1) % n].y - a[i].y};
106         P v{b[(j + 1) % m].x - b[j].x, b[(j + 1) % m].y - b[j].y};
107         I c = G::cross({0, 0}, u, v);
108         if (c >= 0)
109             ++i;
110         if (c <= 0)
111             ++j;
112     }
113     return result;
114 }
115 };
```

50.2 整数三维共线、共面与方向

坐标绝对值 $\leq 10^9$; $\text{orient}(a,b,c,d)$ 为有向六倍四面体体积, 0 表示共面; collinear 判三点共线, on_segment 支持退化线段。int128 精确判定, 不包含三维凸包构造。

```

1 struct IntegerGeometry3D
2 {
3     using I = __int128_t;
4
5     // |coordinates|<=1e9: triple products fit signed int128.
6     struct Point
7     {
8         long long x, y, z;
9     };
10
11     struct Vector
12     {
13         I x, y, z;
14     };
15
16     static Vector diff(Point a, Point b)
17     {
18         return {I(a.x) - b.x, I(a.y) - b.y, I(a.z) - b.z};
19     }
20
21     static Vector cross(Vector a, Vector b)
22     {
23         return {a.y * b.z - a.z * b.y, a.z * b.x - a.x * b.z, a.x * b.y - a.y * b.x};
24     }
25
26     static I dot(Vector a, Vector b)
27     {
28         return a.x * b.x + a.y * b.y + a.z * b.z;
29     }
30
31     static I orient(Point a, Point b, Point c, Point d)
32     {
33         return dot(cross(diff(b, a), diff(c, a)), diff(d, a));
34     }
35
36     static bool collinear(Point a, Point b, Point c)
37     {
38         auto v = cross(diff(b, a), diff(c, a));
39         return !v.x && !v.y && !v.z;
40     }
41
42     static bool on_segment(Point p, Point a, Point b)
43     {
44         return collinear(p, a, b) && dot(diff(a, p), diff(b, p)) <= 0;
45     }
46 };

```

第五十一章 凸包支撑点查询

51.1 支撑点查询恢复下凸包

静态 `build(left,right,oracle)` 返回从左到右的严格下凸包顶点, `Point` 为 $\{x,y\}$ 。有限非空整数点集无需显式给出; 回调 `oracle(a,b)` 必须精确返回使 $a*x+b*y$ 最大的可行点, 允许任意最优解。 `left/right` 必须分别为最小/最大横坐标中纵坐标最低的点; 全体同横坐标时二者必须相同。所有坐标绝对值不超过 10^{18} , 方向差使用 `long long`, 叉积先提升至 `int128`; 回调内部也须保证数值精确且无溢出。对弦 $a \rightarrow b$ 查询方向 $(b.y-a.y, a.x-b.x)$, 严格位于弦下方则递归分治, 否则确认边; 输出时删除共线中间点。若分治找到 K 个严格改善支撑点, 查询次数为 $2K+1$ (同横坐标时为 0), 额外时间、空间和最坏递归深度为 $O(K+1)$ 。 K 包括可能返回的共线非顶点, 不将其等同于最终顶点数。不能直接传浮点近似优化器。二分图闭包组合: 点为 $(|S|, |N(S)|)$, 左点赋权 a 、右点赋权 b , 左 $\rightarrow$ 邻点依赖; 下凸包方向满足 $a \geq 0, b < 0$, 最优闭包的右点恰为邻集。端点为 $(0,0)$ 和 $(\text{左点数}, \text{全部左点的邻点数})$, 有孤立右点时后者不等于全部右点数。该组合已做子集枚举对拍, 尚不代表关联赛题整题通过。

```
1 struct SupportHull
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     struct Point
7     {
8         ll x, y;
9     };
10
11     static I cross(Point a, Point b, Point c)
12     {
13         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
14     }
```

```

16 // oracle(a,b) maximizes a*x+b*y exactly; |coordinates| <= 1e18.
17 // left/right: lowest points at the minimum/maximum x.
18 template <class Oracle>
19 static vector<Point> build(Point left, Point right, Oracle &&oracle)
20 {
21     assert(left.x <= right.x);
22     vector<Point> hull{left};
23     auto dfs = [&](auto &&self, Point a, Point b) -> void
24     {
25         Point c = oracle(b.y - a.y, a.x - b.x);
26         I turn = cross(a, b, c);
27         assert(turn <= 0);
28         if (turn < 0)
29         {
30             assert(a.x < c.x && c.x < b.x);
31             self(self, a, c);
32             self(self, c, b);
33         }
34         else
35         {
36             while (hull.size() >= 2 &&
37                 cross(hull[hull.size() - 2], hull.back(), b) == 0)
38                 hull.pop_back();
39             hull.push_back(b);
40         }
41     };
42     if (left.x < right.x)
43         dfs(dfs, left, right);
44     else
45         assert(left.y == right.y);
46     return hull;
47 }
48 };

```

第五十二章 动态树

52.1 Link-cut tree: 动态森林与路径异或

构造 LinkCutTree(n), 点号 1..n, 初始权值为零。set(x,value) 修改点权; link(x,y) 仅在不同连通块之间连边并返回是否成功; cut(x,y) 只删除真实存在的边, 否则返回 false。connected 查询连通性; query(x,y) 返回路径点权异或, 要求两点连通, 支持 x=y。make_root/find_root 维护与查询当前表示树根。每次操作均摊 $O(\log n)$, $O(n)$ 空间, splay 下传使用显式栈, 支持长链。权值为 unsigned long long。仅维护路径异或, 不包含虚子树聚合或动态直径。

```
1 struct LinkCutTree
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int ch[2] {}, fa = 0;
8         U val = 0, sum = 0;
9         bool rev = false;
10    };
11
12    vector<Node> a;
13    vector<int> st;
14
15    LinkCutTree(int n) : a(n + 1), st(n + 1)
16    {
17    }
18
19    bool is_root(int x) const
20    {
21        int f = a[x].fa;
22        return a[f].ch[0] != x && a[f].ch[1] != x;
23    }
24
25    void pull(int x)
26    {
27        a[x].sum = a[a[x].ch[0]].sum ^ a[x].val ^ a[a[x].ch[1]].sum;
28    }
29
30    void reverse(int x)
31    {
32        if (!x)
33            return;
34        swap(a[x].ch[0], a[x].ch[1]);
35        a[x].rev = !a[x].rev;
36    }
37
38    void push(int x)
39    {
40        if (!a[x].rev)
41            return;
42        reverse(a[x].ch[0]);
```



```

43     reverse(a[x].ch[1]);
44     a[x].rev = false;
45 }
46
47 void rotate(int x)
48 {
49     int y = a[x].fa, z = a[y].fa;
50     int k = a[y].ch[1] == x;
51     int w = a[x].ch[k ^ 1];
52     if (!is_root(y))
53         a[z].ch[a[z].ch[1] == y] = x;
54     a[x].fa = z;
55     a[y].ch[k] = w;
56     if (w)
57         a[w].fa = y;
58     a[x].ch[k ^ 1] = y;
59     a[y].fa = x;
60     pull(y);
61     pull(x);
62 }
63
64 void splay(int x)
65 {
66     int top = 0, y = x;
67     st[top++] = y;
68     while (!is_root(y))
69     {
70         y = a[y].fa;
71         st[top++] = y;
72     }
73     while (top)
74         push(st[--top]);
75     while (!is_root(x))
76     {
77         int y = a[x].fa, z = a[y].fa;
78         if (!is_root(y))
79         {
80             bool same = (a[y].ch[1] == x) == (a[z].ch[1] == y);
81             rotate(same ? y : x);
82         }
83         rotate(x);
84     }
85 }
86
87 void access(int x)
88 {
89     int last = 0;
90     for (int y = x; y; y = a[y].fa)
91     {
92         splay(y);
93         // The old right child's fa remains y: it becomes a path-parent.
94         a[y].ch[1] = last;
95         if (last)
96             a[last].fa = y;
97         pull(y);
98         last = y;

```

```

99     }
100     splay(x);
101 }
102
103 void make_root(int x)
104 {
105     access(x);
106     reverse(x);
107 }
108
109 int find_root(int x)
110 {
111     access(x);
112     push(x);
113     while (a[x].ch[0])
114     {
115         x = a[x].ch[0];
116         push(x);
117     }
118     splay(x);
119     return x;
120 }
121
122 bool connected(int x, int y)
123 {
124     return x == y || find_root(x) == find_root(y);
125 }
126
127 bool link(int x, int y)
128 {
129     make_root(x);
130     if (find_root(y) == x)
131         return false;
132     a[x].fa = y;
133     return true;
134 }
135
136 bool cut(int x, int y)
137 {
138     make_root(x);
139     access(y);
140     if (a[y].ch[0] != x || a[x].ch[1])
141         return false;
142     a[y].ch[0] = 0;
143     a[x].fa = 0;
144     pull(y);
145     return true;
146 }
147
148 void set(int x, U value)
149 {
150     access(x);
151     a[x].val = value;
152     pull(x);

```

```
153     }
154
155     // x and y must be connected. This is a path aggregate, not a subtree sum.
156     U query(int x, int y)
157     {
158         make_root(x);
159         access(y);
160         return a[y].sum;
161     }
162 };
```

第五十三章 随机平衡树

53.1 FHQ Treap: 可重集合与顺序统计

insert(x) 插入一个副本, erase(x) 删除一个副本并返回是否存在; rank(x) 为严格小于 x 的元素数加一, x 不必存在; kth(k) 要求 $1 \leq k \leq \text{size}()$ 。prev/next 为严格前驱/后继, 不存在时返回 nullopt。支持重复值、LLONG_MIN 和 LLONG_MAX, 分裂不使用 x-1。随机优先级下单次期望 $O(\log n)$, 并非确定性最坏界。节点池不回收已删除节点, 空间按累计插入次数计算; 可在构造时指定随机种子。

```
1 struct OrderedTreap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0;
9         unsigned long long pri = 0;
10    };
11
12    vector<Node> a{Node{}};
13    mt19937_64 rng;
14    int root = 0;
15
16    OrderedTreap(unsigned long long seed = 712367821) : rng(seed)
17    {
18    }
19
20    void pull(int p)
21    {
22        a[p].siz = a[a[p].l].siz + a[a[p].r].siz + 1;
23    }
24
25    int node(ll x)
26    {
27        int p = a.size();
28        a.push_back({0, 0, 1, x, rng()});
29        return p;
30    }
31
32    // left: val < x, or val <= x when equal is true. No x-1 overflow.
33    void split(int p, ll x, int &l, int &r, bool equal = false)
34    {
35        if (!p)
36        {
37            l = 0;
38            r = 0;
39            return;
40        }
41        if (a[p].val < x || (equal && a[p].val == x))
42        {
43            l = p;
```

```

44         split(a[p].r, x, a[p].r, r, equal);
45     }
46     else
47     {
48         r = p;
49         split(a[p].l, x, l, a[p].l, equal);
50     }
51     pull(p);
52 }
53
54 int merge(int l, int r)
55 {
56     if (!l || !r)
57         return l ? l : r;
58     if (a[l].pri > a[r].pri)
59     {
60         a[l].r = merge(a[l].r, r);
61         pull(l);
62         return l;
63     }
64     a[r].l = merge(l, a[r].l);
65     pull(r);
66     return r;
67 }
68
69 void insert(ll x)
70 {
71     int p = node(x), l, r;
72     split(root, x, l, r);
73     root = merge(merge(l, p), r);
74 }
75
76 bool erase(ll x)
77 {
78     int l, mid, r;
79     split(root, x, l, mid);
80     split(mid, x, mid, r, true);
81     bool found = mid != 0;
82     if (mid)
83         mid = merge(a[mid].l, a[mid].r);
84     root = merge(l, merge(mid, r));
85     return found;
86 }
87
88 int size() const
89 {
90     return a[root].siz;
91 }
92
93 int less(ll x, bool equal = false) const
94 {
95     int p = root, ans = 0;
96     while (p)
97     {
98         if (a[p].val < x || (equal && a[p].val == x))
99             {

```

```

100         ans += a[a[p].l].siz + 1;
101         p = a[p].r;
102     }
103     else
104         p = a[p].l;
105     }
106     return ans;
107 }
108
109 int rank(ll x) const
110 {
111     return less(x) + 1;
112 }
113
114 ll kth(int k) const
115 {
116     assert(1 <= k && k <= size());
117     int p = root;
118     while (true)
119     {
120         int left = a[a[p].l].siz;
121         if (k == left + 1)
122             return a[p].val;
123         if (k <= left)
124             p = a[p].l;
125         else
126         {
127             k -= left + 1;
128             p = a[p].r;
129         }
130     }
131 }
132
133 optional<ll> prev(ll x) const
134 {
135     int k = less(x);
136     if (!k)
137         return nullopt;
138     return kth(k);
139 }
140
141 optional<ll> next(ll x) const
142 {
143     int k = less(x, true);
144     if (k == size())
145         return nullopt;
146     return kth(k + 1);
147 }
148 };

```

53.2 隐式 Treap：序列翻转、加值与区间和

`insert(k,value)` 在前 k 个元素后插入, $0 \leq k \leq \text{size}()$; `erase(l,r)`、`reverse(l,r)`、`add(l,r,value)`、`query(l,r)` 均使用 1-based 非空闭区间。`values()` 返回当前完整序列, $O(n)$, 使用递归中序遍历。其余操作在随机优先级下期望 $O(\log$

n); 翻转与加法懒标记同时维护。所有值、和、标记及 $\text{value} \times \text{长度}$ 中间结果必须落在 `long long` 范围内。删除不回收节点，内存按累计插入数计算。

```

1 struct SequenceTreap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0, sum = 0, tag = 0;
9         unsigned long long pri = 0;
10        bool rev = false;
11    };
12
13    vector<Node> a{Node{}};
14    mt19937_64 rng;
15    int root = 0;
16
17    SequenceTreap(unsigned long long seed = 712367821) : rng(seed)
18    {
19    }
20
21    int size() const
22    {
23        return a[root].siz;
24    }
25
26    void pull(int p)
27    {
28        a[p].siz = a[a[p].l].siz + a[a[p].r].siz + 1;
29        a[p].sum = a[a[p].l].sum + a[p].val + a[a[p].r].sum;
30    }
31
32    void apply_add(int p, ll x)
33    {
34        if (!p)
35            return;
36        a[p].val += x;
37        a[p].sum += x * a[p].siz;
38        a[p].tag += x;
39    }
40
41    void apply_reverse(int p)
42    {
43        if (!p)
44            return;
45        swap(a[p].l, a[p].r);
46        a[p].rev = !a[p].rev;
47    }
48
49    void push(int p)
50    {
51        if (a[p].rev)
52        {
53            apply_reverse(a[p].l);

```

```

54         apply_reverse(a[p].r);
55         a[p].rev = false;
56     }
57     if (a[p].tag)
58     {
59         apply_add(a[p].l, a[p].tag);
60         apply_add(a[p].r, a[p].tag);
61         a[p].tag = 0;
62     }
63 }
64
65 // Split into the first k elements and the rest.
66 void split(int p, int k, int &l, int &r)
67 {
68     if (!p)
69     {
70         l = 0;
71         r = 0;
72         return;
73     }
74     push(p);
75     int left = a[a[p].l].siz;
76     if (k > left)
77     {
78         l = p;
79         split(a[p].r, k - left - 1, a[p].r, r);
80     }
81     else
82     {
83         r = p;
84         split(a[p].l, k, l, a[p].l);
85     }
86     pull(p);
87 }
88
89 int merge(int l, int r)
90 {
91     if (!l || !r)
92         return l ? l : r;
93     if (a[l].pri > a[r].pri)
94     {
95         push(l);
96         a[l].r = merge(a[l].r, r);
97         pull(l);
98         return l;
99     }
100    push(r);
101    a[r].l = merge(l, a[r].l);
102    pull(r);
103    return r;
104 }
105
106 // Insert after the first k elements, 0 <= k <= size().
107 void insert(int k, ll value)
108 {
109     assert(0 <= k && k <= size());

```



```

110     int p = a.size(), l, r;
111     a.push_back({0, 0, 1, value, value, 0, rng(), false});
112     split(root, k, l, r);
113     root = merge(merge(l, p), r);
114 }
115
116 // All range operations use 1-based inclusive intervals.
117 void erase(int l, int r)
118 {
119     assert(1 <= l && l <= r && r <= size());
120     int x, y, z;
121     split(root, r, y, z);
122     split(y, l - 1, x, y);
123     root = merge(x, z);
124 }
125
126 void reverse(int l, int r)
127 {
128     assert(1 <= l && l <= r && r <= size());
129     int x, y, z;
130     split(root, r, y, z);
131     split(y, l - 1, x, y);
132     apply_reverse(y);
133     root = merge(merge(x, y), z);
134 }
135
136 void add(int l, int r, ll value)
137 {
138     assert(1 <= l && l <= r && r <= size());
139     int x, y, z;
140     split(root, r, y, z);
141     split(y, l - 1, x, y);
142     apply_add(y, value);
143     root = merge(merge(x, y), z);
144 }
145
146 ll query(int l, int r)
147 {
148     assert(1 <= l && l <= r && r <= size());
149     int x, y, z;
150     split(root, r, y, z);
151     split(y, l - 1, x, y);
152     ll ans = a[y].sum;
153     root = merge(merge(x, y), z);
154     return ans;
155 }
156
157 void collect(int p, vector<ll> &out)
158 {
159     if (!p)
160         return;
161     push(p);
162     collect(a[p].l, out);
163     out.push_back(a[p].val);

```

```
164         collect(a[p].r, out);
165     }
166
167     vector<ll> values()
168     {
169         vector<ll> answer;
170         collect(root, answer);
171         return answer;
172     }
173 };
```

第五十四章 伸展树

54.1 Splay: 可重集合与顺序统计

insert/erase/rank/kth/prev/next 与有序 Treap 同义, 严格前驱后继不存在返回 nullopt。重复键计数存储; 失败查询也伸展最后访问节点, 单次均摊 $O(\log n)$ 。不回收节点, 空间按累计新建节点数计算。支持 long long 全范围; 不包含序列 Splay、虚子树信息或矩阵动态 DP。

```
1 struct OrderedSplay
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int ch[2]{}, fa = 0, cnt = 0, siz = 0;
8         ll val = 0;
9     };
10
11     vector<Node> a{Node{}};
12     int root = 0;
13
14     int size() const
15     {
16         return a[root].siz;
17     }
18
19     void pull(int x)
20     {
21         a[x].siz = a[a[x].ch[0]].siz + a[x].cnt + a[a[x].ch[1]].siz;
22     }
23
24     int node(ll value, int parent)
25     {
26         int x = a.size();
27         a.push_back({{0, 0}, parent, 1, 1, value});
28         return x;
29     }
30
31     void rotate(int x)
32     {
33         int y = a[x].fa, z = a[y].fa;
34         int k = a[y].ch[1] == x;
35         int w = a[x].ch[k ^ 1];
36         if (z)
37             a[z].ch[a[z].ch[1] == y] = x;
38         else
39             root = x;
40         a[x].fa = z;
41         a[y].ch[k] = w;
42         if (w)
43             a[w].fa = y;
44         a[x].ch[k ^ 1] = y;
```

```

45     a[y].fa = x;
46     pull(y);
47     pull(x);
48 }
49
50 void splay(int x)
51 {
52     while (a[x].fa)
53     {
54         int y = a[x].fa, z = a[y].fa;
55         if (z)
56         {
57             bool same = (a[y].ch[1] == x) == (a[z].ch[1] == y);
58             rotate(same ? y : x);
59         }
60         rotate(x);
61     }
62 }
63
64 // Splay the last visited node even when the value does not exist.
65 int find(ll value)
66 {
67     int x = root, last = 0;
68     while (x && a[x].val != value)
69     {
70         last = x;
71         x = a[x].ch[value > a[x].val];
72     }
73     if (x)
74         splay(x);
75     else if (last)
76         splay(last);
77     return x;
78 }
79
80 void insert(ll value)
81 {
82     int x = root, parent = 0;
83     while (x && a[x].val != value)
84     {
85         parent = x;
86         x = a[x].ch[value > a[x].val];
87     }
88     if (x)
89     {
90         ++a[x].cnt;
91         pull(x);
92     }
93     else
94     {
95         x = node(value, parent);
96         if (parent)
97             a[parent].ch[value > a[parent].val] = x;
98         else
99             root = x;
100     }

```

```

101     splay(x);
102 }
103
104 bool erase(ll value)
105 {
106     int x = find(value);
107     if (!x)
108         return false;
109     if (a[x].cnt > 1)
110     {
111         --a[x].cnt;
112         pull(x);
113         return true;
114     }
115     int l = a[x].ch[0], r = a[x].ch[1];
116     if (l)
117         a[l].fa = 0;
118     if (r)
119         a[r].fa = 0;
120     if (!l)
121     {
122         root = r;
123         return true;
124     }
125     root = l;
126     x = l;
127     while (a[x].ch[1])
128         x = a[x].ch[1];
129     splay(x);
130     a[x].ch[1] = r;
131     if (r)
132         a[r].fa = x;
133     pull(x);
134     return true;
135 }
136
137 int rank(ll value)
138 {
139     int x = root, last = 0, ans = 1;
140     while (x)
141     {
142         last = x;
143         if (value <= a[x].val)
144             x = a[x].ch[0];
145         else
146         {
147             ans += a[a[x].ch[0]].siz + a[x].cnt;
148             x = a[x].ch[1];
149         }
150     }
151     if (last)
152         splay(last);
153     return ans;
154 }
155
156 ll kth(int k)

```

```

157 {
158     assert(1 <= k && k <= size());
159     int x = root;
160     while (true)
161     {
162         int left = a[a[x].ch[0]].siz;
163         if (k <= left)
164             x = a[x].ch[0];
165         else if (k <= left + a[x].cnt)
166         {
167             splay(x);
168             return a[x].val;
169         }
170         else
171         {
172             k -= left + a[x].cnt;
173             x = a[x].ch[1];
174         }
175     }
176 }
177
178 optional<ll> prev(ll value)
179 {
180     int x = root, last = 0, answer = 0;
181     while (x)
182     {
183         last = x;
184         if (a[x].val < value)
185         {
186             answer = x;
187             x = a[x].ch[1];
188         }
189         else
190             x = a[x].ch[0];
191     }
192     if (last)
193         splay(last);
194     if (!answer)
195         return nullopt;
196     splay(answer);
197     return a[answer].val;
198 }
199
200 optional<ll> next(ll value)
201 {
202     int x = root, last = 0, answer = 0;
203     while (x)
204     {
205         last = x;
206         if (a[x].val > value)
207         {
208             answer = x;
209             x = a[x].ch[0];
210         }
211         else

```

```
212         x = a[x].ch[1];
213     }
214     if (last)
215         splay(last);
216     if (!answer)
217         return nullopt;
218     splay(answer);
219     return a[answer].val;
220 }
221 };
```

第五十五章 状态序列维护

55.1 状态序列 Treap：分组区间 GCD

build(v) 用数值与状态组成的序列替换当前内容。数值为无符号 64 位，状态为 0 或 1。insert(k,value,tag) 在前 k 项后插入。erase(l,r)、query(l,r,tag) 使用 1-based 非空闭区间；set(k,value) 修改值，toggle(k) 翻转状态。query 返回选中状态的 GCD，无对应元素为 nullopt，实际全零则返回 0。回收删除节点，空间按峰值长度计算。建树 $O(n)$ 次 GCD，随机优先级下单次期望 $O(\log n)$ 次 GCD；删除另需 $O(\text{删除长度})$ 回收。一次 GCD 为 $O(\log U)$ 。覆盖 kuangbin 3.7 的序列操作，尚无该原题 OJ 记录。建树后的聚合与删除节点回收均使用递归遍历；笛卡尔树的构造单调栈保留。

```
1 struct GcdSequenceTreap
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0, tag = 0;
8         int cnt[2]{};
9         U val = 0, g[2]{}, pri = 0;
10    };
11
12    vector<Node> a{Node{}};
13    vector<int> recycled;
14    mt19937_64 rng;
15    int root = 0;
16
17    GcdSequenceTreap(U seed = 712367821) : rng(seed)
18    {
19    }
20
21    int size() const
22    {
23        return a[root].siz;
24    }
25
26    int node(U value, int tag)
27    {
28        assert(tag == 0 || tag == 1);
29        int p;
30        if (recycled.empty())
31        {
32            p = a.size();
33            a.push_back(Node{});
34        }
35        else
36        {
37            p = recycled.back();
38            recycled.pop_back();
39            a[p] = Node{};
40        }
41        a[p].siz = 1;
```



```

42     a[p].val = value;
43     a[p].tag = tag;
44     a[p].cnt[tag] = 1;
45     a[p].g[tag] = value;
46     a[p].pri = rng();
47     return p;
48 }
49
50 void pull(int p)
51 {
52     int l = a[p].l, r = a[p].r;
53     a[p].siz = a[l].siz + a[r].siz + 1;
54     for (int t = 0; t < 2; t++)
55     {
56         a[p].cnt[t] = a[l].cnt[t] + a[r].cnt[t] + (a[p].tag == t);
57         a[p].g[t] = gcd(a[l].g[t], a[r].g[t]);
58         if (a[p].tag == t)
59             a[p].g[t] = gcd(a[p].g[t], a[p].val);
60     }
61 }
62
63 void split(int p, int k, int &l, int &r)
64 {
65     if (!p)
66     {
67         l = 0;
68         r = 0;
69         return;
70     }
71     int left = a[a[p].l].siz;
72     if (k > left)
73     {
74         l = p;
75         split(a[p].r, k - left - 1, a[p].r, r);
76     }
77     else
78     {
79         r = p;
80         split(a[p].l, k, l, a[p].l);
81     }
82     pull(p);
83 }
84
85 int merge(int l, int r)
86 {
87     if (!l || !r)
88         return l ? l : r;
89     if (a[l].pri > a[r].pri)
90     {
91         a[l].r = merge(a[l].r, r);
92         pull(l);
93         return l;
94     }
95     a[r].l = merge(l, a[r].l);
96     pull(r);
97     return r;

```

```

98     }
99
100    void rebuild(int p)
101    {
102        if (!p)
103            return;
104        rebuild(a[p].l);
105        rebuild(a[p].r);
106        pull(p);
107    }
108
109    void release(int p)
110    {
111        if (!p)
112            return;
113        release(a[p].l);
114        release(a[p].r);
115        recycled.push_back(p);
116    }
117
118    // O(n) GCD calls; replaces the current sequence.
119    void build(const vector<pair<U, int>> &v)
120    {
121        a.assign(1, Node{});
122        recycled.clear();
123        root = 0;
124        vector<int> st;
125        for (auto [value, tag] : v)
126        {
127            int p = node(value, tag), last = 0;
128            while (!st.empty() && a[st.back()].pri < a[p].pri)
129            {
130                last = st.back();
131                st.pop_back();
132            }
133            a[p].l = last;
134            if (!st.empty())
135                a[st.back()].r = p;
136            st.push_back(p);
137        }
138        if (st.empty())
139            return;
140        root = st.front();
141        rebuild(root);
142    }
143
144    // Insert after the first k entries. Range operations below are 1-based.
145    void insert(int k, U value, int tag)
146    {
147        assert(0 <= k && k <= size());
148        int p = node(value, tag), l, r;
149        split(root, k, l, r);
150        root = merge(merge(l, p), r);
151    }
152
153    void erase(int l, int r)

```

```

154     {
155         assert(1 <= l && l <= r && r <= size());
156         int x, y, z;
157         split(root, r, y, z);
158         split(y, l - 1, x, y);
159         root = merge(x, z);
160         release(y);
161     }
162
163     void set(int k, U value)
164     {
165         assert(1 <= k && k <= size());
166         int x, y, z;
167         split(root, k, y, z);
168         split(y, k - 1, x, y);
169         a[y].val = value;
170         pull(y);
171         root = merge(merge(x, y), z);
172     }
173
174     void toggle(int k)
175     {
176         assert(1 <= k && k <= size());
177         int x, y, z;
178         split(root, k, y, z);
179         split(y, k - 1, x, y);
180         a[y].tag ^= 1;
181         pull(y);
182         root = merge(merge(x, y), z);
183     }
184
185     optional<U> query(int l, int r, int tag)
186     {
187         assert(1 <= l && l <= r && r <= size());
188         assert(tag == 0 || tag == 1);
189         int x, y, z;
190         split(root, r, y, z);
191         split(y, l - 1, x, y);
192         optional<U> answer;
193         if (a[y].cnt[tag])
194             answer = a[y].g[tag];
195         root = merge(merge(x, y), z);
196         return answer;
197     }
198 };

```

第五十六章 一般图匹配

56.1 一般图最大基数匹配：带花树

Blossom(n), 点号 0..n-1; add(u,v) 加无向边, 忽略自环, 允许重边。solve() 从空匹配重新计算, 返回匹配边数; mate[u] 为配对点, 未匹配为 -1。结果支持直接输出方案。缩花维护交错树, 单次增广 $O(n^2+m)$, 总计 $O(n^3+nm)$, 简单图为 $O(n^3)$; 空间 $O(n+m)$ 。重复调用 solve 或加边后重算均可。不处理边权, 不能代替最大权匹配。

```
1 struct Blossom
2 {
3     int n;
4     vector<vector<int>> g;
5     vector<int> mate, pre, base, q;
6     vector<bool> outer, flower, seen;
7
8     Blossom(int n)
9         : n(n), g(n), mate(n, -1), pre(n), base(n), outer(n), flower(n), seen(n)
10    {
11    }
12
13    void add(int u, int v)
14    {
15        assert(0 <= u && u < n && 0 <= v && v < n);
16        if (u == v)
17            return;
18        g[u].push_back(v);
19        g[v].push_back(u);
20    }
21
22    int lca(int u, int v)
23    {
24        fill(seen.begin(), seen.end(), false);
25        while (true)
26        {
27            u = base[u];
28            seen[u] = true;
29            if (mate[u] == -1)
30                break;
31            u = pre[mate[u]];
32        }
33        while (!seen[base[v]])
34            v = pre[mate[base[v]]];
35        return base[v];
36    }
37
38    void mark(int u, int b, int v)
39    {
40        while (base[u] != b)
41        {
42            flower[base[u]] = true;
43            flower[base[mate[u]]] = true;
44            pre[u] = v;
```

```

45         v = mate[u];
46         u = pre[v];
47     }
48 }
49
50 bool augment(int root)
51 {
52     fill(outer.begin(), outer.end(), false);
53     fill(pre.begin(), pre.end(), -1);
54     iota(base.begin(), base.end(), 0);
55     q.assign(1, root);
56     outer[root] = true;
57     for (int head = 0; head < (int)q.size(); head++)
58     {
59         int u = q[head];
60         for (int v : g[u])
61         {
62             if (base[u] == base[v] || mate[u] == v)
63                 continue;
64             if (v == root || (mate[v] != -1 && pre[mate[v]] != -1))
65             {
66                 int b = lca(u, v);
67                 fill(flower.begin(), flower.end(), false);
68                 mark(u, b, v);
69                 mark(v, b, u);
70                 for (int x = 0; x < n; x++)
71                 {
72                     if (!flower[base[x]])
73                         continue;
74                     base[x] = b;
75                     if (!outer[x])
76                     {
77                         outer[x] = true;
78                         q.push_back(x);
79                     }
80                 }
81             }
82             else if (pre[v] == -1)
83             {
84                 pre[v] = u;
85                 if (mate[v] == -1)
86                 {
87                     while (v != -1)
88                     {
89                         int p = pre[v];
90                         int next = mate[p];
91                         mate[v] = p;
92                         mate[p] = v;
93                         v = next;
94                     }
95                     return true;
96                 }
97                 int x = mate[v];
98                 outer[x] = true;
99                 q.push_back(x);
100             }

```

```
101         }
102     }
103     return false;
104 }
105
106 // Rebuild the maximum matching; mate[u] is -1 for an unmatched vertex.
107 int solve()
108 {
109     fill(mate.begin(), mate.end(), -1);
110     int answer = 0;
111     for (int u = 0; u < n; u++)
112     {
113         if (mate[u] == -1 && augment(u))
114             answer++;
115     }
116     return answer;
117 }
118 };
```

第五十七章 线性递推

57.1 Berlekamp–Massey：最短线性递推

`berlekamp_massey(s,p)` 要求 p 为素数, $2 \leq p \leq \text{INT_MAX}$, 输入均在 $[0,p)$; 返回 c , 使 $s[i] = \sum_{j=1..k} c[j-1]s[i-j]$ ($j=1..k, i \geq k$)。空序列或全零序列返回空系数。有限前缀可能对应多种未来, 不能无条件外推; 若已知原序列满足阶数至多 d 的齐次线性递推, 提供至少 $2d$ 项才能恢复可外推的递推。模运算次数 $O(Nk+N \log p)$, 后项来自逐次求逆; 空间 $O(N)$ 。不适用于合数模数或浮点拟合。

```
1 // Prime p; normalized input. s[i] = sum(c[j-1] * s[i-j]), i >= c.size().
2 vector<int> berlekamp_massey(const vector<int> &s, int p = 998244353)
3 {
4     assert(p >= 2);
5     vector<int> c{1}, b{1};
6     int len = 0, shift = 1, last = 1;
7     for (int i = 0; i < (int)s.size(); i++)
8     {
9         assert(0 <= s[i] && s[i] < p);
10        long long d = s[i];
11        for (int j = 1; j <= len; j++)
12            d = (d + 1LL * c[j] * s[i - j]) % p;
13        if (d == 0)
14        {
15            shift++;
16            continue;
17        }
18        long long inv = 1, x = last;
19        for (int e = p - 2; e; e >>= 1)
20        {
21            if (e & 1)
22                inv = inv * x % p;
23            x = x * x % p;
24        }
25        long long factor = d * inv % p;
26        auto old = c;
27        if (c.size() < b.size() + shift)
28            c.resize(b.size() + shift);
29        for (int j = 0; j < (int)b.size(); j++)
30            c[j + shift] = (c[j + shift] - factor * b[j] % p + p) % p;
31        if (2 * len <= i)
32        {
33            len = i + 1 - len;
34            b = old;
35            last = d;
36            shift = 1;
37        }
38    }
```

```
38         else
39             shift++;
40     }
41     c.resize(len + 1);
42     c.erase(c.begin());
43     for (int &x : c)
44         x = x == 0 ? 0 : p - x;
45     return c;
46 }
```


57.2 特征多项式求线性递推第 k 项

`recurrence_nth(init,c,n,p)` 使用 0-based 下标, n 为 unsigned long long。init 与 c 长度均为阶数 k ; $a[i] = \sum_{j=1}^k c[j-1]a[i-j]$ 。 $2 \leq p \leq \text{INT_MAX}$, 数值均在 $[0,p)$, 本函数允许合数模数。空系数约定全零序列。计算 x^n 对特征多项式取余再与初值点乘, $O(k^2 \log(n+1))$ 时间、 $O(k)$ 空间; 不依赖 BM 或其他模板。较大阶数需快速多项式或 Bostan-Mori 版本, 当前函数不覆盖该复杂度需求。

```

1 // init has exactly k entries; c[j-1] multiplies a[n-j]. Modulus need not be prime.
2 int recurrence_nth(const vector<int> &init,
3                   const vector<int> &c,
4                   unsigned long long n,
5                   int p = 998244353)
6 {
7     assert(p >= 2 && init.size() == c.size());
8     int k = c.size();
9     if (!k)
10         return 0;
11     for (int x : init)
12         assert(0 <= x && x < p);
13     for (int x : c)
14         assert(0 <= x && x < p);
15     if (n < (unsigned)k)
16         return init[n];
17     auto multiply = [&](const vector<int> &a, const vector<int> &b)
18     {
19         vector<int> t(2 * k - 1);
20         for (int i = 0; i < k; i++)
21         {
22             for (int j = 0; j < k; j++)
23                 t[i + j] = (t[i + j] + 1LL * a[i] * b[j]) % p;
24         }
25         for (int i = 2 * k - 2; i >= k; i--)
26         {
27             for (int j = 1; j <= k; j++)
28                 t[i - j] = (t[i - j] + 1LL * t[i] * c[j - 1]) % p;
29         }
30         t.resize(k);
31         return t;
32     };
33     vector<int> answer(k), x(k);
34     answer[0] = 1;
35     if (k == 1)
36         x[0] = c[0];
37     else
38         x[1] = 1;
39     while (n)
40     {
41         if (n & 1)

```

```
42         answer = multiply(answer, x);
43     x = multiply(x, x);
44     n >>= 1;
45 }
46 long long value = 0;
47 for (int i = 0; i < k; i++)
48     value = (value + 1LL * answer[i] * init[i]) % p;
49 return value;
50 }
```

第二部分

传统接口风格

第五十八章 Tarjan 与缩点

58.1 Tarjan 强连通分量与缩点

算法约束见动态版本第 2 页。本节代码独立可抄；采用下列实际接口名。

声明 Tarjan_SCC<N>, 先 Init(n), Insert 加边, Run 求分量, Dag 输出缩点图。bel 与 cnt 可直接读取；大对象放在全局或 static。

```
1 template <int N> struct Tarjan_SCC
2 {
3     int n, timer = 0, cnt = 0;
4     vector<int> g[N + 1], st;
5     int dfn[N + 1], low[N + 1], bel[N + 1];
6
7     void Init(int vertices)
8     {
9         assert(0 <= vertices && vertices <= N);
10        n = vertices;
11        for ( int u = 1; u <= n; u++ )
12            g[u].clear();
13    }
14
15    void Insert(int u, int v)
16    {
17        assert(1 <= u && u <= n && 1 <= v && v <= n);
18        g[u].push_back(v);
19    }
20
21    void Dfs(int u)
22    {
23        dfn[u] = low[u] = ++timer;
24        st.push_back(u);
25        for ( int v : g[u] )
26        {
27            if ( !dfn[v] )
28            {
29                Dfs(v);
30                low[u] = min(low[u], low[v]);
31            }
32            else if ( !bel[v] )
33                low[u] = min(low[u], dfn[v]);
34        }
35        if ( low[u] == dfn[u] )
36        {
37            ++cnt;
38            while ( true )
39            {
40                int v = st.back();
41                st.pop_back();
42                bel[v] = cnt;
43                if ( v == u )
44                    break;
```

```

45     }
46 }
47 }
48
49 // IDs 1..cnt are in reverse topological order. Recursive DFS needs O(n) stack.
50 void Run()
51 {
52     timer = 0;
53     cnt = 0;
54     fill(dfn, dfn + n + 1, 0);
55     fill(low, low + n + 1, 0);
56     fill-bel, bel + n + 1, 0);
57     st.clear();
58     for ( int u = 1; u <= n; u++ )
59     {
60         if ( !dfn[u] )
61             Dfs(u);
62     }
63 }
64
65 // Retains parallel edges. Use after Run().
66 vector<vector<int>> Dag() const
67 {
68     vector<vector<int>> result(cnt + 1);
69     for ( int u = 1; u <= n; u++ )
70     {
71         for ( int v : g[u] )
72         {
73             if ( bel[u] != bel[v] )
74                 result[bel[u]].push_back(bel[v]);
75         }
76     }
77     return result;
78 }
79 };

```

第五十九章 双连通分量与圆方树

59.1 Tarjan 点双、边双与圆方树

算法约束见动态版本第 4 页。本节代码独立可抄；采用下列实际接口名。

声明 `Biconnected_Graph<N>`, `Init(n)` 初始化, `Insert` 加边, `Run` 分解, `Block_Forest` 和 `Bridge_Forest` 输出两种森林。 `blocks`、`cut`、`bridge`、`bel`、`cnt` 可直接读取。大对象放在全局或 `static`。

```
1 template <int N> struct Biconnected_Graph
2 {
3     int n, timer = 0, cnt = 0;
4     vector<pair<int, int>> g[N + 1], edges;
5     vector<vector<int>> blocks;
6     vector<int> bridge, st;
7     int dfn[N + 1], low[N + 1], cut[N + 1], bel[N + 1];
8
9     void Init(int vertices)
10    {
11        assert(0 <= vertices && vertices <= N);
12        n = vertices;
13        for ( int u = 1; u <= n; u++ )
14            g[u].clear();
15        edges.clear();
16    }
17
18    int Insert(int u, int v)
19    {
20        assert(1 <= u && u <= n && 1 <= v && v <= n);
21        int id = edges.size();
22        edges.push_back({u, v});
23        if ( u != v )
24        {
25            g[u].push_back({v, id});
26            g[v].push_back({u, id});
27        }
28        return id;
29    }
30
31    void Dfs(int u, int parent_edge)
32    {
33        dfn[u] = low[u] = ++timer;
34        st.push_back(u);
35        int children = 0;
36        for ( auto [v, id] : g[u] )
37        {
38            if ( id == parent_edge )
39                continue;
40            if ( !dfn[v] )
41            {
42                children++;
43                Dfs(v, id);
44                low[u] = min(low[u], low[v]);
```

```

45         if ( low[v] > dfn[u] )
46             bridge[id] = 1;
47         if ( low[v] >= dfn[u] )
48         {
49             if ( parent_edge != -1 || children > 1 )
50                 cut[u] = 1;
51             blocks.push_back({u});
52             while ( true )
53             {
54                 int x = st.back();
55                 st.pop_back();
56                 blocks.back().push_back(x);
57                 if ( x == v )
58                     break;
59             }
60         }
61     }
62     else
63         low[u] = min(low[u], dfn[v]);
64 }
65 if ( parent_edge == -1 && children == 0 )
66     blocks.push_back({u});
67 }
68
69 void Paint(int u)
70 {
71     bel[u] = cnt;
72     for ( auto [v, id] : g[u] )
73     {
74         if ( !bridge[id] && !bel[v] )
75             Paint(v);
76     }
77 }
78
79 void Run()
80 {
81     timer = 0;
82     cnt = 0;
83     fill(dfn, dfn + n + 1, 0);
84     fill(low, low + n + 1, 0);
85     fill(cut, cut + n + 1, 0);
86     fill(bel, bel + n + 1, 0);
87     bridge.assign(edges.size(), 0);
88     blocks.clear();
89     st.clear();
90     for ( int u = 1; u <= n; u++ )
91     {
92         if ( !dfn[u] )
93         {
94             Dfs(u, -1);
95             st.pop_back();
96         }
97     }
98     for ( int u = 1; u <= n; u++ )
99     {
100         if ( bel[u] )

```

```

101         continue;
102         ++cnt;
103         Paint(u);
104     }
105 }
106
107 // Original vertices 1..n; block i has node n+i+1. Includes isolated vertices.
108 vector<vector<int>> Block_Forest() const
109 {
110     vector<vector<int>> tree(n + blocks.size() + 1);
111     for ( int i = 0; i < (int)blocks.size(); i++ )
112     {
113         int b = n + i + 1;
114         for ( int u : blocks[i] )
115         {
116             tree[u].push_back(b);
117             tree[b].push_back(u);
118         }
119     }
120     return tree;
121 }
122
123 // Edge-connected component IDs 1..cnt; forest edges carry original edge IDs.
124 vector<vector<pair<int, int>>> Bridge_Forest() const
125 {
126     vector<vector<pair<int, int>>> tree(cnt + 1);
127     for ( int id = 0; id < (int)edges.size(); id++ )
128     {
129         if ( !bridge[id] )
130             continue;
131         auto [u, v] = edges[id];
132         u = bel[u];
133         v = bel[v];
134         tree[u].push_back({v, id});
135         tree[v].push_back({u, id});
136     }
137     return tree;
138 }
139 };

```


第六十章 奇环顶点判定

60.1 属于简单奇环的全部顶点

算法约束见动态版本第 7 页。本节代码独立可抄；采用下列实际接口名。

```
1 // After Run(). Simple odd cycles have length >= 3; self-loops are ignored.
2 template <int N> vector<int> Odd_Cycle_Vertices(const Biconnected_Graph<N> &graph)
3 {
4     int n = graph.n;
5     auto forest = graph.Block_Forest();
6     vector<int> parent(forest.size()), depth(forest.size());
7     auto root_tree = [&](auto &&self, int u, int p) -> void
8     {
9         parent[u] = p;
10        for ( int v : forest[u] )
11            if ( v != p )
12            {
13                depth[v] = depth[u] + 1;
14                self(self, v, u);
15            }
16    };
17    for ( int u = 1; u <= n; u++ )
18        if ( !depth[u] )
19        {
20            depth[u] = 1;
21            root_tree(root_tree, u, 0);
22        }
```

```

23     vector<vector<pair<int, int>>> edges(graph.blocks.size());
24     for ( auto [u, v] : graph.edges )
25         if ( u != v )
26         {
27             int block = parent[depth[u] >= depth[v] ? u : v] - n - 1;
28             assert(0 <= block && block < (int)edges.size());
29             edges[block].push_back({u, v});
30         }
31     vector<vector<int>> g(n + 1);
32     vector<int> color(n + 1), answer(n + 1);
33     for ( int block = 0; block < (int)edges.size(); block++ )
34     {
35         for ( int u : graph.blocks[block] )
36         {
37             g[u].clear();
38             color[u] = 0;
39         }
40         for ( auto [u, v] : edges[block] )
41         {
42             g[u].push_back(v);
43             g[v].push_back(u);
44         }
45         bool bipartite = true;
46         auto paint = [&](auto &&self, int u) -> void
47         {
48             for ( int v : g[u] )
49                 if ( !color[v] )
50                 {
51                     color[v] = -color[u];
52                     self(self, v);
53                 }
54             else if ( color[v] == color[u] )
55                 bipartite = false;
56         };
57         int root = graph.blocks[block][0];
58         color[root] = 1;
59         paint(paint, root);
60         if ( !bipartite )
61             for ( int u : graph.blocks[block] )
62                 answer[u] = 1;
63     }
64     return answer;
65 }

```

第六十一章 删点连通性

61.1 由点双分解计算删点连通块数

算法约束见动态版本第 9 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Call after Run(). Each vertex is deleted independently from the original graph.
2 template <int N>
3 inline pair<int, vector<int>> Removal_Components(const Biconnected_Graph<N> &graph)
4 {
5     int before = graph.cnt - count(graph.bridge.begin(), graph.bridge.end(), 1);
6     vector<int> after(graph.n + 1, before - 1);
7     after[0] = 0;
8     for ( const auto &block : graph.blocks )
9         if ( block.size() > 1 )
10             for ( int u : block )
11                 after[u]++;
12     return {before, after};
13 }
```

第六十二章 边双缩点与边定向

62.1 边双缩点的大小与内部边数

算法约束见动态版本第 10 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Edge_Compression
2 {
3     int n;
4     vector<int> size, inside;
5     vector<array<int, 3>> edges;
6
7     template <int N>
8     Edge_Compression(const Biconnected_Graph<N> &graph)
9         : n(graph.cnt), size(n + 1), inside(n + 1)
10    {
11        for ( int u = 1; u <= graph.n; u++ )
12            size[graph.bel[u]]++;
13        for ( int id = 0; id < (int)graph.edges.size(); id++ )
14        {
15            auto [u, v] = graph.edges[id];
16            u = graph.bel[u];
17            v = graph.bel[v];
18            if ( u == v )
19                inside[u]++;
20            else
21                edges.push_back({u, v, id});
22        }
23    }
24 };
```

62.2 无向边定向与强连通定向

算法约束见动态版本第 11 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Tree edges go down; other edges go to ancestors. Handles each logical edge once.
2 inline vector<pair<int, int>> Orient_Edges(const Low_Link &graph)
3 {
4     vector<pair<int, int>> direction(graph.edges);
5     vector<int> seen(graph.n + 1);
6     auto dfs = [&](auto &&self, int u) -> void
7     {
8         seen[u] = 1;
9         for ( auto [v, id] : graph.g[u] )
10         {
11             if ( direction[id].first )
12                 continue;
13             direction[id] = {u, v};
14             if ( !seen[v] )
15                 self(self, v);
16         }
17     };
18     for ( int u = 1; u <= graph.n; u++ )
19         if ( !seen[u] )
20             dfs(dfs, u);
21     return direction;
22 }
```

第六十三章 桥树最少加边

63.1 连通无向图最少加边方案

算法约束见动态版本第 12 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Connected, nonempty graph after Run(); new parallel edges are allowed.
2 template <int N>
3 inline vector<pair<int, int>> Bridge_Augmentation(const Biconnected_Graph<N> &graph)
4 {
5     assert(graph.n > 0 && graph.cnt > 0);
6     int bridges = count(graph.bridge.begin(), graph.bridge.end(), 1);
7     assert(graph.cnt - bridges == 1);
8     if ( graph.cnt == 1 )
9         return {};
10    auto tree = graph.Bridge_Forest();
11    vector<int> representative(graph.cnt + 1);
12    for ( int u = 1; u <= graph.n; u++ )
13        representative[graph.bel[u]] = u;
14    vector<int> leaves;
15    auto dfs = [&](auto &&self, int u, int parent) -> void
16    {
17        if ( tree[u].size() == 1 )
18            leaves.push_back(u);
19        for ( auto [v, id] : tree[u] )
20            if ( v != parent )
21                self(self, v, u);
22    };
23    dfs(dfs, 1, 0);
24    if ( leaves.size() % 2 )
25        leaves.push_back(leaves[0]);
26    int half = leaves.size() / 2;
27    vector<pair<int, int>> added;
28    for ( int i = 0; i < half; i++ )
29        added.push_back({representative[leaves[i]], representative[leaves[i + half]]});
30    return added;
31 }
```

第六十四章 数据结构

64.1 并查集

算法约束见动态版本第 13 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```
1 template <int N> struct Disjoint_Set
2 {
3     int n, fa[N + 1], siz[N + 1];
4
5     void Init(int size)
6     {
7         assert(size <= N);
8         n = size;
9         iota(fa, fa + n + 1, 0);
10        fill(siz, siz + n + 1, 1);
11    }
12
13    int Find(int x)
14    {
15        int root = x;
16        while ( root != fa[root] )
17            root = fa[root];
18        while ( x != root )
19        {
20            int next = fa[x];
21            fa[x] = root;
22            x = next;
23        }
24        return root;
25    }
26
27    bool Merge(int x, int y)
28    {
29        x = Find(x);
30        y = Find(y);
31        if ( x == y )
32            return false;
33        if ( siz[x] < siz[y] )
34            swap(x, y);
35        fa[y] = x;
36        siz[x] += siz[y];
37        return true;
38    }
39 };
```

64.2 可撤销并查集

算法约束见动态版本第 14 页。本节代码独立可抄；采用下列实际接口名。

```

1  template <int N> struct Rollback_DSU
2  {
3      int fa[N + 1], siz[N + 1];
4      vector<pair<int, int>> history;
5
6      void Init(int n)
7      {
8          assert(n <= N);
9          history.clear();
10         iota(fa, fa + n + 1, 0);
11         fill(siz, siz + n + 1, 1);
12     }
13
14     int Find(int x) const
15     {
16         while ( fa[x] != x )
17             x = fa[x];
18         return x;
19     }
20
21     int Snapshot() const
22     {
23         return (int)history.size();
24     }
25
26     bool Merge(int x, int y)
27     {
28         x = Find(x);
29         y = Find(y);
30         if ( x == y )
31             return false;
32         if ( siz[x] < siz[y] )
33             swap(x, y);
34         history.push_back({y, siz[x]});
35         fa[y] = x;
36         siz[x] += siz[y];
37         return true;
38     }
39
40     void Rollback(int time)
41     {
42         assert(0 <= time && time <= Snapshot());
43         while ( Snapshot() > time )
44         {
45             auto [y, size] = history.back();
46             history.pop_back();
47             siz[fa[y]] = size;
48             fa[y] = y;
49         }
50     }
51 };

```


64.3 树状数组与第 k 小

算法约束见动态版本第 15 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```

1 template <int N, class T = long long> struct Binary_Indexed_Tree
2 {
3     int n;
4     T tree[N + 1];
5
6     void Init(int size)
7     {
8         assert(size <= N);
9         n = size;
10        fill(tree, tree + n + 1, T(0));
11    }
12
13    void Insert(int x, T value)
14    {
15        assert(x >= 1);
16        for ( ; x <= n; x += x & -x )
17            tree[x] += value;
18    }
19
20    T Sum(int x) const
21    {
22        T answer = 0;
23        for ( ; x; x -= x & -x )
24            answer += tree[x];
25        return answer;
26    }
27
28    T Query(int L, int R) const
29    {
30        return Sum(R) - Sum(L - 1);
31    }
32
33    int Kth(T k) const
34    {
35        assert(k > 0);
36        int now = 0, step = 1;
37        while ( step <= n / 2 )
38            step <= 1;
39        for ( ; step; step >>= 1 )
40            if ( now + step <= n && tree[now + step] < k )
41            {
42                now += step;
43                k -= tree[now];
44            }
45        return now + 1;
46    }
47 };

```

64.4 区间加与区间和

算法约束见动态版本第 15 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```

1 template <int N> struct Segment_Tree
2 {
3     int n;
4     long long tree[4 * N + 5], tag[4 * N + 5];
5
6     void Init(int size)
7     {
8         assert(size > 0 && size <= N);
9         n = size;
10        fill(tree, tree + 4 * n + 5, 0);
11        fill(tag, tag + 4 * n + 5, 0);
12    }
13
14    void Update(int now, int L, int R, long long value)
15    {
16        tree[now] += (R - L + 1) * value;
17        tag[now] += value;
18    }
19
20    void Push_Up(int now)
21    {
22        tree[now] = tree[now << 1] + tree[now << 1 | 1];
23    }
24
25    void Push_Down(int now, int L, int R)
26    {
27        int mid = (L + R) >> 1;
28        Update(now << 1, L, mid, tag[now]);
29        Update(now << 1 | 1, mid + 1, R, tag[now]);
30        tag[now] = 0;
31    }
32
33    void Insert(int ql, int qr, long long value)
34    {
35        Insert(1, 1, n, ql, qr, value);
36    }
37
38    void Insert(int now, int L, int R, int ql, int qr, long long value)
39    {
40        if (ql <= L && R <= qr)
41            return Update(now, L, R, value);
42        Push_Down(now, L, R);
43        int mid = (L + R) >> 1;
44        if (ql <= mid)
45            Insert(now << 1, L, mid, ql, qr, value);
46        if (qr > mid)
47            Insert(now << 1 | 1, mid + 1, R, ql, qr, value);
48        Push_Up(now);
49    }
50
51    long long Query(int ql, int qr)

```

```
52 {
53     return Query(1, 1, n, ql, qr);
54 }
55
56 long long Query(int now, int L, int R, int ql, int qr)
57 {
58     if ( ql <= L && R <= qr )
59         return tree[now];
60     Push_Down(now, L, R);
61     int mid = (L + R) >> 1;
62     long long answer = 0;
63     if ( ql <= mid )
64         answer += Query(now << 1, L, mid, ql, qr);
65     if ( qr > mid )
66         answer += Query(now << 1 | 1, mid + 1, R, ql, qr);
67     return answer;
68 }
69 };
```

64.5 64 位异或线性基

算法约束见动态版本第 18 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Xor_Basis
2 {
3     using U = unsigned long long;
4     using Wide = __uint128_t;
5     U base[64]{};
6     vector<U> p;
7     int rank = 0;
8     bool dependent = false, dirty = false;
9
10    void Init()
11    {
12        fill(base, base + 64, 0);
13        p.clear();
14        rank = 0;
15        dependent = false;
16        dirty = false;
17    }
18
19    bool Insert(U x)
20    {
21        for ( int i = 63; i >= 0; i-- )
22            if ( x >> i & 1 )
23            {
24                if ( !base[i] )
25                {
26                    base[i] = x;
27                    rank++;
28                    dirty = true;
29                    return true;
30                }
31                x ^= base[i];
32            }
33        dependent = true;
34        return false;
35    }
36
37    bool Contains(U x) const
38    {
39        for ( int i = 63; i >= 0; i-- )
40            if ( x >> i & 1 )
41                x ^= base[i];
42        return x == 0;
43    }
44
45    U Query(U x = 0) const
46    {
47        for ( int i = 63; i >= 0; i-- )
48            x = max(x, x ^ base[i]);
49        return x;
50    }

```

```

52 void Rebuild()
53 {
54     vector<U> b(base, base + 64);
55     for ( int i = 0; i < 64; i++ )
56         if ( b[i] )
57             for ( int j = i + 1; j < 64; j++ )
58                 if ( b[j] >> i & 1 )
59                     b[j] ^= b[i];
60     p.clear();
61     for ( U x : b )
62         if ( x )
63             p.push_back(x);
64     dirty = false;
65 }
66
67 // k is 1-based; nonempty excludes the empty input subset, not value zero.
68 optional<U> Kth(Wide k, bool nonempty = true)
69 {
70     if ( !k )
71         return nullopt;
72     Wide index = k;
73     if ( !nonempty || dependent )
74         index--;
75     if ( index >= (Wide(1) << rank) )
76         return nullopt;
77     if ( dirty )
78         Rebuild();
79     U answer = 0;
80     for ( int i = 0; i < rank; i++ )
81         if ( index >> i & 1 )
82             answer ^= p[i];
83     return answer;
84 }
85
86 optional<U> Minimum(bool nonempty = true)
87 {
88     return Kth(1, nonempty);
89 }
90
91 void Merge(const Xor_Basis &other)
92 {
93     dependent |= other.dependent;
94     for ( U x : other.base )
95         if ( x )
96             Insert(x);
97 }
98 };

```

第六十五章 带位置线性基

65.1 带位置限制的异或线性基

算法约束见动态版本第 21 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Position_Basis
2 {
3     using U = unsigned long long;
4     U base[64];
5     int pos[64];
6
7     Position_Basis()
8     {
9         Init();
10    }
11
12    void Init()
13    {
14        fill(base, base + 64, 0);
15        fill(pos, pos + 64, -1);
16    }
```

```

18 void Insert(U x, int at)
19 {
20     assert(at >= 0);
21     for ( int i = 63; i >= 0; i-- )
22         if ( x >> i & 1 )
23         {
24             if ( !base[i] )
25             {
26                 base[i] = x;
27                 pos[i] = at;
28                 return;
29             }
30             if ( at > pos[i] )
31             {
32                 swap(base[i], x);
33                 swap(pos[i], at);
34             }
35             x ^= base[i];
36         }
37 }
38
39 bool Contains(U x, long long lower = 0) const
40 {
41     assert(lower >= 0);
42     for ( int i = 63; i >= 0; i-- )
43         if ( (x >> i & 1) && pos[i] >= lower )
44             x ^= base[i];
45     return x == 0;
46 }
47
48 U Query(long long lower = 0, U x = 0) const
49 {
50     assert(lower >= 0);
51     for ( int i = 63; i >= 0; i-- )
52         if ( pos[i] >= lower )
53             x = max(x, x ^ base[i]);
54     return x;
55 }
56
57 int Rank(long long lower = 0) const
58 {
59     assert(lower >= 0);
60     int answer = 0;
61     for ( int i = 0; i < 64; i++ )
62         answer += pos[i] >= lower;
63     return answer;
64 }
65 };

```

第六十六章 线性空间求交

66.1 异或线性空间求交

算法约束见动态版本第 23 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline Xor_Basis Basis_Intersection(const Xor_Basis &a, const Xor_Basis &b)
2 {
3     using U = unsigned long long;
4     U pivot[64], from_b[64]{};
5     copy(a.base, a.base + 64, pivot);
6     Xor_Basis result;
7     for ( U value : b.base )
8     {
9         U x = value, part = value;
10        for ( int i = 63; i >= 0 && x; i-- )
11            if ( x >> i & 1 )
12            {
13                if ( !pivot[i] )
14                {
15                    pivot[i] = x;
16                    from_b[i] = part;
17                    break;
18                }
19                x ^= pivot[i];
20                part ^= from_b[i];
21            }
22        if ( !x && part )
23            result.Insert(part);
24    }
25    return result;
26 }
```


66.2 Zassenhaus：同时求和空间与交空间

算法约束见动态版本第 24 页。本节代码独立可抄；采用下列实际接口名。

```

1 // Return independent generators of (sum space, intersection space).
2 inline pair<Xor_Basis, Xor_Basis> Basis_Sum_Intersection(const Xor_Basis &a,
3                                                         const Xor_Basis &b)
4 {
5     using W = __uint128_t;
6     W pivot[128]{};
7     auto insert = [&](W x)
8     {
9         for ( int i = 127; i >= 0 && x; i-- )
10             if ( x >> i & 1 )
11             {
12                 if ( !pivot[i] )
13                 {
14                     pivot[i] = x;
15                     return;
16                 }
17                 x ^= pivot[i];
18             }
19     };
20     for ( auto x : a.base )
21         insert((W(x) << 64) | x);
22     for ( auto x : b.base )
23         insert(W(x) << 64);
24     Xor_Basis sum, intersection;
25     for ( int i = 0; i < 128; i++ )
26         if ( pivot[i] )
27         {
28             if ( i < 64 )
29                 intersection.Insert((unsigned long long)pivot[i]);
30             else
31                 sum.Insert((unsigned long long)(pivot[i] >> 64));
32         }
33     return {sum, intersection};
34 }
```

第六十七章 线段树常用修改

67.1 线段树：区间乘、加、赋值与模意义求和

算法约束见动态版本第 25 页。本节代码独立可抄；采用下列实际接口名。

声明 `Affine_Segment_Tree<N>` 后 `Init(a,mod)`, `N` 为最大数组长度。`Update(l,r,mul,add)` 修改, `Query(l,r)` 求和；支持重复初始化。大对象放在全局或 `static`。

```
1 template <int N> struct Affine_Segment_Tree
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         ll sum = 0, mul = 1, add = 0;
8     };
9
10    int n, mod;
11    Node t[4 * N + 4];
12
13    void Init(const vector<ll> &a, int modulus)
14    {
15        n = a.size();
16        mod = modulus;
17        assert(0 < n && n <= N && mod > 0);
18        Build(1, 1, n, a);
19    }
20
21    ll Norm(ll x) const
22    {
23        x %= mod;
24        if ( x < 0 )
25            x += mod;
26        return x;
27    }
28
29    void Push_Up(int p)
30    {
31        t[p].sum = (t[p * 2].sum + t[p * 2 + 1].sum) % mod;
32    }
33
34    void Build(int p, int l, int r, const vector<ll> &a)
35    {
36        t[p] = Node{};
37        if ( l == r )
38        {
39            t[p].sum = Norm(a[l - 1]);
40            return;
41        }
42        int m = (l + r) / 2;
43        Build(p * 2, l, m, a);
44        Build(p * 2 + 1, m + 1, r, a);
```

```

45     Push_Up(p);
46 }
47
48 // New transform follows the old pending transform: x -> x*a+b.
49 void Apply(int p, int len, ll a, ll b)
50 {
51     t[p].sum = (t[p].sum * a % mod + len * b % mod) % mod;
52     t[p].mul = t[p].mul * a % mod;
53     t[p].add = (t[p].add * a % mod + b) % mod;
54 }
55
56 void Push_Down(int p, int l, int r)
57 {
58     if ( t[p].mul == 1 && t[p].add == 0 )
59         return;
60     int m = (l + r) / 2;
61     Apply(p * 2, m - l + 1, t[p].mul, t[p].add);
62     Apply(p * 2 + 1, r - m, t[p].mul, t[p].add);
63     t[p].mul = 1;
64     t[p].add = 0;
65 }
66
67 void Update(int l, int r, ll a, ll b)
68 {
69     assert(1 <= l && l <= r && r <= n);
70     Update(1, 1, n, l, r, Norm(a), Norm(b));
71 }
72
73 void Update(int p, int l, int r, int ql, int qr, ll a, ll b)
74 {
75     if ( ql <= l && r <= qr )
76     {
77         Apply(p, r - l + 1, a, b);
78         return;
79     }
80     Push_Down(p, l, r);
81     int m = (l + r) / 2;
82     if ( ql <= m )
83         Update(p * 2, l, m, ql, qr, a, b);
84     if ( qr > m )
85         Update(p * 2 + 1, m + 1, r, ql, qr, a, b);
86     Push_Up(p);
87 }
88
89 ll Query(int l, int r)
90 {
91     assert(1 <= l && l <= r && r <= n);
92     return Query(1, 1, n, l, r);
93 }
94
95 ll Query(int p, int l, int r, int ql, int qr)
96 {
97     if ( ql <= l && r <= qr )
98         return t[p].sum;

```

```
99     Push_Down(p, l, r);
100     int m = (l + r) / 2;
101     ll answer = 0;
102     if ( ql <= m )
103         answer += Query(p * 2, l, m, ql, qr);
104     if ( qr > m )
105         answer += Query(p * 2 + 1, m + 1, r, ql, qr);
106     return answer % mod;
107 }
108 };
```

第六十八章 网络流

68.1 最大流与方案

算法约束见动态版本第 28 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```
1 template <int N, int M> struct Network_Flow
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int to, next;
8         ll cap, initial;
9     } edge[2 * M];
10
11     int n, tot, head[N + 1], deep[N + 1], cur[N + 1], que[N + 1];
12
13     void Init(int size)
14     {
15         assert(size <= N);
16         n = size;
17         tot = 0;
18         fill(head, head + n + 1, -1);
19     }
20
21     int Add_Edge(int u, int v, ll cap)
22     {
23         assert(cap >= 0 && tot + 2 <= 2 * M);
24         int id = tot;
25         edge[tot] = {v, head[u], cap, cap};
26         head[u] = tot++;
27         edge[tot] = {u, head[v], 0, 0};
28         head[v] = tot++;
29         return id;
30     }
31
32     bool Bfs(int s, int t)
33     {
34         fill(deep, deep + n + 1, -1);
35         int L = 0, R = 0;
36         que[R++] = s;
37         deep[s] = 0;
38         while ( L < R )
39         {
40             int now = que[L++];
41             for ( int i = head[now]; i != -1; i = edge[i].next )
42             {
43                 int to = edge[i].to;
44                 if ( edge[i].cap && deep[to] == -1 )
45                 {
```

```

46         deep[to] = deep[now] + 1;
47         que[R++] = to;
48     }
49 }
50 }
51 return deep[t] != -1;
52 }
53
54 ll Dfs(int now, int t, ll flow)
55 {
56     if ( now == t )
57         return flow;
58     for ( int &i = cur[now]; i != -1; i = edge[i].next )
59     {
60         int to = edge[i].to;
61         if ( !edge[i].cap || deep[to] != deep[now] + 1 )
62             continue;
63         ll delta = Dfs(to, t, min(flow, edge[i].cap));
64         if ( delta )
65         {
66             edge[i].cap -= delta;
67             edge[i ^ 1].cap += delta;
68             return delta;
69         }
70     }
71     return 0;
72 }
73
74 ll Max_Flow(int s, int t, ll limit = LLONG_MAX)
75 {
76     assert(s != t && limit >= 0);
77     ll answer = 0;
78     while ( answer < limit && Bfs(s, t) )
79     {
80         copy(head, head + n + 1, cur);
81         while ( answer < limit )
82         {
83             ll delta = Dfs(s, t, limit - answer);
84             if ( !delta )
85                 break;
86             answer += delta;
87         }
88     }
89     return answer;
90 }
91
92 ll Used(int id) const
93 {
94     return edge[id].initial - edge[id].cap;
95 }
96
97 vector<int> Cut(int s)

```

```

98     {
99         Bfs(s, s);
100         vector<int> answer;
101         for ( int i = 1; i <= n; i++ )
102             if ( deep[i] != -1 )
103                 answer.push_back(i);
104         return answer;
105     }
106 };

```

68.2 势能最小费用最大流

算法约束见动态版本第 30 页。本节代码独立可抄；采用下列实际接口名。
使用 Insert 加边、Flow 求流、Used 查看方案。

```

1  struct Min_Cost_Flow
2  {
3      using ll = long long;
4      using i128 = __int128_t;
5
6      struct Edge
7      {
8          int from, to;
9          ll cap, cost, initial;
10     };
11
12     int n;
13     vector<Edge> e;
14     vector<vector<int>>> g;
15
16     Min_Cost_Flow(int n) : n(n), g(n + 1)
17     {
18     }
19
20     int Insert(int u, int v, ll cap, ll cost)
21     {
22         assert(cap >= 0 && cost != LLONG_MIN);
23         int id = (int)e.size();
24         e.push_back({u, v, cap, cost, cap});
25         e.push_back({v, u, 0, -cost, 0});
26         g[u].push_back(id);
27         g[v].push_back(id + 1);
28         return id;
29     }
30
31     ll Used(int id) const
32     {
33         return e[id].initial - e[id].cap;
34     }
35
36     // Negative costs allowed; initial residual graph must have no negative cycle.
37     pair<ll, i128> Flow(int s, int t, ll limit = LLONG_MAX)
38     {
39         assert(s != t && limit >= 0);
40         const i128 inf = i128(1) << 120;

```

```

41     vector<i128> h(n + 1, 0), d(n + 1);
42     vector<int> pre(n + 1);
43     // Super-source Bellman-Ford: feasible potentials for every vertex.
44     for ( int it = 0; it < n; it++ )
45     {
46         bool changed = false;
47         for ( auto &a : e )
48             if ( a.cap && h[a.to] > h[a.from] + a.cost )
49             {
50                 h[a.to] = h[a.from] + a.cost;
51                 changed = true;
52             }
53         if ( !changed )
54             break;
55         if ( it == n - 1 )
56             throw invalid_argument("negative cycle");
57     }
58     ll f = 0;
59     i128 cost = 0;
60     while ( f < limit )
61     {
62         fill(d.begin(), d.end(), inf);
63         d[s] = 0;
64         priority_queue<pair<i128, int>,
65                       vector<pair<i128, int>>,
66                       greater<pair<i128, int>>>
67             q;
68         q.push({0, s});
69         while ( !q.empty() )
70         {
71             auto [du, u] = q.top();
72             q.pop();
73             if ( du != d[u] )
74                 continue;
75             for ( int id : g[u] )
76             {
77                 auto &a = e[id];
78                 i128 nd = du + a.cost + h[u] - h[a.to];
79                 if ( a.cap && nd < d[a.to] )
80                 {
81                     d[a.to] = nd;
82                     pre[a.to] = id;
83                     q.push({nd, a.to});
84                 }
85             }
86         }
87         if ( d[t] == inf )
88             break;
89         for ( int u = 1; u <= n; u++ )
90             if ( d[u] != inf )
91                 h[u] += d[u];
92         ll take = limit - f;
93         for ( int u = t; u != s; u = e[pre[u]].from )
94             take = min(take, e[pre[u]].cap);
95         for ( int u = t; u != s; u = e[pre[u]].from )
96         {

```



```
97         int id = pre[u];
98         e[id].cap -= take;
99         e[id ^ 1].cap += take;
100        cost += i128(take) * e[id].cost;
101    }
102    f += take;
103 }
104 return {f, cost};
105 }
106 };
```

68.3 上下界可行循环流

算法约束见动态版本第 32 页。本节代码独立可抄；采用下列实际接口名。

```

1  template <int N, int M> struct Bounded_Circulation
2  {
3      int n, tot;
4      Network_Flow<N + 2, M + N> graph;
5      long long balance[N + 1], lower[M];
6      int id[M];
7      bool solved, feasible;
8
9      void Init(int size)
10     {
11         assert(size <= N);
12         n = size;
13         tot = 0;
14         solved = feasible = false;
15         graph.Init(n + 2);
16         fill(balance, balance + n + 1, 0);
17     }
18
19     int Insert(int u, int v, long long L, long long R)
20     {
21         assert(!solved && 0 <= L && L <= R && tot < M);
22         id[tot] = graph.Add_Edge(u, v, R - L);
23         lower[tot] = L;
24         balance[u] -= L;
25         balance[v] += L;
26         return tot++;
27     }
28
29     bool Solve()
30     {
31         assert(!solved);
32         solved = true;
33         long long need = 0;
34         for ( int u = 1; u <= n; u++ )
35         {
36             if ( balance[u] > 0 )
37             {
38                 graph.Add_Edge(n + 1, u, balance[u]);
39                 need += balance[u];
40             }
41             else if ( balance[u] < 0 )
42                 graph.Add_Edge(u, n + 2, -balance[u]);
43         }
44         return feasible = (graph.Max_Flow(n + 1, n + 2) == need);
45     }
46
47     long long Used(int index) const
48     {
49         assert(feasible);
50         return lower[index] + graph.Used(id[index]);
51     }
52 };

```

第六十九章 最大权闭合子图

69.1 最大权闭合子图与选点方案

算法约束见动态版本第 34 页。本节代码独立可抄；采用下列实际接口名。

Maximum_Closure(weight,dependencies,graph) 额外接收 Network_Flow<N,M> 引用，并在每次调用中 Init 重建该网络。N 至少为原点数加 2，M 至少为依赖边数加非零权点数；这是原有向边容量，不是残量边总数。将网络对象放在全局或 static，模板参数可由对象自动推导。

```
1 template <int N, int M>
2 pair<long long, vector<int>> Maximum_Closure(const vector<long long> &weight,
3                                             const vector<pair<int, int>> &dependencies,
4                                             Network_Flow<N, M> &graph)
5 {
6     using I = __int128_t;
7     int n = weight.size(), s = n + 1, t = n + 2;
8     I positive = 0;
9     for ( long long w : weight )
10         if ( w > 0 )
11             positive += w;
12     assert(positive < LLONG_MAX);
13     long long inf = (long long)positive + 1;
14     graph.Init(n + 2);
15     for ( int u = 1; u <= n; u++ )
16     {
17         long long w = weight[u - 1];
18         if ( w > 0 )
19             graph.Add_Edge(s, u, w);
20         else if ( w < 0 )
21             graph.Add_Edge(u, t, (long long)min(-I(w), I(inf)));
22     }
23     for ( auto [u, v] : dependencies )
24     {
25         assert(1 <= u && u <= n && 1 <= v && v <= n);
26         graph.Add_Edge(u, v, inf);
27     }
28     long long answer = (long long)positive - graph.Max_Flow(s, t);
29     vector<int> selected;
30     for ( int u : graph.Cut(s) )
31         if ( u <= n )
32             selected.push_back(u);
33     return {answer, selected};
34 }
```

第七十章 图论

70.1 非负权最短路

算法约束见动态版本第 35 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Shortest_Path
2 {
3     using ll = long long;
4     static constexpr ll inf = LLONG_MAX;
5     int n;
6     vector<vector<pair<int, ll>>> g;
7     vector<ll> dis;
8     vector<int> pre;
9
10    Shortest_Path(int n) : n(n), g(n + 1)
11    {
12    }
13
14    void Insert(int u, int v, ll w)
15    {
16        assert(w >= 0);
17        g[u].push_back({v, w});
18    }
19
20    void Run(int s)
21    {
22        dis.assign(n + 1, inf);
23        pre.assign(n + 1, -1);
24        dis[s] = 0;
25        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<pair<ll, int>>> q;
26        q.push({0, s});
27        while ( !q.empty() )
28        {
29            auto [d, u] = q.top();
30            q.pop();
31            if ( d != dis[u] )
32                continue;
33            for ( auto [v, w] : g[u] )
34                if ( w < inf - d && d + w < dis[v] )
35                {
36                    dis[v] = d + w;
37                    pre[v] = u;
38                    q.push({dis[v], v});
39                }
40        }
41    }
42
43    vector<int> Path(int t) const
44    {
```

```

45     if ( dis[t] == inf )
46         return {};
47     vector<int> ans;
48     for ( ; t != -1; t = pre[t] )
49         ans.push_back(t);
50     reverse(ans.begin(), ans.end());
51     return ans;
52 }
53 };

```

70.2 强连通分量

算法约束见动态版本第 36 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Strong_Component
2 {
3     int n, cnt = 0;
4     vector<vector<int>> g, rg;
5     vector<int> bel;
6
7     Strong_Component(int n) : n(n), g(n + 1), rg(n + 1)
8     {
9     }
10
11     void Insert(int u, int v)
12     {
13         g[u].push_back(v);
14         rg[v].push_back(u);
15     }
16
17     void Dfs1(int u, vector<int> &vis, vector<int> &order)
18     {
19         vis[u] = 1;
20         for ( int v : g[u] )
21         {
22             if ( !vis[v] )
23                 Dfs1(v, vis, order);
24         }
25         order.push_back(u);
26     }
27
28     void Dfs2(int u)
29     {
30         bel[u] = cnt;
31         for ( int v : rg[u] )
32         {
33             if ( !bel[v] )
34                 Dfs2(v);
35         }
36     }
37
38     // Recursive Kosaraju; component IDs are in topological order, 1..cnt.
39     void Run()
40     {
41         vector<int> vis(n + 1), order;

```

```

42     cnt = 0;
43     bel.assign(n + 1, 0);
44     for ( int s = 1; s <= n; s++ )
45     {
46         if ( !vis[s] )
47             Dfs1(s, vis, order);
48     }
49     reverse(order.begin(), order.end());
50     for ( int s : order )
51     {
52         if ( !bel[s] )
53         {
54             ++cnt;
55             Dfs2(s);
56         }
57     }
58 }
59 };

```

70.3 2-SAT 与任意可行方案

算法约束见动态版本第 37 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Two_SAT
2 {
3     int n;
4     Strong_Component g;
5     vector<int> ans;
6
7     Two_SAT(int n) : n(n), g(2 * n)
8     {
9     }
10
11     int id(int x, bool value) const
12     {
13         return 2 * x - 1 + value;
14     }
15
16     void Insert(int x, bool a, int y, bool b)
17     {
18         g.Insert(id(x, !a), id(y, b));
19         g.Insert(id(y, !b), id(x, a));
20     }
21
22     bool Solve()
23     {
24         g.Run();
25         ans.assign(n + 1, 0);

```

```

26     for ( int x = 1; x <= n; x++ )
27     {
28         if ( g.bel[id(x, 0)] == g.bel[id(x, 1)] )
29             return false;
30         ans[x] = g.bel[id(x, 1)] > g.bel[id(x, 0)];
31     }
32     return true;
33 }
34 };

```

70.4 Hopcroft–Karp 最大匹配与最小点覆盖

算法约束见动态版本第 38 页。本节代码独立可抄；采用下列实际接口名。

声明 `Bipartite_Matching<N,M>` 后 `Init(n,m)`，两侧容量分别为 `N`、`M`。Insert 加边，Solve 求匹配，Cover 返回最小点覆盖；`l`、`r` 保存方案。大对象放在全局或 `static`。

```

1  template <int N, int M> struct Bipartite_Matching
2  {
3      int n, m;
4      vector<int> g[N + 1];
5      int l[N + 1], r[M + 1], dep[N + 1], cur[N + 1], que[N + 1];
6
7      void Init(int left_size, int right_size)
8      {
9          assert(0 <= left_size && left_size <= N && 0 <= right_size && right_size <= M);
10         n = left_size;
11         m = right_size;
12         fill(l, l + n + 1, 0);
13         fill(r, r + m + 1, 0);
14         for ( int u = 1; u <= n; u++ )
15             g[u].clear();
16     }
17
18     void Insert(int u, int v)
19     {
20         assert(1 <= u && u <= n && 1 <= v && v <= m);
21         g[u].push_back(v);
22     }
23
24     bool Bfs()
25     {
26         int head = 0, tail = 0;
27         fill(dep, dep + n + 1, n + 1);
28         fill(cur, cur + n + 1, 0);
29         for ( int u = 1; u <= n; u++ )
30         {
31             if ( !l[u] )
32             {
33                 dep[u] = 0;
34                 que[tail++] = u;
35             }
36         }
37         while ( head < tail )
38         {
39             int u = que[head++];

```

```

40         if ( dep[u] >= dep[0] )
41             continue;
42         for ( int v : g[u] )
43         {
44             int next = r[v];
45             if ( dep[next] != n + 1 )
46                 continue;
47             dep[next] = dep[u] + 1;
48             if ( next )
49                 que[tail++] = next;
50         }
51     }
52     return dep[0] != n + 1;
53 }
54
55 bool Dfs(int u)
56 {
57     if ( !u )
58         return true;
59     for ( int &i = cur[u]; i < (int)g[u].size(); i++ )
60     {
61         int v = g[u][i];
62         if ( dep[r[v]] == dep[u] + 1 && Dfs(r[v]) )
63         {
64             l[u] = v;
65             r[v] = u;
66             return true;
67         }
68     }
69     dep[u] = n + 1;
70     return false;
71 }
72
73 // Retains an existing matching; edges may be added between calls.
74 int Solve()
75 {
76     int answer = 0;
77     for ( int u = 1; u <= n; u++ )
78         answer += (l[u] != 0);
79     while ( Bfs() )
80     {
81         for ( int u = 1; u <= n; u++ )
82         {
83             if ( !l[u] && Dfs(u) )
84                 answer++;
85         }
86     }
87     return answer;
88 }
89
90 // Call after Solve(), before adding more edges. Returns left/right vertex IDs.
91 pair<vector<int>, vector<int>> Cover() const
92 {
93     vector<int> left, right;
94     for ( int u = 1; u <= n; u++ )
95     {

```



```

96         if ( dep[u] == n + 1 )
97             left.push_back(u);
98     }
99     for ( int v = 1; v <= m; v++ )
100     {
101         if ( r[v] && dep[r[v]] != n + 1 )
102             right.push_back(v);
103     }
104     return {left, right};
105 }
106 };

```

70.5 割点与桥

算法约束见动态版本第 40 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Low_Link
2 {
3     int n, timer = 0, components = 0;
4     vector<vector<pair<int, int>>> g;
5     vector<int> dfn, low, cut, bridge, delta;
6     int edges = 0;
7
8     Low_Link(int n) : n(n), g(n + 1)
9     {
10    }
11
12    int Insert(int u, int v)
13    {
14        g[u].push_back({v, edges});
15        g[v].push_back({u, edges});
16        return edges++;
17    }

```

```

19 void Dfs(int u, int pe)
20 {
21     dfn[u] = low[u] = ++timer;
22     int children = 0;
23     for ( auto [v, id] : g[u] )
24     {
25         if ( id == pe )
26             continue;
27         if ( !dfn[v] )
28         {
29             ++children;
30             Dfs(v, id);
31             low[u] = min(low[u], low[v]);
32             if ( low[v] > dfn[u] )
33                 bridge[id] = 1;
34             if ( pe != -1 && low[v] >= dfn[u] )
35             {
36                 cut[u] = 1;
37                 delta[u]++;
38             }
39         }
40         else
41             low[u] = min(low[u], dfn[v]);
42     }
43     if ( pe == -1 && children > 1 )
44         cut[u] = 1;
45     if ( pe == -1 )
46         delta[u] = children - 1;
47 }
48
49 void Run()
50 {
51     timer = 0;
52     components = 0;
53     dfn.assign(n + 1, 0);
54     low = dfn;
55     cut = dfn;
56     delta = dfn;
57     bridge.assign(edges, 0);
58     for ( int u = 1; u <= n; u++ )
59         if ( !dfn[u] )
60         {
61             components++;
62             Dfs(u, -1);
63         }
64 }
65 };

```

第七十一章 字典序最小 2-SAT

71.1 递归染色与回滚求最小赋值

算法约束见动态版本第 42 页。本节代码独立可抄；采用下列实际接口名。

声明 Lex_Two_SAT<N>, Init(n) 清空子句; Add、Force、Solve 与动态版对应, answer[1..n] 为答案。内部 Id 的字面量编号从 0 开始, 大对象放全局或 static。

```
1 template <int N> struct Lex_Two_SAT
2 {
3     int n;
4     vector<int> g[2 * N], history, answer;
5     int marked[2 * N];
6
7     void Init(int variables)
8     {
9         assert(0 <= variables && variables <= N);
10        n = variables;
11        for ( int u = 0; u < 2 * n; u++ )
12            g[u].clear();
13        history.clear();
14        answer.clear();
15    }
16
17    int Id(int x, bool value) const
18    {
19        assert(1 <= x && x <= n);
20        return 2 * (x - 1) + value;
21    }
22
23    // (x == a) OR (y == b). Internal literal IDs are 0-based.
24    void Add(int x, bool a, int y, bool b)
25    {
26        int u = Id(x, a), v = Id(y, b);
27        g[u ^ 1].push_back(v);
28        g[v ^ 1].push_back(u);
29        answer.clear();
30    }
31
32    void Force(int x, bool value)
33    {
34        Add(x, value, x, value);
35    }
```

```

37  bool Paint(int u)
38  {
39      if ( marked[u ^ 1] )
40          return false;
41      if ( marked[u] )
42          return true;
43      marked[u] = 1;
44      history.push_back(u);
45      for ( int v : g[u] )
46          if ( !Paint(v) )
47              return false;
48      return true;
49  }
50
51  bool Solve()
52  {
53      fill(marked, marked + 2 * n, 0);
54      history.clear();
55      answer.clear();
56      for ( int x = 1; x <= n; x++ )
57      {
58          int u = Id(x, false);
59          if ( marked[u] || marked[u ^ 1] )
60              continue;
61          int checkpoint = history.size();
62          if ( !Paint(u) )
63          {
64              while ( (int)history.size() > checkpoint )
65              {
66                  marked[history.back()] = 0;
67                  history.pop_back();
68              }
69              if ( !Paint(u ^ 1) )
70                  return false;
71          }
72      }
73      answer.assign(n + 1, 0);
74      for ( int x = 1; x <= n; x++ )
75          answer[x] = marked[Id(x, true)];
76      return true;
77  }
78  };

```

第七十二章 有向欧拉路

72.1 有向图字典序最小欧拉路

算法约束见动态版本第 44 页。本节代码独立可抄；采用下列实际接口名。

声明 Directed_Euler<N>, Init(n) 清空图, Insert 加边并返回从 0 开始的边编号, Run 求解（参数默认值同动态版）。vertices、edge_ids、edges 可直接读取；N 为最大点数，大对象放全局或 static。

```
1 template <int N> struct Directed_Euler
2 {
3     int n;
4     vector<pair<int, int>> g[N + 1], edges;
5     vector<int> vertices, edge_ids;
6     int in[N + 1], out[N + 1], next[N + 1];
7
8     void Init(int count)
9     {
10         assert(0 < count && count <= N);
11         n = count;
12         for ( int u = 1; u <= n; u++ )
13         {
14             g[u].clear();
15             in[u] = 0;
16             out[u] = 0;
17         }
18         edges.clear();
19         vertices.clear();
20         edge_ids.clear();
21     }
```

```
23  int Insert(int u, int v)
24  {
25      assert(1 <= u && u <= n && 1 <= v && v <= n);
26      int id = edges.size();
27      edges.push_back({u, v});
28      g[u].push_back({v, id});
29      out[u]++;
30      in[v]++;
31      vertices.clear();
32      edge_ids.clear();
33      return id;
34  }
35
36  void Dfs(int u, int incoming)
37  {
38      while ( next[u] < (int)g[u].size() )
39      {
40          auto [v, id] = g[u][next[u]++];
41          Dfs(v, id);
42      }
43      vertices.push_back(u);
44      if ( incoming != -1 )
45          edge_ids.push_back(incoming);
46  }
```

```

48 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
49 bool Run(int start = 0, bool lexicographic = true)
50 {
51     assert(0 <= start && start <= n);
52     vertices.clear();
53     edge_ids.clear();
54     int source = 0, sink = 0;
55     for ( int u = 1; u <= n; u++ )
56     {
57         int delta = out[u] - in[u];
58         if ( delta == 1 && !source )
59             source = u;
60         else if ( delta == -1 && !sink )
61             sink = u;
62         else if ( delta != 0 )
63             return false;
64     }
65     if ( bool(source) != bool(sink) )
66         return false;
67     if ( source )
68     {
69         if ( start && start != source )
70             return false;
71         start = source;
72     }
73     if ( !start )
74     {
75         start = 1;
76         for ( int u = 1; u <= n; u++ )
77             if ( out[u] )
78             {
79                 start = u;
80                 break;
81             }
82     }
83     if ( lexicographic )
84         for ( int u = 1; u <= n; u++ )
85             sort(g[u].begin(), g[u].end());
86     fill(next, next + n + 1, 0);
87     Dfs(start, -1);
88     if ( edge_ids.size() != edges.size() )
89     {
90         vertices.clear();
91         edge_ids.clear();
92         return false;
93     }
94     reverse(vertices.begin(), vertices.end());
95     reverse(edge_ids.begin(), edge_ids.end());
96     return true;
97 }
98 };

```

第七十三章 字典序单词链

73.1 字典序最小单词链

算法约束见动态版本第 46 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Lowercase nonempty words. Returns original 0-based IDs; duplicates stay distinct.
2 inline optional<vector<int>> Word_Chain(const vector<string> &words)
3 {
4     int n = words.size();
5     vector<int> order(n);
6     iota(order.begin(), order.end(), 0);
7     for ( const auto &word : words )
8     {
9         assert(!word.empty());
10        for ( char c : word )
11            assert('a' <= c && c <= 'z');
12    }
13    sort(order.begin(),
14         order.end(),
15         [&](int a, int b)
16         {
17             if ( words[a] != words[b] )
18                 return words[a] < words[b];
19             return a < b;
20         });
21    Directed_Euler<26> graph;
22    graph.Init(26);
23    for ( int id : order )
24        graph.Insert(words[id].front() - 'a' + 1, words[id].back() - 'a' + 1);
25    // Preserve word order in each adjacency list; do not sort by destination.
26    if ( !graph.Run(0, false) )
27        return nullopt;
28    vector<int> result;
29    for ( int id : graph.edge_ids )
30        result.push_back(order[id]);
31    return result;
32 }
```


第七十四章 无向欧拉路

74.1 无向图欧拉路与边方向

算法约束见动态版本第 47 页。本节代码独立可抄；采用下列实际接口名。

声明 Undirected_Euler<N>, Init(n) 清空图, Insert 返回从 0 开始的逻辑边编号, Run 参数与动态版一致。
vertices、edge_ids 和原边 edges 可直接读取；N 为最大点数，大对象放全局或 static。

```
1 template <int N> struct Undirected_Euler
2 {
3     int n;
4     vector<pair<int, int>> g[N + 1], edges;
5     vector<int> used, vertices, edge_ids;
6     int next[N + 1];
7
8     void Init(int count)
9     {
10         assert(0 < count && count <= N);
11         n = count;
12         for ( int u = 1; u <= n; u++ )
13         {
14             g[u].clear();
15         }
16         edges.clear();
17         vertices.clear();
18         edge_ids.clear();
19     }
```

```
21  int Insert(int u, int v)
22  {
23      assert(1 <= u && u <= n && 1 <= v && v <= n);
24      int id = edges.size();
25      edges.push_back({u, v});
26      g[u].push_back({v, id});
27      g[v].push_back({u, id});
28      vertices.clear();
29      edge_ids.clear();
30      return id;
31  }
32
33  void Dfs(int u, int incoming)
34  {
35      while ( next[u] < (int)g[u].size() )
36      {
37          auto [v, id] = g[u][next[u]++];
38          if ( used[id] )
39              continue;
40          used[id] = 1;
41          Dfs(v, id);
42      }
43      vertices.push_back(u);
44      if ( incoming != -1 )
45          edge_ids.push_back(incoming);
46  }
```

```

48 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
49 bool Run(int start = 0, bool lexicographic = true)
50 {
51     assert(0 <= start && start <= n);
52     vertices.clear();
53     edge_ids.clear();
54     vector<int> odd;
55     for ( int u = 1; u <= n; u++ )
56         if ( g[u].size() % 2 )
57             odd.push_back(u);
58     if ( !odd.empty() && odd.size() != 2 )
59         return false;
60     if ( !odd.empty() )
61     {
62         if ( start && start != odd[0] && start != odd[1] )
63             return false;
64         if ( !start )
65             start = odd[0];
66     }
67     if ( !start )
68     {
69         start = 1;
70         for ( int u = 1; u <= n; u++ )
71             if ( !g[u].empty() )
72             {
73                 start = u;
74                 break;
75             }
76     }
77     if ( lexicographic )
78         for ( int u = 1; u <= n; u++ )
79             sort(g[u].begin(), g[u].end());
80     fill(next, next + n + 1, 0);
81     used.assign(edges.size(), 0);
82     Dfs(start, -1);
83     if ( edge_ids.size() != edges.size() )
84     {
85         vertices.clear();
86         edge_ids.clear();
87         return false;
88     }
89     reverse(vertices.begin(), vertices.end());
90     reverse(edge_ids.begin(), edge_ids.end());
91     return true;
92 }
93 };

```

第七十五章 混合图欧拉定向

75.1 欧拉回路与指定端点定向

算法约束见动态版本第 49 页。本节代码独立可抄；采用下列实际接口名。

调用时额外传入 `Network_Flow<N,M>` 工作区作为临时最大流图；`N` 至少为 `n+2`，`M` 至少为可反转边数 `+n`。
大流对象放全局或 `static`。

```
1 // Edge = {u,v,type}: type 1 is fixed u->v; type 0 may be reversed.
2 // start=finish=0 requests a circuit; positive endpoints request a specified trail.
3 template <int N, int M>
4 optional<vector<pair<int, int>>>
5 Mixed_Euler_Orientation(int n,
6     const vector<array<int, 3>> &edges,
7     Network_Flow<N, M> &flow,
8     int start = 0,
9     int finish = 0)
10 {
11     assert(n > 0 && 0 <= start && start <= n && 0 <= finish && finish <= n);
12     assert(bool(start) == bool(finish));
13     vector<int> parent(n + 1), id;
14     vector<long long> degree(n + 1), balance(n + 1);
15     iota(parent.begin(), parent.end(), 0);
16     auto find = [&](auto &&self, int u) -> int
17     {
18         if ( parent[u] != u )
19             parent[u] = self(self, parent[u]);
20         return parent[u];
21     };
22     flow.Init(n + 2);
23     vector<pair<int, int>> direction;
24     for ( auto [u, v, type] : edges )
25     {
26         assert(1 <= u && u <= n && 1 <= v && v <= n);
27         assert(type == 0 || type == 1);
28         parent[find(find, u)] = find(find, v);
29         degree[u]++;
30         degree[v]++;
31         balance[u]++;
32         balance[v]--;
33         id.push_back(type == 0 ? flow.Add_Edge(u, v, 1) : -1);
34         direction.push_back({u, v});
35     }
```

```

36     if ( !edges.empty() )
37     {
38         int root = find(find, edges[0][0]);
39         for ( int u = 1; u <= n; u++ )
40             if ( degree[u] && find(find, u) != root )
41                 return nullopt;
42         if ( start && (!degree[start] || !degree[finish]) )
43             return nullopt;
44     }
45     if ( start )
46     {
47         balance[start]--;
48         balance[finish]++;
49     }
50     int source = n + 1, sink = n + 2;
51     long long need = 0;
52     for ( int u = 1; u <= n; u++ )
53     {
54         if ( balance[u] % 2 )
55             return nullopt;
56         if ( balance[u] > 0 )
57         {
58             flow.Add_Edge(source, u, balance[u] / 2);
59             need += balance[u] / 2;
60         }
61         else if ( balance[u] < 0 )
62             flow.Add_Edge(u, sink, -balance[u] / 2);
63     }
64     if ( flow.Max_Flow(source, sink) != need )
65         return nullopt;
66     for ( int i = 0; i < (int)edges.size(); i++ )
67         if ( id[i] != -1 && flow.Used(id[i]) )
68             swap(direction[i].first, direction[i].second);
69     return direction;
70 }

```

75.2 任意起终点的混合欧拉路径

算法约束见动态版本第 51 页。本节代码独立可抄；采用下列实际接口名。

调用时额外传入 `Network_Flow<N,M>` 工作区作为临时最大流图；`N` 至少为 `n+2`，`M` 至少为可反转边数 `+n`。
大流对象放全局或 `static`。

```

1 // Any endpoints: zero odd degrees requests a circuit; two odd degrees tries both
2 // orders.
3 template <int N, int M>
4 optional<vector<pair<int, int>>>
5 Mixed_Euler_Trail(int n, const vector<array<int, 3>> &edges, Network_Flow<N, M> &flow)
6 {
7     assert(n > 0);
8     vector<long long> degree(n + 1);
9     vector<int> odd;
10    for ( auto [u, v, type] : edges )
11    {
12        assert(1 <= u && u <= n && 1 <= v && v <= n);
13        degree[u]++;
14        degree[v]++;
15    }
16    for ( int u = 1; u <= n; u++ )
17        if ( degree[u] % 2 )
18            odd.push_back(u);
19    if ( odd.empty() )
20        return Mixed_Euler_Orientation(n, edges, flow);
21    if ( odd.size() != 2 )
22        return nullopt;
23    auto answer = Mixed_Euler_Orientation(n, edges, flow, odd[0], odd[1]);
24    if ( answer )
25        return answer;
26    return Mixed_Euler_Orientation(n, edges, flow, odd[1], odd[0]);
27 }

```

第七十六章 图上异或行走

76.1 无向图最大异或行走

算法约束见动态版本第 52 页。本节代码独立可抄；采用下列实际接口名。

声明 `Xor_Walk<N>`，`Init(n)` 清空，`Insert` 加边，`Build(root)` 预处理指定连通块，`Query(u,v)` 返回可选最大值。N 为最大点号，大对象置于全局或 `static`；空间 $O(N+m+64)$ 。DFS 保持递归。

```
1 template <int N> struct Xor_Walk
2 {
3     using U = unsigned long long;
4     int n = 0;
5     bool built = false;
6     vector<pair<int, U>> g[N + 1];
7     int vis[N + 1];
8     U dist[N + 1];
9     Xor_Basis cycles;
10
11     void Init(int size)
12     {
13         assert(0 <= size && size <= N);
14         for ( int u = 1; u <= n; u++ )
15             g[u] = vector<pair<int, U>>();
16         n = size;
17         cycles.Init();
18         built = false;
19     }
```

```

21 void Insert(int u, int v, U w)
22 {
23     assert(1 <= u && u <= n && 1 <= v && v <= n);
24     g[u].push_back({v, w});
25     g[v].push_back({u, w});
26     built = false;
27 }
28
29 void Dfs(int u)
30 {
31     vis[u] = 1;
32     for ( auto [v, w] : g[u] )
33     {
34         if ( !vis[v] )
35         {
36             dist[v] = dist[u] ^ w;
37             Dfs(v);
38         }
39         else
40             cycles.Insert(dist[u] ^ dist[v] ^ w);
41     }
42 }
43
44 void Build(int root = 1)
45 {
46     assert(1 <= root && root <= n);
47     fill(vis, vis + n + 1, 0);
48     fill(dist, dist + n + 1, 0);
49     cycles.Init();
50     Dfs(root);
51     built = true;
52 }
53
54 optional<U> Query(int u, int v) const
55 {
56     assert(built && 1 <= u && u <= n && 1 <= v && v <= n);
57     if ( !vis[u] || !vis[v] )
58         return nullopt;
59     return cycles.Query(dist[u] ^ dist[v]);
60 }
61 };

```


第七十七章 树上算法

77.1 重链剖分与 LCA

算法约束见动态版本第 54 页。本节代码独立可抄；采用下列实际接口名。

声明 Heavy_Light-Decomposition<N> 后 Init(n), Insert 加边, Build(root) 剖分; Lca 与 Path 对应动态接口。大对象放在全局或 static。

```
1 template <int N> struct Heavy_Light-Decomposition
2 {
3     int n, timer = 0;
4     vector<int> g[N + 1];
5     int fa[N + 1], dep[N + 1], siz[N + 1], son[N + 1];
6     int top[N + 1], dfn[N + 1], rk[N + 1];
7
8     void Init(int vertices)
9     {
10         assert(0 < vertices && vertices <= N);
11         n = vertices;
12         for ( int u = 1; u <= n; u++ )
13             g[u].clear();
14     }
15
16     void Insert(int u, int v)
17     {
18         g[u].push_back(v);
19         g[v].push_back(u);
20     }
21
22     void Dfs1(int u, int p)
23     {
24         fa[u] = p;
25         siz[u] = 1;
26         son[u] = 0;
27         for ( int v : g[u] )
28         {
29             if ( v == p )
30                 continue;
31             dep[v] = dep[u] + 1;
32             Dfs1(v, u);
33             siz[u] += siz[v];
34             if ( !son[u] || siz[v] > siz[son[u]] )
35                 son[u] = v;
36         }
37     }
38
39     void Dfs2(int u, int t)
40     {
41         top[u] = t;
42         dfn[u] = ++timer;
43         rk[timer] = u;
44         if ( son[u] )
```

```

45     Dfs2(son[u], t);
46     for ( int v : g[u] )
47     {
48         if ( v != fa[u] && v != son[u] )
49             Dfs2(v, v);
50     }
51 }
52
53 // Connected tree only. Recursive DFS; subtree interval is [dfn, dfn+siz-1].
54 void Build(int root = 1)
55 {
56     assert(1 <= root && root <= n);
57     timer = 0;
58     dep[root] = 0;
59     Dfs1(root, root);
60     Dfs2(root, root);
61     assert(timer == n);
62 }
63
64 int Lca(int u, int v) const
65 {
66     while ( top[u] != top[v] )
67     {
68         if ( dep[top[u]] < dep[top[v]] )
69             swap(u, v);
70         u = fa[top[u]];
71     }
72     return dep[u] < dep[v] ? u : v;
73 }
74
75 // Commutative operations only; edge=true excludes the LCA's position.
76 template <class F> void Path(int u, int v, F work, bool edge = false) const
77 {
78     while ( top[u] != top[v] )
79     {
80         if ( dep[top[u]] < dep[top[v]] )
81             swap(u, v);
82         work(dfn[top[u]], dfn[u]);
83         u = fa[top[u]];
84     }
85     if ( dep[u] > dep[v] )
86         swap(u, v);
87     if ( dfn[u] + edge <= dfn[v] )
88         work(dfn[u] + edge, dfn[v]);
89 }
90 };

```

第七十八章 离线最近公共祖先

78.1 离线 Tarjan LCA

算法约束见动态版本第 56 页。本节代码独立可抄；采用下列实际接口名。

声明 Offline_LCA<N>, Init(n) 清空树与查询；Insert 加无向树边，Add_Query 返回从 0 开始的查询编号，Run(root) 统一计算，answer[id] 为答案。大对象放全局或 static。

```
1 template <int N> struct Offline_LCA
2 {
3     int n;
4     vector<int> g[N + 1], answer;
5     vector<pair<int, int>> queries[N + 1];
6     int parent[N + 1], size[N + 1], ancestor[N + 1], done[N + 1];
7
8     void Init(int vertices)
9     {
10         assert(0 <= vertices && vertices <= N);
11         n = vertices;
12         for ( int u = 1; u <= n; u++ )
13         {
14             g[u].clear();
15             queries[u].clear();
16         }
17         answer.clear();
18     }
19
20     void Insert(int u, int v)
21     {
22         assert(1 <= u && u <= n && 1 <= v && v <= n);
23         g[u].push_back(v);
24         g[v].push_back(u);
25     }
26
27     int Add_Query(int u, int v)
28     {
29         assert(1 <= u && u <= n && 1 <= v && v <= n);
30         int id = answer.size();
31         answer.push_back(0);
32         queries[u].push_back({v, id});
33         queries[v].push_back({u, id});
34         return id;
35     }
```

```

37     int Find(int u)
38     {
39         if ( parent[u] != u )
40             parent[u] = Find(parent[u]);
41         return parent[u];
42     }
43
44     void Merge(int u, int v)
45     {
46         u = Find(u);
47         v = Find(v);
48         if ( size[u] < size[v] )
49             swap(u, v);
50         parent[v] = u;
51         size[u] += size[v];
52     }
53
54     void Dfs(int u, int p)
55     {
56         for ( int v : g[u] )
57             if ( v != p )
58             {
59                 Dfs(v, u);
60                 Merge(u, v);
61                 ancestor[Find(u)] = u;
62             }
63         done[u] = 1;
64         for ( auto [v, id] : queries[u] )
65             if ( done[v] )
66                 answer[id] = ancestor[Find(v)];
67     }
68
69     // Input must be a nonempty tree. Query IDs are 0-based.
70     void Run(int root = 1)
71     {
72         assert(1 <= root && root <= n);
73         iota(parent, parent + n + 1, 0);
74         iota(ancestor, ancestor + n + 1, 0);
75         fill(size, size + n + 1, 1);
76         fill(done, done + n + 1, 0);
77         fill(answer.begin(), answer.end(), 0);
78         Dfs(root, 0);
79     }
80 };

```

第七十九章 在线最近公共祖先

79.1 欧拉序与 ST 表 LCA

算法约束见动态版本第 58 页。本节代码独立可抄；采用下列实际接口名。

声明 Euler_LCA<N,LOG>, Init(n) 清空树；Insert 加边，Build(root) 预处理，Lca 与 Distance 在线查询。LOG 层必须满足 2 的 LOG 次方大于 $2n-1$ ，默认 20 支持 50 万点；st 固定容量为 LOG 乘以 2N。大对象放全局或 static。

```
1 template <int N, int LOG = 20> struct Euler_LCA
2 {
3     static_assert(1 <= LOG && LOG <= 31);
4     int n;
5     bool ready = false;
6     vector<int> g[N + 1], euler;
7     int first[N + 1], depth[N + 1], lg[2 * N + 1];
8     int st[LOG][2 * N];
9
10    void Init(int vertices)
11    {
12        assert(0 < vertices && vertices <= N);
13        assert((1LL << LOG) > 2LL * vertices - 1);
14        n = vertices;
15        for ( int u = 1; u <= n; u++ )
16            g[u].clear();
17        euler.clear();
18        ready = false;
19    }
20
21    void Insert(int u, int v)
22    {
23        assert(1 <= u && u <= n && 1 <= v && v <= n);
24        g[u].push_back(v);
25        g[v].push_back(u);
26        ready = false;
27    }
28
29    int Shallower(int u, int v) const
30    {
31        return depth[u] <= depth[v] ? u : v;
32    }
```

```

34 void Dfs(int u, int parent)
35 {
36     first[u] = euler.size();
37     euler.push_back(u);
38     for ( int v : g[u] )
39         if ( v != parent )
40             {
41                 depth[v] = depth[u] + 1;
42                 Dfs(v, u);
43                 euler.push_back(u);
44             }
45 }
46
47 // Input must be a nonempty tree. Rebuilding may change the root.
48 void Build(int root = 1)
49 {
50     assert(1 <= root && root <= n);
51     euler.clear();
52     depth[root] = 0;
53     Dfs(root, 0);
54     int m = euler.size();
55     lg[0] = 0;
56     lg[1] = 0;
57     for ( int i = 2; i <= m; i++ )
58         lg[i] = lg[i / 2] + 1;
59     copy(euler.begin(), euler.end(), st[0]);
60     for ( int k = 1; k <= lg[m]; k++ )
61     {
62         for ( int i = 0; i + (1 << k) <= m; i++ )
63             st[k][i] = Shallower(st[k - 1][i], st[k - 1][i + (1 << (k - 1))]);
64     }
65     ready = true;
66 }
67
68 int Lca(int u, int v) const
69 {
70     assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
71     int l = first[u], r = first[v];
72     if ( l > r )
73         swap(l, r);
74     int k = lg[r - l + 1];
75     return Shallower(st[k][l], st[k][r - (1 << k) + 1]);
76 }
77
78 int Distance(int u, int v) const
79 {
80     int z = Lca(u, v);
81     return depth[u] + depth[v] - 2 * depth[z];
82 }
83 };

```

第八十章 倍增祖先与路径最值

80.1 倍增 LCA、祖先与最大边权

算法约束见动态版本第 60 页。本节代码独立可抄；采用下列实际接口名。

声明 `Lifting_LCA<N,LOG>`, `Init(n)` 清空树；`Insert`、`Build`、`Jump`、`Lca`、`Distance`、`Max_Edge` 对应动态接口。LOG 层要求 2 的 LOG 次方大于 n，默认 20；大对象放全局或 static。

```
1 template <int N, int LOG = 20> struct Lifting_LCA
2 {
3     using ll = long long;
4     static constexpr ll LOW = numeric_limits<ll>::lowest();
5     static_assert(1 <= LOG && LOG <= 31);
6     int n;
7     bool ready = false;
8     vector<pair<int, ll>> g[N + 1];
9     int up[LOG][N + 1], depth[N + 1];
10    ll mx[LOG][N + 1];
11
12    void Init(int vertices)
13    {
14        assert(0 < vertices && vertices <= N && (1LL << LOG) > vertices);
15        n = vertices;
16        for (int u = 1; u <= n; u++)
17            g[u].clear();
18        for (int k = 0; k < LOG; k++)
19        {
20            up[k][0] = 0;
21            mx[k][0] = LOW;
22        }
23        ready = false;
24    }
25
26    void Insert(int u, int v, ll w = 1)
27    {
28        assert(1 <= u && u <= n && 1 <= v && v <= n);
29        g[u].push_back({v, w});
30        g[v].push_back({u, w});
31        ready = false;
32    }
```

```

34 void Dfs(int u, int parent, ll weight)
35 {
36     up[0][u] = parent;
37     mx[0][u] = weight;
38     for ( int k = 1; k < LOG; k++ )
39     {
40         int p = up[k - 1][u];
41         up[k][u] = up[k - 1][p];
42         mx[k][u] = max(mx[k - 1][u], mx[k - 1][p]);
43     }
44     for ( auto [v, w] : g[u] )
45         if ( v != parent )
46         {
47             depth[v] = depth[u] + 1;
48             Dfs(v, u, w);
49         }
50 }
51
52 // Nonempty tree only. Node 0 means no ancestor.
53 void Build(int root = 1)
54 {
55     assert(1 <= root && root <= n);
56     depth[root] = 0;
57     Dfs(root, 0, LOW);
58     ready = true;
59 }
60
61 int Jump(int u, long long steps) const
62 {
63     assert(ready && 1 <= u && u <= n && steps >= 0);
64     if ( steps > depth[u] )
65         return 0;
66     for ( int k = 0; k < LOG; k++ )
67         if ( steps >> k & 1 )
68             u = up[k][u];
69     return u;
70 }

```



```

72  int Lca(int u, int v) const
73  {
74      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
75      if ( depth[u] < depth[v] )
76          swap(u, v);
77      u = Jump(u, depth[u] - depth[v]);
78      if ( u == v )
79          return u;
80      for ( int k = LOG - 1; k >= 0; k-- )
81          if ( up[k][u] != up[k][v] )
82              {
83                  u = up[k][u];
84                  v = up[k][v];
85              }
86      return up[0][u];
87  }
88
89  int Distance(int u, int v) const
90  {
91      return depth[u] + depth[v] - 2 * depth[Lca(u, v)];
92  }
93
94  optional<ll> Max_Edge(int u, int v) const
95  {
96      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
97      if ( u == v )
98          return nullopt;
99      if ( depth[u] < depth[v] )
100          swap(u, v);
101      ll answer = LOW;
102      int delta = depth[u] - depth[v];
103      for ( int k = 0; k < LOG; k++ )
104          if ( delta >> k & 1 )
105              {
106                  answer = max(answer, mx[k][u]);
107                  u = up[k][u];
108              }
109      if ( u == v )
110          return answer;
111      for ( int k = LOG - 1; k >= 0; k-- )
112          if ( up[k][u] != up[k][v] )
113              {
114                  answer = max(answer, max(mx[k][u], mx[k][v]));
115                  u = up[k][u];
116                  v = up[k][v];
117              }
118      return max(answer, max(mx[0][u], mx[0][v]));
119  }
120 };

```

第八十一章 树上路径交

81.1 两条树路径的交集

算法约束见动态版本第 63 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Path_Intersection_Result
2 {
3     int u, v, vertices;
4 };
5
6 // Same nonempty tree and root for depth[] and lca(). Empty result is {0,0,0}.
7 template <class Depth, class Lca>
8 Path_Intersection_Result
9 Path_Intersection(int a, int b, int c, int d, const Depth &depth, Lca lca)
10 {
11     int z = lca(a, b);
12     auto project = [&](int x)
13     {
14         int p = z;
15         for ( int v : {lca(a, x), lca(b, x)} )
16             if ( depth[v] > depth[p] )
17                 p = v;
18         return p;
19     };
20     auto distance = [&](int u, int v)
21     {
22         return 1LL * depth[u] + depth[v] - 2LL * depth[lca(u, v)];
23     };
24     int p = project(c), q = project(d);
25     if ( p == q && distance(c, p) + distance(p, d) != distance(c, d) )
26         return {0, 0, 0};
27     return {p, q, int(distance(p, q) + 1)};
28 }
```

第八十二章 点分治

82.1 点分治：树上距离点对计数

算法约束见动态版本第 64 页。本节代码独立可抄；采用下列实际接口名。

声明 Centroid_Pairs<N>, Init(n) 初始化, Insert 加边, Build 预处理; Count_Leq 和 Count_Exact 返回计数, parent 保存点分树父节点。大对象放在全局或 static。

```
1 template <int N> struct Centroid_Pairs
2 {
3     using ll = long long;
4     int n;
5     vector<pair<int, ll>> g[N + 1];
6     int siz[N + 1], removed[N + 1], parent[N + 1];
7     vector<ll> all[N + 1];
8     vector<vector<ll>> part[N + 1];
9
10    void Init(int vertices)
11    {
12        assert(0 <= vertices && vertices <= N);
13        n = vertices;
14        for ( int u = 1; u <= n; u++ )
15            g[u].clear();
16    }
17
18    void Insert(int u, int v, ll w)
19    {
20        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
21        g[u].push_back({v, w});
22        g[v].push_back({u, w});
23    }
24
25    void Size_Dfs(int u, int p)
26    {
27        siz[u] = 1;
28        for ( auto [v, w] : g[u] )
29        {
30            if ( v == p || removed[v] )
31                continue;
32            Size_Dfs(v, u);
33            siz[u] += siz[v];
34        }
35    }
36
37    int Centroid(int u, int p, int total)
38    {
39        for ( auto [v, w] : g[u] )
40        {
41            if ( v != p && !removed[v] && siz[v] > total / 2 )
42                return Centroid(v, u, total);
43        }
44        return u;
```

```

45     }
46
47     void Collect(int u, int p, ll d, vector<ll> &out)
48     {
49         out.push_back(d);
50         for ( auto [v, w] : g[u] )
51         {
52             if ( v == p || removed[v] )
53                 continue;
54             assert(d <= numeric_limits<ll>::max() - w);
55             Collect(v, u, d + w, out);
56         }
57     }
58
59     void Decompose(int entry, int p)
60     {
61         Size_Dfs(entry, 0);
62         int c = Centroid(entry, 0, siz[entry]);
63         removed[c] = 1;
64         parent[c] = p;
65         all[c] = {0};
66         for ( auto [v, w] : g[c] )
67         {
68             if ( removed[v] )
69                 continue;
70             part[c].push_back({});
71             auto &d = part[c].back();
72             Collect(v, c, w, d);
73             sort(d.begin(), d.end());
74             all[c].insert(all[c].end(), d.begin(), d.end());
75         }
76         sort(all[c].begin(), all[c].end());
77         for ( auto [v, w] : g[c] )
78         {
79             if ( !removed[v] )
80                 Decompose(v, c);
81         }
82     }
83
84     // Input must be a forest. Rebuild after changing edges.
85     void Build()
86     {
87         fill(removed, removed + n + 1, 0);
88         for ( int u = 1; u <= n; u++ )
89         {
90             all[u].clear();
91             part[u].clear();
92         }
93         for ( int u = 1; u <= n; u++ )
94         {
95             if ( !removed[u] )
96                 Decompose(u, 0);
97         }
98     }
99
100     ll Pairs(const vector<ll> &d, ll k) const

```

```

101 {
102     int l = 0, r = (int)d.size() - 1;
103     ll answer = 0;
104     while ( l < r )
105     {
106         if ( d[l] <= k - d[r] )
107         {
108             answer += r - l;
109             l++;
110         }
111         else
112             r--;
113     }
114     return answer;
115 }
116
117 // Unordered Pairs of distinct connected vertices. Nonnegative edge weights.
118 ll Count_Leq(ll k) const
119 {
120     if ( k < 0 )
121         return 0;
122     ll answer = 0;
123     for ( int c = 1; c <= n; c++ )
124     {
125         answer += Pairs(all[c], k);
126         for ( const auto &d : part[c] )
127             answer -= Pairs(d, k);
128     }
129     return answer;
130 }
131
132 ll Count_Exact(ll k) const
133 {
134     if ( k < 0 )
135         return 0;
136     return Count_Leq(k) - (k ? Count_Leq(k - 1) : 0);
137 }
138 };

```

第八十三章 树上启发式合并

83.1 树上启发式合并：子树众数颜色和

算法约束见动态版本第 67 页。本节代码独立可抄；采用下列实际接口名。

声明 Subtree_Colors<N>, Init(n) 初始化, Insert 加边, Run(colors,root) 计算; answer 与 max_freq 为最终结果。Add_Color 为内部统计函数。大对象放在全局或 static。

```
1 template <int N> struct Subtree_Colors
2 {
3     int n, best = 0;
4     long long sum = 0;
5     vector<int> g[N + 1], order, value, freq;
6     int siz[N + 1], heavy[N + 1], tin[N + 1], tout[N + 1];
7     int color[N + 1], max_freq[N + 1];
8     long long answer[N + 1];
9
10    void Init(int vertices)
11    {
12        assert(0 <= vertices && vertices <= N);
13        n = vertices;
14        for ( int u = 1; u <= n; u++ )
15            g[u].clear();
16    }
17
18    void Insert(int u, int v)
19    {
20        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v);
21        g[u].push_back(v);
22        g[v].push_back(u);
23    }
24
25    void Prepare(int u, int p)
26    {
27        siz[u] = 1;
28        tin[u] = order.size();
29        order.push_back(u);
30        for ( int v : g[u] )
31        {
32            if ( v == p )
33                continue;
34            Prepare(v, u);
35            siz[u] += siz[v];
36            if ( siz[v] > siz[heavy[u]] )
37                heavy[u] = v;
38        }
39        tout[u] = (int)order.size() - 1;
40    }
41
42    void Add_Color(int u)
43    {
44        int c = color[u];
```

```

45     freq[c]++;
46     if ( freq[c] > best )
47     {
48         best = freq[c];
49         sum = value[c];
50     }
51     else if ( freq[c] == best )
52         sum += value[c];
53 }
54
55 void Dfs(int u, int p, bool keep)
56 {
57     for ( int v : g[u] )
58     {
59         if ( v != p && v != heavy[u] )
60             Dfs(v, u, false);
61     }
62     if ( heavy[u] )
63         Dfs(heavy[u], u, true);
64     for ( int v : g[u] )
65     {
66         if ( v == p || v == heavy[u] )
67             continue;
68         for ( int i = tin[v]; i <= tout[v]; i++ )
69             Add_Color(order[i]);
70     }
71     Add_Color(u);
72     max_freq[u] = best;
73     answer[u] = sum;
74     if ( !keep )
75     {
76         for ( int i = tin[u]; i <= tout[u]; i++ )
77             freq[color[order[i]]]--;
78         best = 0;
79         sum = 0;
80     }
81 }
82
83 // A connected tree, colors[1..n]. Sum all colors tied for maximum frequency.
84 void Run(const vector<int> &colors, int root = 1)
85 {
86     assert((int)colors.size() == n + 1);
87     fill(siz, siz + n + 1, 0);
88     fill(heavy, heavy + n + 1, 0);
89     fill(tin, tin + n + 1, 0);
90     fill(tout, tout + n + 1, 0);
91     fill(color, color + n + 1, 0);
92     fill(max_freq, max_freq + n + 1, 0);
93     fill(answer, answer + n + 1, 0);
94     order.clear();
95     value.assign(colors.begin() + 1, colors.end());
96     sort(value.begin(), value.end());
97     value.erase(unique(value.begin(), value.end()), value.end());
98     freq.assign(value.size(), 0);
99     best = 0;
100    sum = 0;

```

```
101     if ( !n )
102         return;
103     assert(1 <= root && root <= n);
104     for ( int u = 1; u <= n; u++ )
105         color[u] =
106             lower_bound(value.begin(), value.end(), colors[u]) - value.begin();
107     Prepare(root, 0);
108     Dfs(root, 0, false);
109 }
110 };
```


第八十四章 虚树

84.1 虚树：关键点压缩与路径信息

算法约束见动态版本第 70 页。本节代码独立可抄；采用下列实际接口名。

声明 `Virtual_Tree<N,LOG>`，要求 2 的 LOG 次方大于实际点数，默认 LOG=20。Init(n)、Insert、Prepare、Build 对应初始化、加边、预处理和查询；Lca、Ancestor、Path 为对应辅助接口。大对象放在全局或 static。

```
1 template <int N, int LOG = 20> struct Virtual_Tree
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int from, to;
8         ll distance, minimum;
9     };
10
11     struct Result
12     {
13         vector<int> vertices;
14         vector<Edge> edges;
15     };
16
17     int n, log = 1, timer = 0;
18     vector<pair<int, ll>> g[N + 1];
19     int up[LOG][N + 1], depth[N + 1], tin[N + 1], tout[N + 1];
20     ll mn[LOG][N + 1], dist[N + 1];
21
22     void Init(int vertices)
23     {
24         assert(0 < vertices && vertices <= N);
25         n = vertices;
26         for (int u = 1; u <= n; u++)
27             g[u].clear();
28     }
29
30     void Insert(int u, int v, ll w = 1)
31     {
32         assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
33         g[u].push_back({v, w});
34         g[v].push_back({u, w});
35     }
36
37     void Dfs(int u, int p)
38     {
39         tin[u] = ++timer;
40         up[0][u] = p;
41         for (int j = 1; j < log; j++)
42         {
43             up[j][u] = up[j - 1][up[j - 1][u]];
44             mn[j][u] = min(mn[j - 1][u], mn[j - 1][up[j - 1][u]]);
```

```

45     }
46     for ( auto [v, w] : g[u] )
47     {
48         if ( v == p )
49             continue;
50         assert(dist[u] <= numeric_limits<ll>::max() - w);
51         depth[v] = depth[u] + 1;
52         dist[v] = dist[u] + w;
53         mn[0][v] = w;
54         Dfs(v, u);
55     }
56     tout[u] = timer;
57 }
58
59 void Prepare(int root = 1)
60 {
61     assert(1 <= root && root <= n);
62     log = 1;
63     while ( (1LL << log) <= n )
64         log++;
65     assert(log <= LOG);
66     for ( int j = 0; j < log; j++ )
67     {
68         fill(up[j], up[j] + n + 1, 0);
69         fill(mn[j], mn[j] + n + 1, numeric_limits<ll>::max());
70     }
71     fill(depth, depth + n + 1, 0);
72     fill(tin, tin + n + 1, 0);
73     fill(tout, tout + n + 1, 0);
74     fill(dist, dist + n + 1, 0);
75     timer = 0;
76     Dfs(root, root);
77 }
78
79 bool Ancestor(int u, int v) const
80 {
81     return tin[u] <= tin[v] && tout[v] <= tout[u];
82 }
83
84 int Lca(int u, int v) const
85 {
86     if ( Ancestor(u, v) )
87         return u;
88     if ( Ancestor(v, u) )
89         return v;
90     for ( int j = log - 1; j >= 0; j-- )
91     {
92         if ( !Ancestor(up[j][u], v) )
93             u = up[j][u];
94     }
95     return up[0][u];
96 }
97
98 Edge Path(int u, int v) const
99 {
100     assert(u != v && Ancestor(u, v));

```

```

101     ll minimum = numeric_limits<ll>::max();
102     int x = v, steps = depth[v] - depth[u];
103     for ( int j = 0; j < log; j++ )
104     {
105         if ( steps >> j & 1 )
106         {
107             minimum = min(minimum, mn[j][x]);
108             x = up[j][x];
109         }
110     }
111     return {u, v, dist[v] - dist[u], minimum};
112 }
113
114 // Uses original vertex IDs. Empty input gives an empty result.
115 Result Build(vector<int> keys) const
116 {
117     if ( keys.empty() )
118         return {};
119     for ( int u : keys )
120         assert(1 <= u && u <= n);
121     auto cmp = [&](int u, int v)
122     {
123         return tin[u] < tin[v];
124     };
125     sort(keys.begin(), keys.end(), cmp);
126     keys.erase(unique(keys.begin(), keys.end()), keys.end());
127     int k = keys.size();
128     for ( int i = 1; i < k; i++ )
129         keys.push_back(Lca(keys[i - 1], keys[i]));
130     sort(keys.begin(), keys.end(), cmp);
131     keys.erase(unique(keys.begin(), keys.end()), keys.end());
132     Result result{keys, {}};
133     vector<int> st;
134     for ( int u : keys )
135     {
136         while ( !st.empty() && !Ancestor(st.back(), u) )
137             st.pop_back();
138         if ( !st.empty() )
139             result.edges.push_back(Path(st.back(), u));
140         st.push_back(u);
141     }
142     return result;
143 }
144 };

```

第八十五章 字符串

85.1 前缀函数

算法约束见动态版本第 73 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline vector<int> Prefix_Function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> p(n);
5     for ( int i = 1; i < n; i++ )
6     {
7         int j = p[i - 1];
8         while ( j && s[i] != s[j] )
9             j = p[j - 1];
10        p[i] = j + (s[i] == s[j]);
11    }
12    return p;
13 }
```

85.2 KMP 匹配

算法约束见动态版本第 73 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline vector<int> Kmp_Match(const string &s, const string &t)
2 {
3     assert(!t.empty());
4     auto p = Prefix_Function(t);
5     vector<int> ans;
6     for ( int i = 0, j = 0; i < (int)s.size(); i++ )
7     {
8         while ( j && s[i] != t[j] )
9             j = p[j - 1];
10        j += (s[i] == t[j]);
11        if ( j == (int)t.size() )
12        {
13            ans.push_back(i - j + 1);
14            j = p[j - 1];
15        }
16    }
17    return ans;
18 }
```

85.3 Z 函数

算法约束见动态版本第 74 页。本节代码独立可抄；采用下列实际接口名。

```

1 inline vector<int> Z_Function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> a(n);
5     if ( n )
6         a[0] = n;
7     for ( int i = 1, l = 0, r = 0; i < n; i++ )
8     {
9         if ( i < r )
10            a[i] = min(r - i, a[i - l]);
11        while ( i + a[i] < n && s[a[i]] == s[i + a[i]] )
12            ++a[i];
13        if ( i + a[i] > r )
14        {
15            l = i;
16            r = i + a[i];
17        }
18    }
19    return a;
20 }
```

85.4 Manacher 回文半径

算法约束见动态版本第 74 页。本节代码独立可抄；采用下列实际接口名。

```

1 // odd[i] includes the center; even[i] is centered before i.
2 inline pair<vector<int>, vector<int>> Manacher(const string &s)
3 {
4     int n = (int)s.size();
5     vector<int> odd(n), even(n);
6     for ( int i = 0, l = 0, r = -1; i < n; i++ )
7     {
8         int k = i > r ? 1 : min(odd[l + r - i], r - i + 1);
9         while ( i - k >= 0 && i + k < n && s[i - k] == s[i + k] )
10            ++k;
11        odd[i] = k--;
12        if ( i + k > r )
13        {
14            l = i - k;
15            r = i + k;
16        }
17    }
18    for ( int i = 0, l = 0, r = -1; i < n; i++ )
19    {
20        int k = i > r ? 0 : min(even[l + r - i + 1], r - i + 1);
21        while ( i - k - 1 >= 0 && i + k < n && s[i - k - 1] == s[i + k] )
22            ++k;
```

```

23     even[i] = k--;
24     if ( i + k > r )
25     {
26         l = i - k - 1;
27         r = i + k;
28     }
29 }
30 return {odd, even};
31 }

```

85.5 最小循环表示

算法约束见动态版本第 75 页。本节代码独立可抄；采用下列实际接口名。

```

1 inline int Minimum_Rotation(const string &s)
2 {
3     int n = (int)s.size(), i = 0, j = 1, k = 0;
4     if ( !n )
5         return 0;
6     while ( i < n && j < n && k < n )
7     {
8         unsigned char a = s[(i + k) % n], b = s[(j + k) % n];
9         if ( a == b )
10            {
11                ++k;
12                continue;
13            }
14         if ( a > b )
15            {
16                i += k + 1;
17                if ( i == j )
18                    ++i;
19            }
20         else
21            {
22                j += k + 1;
23                if ( i == j )
24                    ++j;
25            }
26         k = 0;
27     }
28     return min(i, j);
29 }

```

85.6 AC 自动机出现次数

算法约束见动态版本第 75 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct AC_Automaton
2 {
3     struct Node
4     {
5         array<int, 26> go{};
6         int fail = 0;

```

```

7      };
8
9      vector<Node> a{Node{}};
10     vector<int> order;
11     bool built = false;
12
13     int Insert(const string &s)
14     {
15         assert(!built && !s.empty());
16         int u = 0;
17         for ( char ch : s )
18         {
19             int c = ch - 'a';
20             assert(0 <= c && c < 26);
21             if ( !a[u].go[c] )
22             {
23                 int v = (int)a.size();
24                 a[u].go[c] = v;
25                 a.push_back(Node{});
26             }
27             u = a[u].go[c];
28         }
29         return u;
30     }
31
32     void Build()
33     {
34         assert(!built);
35         built = true;
36         queue<int> q;
37         for ( int v : a[0].go )
38             if ( v )
39                 q.push(v);
40         while ( !q.empty() )
41         {
42             int u = q.front();
43             q.pop();
44             order.push_back(u);
45             for ( int c = 0; c < 26; c++ )
46             {
47                 int v = a[u].go[c];
48                 if ( v )
49                 {
50                     a[v].fail = a[a[u].fail].go[c];
51                     q.push(v);
52                 }
53                 else
54                     a[u].go[c] = a[a[u].fail].go[c];
55             }
56         }
57     }
58
59     vector<long long> count(const string &s) const
60     {
61         assert(built);
62         vector<long long> ans(a.size());

```

```
63     int u = 0;
64     for ( char ch : s )
65     {
66         u = a[u].go[ch - 'a'];
67         ++ans[u];
68     }
69     for ( int i = (int)order.size() - 1; i >= 0; i-- )
70         ans[a[order[i]].fail] += ans[order[i]];
71     return ans;
72 }
73 };
```


85.7 倍增后缀数组与 LCP

算法约束见动态版本第 78 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Suffix_Array
2 {
3     vector<int> sa, rk, lcp;
4
5     Suffix_Array(const string &s)
6     {
7         vector<int> a;
8         a.reserve(s.size());
9         for ( unsigned char c : s )
10             a.push_back(c);
11         Init(a, 256);
12     }
13
14     Suffix_Array(const vector<int> &s, int alphabet)
15     {
16         Init(s, alphabet);
17     }
18
19     void Init(const vector<int> &s, int alphabet)
20     {
21         assert(s.size() < INT_MAX);
22         assert(1 <= alphabet && alphabet < INT_MAX);
23         int n = (int)s.size();
24         sa.resize(n);
25         rk.resize(n);
26         lcp.assign(n, 0);
27         if ( !n )
28             return;
29         vector<int> order(n), old(n), cnt(max(n + 1, alphabet + 1));
30         int m = alphabet;
31         for ( int i = 0; i < n; i++ )
32         {
33             assert(0 <= s[i] && s[i] < alphabet);
34             rk[i] = s[i] + 1;
35             cnt[rk[i]]++;
36         }
37         for ( int i = 1; i <= m; i++ )
38             cnt[i] += cnt[i - 1];
39         for ( int i = n - 1; i >= 0; i-- )
40             sa[--cnt[rk[i]]] = i;

```

```

41     for ( int k = 1; k < n; k *= 2 )
42     {
43         int p = 0;
44         for ( int i = n - k; i < n; i++ )
45             order[p++] = i;
46         for ( int i : sa )
47             if ( i >= k )
48                 order[p++] = i - k;
49         fill(cnt.begin(), cnt.begin() + m + 1, 0);
50         for ( int x : rk )
51             cnt[x]++;
52         for ( int i = 1; i <= m; i++ )
53             cnt[i] += cnt[i - 1];
54         for ( int i = n - 1; i >= 0; i-- )
55             sa[--cnt[rk[order[i]]]] = order[i];
56         old.swap(rk);
57         auto second = [&](int i)
58         {
59             return i < n - k ? old[i + k] : 0;
60         };
61         m = 1;
62         rk[sa[0]] = m;
63         for ( int i = 1; i < n; i++ )
64         {
65             int a = sa[i - 1], b = sa[i];
66             if ( old[a] != old[b] || second(a) != second(b) )
67                 m++;
68             rk[b] = m;
69         }
70         if ( m == n )
71             break;
72     }
73     for ( int i = 0; i < n; i++ )
74         rk[sa[i]] = i;
75     for ( int i = 0, k = 0; i < n; i++ )
76     {
77         if ( !rk[i] )
78         {
79             k = 0;
80             continue;
81         }
82         int j = sa[rk[i] - 1];
83         while ( k < n - i && k < n - j && s[i + k] == s[j + k] )
84             ++k;
85         lcp[rk[i]] = k;
86         if ( k )
87             --k;
88     }
89 }
90 };

```

85.8 后缀自动机

算法约束见动态版本第 80 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Suffix_Automaton
2 {
3     struct Node
4     {
5         array<int, 26> go{};
6         int link = -1, len = 0;
7         long long occ = 0;
8     };
9
10    vector<Node> a{Node{}};
11    int last = 0;
12
13    void Extend(int c)
14    {
15        assert(0 <= c && c < 26);
16        int cur = (int)a.size();
17        a.push_back(Node{});
18        a[cur].len = a[last].len + 1;
19        a[cur].occ = 1;
20        int p = last;
21        while ( p != -1 && !a[p].go[c] )
22        {
23            a[p].go[c] = cur;
24            p = a[p].link;
25        }
26        if ( p == -1 )
27            a[cur].link = 0;
28        else
29        {
30            int q = a[p].go[c];
31            if ( a[p].len + 1 == a[q].len )
32                a[cur].link = q;
33            else
34            {
35                int clone = (int)a.size();
36                Node copy = a[q];
37                a.push_back(copy);
38                a[clone].len = a[p].len + 1;
39                a[clone].occ = 0;
40                while ( p != -1 && a[p].go[c] == q )
41                {
42                    a[p].go[c] = clone;
43                    p = a[p].link;
44                }
45                a[q].link = a[cur].link = clone;
46            }
47        }
48        last = cur;
49    }
50
51    vector<long long> Counts() const
52    {

```

```
53     vector<int> bucket(a[last].len + 1), order(a.size());
54     vector<long long> ans(a.size());
55     for ( auto &v : a )
56         ++bucket[v.len];
57     partial_sum(bucket.begin(), bucket.end(), bucket.begin());
58     for ( int i = 0; i < (int)a.size(); i++ )
59     {
60         order[--bucket[a[i].len]] = i;
61         ans[i] = a[i].occ;
62     }
63     for ( int i = (int)order.size() - 1; i > 0; i-- )
64         ans[a[order[i]].link] += ans[order[i]];
65     return ans;
66 }
67
68 long long Distinct() const
69 {
70     long long ans = 0;
71     for ( int i = 1; i < (int)a.size(); i++ )
72         ans += a[i].len - a[a[i].link].len;
73     return ans;
74 }
75 };
```

第八十六章 后缀查询与重复子串

86.1 后缀 LCP 与子串比较

算法约束见动态版本第 82 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Suffix_LCP
2 {
3     int n;
4     vector<int> rk, lg;
5     vector<vector<int>> st;
6
7     Suffix_LCP(const Suffix_Array &suffix)
8     {
9         Init(suffix);
10    }
11
12    void Init(const Suffix_Array &suffix)
13    {
14        n = suffix.sa.size();
15        rk = suffix.rk;
16        lg.assign(n + 1, 0);
17        for ( int i = 2; i <= n; i++ )
18            lg[i] = lg[i / 2] + 1;
19        st.assign(lg[n] + 1, {});
20        st[0] = suffix.lcp;
21        for ( int k = 1; k <= lg[n]; k++ )
22        {
23            int len = 1 << k;
24            st[k].resize(n - len + 1);
25            for ( int i = 0; i <= n - len; i++ )
26                st[k][i] = min(st[k - 1][i], st[k - 1][i + len / 2]);
27        }
28    }
```

```

30 int Query(int x, int y) const
31 {
32     assert(0 <= x && x <= n && 0 <= y && y <= n);
33     if ( x == y || x == n || y == n )
34         return n - max(x, y);
35     int a = rk[x], b = rk[y];
36     if ( a > b )
37         swap(a, b);
38     int k = lg[b - a];
39     return min(st[k][a + 1], st[k][b - (1 << k) + 1]);
40 }
41
42 // Compare [l1,r1) and [l2,r2); return -1, 0 or 1.
43 int Compare(int l1, int r1, int l2, int r2) const
44 {
45     assert(0 <= l1 && l1 <= r1 && r1 <= n);
46     assert(0 <= l2 && l2 <= r2 && r2 <= n);
47     int a = r1 - l1, b = r2 - l2;
48     if ( Query(l1, l2) >= min(a, b) )
49         return (a > b) - (a < b);
50     return rk[l1] < rk[l2] ? -1 : 1;
51 }
52 };

```

86.2 两个前缀的最长公共后缀

算法约束见动态版本第 83 页。本节代码独立可抄；采用下列实际接口名。

```

1 // reversed must be built on the reversed original sequence.
2 inline int Prefix_Lcs(const Suffix_LCP &reversed, int x, int y)
3 {
4     assert(0 <= x && x <= reversed.n && 0 <= y && y <= reversed.n);
5     return reversed.Query(reversed.n - x, reversed.n - y);
6 }

```

86.3 重复子串 AA 的起止位置计数

算法约束见动态版本第 84 页。本节代码独立可抄；采用下列实际接口名。

```

1 // Count nonempty squares AA by their starting and ending positions.
2 inline pair<vector<long long>, vector<long long>>
3 Square_Counts(const Suffix_LCP &forward, const Suffix_LCP &reversed)
4 {
5     int n = forward.n;
6     assert(reversed.n == n);
7     vector<long long> start(n + 1), finish(n + 1);
8     for ( int len = 1; len <= n / 2; len++ )
9         for ( int i = 0; i < n - len; i += len )
10            {
11                int j = i + len;
12                int left = min(len - 1, Prefix_Lcs(reversed, i, j));
13                int right = min(len, forward.Query(i, j));
14                if ( left + right < len )
15                    continue;
16                int l = i - left, r = i + right - len;
17                start[l]++;
18                start[r + 1]--;
19                finish[l + 2 * len - 1]++;
20                finish[r + 2 * len]--;
21            }
22     for ( int i = 1; i < n; i++ )
23     {
24         start[i] += start[i - 1];
25         finish[i] += finish[i - 1];
26     }
27     start.resize(n);
28     finish.resize(n);
29     return {start, finish};
30 }
```

第八十七章 回文树

87.1 回文树 PAM: 回文后缀与出现次数

算法约束见动态版本第 85 页。本节代码独立可抄；采用下列实际接口名。

声明 `Palindromic_Tree<N>` 后调用 `Init()`；`N` 为最大追加字符数，内部另留两个根节点。`Insert`、`Distinct`、`Occurrences` 对应动态版接口；`t`、`last`、`total` 可直接读。大对象放在全局或 `static`。

```
1 template <int N> struct Palindromic_Tree
2 {
3     struct Node
4     {
5         int next[26]{};
6         int len = 0, fail = 0, suffixes = 0, end = -1;
7         long long hits = 0;
8     };
9
10    Node t[N + 2];
11    int tot = 1;
12    vector<int> s;
13    int last = 0;
14    long long total = 0;
15
16    void Init()
17    {
18        tot = 1;
19        t[0] = Node{};
20        t[1] = Node{};
21        t[0].fail = 1;
22        t[1].len = -1;
23        s.assign(1, -1);
24        last = 0;
25        total = 0;
26    }
27
28    int Get_Fail(int u) const
29    {
30        int pos = s.size() - 1;
31        while ( s[pos - t[u].len - 1] != s[pos] )
32            u = t[u].fail;
33        return u;
34    }
35
36    // Append a lowercase letter; return the longest palindromic suffix node.
37    int Insert(char ch)
38    {
39        int c = ch - 'a';
40        assert(0 <= c && c < 26);
41        assert(s.size() <= N);
42        s.push_back(c);
43        int u = Get_Fail(last);
44        if ( !t[u].next[c] )
```



```

45     {
46         int v = ++tot;
47         t[v] = Node{};
48         t[v].len = t[u].len + 2;
49         t[v].fail = t[Get_Fail(t[u].fail)].next[c];
50         t[v].suffixes = t[t[v].fail].suffixes + 1;
51         t[v].end = (int)s.size() - 2;
52         t[u].next[c] = v;
53     }
54     last = t[u].next[c];
55     t[last].hits++;
56     total += t[last].suffixes;
57     return last;
58 }
59
60 int Distinct() const
61 {
62     return tot - 1;
63 }
64
65 // Copy-and-propagate: repeated calls and later appends remain valid.
66 vector<long long> Occurrences() const
67 {
68     vector<long long> count(tot + 1);
69     for ( int u = 2; u <= tot; u++ )
70         count[u] = t[u].hits;
71     for ( int u = tot; u >= 2; u-- )
72         count[t[u].fail] += count[u];
73     count[0] = 0;
74     count[1] = 0;
75     return count;
76 }
77 };

```

第八十八章 数论

88.1 模运算、素性、CRT 与类欧几里德

算法约束见动态版本第 87 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Number_Theory
2 {
3     using ll = long long;
4     using ull = unsigned long long;
5     using i128 = __int128_t;
6     using u128 = __uint128_t;
7
8     static ull Mul(ull a, ull b, ull m)
9     {
10         return u128(a) * b % m;
11     }
12
13     static ull Power(ull a, ull b, ull m)
14     {
15         assert(m);
16         ull r = 1 % m;
17         a %= m;
18         for ( ; b; b >>= 1, a = Mul(a, a, m) )
19             if ( b & 1 )
20                 r = Mul(r, a, m);
21         return r;
22     }
23
24     static bool Prime(ull n)
25     {
26         if ( n < 2 )
27             return false;
28         for ( ull p : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37} )
29             if ( n % p == 0 )
30                 return n == p;
31         ull d = n - 1;
32         int s = 0;
33         while ( !(d & 1) )
34         {
35             d >>= 1;
36             ++s;
37         }
38         for ( ull a :
39             {2ULL, 325ULL, 9375ULL, 28178ULL, 450775ULL, 9780504ULL, 1795265022ULL} )
40         {
41             if ( a % n == 0 )
42                 continue;
43             ull x = Power(a, d, n);
44             if ( x == 1 || x == n - 1 )
45                 continue;
46             bool ok = false;
```

```

47     for ( int j = 1; j < s; j++ )
48     {
49         x = Mul(x, x, n);
50         if ( x == n - 1 )
51         {
52             ok = true;
53             break;
54         }
55     }
56     if ( !ok )
57         return false;
58 }
59 return true;
60 }
61
62 static i128 Exgcd(i128 a, i128 b, i128 &x, i128 &y)
63 {
64     if ( !b )
65     {
66         x = 1;
67         y = 0;
68         return a;
69     }
70     i128 d = Exgcd(b, a % b, y, x);
71     y -= a / b * x;
72     return d;
73 }
74
75 static ll Inverse(ll a, ll m)
76 {
77     assert(m > 0);
78     a %= m;
79     if ( a < 0 )
80         a += m;
81     i128 x, y;
82     if ( Exgcd(a, m, x, y) != 1 )
83         return -1;
84     return (x % m + m) % m;
85 }
86
87 // x = r (mod m); false means inconsistent. Throws if lcm exceeds int64.
88 static bool Crt(ll &r, ll &m, ll b, ll n)
89 {
90     assert(m > 0 && n > 0);
91     r %= m;
92     if ( r < 0 )
93         r += m;
94     b %= n;
95     if ( b < 0 )
96         b += n;
97     i128 x, y;
98     ll g = (ll)Exgcd(m, n, x, y);
99     i128 diff = i128(b) - r;
100     if ( diff % g )
101         return false;
102     i128 q = n / g, k = (diff / g * x % q + q) % q, mod = i128(m) * q;

```

```

103     if ( mod > LLONG_MAX )
104         throw overflow_error("CRT modulus");
105     r = (r + i128(m) * k) % mod;
106     m = (ll)mod;
107     return true;
108 }
109
110 // sum_{i=0}^{n-1} floor((a*i+b)/m), signed a,b; 0<=n<=1e9, 1<=m<=1e9.
111 static i128 Floor_Sum(ll n, ll m, ll a, ll b)
112 {
113     assert(n >= 0 && m > 0);
114     i128 ans = 0;
115     auto norm = [&](ll &v)
116     {
117         ll q = v / m;
118         if ( v % m < 0 )
119             --q;
120         v = (ll)(i128(v) - i128(q) * m);
121         return q;
122     };
123     ans += i128(norm(a)) * n * (n - 1) / 2;
124     ans += i128(norm(b)) * n;
125     while ( true )
126     {
127         ans += i128(n) * (n - 1) / 2 * (a / m);
128         a %= m;
129         ans += i128(n) * (b / m);
130         b %= m;
131         i128 y = i128(a) * n + b;
132         if ( y < m )
133             break;
134         n = (ll)(y / m);
135         b = (ll)(y % m);
136         swap(a, m);
137     }
138     return ans;
139 }
140 };

```

88.2 Pollard-Rho 因数分解

算法约束见动态版本第 90 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Pollard_Rho
2 {
3     using ull = unsigned long long;
4     using u128 = __uint128_t;
5     mt19937_64 rng;
6
7     Pollard_Rho(ull seed = 712367821) : rng(seed)
8     {
9     }
10
11     ull Rho(ull n)
12     {
13         if ( n % 2 == 0 )
14             return 2;
15         for ( ;; )
16         {
17             ull c = rng() % (n - 1) + 1, x = rng() % n, y = x, d = 1;
18             auto f = [&](ull v)
19             {
20                 return (Number_Theory::Mul(v, v, n) + u128(c)) % n;
21             };
22             // Retry bounded attempts; Las Vegas: only exact divisors returned.
23             for ( int i = 0; i < 200000 && d == 1; i++ )
24             {
25                 x = f(x);
26                 y = f(f(y));
27                 d = gcd(x > y ? x - y : y - x, n);
28             }
29             if ( 1 < d && d < n )
30                 return d;
31         }
32     }
33
34     void Split(ull n, vector<ull> &a)
35     {
36         if ( n == 1 )
37             return;
38         if ( Number_Theory::Prime(n) )
39         {
40             a.push_back(n);
41             return;
42         }
43         ull d = Rho(n);
44         Split(d, a);
45         Split(n / d, a);
46     }

```

```

47
48     vector<ull> Factor(ull n)
49     {
50         assert(n >= 1);
51         vector<ull> a;
52         Split(n, a);
53         sort(a.begin(), a.end());
54         return a;
55     }
56 };

```

88.3 线性筛、欧拉函数与 Möbius 函数

算法约束见动态版本第 91 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Linear_Sieve
2 {
3     vector<int> Prime, lp, phi, mu;
4
5     Linear_Sieve(int n) : lp(n + 1), phi(n + 1), mu(n + 1)
6     {
7         if ( n )
8             phi[1] = mu[1] = 1;
9         for ( int i = 2; i <= n; i++ )
10            {
11                if ( !lp[i] )
12                {
13                    lp[i] = i;
14                    Prime.push_back(i);
15                    phi[i] = i - 1;
16                    mu[i] = -1;
17                }
18                for ( int p : Prime )
19                {
20                    if ( p > n / i )
21                        break;
22                    int j = i * p;
23                    lp[j] = p;
24                    if ( i % p == 0 )
25                    {
26                        phi[j] = phi[i] * p;
27                        mu[j] = 0;
28                        break;
29                    }
30                    phi[j] = phi[i] * (p - 1);
31                    mu[j] = -mu[i];
32                }
33            }
34     }
35 };

```

88.4 静态模整数

算法约束见动态版本第 92 页。本节代码独立可抄；采用下列实际接口名。

```

1 template <int mod> struct Mod_Int
2 {
3     int v;
4
5     Mod_Int(long long x = 0) : v((x % mod + mod) % mod)
6     {
7     }
8
9     Mod_Int operator+(Mod_Int b) const
10    {
11        return Mod_Int((long long)v + b.v);
12    }
13
14    Mod_Int operator-(Mod_Int b) const
15    {
16        return Mod_Int((long long)v - b.v);
17    }
18
19    Mod_Int operator*(Mod_Int b) const
20    {
21        return Mod_Int(1LL * v * b.v);
22    }
23
24    Mod_Int Pow(long long e) const
25    {
26        assert(e >= 0);
27        Mod_Int a = *this, r = 1;
28        for ( ; e; e >>= 1, a = a * a )
29            if ( e & 1 )
30                r = r * a;
31        return r;
32    }
33
34    Mod_Int Inv() const
35    {
36        assert(v);
37        return Pow(mod - 2);
38    } // prime modulus
39
40    Mod_Int operator/(Mod_Int b) const
41    {
42        return *this * b.Inv();
43    }
44 };

```

88.5 阶乘组合数

算法约束见动态版本第 93 页。本节代码独立可抄；采用下列实际接口名。

Combination<prime>(n=0) 构造，Init(n) 扩大上限；Choose、Permute 对应组合数和排列数，fac、ifac 可只读访问。

```

1  template <int mod> struct Combination
2  {
3      using Z = Mod_Int<mod>;
4      vector<Z> fac{Z(1)}, ifac{Z(1)};
5
6      Combination(int n = 0)
7      {
8          Init(n);
9      }
10
11     void Init(int n)
12     {
13         assert(0 <= n && n < mod);
14         int old = (int)fac.size() - 1;
15         if ( n <= old )
16             return;
17         fac.resize(n + 1);
18         ifac.resize(n + 1);
19         for ( int i = old + 1; i <= n; i++ )
20             fac[i] = fac[i - 1] * i;
21         ifac[n] = fac[n].Inv();
22         for ( int i = n; i > old; i-- )
23             ifac[i - 1] = ifac[i] * i;
24     }
25
26     Z Choose(int n, int k) const
27     {
28         if ( k < 0 || k > n )
29             return 0;
30         assert(n < (int)fac.size());
31         return fac[n] * ifac[k] * ifac[n - k];
32     }
33
34     Z Permute(int n, int k) const
35     {
36         if ( k < 0 || k > n )
37             return 0;
38         assert(n < (int)fac.size());
39         return fac[n] * ifac[n - k];
40     }
41 };

```


第八十九章 批量模逆元

89.1 任意数组的批量逆元

算法约束见动态版本第 94 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline optional<vector<long long>> Batch_Inverse(vector<long long> a, long long mod)
2 {
3     assert(mod > 0);
4     int n = a.size();
5     vector<long long> prefix(n + 1, 1 % mod);
6     for ( int i = 0; i < n; i++ )
7     {
8         a[i] %= mod;
9         if ( a[i] < 0 )
10             a[i] += mod;
11         prefix[i + 1] = __int128_t(prefix[i]) * a[i] % mod;
12     }
13     long long inverse = Number_Theory::Inverse(prefix[n], mod);
14     if ( inverse == -1 )
15         return nullopt;
16     for ( int i = n - 1; i >= 0; i-- )
17     {
18         long long value = a[i];
19         a[i] = __int128_t(inverse) * prefix[i] % mod;
20         inverse = __int128_t(inverse) * value % mod;
21     }
22     return a;
23 }
```

第九十章 混合进制中国剩余定理

90.1 Garner：大周期 CRT 与混合进制

算法约束见动态版本第 95 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Pairwise coprime positive moduli. Digits use the input order.
2 inline vector<long long> Garner_Digits(const vector<long long> &b,
3                                         const vector<long long> &moduli)
4 {
5     using ll = long long;
6     using i128 = __int128_t;
7     int n = b.size();
8     assert(moduli.size() == b.size());
9     vector<ll> value(n), coefficient(n, 1), digit(n);
10    for ( ll m : moduli )
11        assert(m > 0);
12    for ( int i = 0; i < n; i++ )
13    {
14        ll m = moduli[i];
15        ll inverse = Number_Theory::Inverse(coefficient[i], m);
16        if ( inverse == -1 )
17            throw invalid_argument("Garner moduli are not coprime");
18        i128 delta = (i128(b[i]) - value[i]) % m;
19        if ( delta < 0 )
20            delta += m;
21        digit[i] = delta * inverse % m;
22        for ( int j = i + 1; j < n; j++ )
23        {
24            value[j] = (value[j] + i128(coefficient[j]) * digit[i]) % moduli[j];
25            coefficient[j] = i128(coefficient[j]) * m % moduli[j];
26        }
27    }
28    return digit;
29 }
30
31 // Return the least nonnegative CRT solution modulo target, not all solutions.
32 inline long long
33 Garner(const vector<long long> &b, const vector<long long> &moduli, long long target)
34 {
35     assert(target > 0);
36     auto digit = Garner_Digits(b, moduli);
37     long long answer = 0;
38     for ( int i = (int)digit.size() - 1; i >= 0; i-- )
39         answer = (__int128_t(answer) * moduli[i] + digit[i]) % target;
40     return answer;
41 }
```

第九十一章 乘法阶与原根

91.1 乘法阶与原根

算法约束见动态版本第 97 页。本节代码独立可抄；采用下列实际接口名。

构造 Primitive_Root(n), Init、Order、Is_Root、Minimum、All 对应动态接口；Factors 为 phi 的不同素因子，依赖 Number_Theory::Power。

```
1 struct Primitive_Root
2 {
3     int mod, phi;
4     bool exists;
5     vector<int> Factors;
6
7     static vector<int> Factor(int n)
8     {
9         vector<int> result;
10        for ( int p = 2; p <= n / p; p++ )
11            if ( n % p == 0 )
12            {
13                result.push_back(p);
14                while ( n % p == 0 )
15                    n /= p;
16            }
17        if ( n > 1 )
18            result.push_back(n);
19        return result;
20    }
21
22    Primitive_Root(int n)
23    {
24        Init(n);
25    }
```

```
27 void Init(int n)
28 {
29     assert(n >= 2);
30     mod = n;
31     phi = n;
32     for ( int p : Factor(n) )
33         phi -= phi / p;
34     int odd = n;
35     if ( odd % 2 == 0 )
36         odd /= 2;
37     exists = n == 2 || n == 4 || (odd % 2 && Factor(odd).size() == 1);
38     Factors = Factor(phi);
39 }
40
41 int Order(long long a) const
42 {
43     a %= mod;
44     if ( a < 0 )
45         a += mod;
46     if ( gcd(a, (long long)mod) != 1 )
47         return 0;
48     int result = phi;
49     for ( int p : Factors )
50         while ( result % p == 0 && Number_Theory::Power(a, result / p, mod) == 1 )
51             result /= p;
52     return result;
53 }
```

```

55     bool Is_Root(long long a) const
56     {
57         if ( !exists )
58             return false;
59         a %= mod;
60         if ( a < 0 )
61             a += mod;
62         if ( gcd(a, (long long)mod) != 1 )
63             return false;
64         for ( int p : Factors )
65             if ( Number_Theory::Power(a, phi / p, mod) == 1 )
66                 return false;
67         return true;
68     }
69
70     int Minimum() const
71     {
72         if ( !exists )
73             return -1;
74         for ( int g = 1; g < mod; g++ )
75             if ( Is_Root(g) )
76                 return g;
77         return -1;
78     }
79
80     vector<int> All() const
81     {
82         int g = Minimum();
83         vector<int> roots;
84         if ( g == -1 )
85             return roots;
86         long long value = 1;
87         for ( int k = 1; k <= phi; k++ )
88         {
89             value = value * g % mod;
90             if ( gcd(k, phi) == 1 )
91                 roots.push_back(value);
92         }
93         sort(roots.begin(), roots.end());
94         return roots;
95     }
96 };

```

第九十二章 Carmichael 函数

92.1 Carmichael: 模乘法群的公共指数

算法约束见动态版本第 100 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline int Carmichael(int n)
2 {
3     assert(n >= 1);
4     long long answer = 1;
5     for ( int p = 2; p <= n / p; p++ )
6         if ( n % p == 0 )
7         {
8             int power = 1;
9             while ( n % p == 0 )
10             {
11                 n /= p;
12                 power *= p;
13             }
14             int value = power / p * (p - 1);
15             if ( p == 2 && power >= 8 )
16                 value /= 2;
17             answer = answer / gcd(answer, (long long)value) * value;
18         }
19     if ( n > 1 )
20         answer = answer / gcd(answer, (long long)(n - 1)) * (n - 1);
21     return answer;
22 }
```

第九十三章 素数模数组合数

93.1 Lucas：小素数模的大下标组合数

算法约束见动态版本第 101 页。本节代码独立可抄；采用下列实际接口名。

声明 Lucas_Theorem<N>, Init(p) 重建素数模数下的表, Choose(n,k) 查询。要求 p 不超过 N；大对象放全局或 static。传统版空间 $O(N)$ 。

```
1 template <int N> struct Lucas_Theorem
2 {
3     int mod;
4     int fac[N + 1], ifac[N + 1];
5
6     void Init(int p)
7     {
8         assert(2 <= p && p <= N);
9         mod = p;
10        fac[0] = 1;
11        for ( int i = 1; i < p; i++ )
12            fac[i] = 1LL * fac[i - 1] * i % p;
13        long long a = fac[p - 1], r = 1;
14        for ( int e = p - 2; e; e >>= 1 )
15        {
16            if ( e & 1 )
17                r = r * a % p;
18            a = a * a % p;
19        }
20        ifac[p - 1] = r;
21        for ( int i = p - 1; i; i-- )
22            ifac[i - 1] = 1LL * ifac[i] * i % p;
23    }
24
25    int Choose(unsigned long long n, unsigned long long k) const
26    {
27        if ( k > n )
28            return 0;
29        long long answer = 1;
30        while ( n || k )
31        {
32            int a = n % mod, b = k % mod;
33            if ( b > a )
34                return 0;
35            answer = answer * fac[a] % mod;
36            answer = answer * ifac[b] % mod;
37            answer = answer * ifac[a - b] % mod;
38            n /= mod;
39            k /= mod;
40        }
41        return answer;
42    }
43 };
```

第九十四章 合数模数组合数

94.1 合数模数组合数（扩展 Lucas）

算法约束见动态版本第 103 页。本节代码独立可抄；采用下列实际接口名。

声明 `Ex_Lucas<N>` 后 `Init(mod)`，要求 `mod` 不超过 `N`；`Choose` 为查询接口。`fac` 是共用静态前缀积数组，各素数幂通过 `start` 分段；各段总长度不超过 `mod+1`，所以 `N` 限制模数即可。`Init` 可切换模数并清空旧预处理，大对象放在全局或 `static`。

```
1 template <int N> struct Ex_Lucas
2 {
3     using ull = unsigned long long;
4     using ll = long long;
5
6     struct Factor
7     {
8         int p, q, exponent, start, crt;
9     };
10
11     int mod;
12     int fac[N + 1], used;
13     vector<Factor> factors;
14
15     static ll Power(ll a, ull b, int m)
16     {
17         ll result = 1;
18         while ( b )
19         {
20             if ( b & 1 )
21                 result = result * a % m;
22             a = a * a % m;
23             b >>= 1;
24         }
25         return result;
26     }
27
28     void Init(int m)
29     {
30         assert(1 <= m && m <= N);
31         mod = m;
32         used = 0;
33         factors.clear();
34         int x = m;
35         for ( int p = 2; p <= x / p; p++ )
36         {
37             if ( x % p )
38                 continue;
39             int q = 1, e = 0;
40             do
41             {
42                 x /= p;
43                 q *= p;
```



```

44         e++;
45     } while ( x % p == 0 );
46     Add_Factor(p, q, e);
47 }
48 if ( x > 1 )
49     Add_Factor(x, x, 1);
50 }
51
52 void Add_Factor(int p, int q, int e)
53 {
54     int start = used;
55     fac[used++] = 1;
56     for ( int i = 1; i < q; i++ )
57     {
58         fac[used] = ll(fac[used - 1]) * (i % p ? i : 1) % q;
59         used++;
60     }
61     fac[used] = fac[used - 1];
62     used++;
63     int phi = q - q / p;
64     int crt = ll(mod / q) * Power(mod / q % q, phi - 1, q) % mod;
65     factors.push_back({p, q, e, start, crt});
66 }
67
68 static ull Valuation(ull n, int p)
69 {
70     ull result = 0;
71     while ( n )
72     {
73         n /= p;
74         result += n;
75     }
76     return result;
77 }
78
79 ll Unit(ull n, const Factor &f) const
80 {
81     ll result = 1;
82     while ( n )
83     {
84         result = result * Power(fac[f.start + f.q], n / f.q, f.q) % f.q;
85         result = result * fac[f.start + n % f.q] % f.q;
86         n /= f.p;
87     }
88     return result;
89 }

```

```
91     int Choose(ull n, ull k) const
92     {
93         if ( k > n || mod == 1 )
94             return 0;
95         ll answer = 0;
96         for ( const auto &f : factors )
97         {
98             ull e = Valuation(n, f.p) - Valuation(k, f.p) - Valuation(n - k, f.p);
99             if ( e >= ull(f.exponent) )
100                 continue;
101             ll denominator = Unit(k, f) * Unit(n - k, f) % f.q;
102             ll residue =
103                 Unit(n, f) * Power(denominator, f.q - f.q / f.p - 1, f.q) % f.q;
104             residue = residue * Power(f.p, e, f.q) % f.q;
105             answer = (answer + residue * f.crt) % mod;
106         }
107         return answer;
108     }
109 };
```

第九十五章 二次剩余

95.1 素数模平方根 (Cipolla)

算法约束见动态版本第 106 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline vector<long long> Mod_Sqrt(long long n, int p)
2 {
3     using ll = long long;
4     assert(p >= 2);
5     n = (n % p + p) % p;
6     if ( n == 0 || p == 2 )
7         return {n};
8     auto Power = [p](ll a, ll b)
9     {
10         ll result = 1;
11         while ( b )
12         {
13             if ( b & 1 )
14                 result = result * a % p;
15             a = a * a % p;
16             b >>= 1;
17         }
18         return result;
19     };
20     if ( Power(n, (p - 1) / 2) != 1 )
21         return {};
22     ll answer;
23     if ( p % 4 == 3 )
24         answer = Power(n, (ll(p) + 1) / 4);
25     else
26     {
27         static mt19937_64 rng(712367);
28         ll a, w;
29         do
30         {
31             a = rng() % p;
32             w = (a * a % p - n + p) % p;
33         } while ( Power(w, (p - 1) / 2) != p - 1 );
34         using P = pair<ll, ll>;
35         auto Mul = [p, w](P x, P y)
36         {
37             ll real = (x.first * y.first + x.second * y.second % p * w) % p;
38             ll imag = (x.first * y.second + x.second * y.first) % p;
39             return P{real, imag};
40         };
41         P result{1, 0}, base{a, 1};
42         ll exponent = (ll(p) + 1) / 2;
43         while ( exponent )
44         {
45             if ( exponent & 1 )
46                 result = Mul(result, base);
```

```
47         base = Mul(base, base);
48         exponent >>= 1;
49     }
50     answer = result.first;
51 }
52 ll other = p - answer;
53 if ( answer > other )
54     swap(answer, other);
55 return {answer, other};
56 }
```

第九十六章 多项式插值

96.1 拉格朗日插值与连续点优化

算法约束见动态版本第 108 页。本节代码独立可抄；采用下列实际接口名。

声明 Lagrange_Interpolation<N>; Init、Consecutive、Query 对应动态接口，N 为最大样本数。坐标、权值与工作数组均为静态数组，大对象放在全局或 static；Query 会改写工作数组。

```
1 template <int N> struct Lagrange_Interpolation
2 {
3     using ll = long long;
4     int n, mod;
5     ll x[N], w[N], invfac[N], pre[N + 1];
6
7     ll Norm(ll a) const
8     {
9         a %= mod;
10        if ( a < 0 )
11            a += mod;
12        return a;
13    }
14
15    ll Power(ll a, int b) const
16    {
17        ll result = 1;
18        while ( b )
19        {
20            if ( b & 1 )
21                result = result * a % mod;
22            a = a * a % mod;
23            b >>= 1;
24        }
25        return result;
26    }
27
28    void Init(const vector<ll> &xs, const vector<ll> &ys, int p)
29    {
30        n = ys.size();
31        mod = p;
32        assert(p >= 2 && n <= p && xs.size() == ys.size());
33        assert(n <= N);
34        for ( int i = 0; i < n; i++ )
35            x[i] = Norm(xs[i]);
36        for ( int i = 0; i < n; i++ )
37        {
38            ll denominator = 1;
39            for ( int j = 0; j < n; j++ )
40                if ( i != j )
41                    denominator = denominator * Norm(x[i] - x[j]) % mod;
42            assert(denominator != 0);
43            w[i] = Norm(ys[i]) * Power(denominator, mod - 2) % mod;
44        }
```

```

45     }
46
47     void Consecutive(const vector<ll> &ys, int p)
48     {
49         n = ys.size();
50         mod = p;
51         assert(p >= 2 && n <= p);
52         assert(n <= N);
53         if ( !n )
54             return;
55
56         ll factorial = 1;
57         for ( int i = 1; i < n; i++ )
58             factorial = factorial * i % mod;
59         invfac[n - 1] = Power(factorial, mod - 2);
60         for ( int i = n - 1; i > 0; i-- )
61             invfac[i - 1] = invfac[i] * i % mod;
62         for ( int i = 0; i < n; i++ )
63         {
64             x[i] = i;
65             w[i] = Norm(ys[i]) * invfac[i] % mod * invfac[n - 1 - i] % mod;
66             if ( (n - 1 - i) & 1 )
67                 w[i] = Norm(-w[i]);
68         }
69     }
70
71     ll Query(ll k)
72     {
73         k = Norm(k);
74         pre[0] = 1;
75         for ( int i = 0; i < n; i++ )
76             pre[i + 1] = pre[i] * Norm(k - x[i]) % mod;
77         ll answer = 0, suffix = 1;
78         for ( int i = n - 1; i >= 0; i-- )
79         {
80             answer = (answer + w[i] * pre[i] % mod * suffix) % mod;
81             suffix = suffix * Norm(k - x[i]) % mod;
82         }
83         return answer;
84     }
85 };

```

第九十七章 多项式

97.1 NTT、形式幂级数、FWT 与递推

算法约束见动态版本第 110 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Polynomial
2 {
3     static constexpr int mod = 998244353;
4     using Z = Mod_Int<mod>;
5     using Poly = vector<Z>;
6
7     static void Ntt(Poly &a, bool invert = false)
8     {
9         int n = (int)a.size();
10        assert(n > 0 && (n & (n - 1)) == 0 && n <= (1 << 23));
11        for ( int i = 1, j = 0; i < n; i++ )
12        {
13            int bit = n >> 1;
14            for ( ; j & bit; bit >>= 1 )
15                j ^= bit;
16            j ^= bit;
17            if ( i < j )
18                swap(a[i], a[j]);
19        }
20        for ( int len = 2; len <= n; len *= 2 )
21        {
22            Z step = Z(3).Pow((mod - 1) / len);
23            if ( invert )
24                step = step.Inv();
25            for ( int i = 0; i < n; i += len )
26            {
27                Z w = 1;
28                for ( int j = 0; j < len / 2; j++, w = w * step )
29                {
30                    Z u = a[i + j], v = a[i + j + len / 2] * w;
31                    a[i + j] = u + v;
32                    a[i + j + len / 2] = u - v;
33                }
34            }
35        }
36        if ( invert )
37        {
38            Z Inv = Z(n).Inv();
39            for ( auto &x : a )
40                x = x * Inv;
41        }
42    }
43
44    static Poly Multiply(Poly a, Poly b)
45    {
46        if ( a.empty() || b.empty() )
```

```

47     return {};
48     int size = (int)a.size() + (int)b.size() - 1, n = 1;
49     assert(size <= (1 << 23));
50     while ( n < size )
51         n *= 2;
52     a.resize(n);
53     b.resize(n);
54     Ntt(a);
55     Ntt(b);
56     for ( int i = 0; i < n; i++ )
57         a[i] = a[i] * b[i];
58     Ntt(a, true);
59     a.resize(size);
60     return a;
61 }
62
63 static Poly Inverse(const Poly &a, int n)
64 {
65     assert(n > 0 && !a.empty() && a[0].v);
66     Poly b{a[0].Inv()};
67     while ( (int)b.size() < n )
68     {
69         int m = min(n, 2 * (int)b.size());
70         Poly f(a.begin(), a.begin() + min(m, (int)a.size()));
71         auto t = Multiply(f, b);
72         t.resize(m);
73         for ( auto &x : t )
74             x = Z(0) - x;
75         t[0] = t[0] + 2;
76         b = Multiply(b, t);
77         b.resize(m);
78     }
79     return b;
80 }
81
82 static Poly Derivative(const Poly &a)
83 {
84     Poly b;
85     for ( int i = 1; i < (int)a.size(); i++ )
86         b.push_back(a[i] * i);
87     return b;
88 }
89
90 static Poly Integral(const Poly &a)
91 {
92     Poly b(a.size() + 1);
93     vector<Z> Inv(a.size() + 1);
94     if ( !a.empty() )
95         Inv[1] = 1;
96     for ( int i = 2; i <= (int)a.size(); i++ )
97         Inv[i] = Z(mod - mod / i) * Inv[mod % i];
98     for ( int i = 0; i < (int)a.size(); i++ )
99         b[i + 1] = a[i] * Inv[i + 1];
100     return b;
101 }
102

```



```

103 static Poly Log(const Poly &a, int n)
104 {
105     assert(n > 0 && !a.empty() && a[0].v == 1);
106     auto b = Multiply(Derivative(a), Inverse(a, n));
107     b.resize(n - 1);
108     return Integral(b);
109 }
110
111 static Poly Exp(const Poly &a, int n)
112 {
113     assert(n > 0 && (a.empty() || a[0].v == 0));
114     Poly b{1};
115     while ( (int)b.size() < n )
116     {
117         int m = min(n, 2 * (int)b.size());
118         auto t = Log(b, m);
119         for ( int i = 0; i < m; i++ )
120             t[i] = (i < (int)a.size() ? a[i] : Z(0)) - t[i];
121         t[0] = t[0] + 1;
122         b = Multiply(b, t);
123         b.resize(m);
124     }
125     return b;
126 }
127
128 static void Fwt(Poly &a, char op, bool invert = false)
129 {
130     int n = (int)a.size();
131     assert(n > 0 && (n & (n - 1)) == 0);
132     const Z inv2 = (mod + 1) / 2;
133     for ( int len = 1; len < n; len *= 2 )
134         for ( int i = 0; i < n; i += len * 2 )
135             for ( int j = 0; j < len; j++ )
136             {
137                 Z &u = a[i + j], &v = a[i + j + len], x = u, y = v;
138                 if ( op == '^' )
139                 {
140                     u = x + y;
141                     v = x - y;
142                     if ( invert )
143                     {
144                         u = u * inv2;
145                         v = v * inv2;
146                     }
147                 }
148                 else if ( op == '|' )
149                     v = invert ? y - x : y + x;
150                 else
151                 {
152                     assert(op == '&');
153                     u = invert ? x - y : x + y;
154                 }
155             }
156 }
157
158 // c: s[n]=sum c[i-1]*s[n-i], i=1..L.

```

```

159 static Poly Berlekamp_Massey(const Poly &s)
160 {
161     Poly c{1}, b{1};
162     int l = 0, m = 1;
163     Z last = 1;
164     for ( int n = 0; n < (int)s.size(); n++ )
165     {
166         Z d = s[n];
167         for ( int i = 1; i <= l; i++ )
168             d = d + c[i] * s[n - i];
169         if ( !d.v )
170         {
171             ++m;
172             continue;
173         }
174         auto old = c;
175         Z coef = d / last;
176         if ( c.size() < b.size() + m )
177             c.resize(b.size() + m);
178         for ( int j = 0; j < (int)b.size(); j++ )
179             c[j + m] = c[j + m] - coef * b[j];
180         if ( 2 * l <= n )
181         {
182             l = n + 1 - l;
183             b = old;
184             last = d;
185             m = 1;
186         }
187         else
188             ++m;
189     }
190     c.erase(c.begin());
191     for ( auto &x : c )
192         x = Z(0) - x;
193     return c;
194 }
195
196 static Z Recurrence(Poly init, const Poly &c, unsigned long long n)
197 {
198     int k = (int)c.size();
199     if ( n < init.size() )
200         return init[n];
201     if ( !k )
202         return 0;
203     assert((int)init.size() >= k);
204     auto combine = [&](const Poly &a, const Poly &b)
205     {
206         Poly t(2 * k - 1);
207         for ( int i = 0; i < k; i++ )
208             for ( int j = 0; j < k; j++ )
209                 t[i + j] = t[i + j] + a[i] * b[j];
210         for ( int i = 2 * k - 2; i >= k; i-- )
211             for ( int j = 1; j <= k; j++ )
212                 t[i - j] = t[i - j] + t[i] * c[j - 1];
213         t.resize(k);
214         return t;

```

```
215     };
216     Poly a(k), r(k);
217     r[0] = 1;
218     if ( k == 1 )
219         a[0] = c[0];
220     else
221         a[1] = 1;
222     for ( ; n; n >>= 1, a = combine(a, a) )
223         if ( n & 1 )
224             r = combine(r, a);
225     Z ans = 0;
226     for ( int i = 0; i < k; i++ )
227         ans = ans + r[i] * init[i];
228     return ans;
229 }
230 };
```

第九十八章 代数与数论进阶

98.1 模高斯消元、行列式、矩阵树定理

算法约束见动态版本第 115 页。本节代码独立可抄；采用下列实际接口名。

```
1 template <int mod> struct Linear_Algebra
2 {
3     using Z = Mod_Int<mod>;
4     using Matrix = vector<vector<Z>>>;
5
6     struct Solution
7     {
8         bool consistent;
9         int rank;
10        vector<Z> particular;
11        Matrix kernel;
12    };
13
14    // Augmented m x (n+1), prime modulus. Empty system needs explicit n.
15    static Solution Solve(Matrix a, int n)
16    {
17        int m = (int)a.size(), row = 0;
18        vector<int> where(n, -1);
19        for ( const auto &v : a )
20            assert((int)v.size() == n + 1);
21        for ( int col = 0; col < n && row < m; col++ )
22        {
23            int p = row;
24            while ( p < m && !a[p][col].v )
25                ++p;
26            if ( p == m )
27                continue;
28            swap(a[p], a[row]);
29            Z Inv = a[row][col].Inv();
30            for ( int j = col; j <= n; j++ )
31                a[row][j] = a[row][j] * Inv;
32            for ( int i = 0; i < m; i++ )
33                if ( i != row && a[i][col].v )
34                {
35                    Z f = a[i][col];
36                    for ( int j = col; j <= n; j++ )
37                        a[i][j] = a[i][j] - f * a[row][j];
38                }
39            where[col] = row++;
40        }
41        for ( int i = row; i < m; i++ )
42            if ( a[i][n].v )
43                return {false, row, {}, {}};
44        Solution ans{true, row, vector<Z>(n), {}};
45        for ( int j = 0; j < n; j++ )
46            if ( where[j] != -1 )
```

```

47         ans.particular[j] = a[where[j]][n];
48     for ( int j = 0; j < n; j++ )
49         if ( where[j] == -1 )
50         {
51             vector<Z> v(n);
52             v[j] = 1;
53             for ( int k = 0; k < n; k++ )
54                 if ( where[k] != -1 )
55                     v[k] = Z(0) - a[where[k]][j];
56             ans.kernel.push_back(v);
57         }
58     return ans;
59 }
60
61 static Z Determinant(Matrix a)
62 {
63     int n = (int)a.size();
64     Z ans = 1;
65     for ( int i = 0; i < n; i++ )
66     {
67         assert((int)a[i].size() == n);
68         int p = i;
69         while ( p < n && !a[p][i].v )
70             ++p;
71         if ( p == n )
72             return 0;
73         if ( p != i )
74         {
75             swap(a[p], a[i]);
76             ans = Z(0) - ans;
77         }
78         ans = ans * a[i][i];
79         Z Inv = a[i][i].Inv();
80         for ( int j = i + 1; j < n; j++ )
81         {
82             Z f = a[j][i] * Inv;
83             for ( int k = i; k < n; k++ )
84                 a[j][k] = a[j][k] - f * a[i][k];
85         }
86     }
87     return ans;
88 }
89
90 static Matrix Multiply(const Matrix &a, const Matrix &b)
91 {
92     assert(!a.empty() && !b.empty());
93     int n = a.size(), m = b[0].size(), k = b.size();
94     assert((int)a[0].size() == k);
95     Matrix c(n, vector<Z>(m));
96     for ( int i = 0; i < n; i++ )
97         for ( int t = 0; t < k; t++ )
98             for ( int j = 0; j < m; j++ )
99                 c[i][j] = c[i][j] + a[i][t] * b[t][j];
100     return c;
101 }
102

```

```

103 static Matrix Power(Matrix a, unsigned long long e)
104 {
105     int n = a.size();
106     assert(n > 0 && (int)a[0].size() == n);
107     Matrix r(n, vector<Z>(n));
108     for ( int i = 0; i < n; i++ )
109         r[i][i] = 1;
110     for ( ; e; e >>= 1, a = Multiply(a, a) )
111         if ( e & 1 )
112             r = Multiply(r, a);
113     return r;
114 }
115
116 // Undirected multigraph, vertices 0..n-1; loops ignored.
117 static Z Spanning_Trees(int n, const vector<pair<int, int>> &edges)
118 {
119     assert(n > 0);
120     Matrix lap(n, vector<Z>(n));
121     for ( auto [u, v] : edges )
122         if ( u != v )
123         {
124             lap[u][u] = lap[u][u] + 1;
125             lap[v][v] = lap[v][v] + 1;
126             lap[u][v] = lap[u][v] - 1;
127             lap[v][u] = lap[v][u] - 1;
128         }
129     lap.pop_back();
130     for ( auto &row : lap )
131         row.pop_back();
132     return Determinant(lap);
133 }
134 };

```

98.2 杜教筛

算法约束见动态版本第 118 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Du_Jiao
2 {
3     using ll = long long;
4     using I = __int128_t;
5     int limit;
6     vector<ll> pmu, pphi;
7     unordered_map<ll, ll> mmu;
8     unordered_map<ll, I> mphi;
9
10    Du_Jiao(int limit) : limit(limit), pmu(limit + 1), pphi(limit + 1)
11    {
12        assert(limit >= 1);
13        Linear_Sieve s(limit);
14        for ( int i = 1; i <= limit; i++ )
15        {
16            pmu[i] = pmu[i - 1] + s.mu[i];
17            pphi[i] = pphi[i - 1] + s.phi[i];
18        }
19    }
20
21    ll Mertens(ll n)
22    {
23        if ( n <= limit )
24            return pmu[n];
25        if ( mmu.count(n) )
26            return mmu[n];
27        ll ans = 1;
28        for ( ll l = 2, r; l <= n; l = r + 1 )
29        {
30            r = n / (n / l);
31            ans -= (r - l + 1) * Mertens(n / l);
32        }
33        return mmu[n] = ans;
34    }
35
36    I Totient_Sum(ll n)
37    {
38        if ( n <= limit )
39            return pphi[n];
40        if ( mphi.count(n) )
41            return mphi[n];
42        I ans = I(n) * (n + 1) / 2;
43        for ( ll l = 2, r; l <= n; l = r + 1 )
44        {
45            r = n / (n / l);
46            ans -= I(r - l + 1) * Totient_Sum(n / l);
47        }
48        return mphi[n] = ans;
49    }
50 };

```

98.3 扩展 BSGS

算法约束见动态版本第 119 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Discrete_Log
2 {
3     using ll = long long;
4
5     // 1<=m<=1e12; Expected O(sqrt(m)) time, O(sqrt(m)) storage. Smallest x>=0, or -1.
6     static ll Solve(ll a, ll b, ll m)
7     {
8         assert(m >= 1 && m <= 1000000000000LL);
9         a = (a % m + m) % m;
10        b = (b % m + m) % m;
11        if ( m == 1 || b == 1 )
12            return 0;
13        if ( a == 1 )
14            return -1;
15        ll offset = 0, k = 1;
16        for ( ll g; (g = gcd(a, m)) > 1; )
17        {
18            if ( b == k )
19                return offset;
20            if ( b % g )
21                return -1;
22            b /= g;
23            m /= g;
24            k = (__int128)k * (a / g) % m;
25            ++offset;
26        }
27        ll target = (__int128)b * Number_Theory::Inverse(k, m) % m;
28        ll step = sqrtl(m) + 1;
29        unordered_map<ll, ll> baby;
30        baby.reserve(step);
31        ll cur = 1 % m;
32        for ( ll j = 0; j < step; j++ )
33        {
34            if ( !baby.count(cur) )
35                baby[cur] = j;
36            cur = (__int128)cur * a % m;
37        }
38        ll Inv = Number_Theory::Inverse(Number_Theory::Power(a, step, m), m);
39        cur = target;
40        for ( ll i = 0; i <= step; i++ )
41        {
42            auto it = baby.find(cur);
43            if ( it != baby.end() )
44                return offset + i * step + it->second;
45            cur = (__int128)cur * Inv % m;
46        }
47        return -1;
48    }
49 };

```


第九十九章 计算几何

99.1 整数精确二维几何

算法约束见动态版本第 121 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Integer_Geometry
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     // Contract: |coordinate|<=1e12. Differences/products widened before arithmetic.
7     struct Point
8     {
9         ll x = 0, y = 0;
10
11         bool operator<(Point b) const
12         {
13             return tie(x, y) < tie(b.x, b.y);
14         }
15
16         bool operator==(Point b) const
17         {
18             return x == b.x && y == b.y;
19         }
20     };
21
22     static I Cross(Point a, Point b, Point c)
23     {
24         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
25     }
26
27     static I Dot(Point a, Point b, Point c)
28     {
29         return (I(b.x) - a.x) * (I(c.x) - a.x) + (I(b.y) - a.y) * (I(c.y) - a.y);
30     }
31
32     static I Distance_Squared(Point a, Point b)
33     {
34         return Dot(a, b, b);
35     }
36
37     static int Sign(I x)
38     {
39         return (x > 0) - (x < 0);
40     }
41
42     static bool On_Segment(Point p, Point a, Point b)
43     {
44         return Cross(a, b, p) == 0 && Dot(p, a, b) <= 0;
45     }
46 }
```

```

47 static bool Intersect(Point a, Point b, Point c, Point d)
48 {
49     int x = Sign(Cross(a, b, c)), y = Sign(Cross(a, b, d));
50     int Z_Function = Sign(Cross(c, d, a)), w = Sign(Cross(c, d, b));
51     if ( !x && On_Segment(c, a, b) )
52         return true;
53     if ( !y && On_Segment(d, a, b) )
54         return true;
55     if ( !Z_Function && On_Segment(a, c, d) )
56         return true;
57     if ( !w && On_Segment(b, c, d) )
58         return true;
59     return x * y < 0 && Z_Function * w < 0;
60 }
61
62 static vector<Point> Convex_Hull(vector<Point> p)
63 {
64     sort(p.begin(), p.end());
65     p.erase(unique(p.begin(), p.end()), p.end());
66     int n = (int)p.size();
67     if ( n < 3 )
68         return p;
69     vector<Point> h;
70     for ( Point v : p )
71     {
72         while ( h.size() > 1 && Cross(h[h.size() - 2], h.back(), v) <= 0 )
73             h.pop_back();
74         h.push_back(v);
75     }
76     int low = (int)h.size();
77     for ( int i = n - 2; i >= 0; i-- )
78     {
79         while ( (int)h.size() > low && Cross(h[h.size() - 2], h.back(), p[i]) <= 0 )
80             h.pop_back();
81         h.push_back(p[i]);
82     }
83     h.pop_back();
84     return h; // CCW, no repeated endpoint, no collinear interior points.
85 }
86
87 static I Area_Twice(const vector<Point> &p)
88 {
89     I ans = 0;
90     int n = (int)p.size();
91     for ( int i = 0; i < n; i++ )
92         ans += I(p[i].x) * p[(i + 1) % n].y - I(p[i].y) * p[(i + 1) % n].x;
93     return ans;
94 }
95
96 // 0 outside, 1 boundary, 2 inside. Simple polygon; either orientation.
97 static int Contains(const vector<Point> &p, Point q)
98 {
99     int winding = 0, n = (int)p.size();
100     for ( int i = 0; i < n; i++ )
101     {
102         Point a = p[i], b = p[(i + 1) % n];

```

```

103         if ( On_Segment(q, a, b) )
104             return 1;
105         I c = Cross(a, b, q);
106         if ( a.y <= q.y && b.y > q.y && c > 0 )
107             ++winding;
108         if ( a.y > q.y && b.y <= q.y && c < 0 )
109             --winding;
110     }
111     return winding ? 2 : 0;
112 }
113
114 // Input must be the strict CCW hull above. O(log n).
115 static int Convex_Contains(const vector<Point> &p, Point q)
116 {
117     int n = (int)p.size();
118     if ( n < 3 )
119         return Contains(p, q);
120     I a = Cross(p[0], p[1], q), b = Cross(p[0], p[n - 1], q);
121     if ( a < 0 || b > 0 )
122         return 0;
123     if ( !a )
124         return On_Segment(q, p[0], p[1]) ? 1 : 0;
125     if ( !b )
126         return On_Segment(q, p[0], p[n - 1]) ? 1 : 0;
127     int l = 1, r = n - 1;
128     while ( l + 1 < r )
129     {
130         int m = (l + r) / 2;
131         if ( Cross(p[0], p[m], q) >= 0 )
132             l = m;
133         else
134             r = m;
135     }
136     I c = Cross(p[l], p[r], q);
137     return c < 0 ? 0 : c == 0 ? 1 : 2;
138 }
139
140 static I Diameter_Squared(const vector<Point> &p)
141 {
142     int n = (int)p.size();
143     if ( n < 2 )
144         return 0;
145     if ( n == 2 )
146         return Distance_Squared(p[0], p[1]);
147     I ans = 0;
148     int j = 1;
149     for ( int i = 0; i < n; i++ )
150     {
151         int k = (i + 1) % n;
152         while ( Cross(p[i], p[k], p[(j + 1) % n]) > Cross(p[i], p[k], p[j]) )
153             j = (j + 1) % n;
154         ans =
155             max({ans, Distance_Squared(p[i], p[j]), Distance_Squared(p[k], p[j])});
156     }
157     return ans;
158 }

```

```

159
160 struct PolarLess
161 {
162     static int half(Point p)
163     {
164         return p.y < 0 || (p.y == 0 && p.x < 0);
165     }
166
167     bool operator()(Point a, Point b) const
168     {
169         if ( half(a) != half(b) )
170             return half(a) < half(b);
171         I c = Cross({0, 0}, a, b);
172         if ( c )
173             return c > 0;
174         return Distance_Squared({0, 0}, a) < Distance_Squared({0, 0}, b);
175     }
176 };
177 };

```

99.2 浮点点线圆构造

算法约束见动态版本第 124 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Real_Geometry
2 {
3     using R = long double;
4     // Tolerance is explicit, not used in sorting. Requires finite, well-scaled input.
5     R eps;
6
7     Real_Geometry(R eps = 1e-12L) : eps(eps)
8     {
9         assert(eps > 0);
10    }
11
12    struct Point
13    {
14        R x = 0, y = 0;
15
16        Point operator+(Point b) const
17        {
18            return {x + b.x, y + b.y};
19        }
20
21        Point operator-(Point b) const
22        {
23            return {x - b.x, y - b.y};
24        }
25
26        Point operator*(R k) const
27        {
28            return {x * k, y * k};
29        }
30
31        Point operator/(R k) const

```

```

32     {
33         return {x / k, y / k};
34     }
35 };
36
37 struct Circle
38 {
39     Point o;
40     R r;
41 };
42 enum class Kind
43 {
44     none,
45     one,
46     two,
47     infinite,
48     degenerate
49 };
50
51 struct Result
52 {
53     Kind kind;
54     vector<Point> p;
55 };
56
57 static R Dot(Point a, Point b)
58 {
59     return a.x * b.x + a.y * b.y;
60 }
61
62 static R Cross(Point a, Point b)
63 {
64     return a.x * b.y - a.y * b.x;
65 }
66
67 static R norm(Point a)
68 {
69     return hypotl(a.x, a.y);
70 }
71
72 static Point perp(Point a)
73 {
74     return {-a.y, a.x};
75 }
76
77 Point Projection(Point p, Point a, Point b) const
78 {
79     Point v = b - a;
80     R d = Dot(v, v);
81     if ( d == 0 )
82         return a;
83     return a + v * (Dot(p - a, v) / d);
84 }
85
86 R Segment_Distance(Point p, Point a, Point b) const
87 {

```

```

88     Point v = b - a;
89     R d = Dot(v, v);
90     if ( d == 0 )
91         return norm(p - a);
92     R t = clamp(Dot(p - a, v) / d, R(0), R(1));
93     return norm(p - (a + v * t));
94 }
95
96 Result Line_Intersection(Point a, Point b, Point c, Point d) const
97 {
98     Point u = b - a, v = d - c;
99     R nu = norm(u), nv = norm(v);
100    if ( nu == 0 || nv == 0 )
101        return {Kind::degenerate, {}};
102    R det = Cross(u, v);
103    if ( fabsl(det) <= eps * nu * nv )
104        return {fabsl(Cross(c - a, u)) <= eps * nu * max(R(1), norm(c - a))
105                ? Kind::infinite
106                : Kind::none,
107                {}};
108    return {Kind::one, {a + u * (Cross(c - a, v) / det)}};
109 }
110
111 Result Line_Circle(Point a, Point b, Circle c) const
112 {
113     assert(c.r >= 0);
114     Point v = b - a;
115     R len = norm(v);
116     if ( len == 0 )
117         return {Kind::degenerate, {}};
118     Point h = Projection(c.o, a, b);
119     R d = norm(h - c.o), tol = eps * max(d, c.r);
120     if ( d > c.r + tol )
121         return {Kind::none, {}};
122     if ( fabsl(d - c.r) <= tol )
123         return {Kind::one, {h}};
124     R t = sqrtl(max(R(0), (c.r - d) * (c.r + d)));
125     Point w = v * (t / len);
126     return {Kind::two, {h - w, h + w}};
127 }
128
129 Result Circles(Circle a, Circle b) const
130 {
131     assert(a.r >= 0 && b.r >= 0);
132     Point v = b.o - a.o;
133     R d = norm(v);
134     R tol = eps * max({d, a.r, b.r});
135     // Infinite intersections require identical positive-radius circles.
136     if ( d == 0 )
137     {
138         if ( a.r != b.r )
139             return {Kind::none, {}};
140         if ( a.r == 0 )
141             return {Kind::one, {a.o}};
142         return {Kind::infinite, {}};
143     }

```

```

144     if ( a.r == 0 && b.r == 0 )
145         return {Kind::none, {}};
146     if ( a.r == 0 || b.r == 0 )
147     {
148         R radius = max(a.r, b.r);
149         if ( fabsl(d - radius) > tol )
150             return {Kind::none, {}};
151         return {Kind::one, {a.r == 0 ? a.o : b.o}};
152     }
153     if ( d > a.r + b.r + tol || d < fabsl(a.r - b.r) - tol )
154         return {Kind::none, {}};
155     R x = (d * d + (a.r - b.r) * (a.r + b.r)) / (2 * d);
156     Point h = a.o + v * (x / d);
157     // Near-concentric circles must not be mistaken for internal tangency.
158     if ( fabsl(x) > a.r + tol )
159         return {Kind::none, {}};
160     if ( fabsl(fabsl(x) - a.r) <= tol )
161         return {Kind::one, {h}};
162     R y = sqrtl(max(R(0), (a.r - x) * (a.r + x)));
163     Point w = perp(v) * (y / d);
164     return {Kind::two, {h - w, h + w}};
165 }
166
167 R Overlap(Circle a, Circle b) const
168 {
169     assert(a.r >= 0 && b.r >= 0);
170     R d = norm(a.o - b.o), pi = acosl(-1.L);
171     if ( d >= a.r + b.r )
172         return 0;
173     if ( d <= fabsl(a.r - b.r) )
174         return pi * min(a.r, b.r) * min(a.r, b.r);
175     R difference = (a.r - b.r) * (a.r + b.r);
176     R x = acosl(clamp((d * d + difference) / (2 * d * a.r), R(-1), R(1)));
177     R y = acosl(clamp((d * d - difference) / (2 * d * b.r), R(-1), R(1)));
178     auto segment = [] (R angle)
179     {
180         if ( angle < 1e-3L )
181         {
182             R z = angle * angle;
183             return angle * z * (R(2) / 3 + z * (-R(2) / 15 + z * R(4) / 315));
184         }
185         return angle - sinl(2 * angle) / 2;
186     };
187     return a.r * a.r * segment(x) + b.r * b.r * segment(y);
188 }
189 };

```

第一百章 优化与可持久化

100.1 Max-plus 矩阵幂

算法约束见动态版本第 128 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Max_Plus_Matrix
2 {
3     using I = __int128_t;
4     static constexpr I neg = -(I(1) << 120);
5     int n;
6     vector<vector<I>> a;
7
8     Max_Plus_Matrix(int n, bool identity = false) : n(n), a(n, vector<I>(n, neg))
9     {
10         if ( identity )
11             for ( int i = 0; i < n; i++ )
12                 a[i][i] = 0;
13     }
14
15     Max_Plus_Matrix operator*(const Max_Plus_Matrix &b) const
16     {
17         assert(n == b.n);
18         Max_Plus_Matrix c(n);
19         for ( int i = 0; i < n; i++ )
20             for ( int k = 0; k < n; k++ )
21                 if ( a[i][k] != neg )
22                     for ( int j = 0; j < n; j++ )
23                         if ( b.a[k][j] != neg )
24                             c.a[i][j] = max(c.a[i][j], a[i][k] + b.a[k][j]);
25         return c;
26     }
27
28     Max_Plus_Matrix Power(unsigned long long e) const
29     {
30         Max_Plus_Matrix b = *this, r(n, true);
31         for ( ; e >>= 1, b = b * b )
32             if ( e & 1 )
33                 r = r * b;
34         return r;
35     }
36 };
```

100.2 离散李超树与最优直线编号

算法约束见动态版本第 128 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Li_Chao_Tree
2 {
3     using ll = long long;
4     using I = __int128_t;
5 }
```



```

6   struct Line
7   {
8       ll k = 0, b = 0;
9       int id = -1;
10
11       I value(ll x) const
12       {
13           return I(k) * x + b;
14       }
15   };
16
17   struct Node
18   {
19       Line line;
20       int left = -1, right = -1;
21   };
22
23   vector<ll> xs;
24   vector<Node> a{Node{}};
25
26   Li_Chao_Tree(vector<ll> x) : xs(x)
27   {
28       sort(xs.begin(), xs.end());
29       xs.erase(unique(xs.begin(), xs.end()), xs.end());
30       assert(!xs.empty());
31   }
32
33   bool better(Line u, Line v, ll x) const
34   {
35       return v.id == -1 ||
36              (u.id != -1 && pair{u.value(x), u.id} < pair{v.value(x), v.id});
37   }
38
39   void Insert(Line line)
40   {
41       assert(line.id >= 0);
42       Insert(0, 0, (int)xs.size() - 1, line);
43   }
44
45   void Insert(int p, int l, int r, Line line)
46   {
47       int m = (l + r) / 2;
48       if ( better(line, a[p].line, xs[m]) )
49           swap(line, a[p].line);
50       if ( l == r || line.id == -1 )
51           return;
52       bool left = better(line, a[p].line, xs[l]);
53       bool right = better(line, a[p].line, xs[r]);
54       if ( !left && !right )
55           return;
56       int next = left ? a[p].left : a[p].right;
57       if ( next == -1 )
58       {
59           next = (int)a.size();
60           a.push_back(Node{});
61           if ( left )

```

```

62         a[p].left = next;
63     else
64         a[p].right = next;
65 }
66 if ( left )
67     Insert(next, l, m, line);
68 else
69     Insert(next, m + 1, r, line);
70 }
71
72 pair<I, int> Query(ll x) const
73 {
74     int idx = lower_bound(xs.begin(), xs.end(), x) - xs.begin();
75     assert(idx < (int)xs.size() && xs[idx] == x);
76     int p = 0, l = 0, r = (int)xs.size() - 1;
77     Line best;
78     while ( p != -1 )
79     {
80         if ( better(a[p].line, best, x) )
81             best = a[p].line;
82         if ( l == r )
83             break;
84         int m = (l + r) / 2;
85         if ( idx <= m )
86         {
87             p = a[p].left;
88             r = m;
89         }
90         else
91         {
92             p = a[p].right;
93             l = m + 1;
94         }
95     }
96     assert(best.id != -1);
97     return {best.value(x), best.id};
98 }
99 };

```

100.3 主席树：静态区间第 k 小

算法约束见动态版本第 130 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Persistent_Kth
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     vector<Node> t{Node{}};
9     vector<int> root{0};
10    vector<long long> vals;
11
12    Persistent_Kth(const vector<long long> &a) : vals(a)

```

```

13 {
14     sort(vals.begin(), vals.end());
15     vals.erase(unique(vals.begin(), vals.end()), vals.end());
16     for ( long long x : a )
17     {
18         int p = lower_bound(vals.begin(), vals.end(), x) - vals.begin();
19         root.push_back(Insert(root.back(), 0, (int)vals.size() - 1, p));
20     }
21 }
22
23 int Insert(int old, int l, int r, int x)
24 {
25     int p = t.size();
26     Node copy = t[old];
27     t.push_back(copy);
28     ++t[p].sum;
29     if ( l < r )
30     {
31         int m = (l + r) / 2;
32         if ( x <= m )
33         {
34             int v = Insert(t[old].l, l, m, x);
35             t[p].l = v;
36         }
37         else
38         {
39             int v = Insert(t[old].r, m + 1, r, x);
40             t[p].r = v;
41         }
42     }
43     return p;
44 }
45
46 long long Kth(int l, int r, int k) const
47 {
48     assert(1 <= l && l <= r && r < (int)root.size() && 1 <= k && k <= r - l + 1);
49     int u = root[l - 1], v = root[r], L = 0, R = (int)vals.size() - 1;
50     while ( L < R )
51     {
52         int m = (L + R) / 2, c = t[t[v].l].sum - t[t[u].l].sum;
53         if ( k <= c )
54         {
55             u = t[u].l;
56             v = t[v].l;
57             R = m;
58         }
59         else

```

```
60         {
61             k -= c;
62             u = t[u].r;
63             v = t[v].r;
64             L = m + 1;
65         }
66     }
67     return vals[L];
68 }
69 };
```

第一百零一章 可持久化数组

101.1 可持久化数组：历史版本分叉与单点修改

算法约束见动态版本第 133 页。本节代码独立可抄；采用下列实际接口名。

声明 Persistent_Array<NODES>, Init(a) 初始化；NODES 限制节点总数，不是数组长度。建议按 $2n-1$ 加上修改数乘以 $(\text{ceil}(\log_2(n))+1)$ 分配；Copy 不分配节点。Set、Copy、Query 对应动态接口。大对象放在全局或 static，Init 会丢弃全部旧版本。

```
1 template <int NODES> struct Persistent_Array
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0;
8         ll value = 0;
9     };
10
11     int n;
12     Node t[NODES + 1];
13     int tot = 0;
14     vector<int> root;
15
16     void Init(const vector<ll> &a)
17     {
18         n = a.size();
19         assert(n > 0);
20         tot = 0;
21         root.clear();
22         root.push_back(Build(1, n, a));
23     }
24
25     int Build(int l, int r, const vector<ll> &a)
26     {
27         assert(tot < NODES);
28         int p = ++tot;
29         t[p] = Node{};
30         if ( l == r )
31             t[p].value = a[l - 1];
32         else
33         {
34             int m = (l + r) / 2;
35             int left = Build(l, m, a);
36             int right = Build(m + 1, r, a);
37             t[p].l = left;
38             t[p].r = right;
39         }
40         return p;
41     }
42
43     int Modify(int old, int l, int r, int pos, ll value)
```

```

44     {
45         assert(tot < NODES);
46         int p = ++tot;
47         t[p] = t[old];
48         if ( l == r )
49             t[p].value = value;
50         else
51         {
52             int m = (l + r) / 2;
53             if ( pos <= m )
54             {
55                 int child = Modify(t[old].l, l, m, pos, value);
56                 t[p].l = child;
57             }
58             else
59             {
60                 int child = Modify(t[old].r, m + 1, r, pos, value);
61                 t[p].r = child;
62             }
63         }
64         return p;
65     }
66
67     // Create a new version from any existing version. Positions are 1-based.
68     int Set(int version, int pos, ll value)
69     {
70         assert(0 <= version && version < (int)root.size());
71         assert(1 <= pos && pos <= n);
72         int p = Modify(root[version], 1, n, pos, value);
73         root.push_back(p);
74         return (int)root.size() - 1;
75     }
76
77     int Copy(int version)
78     {
79         assert(0 <= version && version < (int)root.size());
80         root.push_back(root[version]);
81         return (int)root.size() - 1;
82     }
83
84     ll Query(int version, int pos) const
85     {
86         assert(0 <= version && version < (int)root.size());
87         assert(1 <= pos && pos <= n);
88         int p = root[version], l = 1, r = n;
89         while ( l < r )
90         {
91             int m = (l + r) / 2;
92             if ( pos <= m )
93             {
94                 p = t[p].l;
95                 r = m;

```

```
96         }
97         else
98         {
99             p = t[p].r;
100             l = m + 1;
101         }
102     }
103     return t[p].value;
104 }
105 };
```

第一百零二章 树上路径第 k 小

102.1 树上路径点权第 k 小

算法约束见动态版本第 136 页。本节代码独立可抄；采用下列实际接口名。

声明 `Tree_Path_Kth<N,NODES,LOG>`, `Init(a)` 清空图并保存权值; `Insert`、`Build`、`Kth`、`Lca` 对应动态接口。

要求 2 的 LOG 次方大于实际点数，默认 LOG=20。若不同权值数为 D，NODES 至少取 n 乘以 $(\text{ceil}(\log_2(D))+1)$ ，另有编号 0 空节点。大对象放在全局或 static；`Init` 与 `Build` 分别清空图和重建版本。

```
1 template <int N, int NODES, int LOG = 20> struct Tree_Path_Kth
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, sum = 0;
8     };
9
10    static_assert(1 <= LOG && LOG <= 31);
11    int n, tot;
12    vector<int> g[N + 1];
13    int up[LOG][N + 1], root[N + 1], dep[N + 1];
14    ll value[N + 1];
15    vector<ll> vals;
16    Node t[NODES + 1];
17
18    void Init(const vector<ll> &a)
19    {
20        n = a.size();
21        assert(0 < n && n <= N && (1LL << LOG) > n);
22        for (int u = 1; u <= n; u++)
23        {
24            value[u] = a[u - 1];
25            g[u].clear();
26        }
27        root[0] = 0;
28        for (int j = 0; j < LOG; j++)
29            up[j][0] = 0;
30    }
31
32    void Insert(int u, int v)
33    {
34        g[u].push_back(v);
35        g[v].push_back(u);
36    }
37
38    int Update(int old, int l, int r, int x)
39    {
40        assert(tot < NODES);
41        int p = ++tot;
42        t[p] = t[old];
43        t[p].sum++;
```



```

44     if ( l == r )
45         return p;
46     int m = (l + r) / 2;
47     if ( x <= m )
48     {
49         int child = Update(t[old].l, l, m, x);
50         t[p].l = child;
51     }
52     else
53     {
54         int child = Update(t[old].r, m + 1, r, x);
55         t[p].r = child;
56     }
57     return p;
58 }
59
60 void Dfs(int u, int p)
61 {
62     up[0][u] = p;
63     for ( int j = 1; j < LOG; j++ )
64         up[j][u] = up[j - 1][up[j - 1][u]];
65     int x = lower_bound(vals.begin(), vals.end(), value[u]) - vals.begin();
66     root[u] = Update(root[p], 0, (int)vals.size() - 1, x);
67     for ( int v : g[u] )
68     {
69         if ( v == p )
70             continue;
71         dep[v] = dep[u] + 1;
72         Dfs(v, u);
73     }
74 }
75
76 void Build(int r = 1)
77 {
78     assert(1 <= r && r <= n);
79     vals.assign(value + 1, value + n + 1);
80     sort(vals.begin(), vals.end());
81     vals.erase(unique(vals.begin(), vals.end()), vals.end());
82     tot = 0;
83     t[0] = Node{};
84     dep[r] = 0;
85     Dfs(r, 0);
86 }
87
88 int Lca(int u, int v) const
89 {
90     if ( dep[u] < dep[v] )
91         swap(u, v);
92     int diff = dep[u] - dep[v];
93     for ( int j = 0; j < LOG; j++ )
94         if ( diff >> j & 1 )
95             u = up[j][u];
96     if ( u == v )
97         return u;
98     for ( int j = LOG - 1; j >= 0; j-- )
99         if ( up[j][u] != up[j][v] )

```

```

100         {
101             u = up[j][u];
102             v = up[j][v];
103         }
104         return up[0][u];
105     }
106
107 ll Kth(int u, int v, int k) const
108 {
109     int z = Lca(u, v);
110     assert(1 <= k && k <= dep[u] + dep[v] - 2 * dep[z] + 1);
111     int a = root[u], b = root[v], c = root[z], d = root[up[0][z]];
112     int l = 0, r = (int)vals.size() - 1;
113     while ( l < r )
114     {
115         int count = t[t[a].l].sum + t[t[b].l].sum - t[t[c].l].sum - t[t[d].l].sum;
116         int m = (l + r) / 2;
117         if ( k <= count )
118         {
119             a = t[a].l;
120             b = t[b].l;
121             c = t[c].l;
122             d = t[d].l;
123             r = m;
124         }
125         else
126         {
127             k -= count;
128             a = t[a].r;
129             b = t[b].r;
130             c = t[c].r;
131             d = t[d].r;
132             l = m + 1;
133         }
134     }
135     return vals[l];
136 }
137 };

```

第一百零三章 动态区间顺序统计

103.1 动态区间第 k 小 (树状数组套权值线段树)

算法约束见动态版本第 139 页。本节代码独立可抄；采用下列实际接口名。

声明 `Dynamic_Kth<N,NODES>` 后 `Init(a,future)`, `Set` 与 `Kth` 为公开操作。N 限制数组长度, NODES 限制分配节点的最大编号, 另有编号 0 空节点。按照上面的 B 分配一定足够; 例如 n 不超过 100000 且 D 不超过 200000 时, B 不超过 16682000。节点回收使此界与修改次数无关。大对象放在全局或 `static`, `Init` 清空旧状态。

```
1 template <int N, int NODES> struct Dynamic_Kth
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, sum = 0;
8     };
9
10    int n, tot, free_head;
11    int bit[N + 1], rank[N + 1];
12    vector<ll> vals;
13    Node t[NODES + 1];
14
15    void Init(const vector<ll> &a, const vector<ll> &future = {})
16    {
17        n = a.size();
18        assert(0 < n && n <= N);
19        tot = 0;
20        free_head = 0;
21        t[0] = Node{};
22        fill(bit, bit + n + 1, 0);
23        vals = future;
24        vals.insert(vals.end(), a.begin(), a.end());
25        sort(vals.begin(), vals.end());
26        vals.erase(unique(vals.begin(), vals.end()), vals.end());
27        for (int i = 1; i <= n; i++)
28        {
29            rank[i] = Id(a[i - 1]);
30            Modify(i, rank[i], 1);
31        }
32    }
33
34    int Id(ll value) const
35    {
36        int p = lower_bound(vals.begin(), vals.end(), value) - vals.begin();
37        assert(p < (int)vals.size() && vals[p] == value);
38        return p;
39    }
40
41    int New_Node()
42    {
43        if ( !free_head )
```

```

44     {
45         assert(tot < NODES);
46         int p = ++tot;
47         t[p] = Node{};
48         return p;
49     }
50     int p = free_head;
51     free_head = t[p].l;
52     t[p] = Node{};
53     return p;
54 }
55
56 int Update(int p, int l, int r, int x, int delta)
57 {
58     if ( !p )
59         p = New_Node();
60     t[p].sum += delta;
61     if ( l < r )
62     {
63         int m = (l + r) / 2;
64         if ( x <= m )
65         {
66             int child = Update(t[p].l, l, m, x, delta);
67             t[p].l = child;
68         }
69         else
70         {
71             int child = Update(t[p].r, m + 1, r, x, delta);
72             t[p].r = child;
73         }
74     }
75     if ( !t[p].sum )
76     {
77         t[p].l = free_head;
78         free_head = p;
79         return 0;
80     }
81     return p;
82 }
83
84 void Modify(int pos, int x, int delta)
85 {
86     for ( int i = pos; i <= n; i += i & -i )
87         bit[i] = Update(bit[i], 0, (int)vals.size() - 1, x, delta);
88 }

```

```

90 void Set(int pos, ll value)
91 {
92     assert(1 <= pos && pos <= n);
93     int x = Id(value);
94     if ( rank[pos] == x )
95         return;
96     Modify(pos, rank[pos], -1);
97     Modify(pos, x, 1);
98     rank[pos] = x;
99 }
100
101 ll Kth(int l, int r, int k) const
102 {
103     assert(1 <= l && l <= r && r <= n && 1 <= k && k <= r - l + 1);
104     vector<int> a, b;
105     for ( int i = r; i; i -= i & -i )
106         a.push_back(bit[i]);
107     for ( int i = l - 1; i; i -= i & -i )
108         b.push_back(bit[i]);
109     int L = 0, R = (int)vals.size() - 1;
110     while ( L < R )
111     {
112         int count = 0;
113         for ( int p : a )
114             count += t[t[p].l].sum;
115         for ( int p : b )
116             count -= t[t[p].l].sum;
117         bool left = k <= count;
118         if ( !left )
119             k -= count;
120         for ( int &p : a )
121             p = left ? t[p].l : t[p].r;
122         for ( int &p : b )
123             p = left ? t[p].l : t[p].r;
124         int m = (L + R) / 2;
125         if ( left )
126             R = m;
127         else
128             L = m + 1;
129     }
130     return vals[L];
131 }
132 };

```

第一百零四章 区间不同数统计

104.1 静态区间不同数计数（前驱位置主席树）

算法约束见动态版本第 142 页。本节代码独立可抄；采用下列实际接口名。

声明 Persistent_Distinct<N,NODES> 后 Init(a), Query(l,r) 为查询。N 限制数组长度；NODES 至少取 $n \times (\lceil \log_2(n) \rceil + 1)$ ，另有编号 0 空节点。n 不超过 1000000 时可取 NODES=21000000。大对象放在全局或 static，Init 可重复调用并清空全部旧版本。

```
1 template <int N, int NODES> struct Persistent_Distinct
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     int n, tot;
9     Node t[NODES + 1];
10    int root[N + 1];
11
12    void Init(const vector<long long> &a)
13    {
14        n = a.size();
15        assert(0 < n && n <= N);
16        tot = 0;
17        t[0] = Node{};
18        root[0] = 0;
19        vector<long long> vals = a;
20        sort(vals.begin(), vals.end());
21        vals.erase(unique(vals.begin(), vals.end()), vals.end());
22        vector<int> last(vals.size());
23        for ( int i = 1; i <= n; i++ )
24        {
25            int x = lower_bound(vals.begin(), vals.end(), a[i - 1]) - vals.begin();
26            root[i] = Insert(root[i - 1], 0, n - 1, last[x]);
27            last[x] = i;
28        }
29    }
30
31    int Insert(int old, int l, int r, int x)
32    {
33        assert(tot < NODES);
34        int p = ++tot;
35        t[p] = t[old];
36        t[p].sum++;
37        if ( l < r )
38        {
39            int m = (l + r) / 2;
40            if ( x <= m )
41            {
42                int child = Insert(t[old].l, l, m, x);
43                t[p].l = child;
```

```

44         }
45         else
46         {
47             int child = Insert(t[old].r, m + 1, r, x);
48             t[p].r = child;
49         }
50     }
51     return p;
52 }
53
54 int Query(int l, int r) const
55 {
56     assert(1 <= l && l <= r && r <= n);
57     int u = root[l - 1], v = root[r];
58     int L = 0, R = n - 1, answer = 0;
59     while ( L < R )
60     {
61         int m = (L + R) / 2;
62         if ( l - 1 <= m )
63         {
64             u = t[u].l;
65             v = t[v].l;
66             R = m;
67         }
68         else
69         {
70             answer += t[t[v].l].sum - t[t[u].l].sum;
71             u = t[u].r;
72             v = t[v].r;
73             L = m + 1;
74         }
75     }
76     return answer + t[v].sum - t[u].sum;
77 }
78 };

```

第一百零五章 图论进阶

105.1 矩形二分图最小权匹配

算法约束见动态版本第 144 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Kuhn_Munkres
2 {
3     using I = __int128_t;
4
5     // Minimum cost injection from n rows to m columns, n<=m. Zero-based matrix.
6     static pair<I, vector<int>> Solve(const vector<vector<long long>> &cost)
7     {
8         int n = cost.size();
9         if ( !n )
10             return {0, {}};
11         int m = cost[0].size();
12         assert(n <= m);
13         vector<I> u(n + 1), v(m + 1);
14         vector<int> p(m + 1), way(m + 1);
15         const I inf = I(1) << 120;
16         for ( int i = 1; i <= n; i++ )
17         {
18             assert((int)cost[i - 1].size() == m);
19             p[0] = i;
20             int j0 = 0;
21             vector<I> dis(m + 1, inf);
22             vector<bool> Used(m + 1);
23             do
24             {
25                 Used[j0] = true;
26                 int i0 = p[j0], j1 = 0;
27                 I delta = inf;
28                 for ( int j = 1; j <= m; j++ )
29                     if ( !Used[j] )
30                     {
31                         I cur = I(cost[i0 - 1][j - 1]) - u[i0] - v[j];
32                         if ( cur < dis[j] )
33                         {
34                             dis[j] = cur;
35                             way[j] = j0;
36                         }
37                         if ( dis[j] < delta )
38                         {
39                             delta = dis[j];
40                             j1 = j;
41                         }
42                     }
43                 for ( int j = 0; j <= m; j++ )
44                     if ( Used[j] )
45                     {
46                         u[p[j]] += delta;
```



```

47         v[j] -= delta;
48     }
49     else
50         dis[j] -= delta;
51     j0 = j1;
52 } while ( p[j0] );
53 do
54 {
55     int j1 = way[j0];
56     p[j0] = p[j1];
57     j0 = j1;
58 } while ( j0 );
59 }
60 vector<int> ans(n);
61 for ( int j = 1; j <= m; j++ )
62     if ( p[j] )
63         ans[p[j] - 1] = j - 1;
64 return {-v[0], ans};
65 }
66 };

```

105.2 有向最小树形图

算法约束见动态版本第 145 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Directed_MST
2 {
3     using I = __int128_t;
4
5     struct Edge
6     {
7         int u, v;
8         I w;
9     };
10
11     // Directed minimum rooted OUT-arborescence cost, 0-based, or nullopt.
12     static optional<I> Solve(int n, int root, vector<Edge> e)
13     {
14         const I inf = I(1) << 120;
15         I ans = 0;
16         while ( true )
17         {
18             vector<I> in(n, inf);
19             vector<int> pre(n, -1), id(n, -1), vis(n, -1);
20             for ( auto a : e )
21                 if ( a.u != a.v && a.w < in[a.v] )
22                 {
23                     in[a.v] = a.w;
24                     pre[a.v] = a.u;
25                 }
26             in[root] = 0;
27             for ( I x : in )
28             {
29                 if ( x == inf )
30                     return nullopt;

```

```

31         ans += x;
32     }
33     int cnt = 0;
34     for ( int i = 0; i < n; i++ )
35     {
36         int u = i;
37         while ( vis[u] != i && id[u] == -1 && u != root )
38         {
39             vis[u] = i;
40             u = pre[u];
41         }
42         if ( u != root && id[u] == -1 )
43         {
44             id[u] = cnt;
45             for ( int v = pre[u]; v != u; v = pre[v] )
46                 id[v] = cnt;
47             ++cnt;
48         }
49     }
50     if ( !cnt )
51         return ans;
52     for ( int i = 0; i < n; i++ )
53         if ( id[i] == -1 )
54             id[i] = cnt++;
55     for ( auto &a : e )
56     {
57         int v = a.v;
58         a.u = id[a.u];
59         a.v = id[a.v];
60         if ( a.u != a.v )
61             a.w -= in[v];
62     }
63     root = id[root];
64     n = cnt;
65 }
66 }
67 };

```

105.3 无向全局最小割

算法约束见动态版本第 147 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Global_Min_Cut
2 {
3     using I = __int128_t;
4
5     // Symmetric nonnegative adjacency matrix; returns weight and one cut side.
6     static pair<I, vector<int>> Solve(vector<vector<I>> w)
7     {
8         int n = w.size();
9         assert(n >= 2);
10        vector<int> active(n);
11        iota(active.begin(), active.end(), 0);
12        vector<vector<int>> group(n);
13        for ( int i = 0; i < n; i++ )
14            group[i] = {i};
15        I best = I(1) << 120;
16        vector<int> side;
17        while ( active.size() > 1 )
18        {
19            vector<I> dis(n);
20            vector<bool> Used(n);
21            int prev = -1;
22            for ( int step = 0; step < (int)active.size(); step++ )
23            {
24                int sel = -1;
25                for ( int v : active )
26                    if ( !Used[v] && (sel == -1 || dis[v] > dis[sel]) )
27                        sel = v;
28                if ( step + 1 == (int)active.size() )
29                {
30                    if ( dis[sel] < best )
31                    {
32                        best = dis[sel];
33                        side = group[sel];
34                    }
35                    for ( int v : active )
36                        if ( v != prev && v != sel )
37                            w[prev][v] = w[v][prev] = w[prev][sel] + w[sel][v];
38                    group[prev].insert(
39                        group[prev].end(), group[sel].begin(), group[sel].end());
40                    active.erase(find(active.begin(), active.end(), sel));
41                    break;
42                }
43                Used[sel] = true;
44                prev = sel;
45                for ( int v : active )
46                    if ( !Used[v] )
47                        dis[v] += w[sel][v];
48            }
49        }
50        return {best, side};
51    }
52 };

```

第一百零六章 带权二分图匹配

106.1 二分图最大权匹配（缺边与可选匹配）

算法约束见动态版本第 149 页。本节代码独立可抄；采用下列实际接口名。

声明 `Weighted_Matching<N,M>` 后 `Init(n,m)`, `N`、`M` 分别限制左右点数。`Insert` 加边, `Solve` 求解; `l`、`r` 保存方案, `Init` 清空原图。权值矩阵是静态数组, 内部增广辅助数组按实际规模分配, 大对象放在全局或 `static`。

```
1 template <int N, int M> struct Weighted_Matching
2 {
3     using I = __int128_t;
4     static constexpr I inf = I(1) << 120;
5     int n, m;
6     I w[N + 1][M + 1];
7     int l[N + 1], r[M + 1];
8
9     void Init(int left_size, int right_size)
10    {
11        assert(0 <= left_size && left_size <= N);
12        assert(0 <= right_size && right_size <= M);
13        n = left_size;
14        m = right_size;
15        for ( int i = 0; i <= n; i++ )
16            fill(w[i], w[i] + m + 1, -inf);
17        fill(l, l + n + 1, 0);
18        fill(r, r + m + 1, 0);
19    }
20
21    void Insert(int x, int y, long long weight)
22    {
23        assert(1 <= x && x <= n && 1 <= y && y <= m);
24        w[x][y] = max(w[x][y], I(weight));
25    }
26
27    optional<I> Solve(bool allow_unmatched = false)
28    {
29        fill(l, l + n + 1, 0);
30        fill(r, r + m + 1, 0);
31        if ( !allow_unmatched && n > m )
32            return nullopt;
33        int cols = m + (allow_unmatched ? n : 0);
34        vector<I> u(n + 1), v(cols + 1);
35        vector<int> p(cols + 1), way(cols + 1);
36        for ( int i = 1; i <= n; i++ )
37        {
38            p[0] = i;
39            int j0 = 0;
40            vector<I> dis(cols + 1, inf);
41            vector<bool> used(cols + 1);
42            do
43            {
44                used[j0] = true;
```

```

45     int x = p[j0], j1 = 0;
46     I delta = inf;
47     for ( int j = 1; j <= cols; j++ )
48     {
49         if ( used[j] )
50             continue;
51         I weight = j <= m ? w[x][j] : 0;
52         if ( weight != -inf )
53         {
54             I cur = -weight - u[x] - v[j];
55             if ( cur < dis[j] )
56             {
57                 dis[j] = cur;
58                 way[j] = j0;
59             }
60         }
61         if ( dis[j] < delta )
62         {
63             delta = dis[j];
64             j1 = j;
65         }
66     }
67     if ( delta == inf )
68         return nullopt;
69     for ( int j = 0; j <= cols; j++ )
70     {
71         if ( used[j] )
72         {
73             u[p[j]] += delta;
74             v[j] -= delta;
75         }
76         else if ( dis[j] != inf )
77             dis[j] -= delta;
78     }
79     j0 = j1;
80     } while ( p[j0] );
81     do
82     {
83         int j1 = way[j0];
84         p[j0] = p[j1];
85         j0 = j1;
86     } while ( j0 );
87 }
88 I answer = 0;
89 for ( int j = 1; j <= m; j++ )
90 {
91     if ( !p[j] )
92         continue;
93     l[p[j]] = j;
94     r[j] = p[j];
95     answer += w[p[j]][j];
96 }
97 return answer;
98 }
99 };

```

第一百零七章 精确几何进阶

107.1 整数最近点对与 Minkowski 和

算法约束见动态版本第 151 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Geometry_Extra
2 {
3     using G = Integer_Geometry;
4     using P = G::Point;
5     using I = G::I;
6
7     // No pair exists for n<2. Exact squared distance, O(n log n).
8     static optional<I> Closest_Pair(vector<P> p)
9     {
10         int n = p.size();
11         if ( n < 2 )
12             return nullopt;
13         sort(p.begin(), p.end());
14         for ( int i = 1; i < n; i++ )
15             if ( p[i] == p[i - 1] )
16                 return I(0);
17         vector<P> tmp(n);
18         auto by_y = [](P a, P b)
19         {
20             return tie(a.y, a.x) < tie(b.y, b.x);
21         };
22         function<I(int, int)> Solve = [&](int l, int r) -> I
23         {
24             if ( r - l <= 3 )
25             {
26                 I best = I(1) << 120;
27                 for ( int i = l; i < r; i++ )
28                     for ( int j = i + 1; j < r; j++ )
29                         best = min(best, G::Distance_Squared(p[i], p[j]));
30                 sort(p.begin() + l, p.begin() + r, by_y);
31                 return best;
32             }
33             int m = (l + r) / 2;
34             long long x = p[m].x;
35             I best = min(Solve(l, m), Solve(m, r));
36             merge(p.begin() + l,
37                 p.begin() + m,
38                 p.begin() + m,
39                 p.begin() + r,
40                 tmp.begin() + l,
41                 by_y);
42             copy(tmp.begin() + l, tmp.begin() + r, p.begin() + l);
43             vector<P> strip;
44             for ( int i = l; i < r; i++ )
45                 if ( (I(p[i].x) - x) * (I(p[i].x) - x) < best )
46                     {
```

```

47         for ( int j = (int)strip.size() - 1; j >= 0; j-- )
48         {
49             I dy = I(p[i].y) - strip[j].y;
50             if ( dy * dy >= best )
51                 break;
52             best = min(best, G::Distance_Squared(p[i], strip[j]));
53         }
54         strip.push_back(p[i]);
55     }
56     return best;
57 };
58 return Solve(0, n);
59 }
60
61 // Inputs strict CCW hulls. Coordinate sums must obey IntegerGeometry's bound.
62 static vector<P> Minkowski(vector<P> a, vector<P> b)
63 {
64     if ( a.empty() || b.empty() )
65         return {};
66     auto plus = [](P u, P v)
67     {
68         return P{u.x + v.x, u.y + v.y};
69     };
70     if ( a.size() < 3 || b.size() < 3 )
71     {
72         vector<P> p;
73         for ( P u : a )
74             for ( P v : b )
75                 p.push_back(plus(u, v));
76         return G::Convex_Hull(p);
77     }
78     auto start = [](vector<P> &p)
79     {
80         auto it = min_element(p.begin(),
81                               p.end(),
82                               [](P u, P v)
83                               {
84                                   return tie(u.y, u.x) < tie(v.y, v.x);
85                               });
86         rotate(p.begin(), it, p.end());
87     };
88     start(a);
89     start(b);
90     int n = a.size(), m = b.size(), i = 0, j = 0;
91     vector<P> result;
92     while ( i < n || j < m )
93     {
94         result.push_back(plus(a[i % n], b[j % m]));
95         if ( i == n )
96         {
97             ++j;
98             continue;
99         }
100         if ( j == m )
101         {
102             ++i;

```

```
103         continue;
104     }
105     P u{a[(i + 1) % n].x - a[i].x, a[(i + 1) % n].y - a[i].y};
106     P v{b[(j + 1) % m].x - b[j].x, b[(j + 1) % m].y - b[j].y};
107     I c = G::Cross({0, 0}, u, v);
108     if ( c >= 0 )
109         ++i;
110     if ( c <= 0 )
111         ++j;
112 }
113 return result;
114 }
115 };
```


107.2 整数三维共线、共面与方向

算法约束见动态版本第 154 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Integer_Geometry_3D
2 {
3     using I = __int128_t;
4
5     // |coordinates|<=1e9: triple products fit signed int128.
6     struct Point
7     {
8         long long x, y, Z_Function;
9     };
10
11     struct Vector
12     {
13         I x, y, Z_Function;
14     };
15
16     static Vector diff(Point a, Point b)
17     {
18         return {I(a.x) - b.x, I(a.y) - b.y, I(a.Z_Function) - b.Z_Function};
19     }
20
21     static Vector Cross(Vector a, Vector b)
22     {
23         return {a.y * b.Z_Function - a.Z_Function * b.y,
24                 a.Z_Function * b.x - a.x * b.Z_Function,
25                 a.x * b.y - a.y * b.x};
26     }
27
28     static I Dot(Vector a, Vector b)
29     {
30         return a.x * b.x + a.y * b.y + a.Z_Function * b.Z_Function;
31     }
32
33     static I orient(Point a, Point b, Point c, Point d)
34     {
35         return Dot(Cross(diff(b, a), diff(c, a)), diff(d, a));
36     }
37
38     static bool collinear(Point a, Point b, Point c)
39     {
40         auto v = Cross(diff(b, a), diff(c, a));
41         return !v.x && !v.y && !v.Z_Function;
42     }
43
44     static bool On_Segment(Point p, Point a, Point b)
45     {
46         return collinear(p, a, b) && Dot(diff(a, p), diff(b, p)) <= 0;
47     }
48 };

```

第一百零八章 凸包支撑点查询

108.1 支撑点查询恢复下凸包

算法约束见动态版本第 155 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Support_Hull
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     struct Point
7     {
8         ll x, y;
9     };
10
11     static I Cross(Point a, Point b, Point c)
12     {
13         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
14     }
```

```

16 // oracle(a,b) maximizes a*x+b*y exactly; |coordinates| <= 1e18.
17 // left/right: lowest points at the minimum/maximum x.
18 template <class Oracle>
19 static vector<Point> Build(Point left, Point right, Oracle &&oracle)
20 {
21     assert(left.x <= right.x);
22     vector<Point> hull{left};
23     auto dfs = [&](auto &&self, Point a, Point b) -> void
24     {
25         Point c = oracle(b.y - a.y, a.x - b.x);
26         I turn = Cross(a, b, c);
27         assert(turn <= 0);
28         if ( turn < 0 )
29         {
30             assert(a.x < c.x && c.x < b.x);
31             self(self, a, c);
32             self(self, c, b);
33         }
34         else
35         {
36             while ( hull.size() >= 2 &&
37                     Cross(hull[hull.size() - 2], hull.back(), b) == 0 )
38                 hull.pop_back();
39             hull.push_back(b);
40         }
41     };
42     if ( left.x < right.x )
43         dfs(dfs, left, right);
44     else
45         assert(left.y == right.y);
46     return hull;
47 }
48 };

```

第一百零九章 动态树

109.1 Link-cut tree: 动态森林与路径异或

算法约束见动态版本第 157 页。本节代码独立可抄；采用下列实际接口名。

声明 `Link_Cut_Tree<N>`，然后 `Init(n)` 初始化；`N` 为最大点号。`Set`、`Link`、`Cut`、`Connected`、`Query` 分别对应动态版的小写接口。大对象放在全局或 `static`。

```
1 template <int N> struct Link_Cut_Tree
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int ch[2]{}, fa = 0;
8         U val = 0, sum = 0;
9         bool rev = false;
10    };
11
12    Node tree[N + 1];
13    int st[N + 1];
14
15    void Init(int n)
16    {
17        assert(0 <= n && n <= N);
18        fill(tree, tree + n + 1, Node{});
19    }
20
21    bool Is_Root(int x) const
22    {
23        int f = tree[x].fa;
24        return tree[f].ch[0] != x && tree[f].ch[1] != x;
25    }
26
27    void Push_Up(int x)
28    {
29        tree[x].sum = tree[tree[x].ch[0]].sum ^ tree[x].val ^ tree[tree[x].ch[1]].sum;
30    }
31
32    void Reverse(int x)
33    {
34        if ( !x )
35            return;
36        swap(tree[x].ch[0], tree[x].ch[1]);
37        tree[x].rev = !tree[x].rev;
38    }
39
40    void Push_Down(int x)
41    {
42        if ( !tree[x].rev )
43            return;
44        Reverse(tree[x].ch[0]);
```

```

45     Reverse(tree[x].ch[1]);
46     tree[x].rev = false;
47 }
48
49 void Rotate(int x)
50 {
51     int y = tree[x].fa, z = tree[y].fa;
52     int k = tree[y].ch[1] == x;
53     int w = tree[x].ch[k ^ 1];
54     if ( !Is_Root(y) )
55         tree[z].ch[tree[z].ch[1] == y] = x;
56     tree[x].fa = z;
57     tree[y].ch[k] = w;
58     if ( w )
59         tree[w].fa = y;
60     tree[x].ch[k ^ 1] = y;
61     tree[y].fa = x;
62     Push_Up(y);
63     Push_Up(x);
64 }
65
66 void Splay(int x)
67 {
68     int top = 0, y = x;
69     st[top++] = y;
70     while ( !Is_Root(y) )
71     {
72         y = tree[y].fa;
73         st[top++] = y;
74     }
75     while ( top )
76         Push_Down(st[--top]);
77     while ( !Is_Root(x) )
78     {
79         int y = tree[x].fa, z = tree[y].fa;
80         if ( !Is_Root(y) )
81         {
82             bool same = (tree[y].ch[1] == x) == (tree[z].ch[1] == y);
83             Rotate(same ? y : x);
84         }
85         Rotate(x);
86     }
87 }
88
89 void Access(int x)
90 {
91     int last = 0;
92     for ( int y = x; y; y = tree[y].fa )
93     {
94         Splay(y);
95         // The old right child's fa remains y: it becomes a path-parent.
96         tree[y].ch[1] = last;
97         if ( last )
98             tree[last].fa = y;
99         Push_Up(y);
100        last = y;

```

```

101     }
102     Splay(x);
103 }
104
105 void Make_Root(int x)
106 {
107     Access(x);
108     Reverse(x);
109 }
110
111 int Find_Root(int x)
112 {
113     Access(x);
114     Push_Down(x);
115     while ( tree[x].ch[0] )
116     {
117         x = tree[x].ch[0];
118         Push_Down(x);
119     }
120     Splay(x);
121     return x;
122 }
123
124 bool Connected(int x, int y)
125 {
126     return x == y || Find_Root(x) == Find_Root(y);
127 }
128
129 bool Link(int x, int y)
130 {
131     Make_Root(x);
132     if ( Find_Root(y) == x )
133         return false;
134     tree[x].fa = y;
135     return true;
136 }
137
138 bool Cut(int x, int y)
139 {
140     Make_Root(x);
141     Access(y);
142     if ( tree[y].ch[0] != x || tree[x].ch[1] )
143         return false;
144     tree[y].ch[0] = 0;
145     tree[x].fa = 0;
146     Push_Up(y);
147     return true;
148 }
149
150 void Set(int x, U value)
151 {
152     Access(x);
153     tree[x].val = value;
154     Push_Up(x);

```

```
155     }
156
157     // x and y must be connected. This is a path aggregate, not a subtree sum.
158     U Query(int x, int y)
159     {
160         Make_Root(x);
161         Access(y);
162         return tree[y].sum;
163     }
164 };
```

第一百一十章 随机平衡树

110.1 FHQ Treap: 可重集合与顺序统计

算法约束见动态版本第 161 页。本节代码独立可抄；采用下列实际接口名。

声明容量模板参数 N，然后 Init() 或 Init(seed)。N 限制累计插入次数，不是当前元素个数；大对象放在全局或 static。接口采用 Insert、Erase、Kth/Query 等命名。

```
1 template <int N> struct Ordered_Treap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0;
9         unsigned long long pri = 0;
10    };
11
12    Node tree[N + 1];
13    int tot = 0;
14    mt19937_64 rng;
15    int root = 0;
16
17    void Init(unsigned long long seed = 712367821)
18    {
19        root = 0;
20        tot = 0;
21        tree[0] = Node{};
22        rng.seed(seed);
23    }
24
25    void Push_Up(int p)
26    {
27        tree[p].siz = tree[tree[p].l].siz + tree[tree[p].r].siz + 1;
28    }
29
30    int New_Node(ll x)
31    {
32        assert(tot < N);
33        int p = ++tot;
34        tree[p] = {0, 0, 1, x, rng()};
35        return p;
36    }
37
38    // left: val < x, or val <= x when equal is true. No x-1 overflow.
39    void Split(int p, ll x, int &l, int &r, bool equal = false)
40    {
41        if ( !p )
42        {
43            l = 0;
44            r = 0;
```



```

45         return;
46     }
47     if ( tree[p].val < x || (equal && tree[p].val == x) )
48     {
49         l = p;
50         Split(tree[p].r, x, tree[p].r, r, equal);
51     }
52     else
53     {
54         r = p;
55         Split(tree[p].l, x, l, tree[p].l, equal);
56     }
57     Push_Up(p);
58 }
59
60 int Merge(int l, int r)
61 {
62     if ( !l || !r )
63         return l ? l : r;
64     if ( tree[l].pri > tree[r].pri )
65     {
66         tree[l].r = Merge(tree[l].r, r);
67         Push_Up(l);
68         return l;
69     }
70     tree[r].l = Merge(l, tree[r].l);
71     Push_Up(r);
72     return r;
73 }
74
75 void Insert(ll x)
76 {
77     int p = New_Node(x), l, r;
78     Split(root, x, l, r);
79     root = Merge(Merge(l, p), r);
80 }
81
82 bool Erase(ll x)
83 {
84     int l, mid, r;
85     Split(root, x, l, mid);
86     Split(mid, x, mid, r, true);
87     bool found = mid != 0;
88     if ( mid )
89         mid = Merge(tree[mid].l, tree[mid].r);
90     root = Merge(l, Merge(mid, r));
91     return found;
92 }
93
94 int Size() const
95 {
96     return tree[root].siz;
97 }
98
99 int Count_Less(ll x, bool equal = false) const
100 {

```

```

101     int p = root, ans = 0;
102     while ( p )
103     {
104         if ( tree[p].val < x || (equal && tree[p].val == x) )
105         {
106             ans += tree[tree[p].l].siz + 1;
107             p = tree[p].r;
108         }
109         else
110             p = tree[p].l;
111     }
112     return ans;
113 }
114
115 int Rank(ll x) const
116 {
117     return Count_Less(x) + 1;
118 }
119
120 ll Kth(int k) const
121 {
122     assert(1 <= k && k <= Size());
123     int p = root;
124     while ( true )
125     {
126         int left = tree[tree[p].l].siz;
127         if ( k == left + 1 )
128             return tree[p].val;
129         if ( k <= left )
130             p = tree[p].l;
131         else
132         {
133             k -= left + 1;
134             p = tree[p].r;
135         }
136     }
137 }
138
139 optional<ll> Predecessor(ll x) const
140 {
141     int k = Count_Less(x);
142     if ( !k )
143         return nullopt;
144     return Kth(k);
145 }
146
147 optional<ll> Successor(ll x) const
148 {
149     int k = Count_Less(x, true);
150     if ( k == Size() )
151         return nullopt;
152     return Kth(k + 1);
153 }
154 };

```

110.2 隐式 Treap：序列翻转、加值与区间和

算法约束见动态版本第 163 页。本节代码独立可抄；采用下列实际接口名。

声明容量模板参数 N，然后 Init() 或 Init(seed)。N 限制累计插入次数，不是当前元素个数；大对象放在全局或 static。接口采用 Insert、Erase、Kth/Query 等命名。

```

1 template <int N> struct Sequence_Treap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0, sum = 0, tag = 0;
9         unsigned long long pri = 0;
10        bool rev = false;
11    };
12
13    Node tree[N + 1];
14    int tot = 0;
15    mt19937_64 rng;
16    int root = 0;
17
18    void Init(unsigned long long seed = 712367821)
19    {
20        root = 0;
21        tot = 0;
22        tree[0] = Node{};
23        rng.seed(seed);
24    }
25
26    int Size() const
27    {
28        return tree[root].siz;
29    }
30
31    void Push_Up(int p)
32    {
33        tree[p].siz = tree[tree[p].l].siz + tree[tree[p].r].siz + 1;
34        tree[p].sum = tree[tree[p].l].sum + tree[p].val + tree[tree[p].r].sum;
35    }
36
37    void Apply_Add(int p, ll x)
38    {
39        if ( !p )
40            return;
41        tree[p].val += x;
42        tree[p].sum += x * tree[p].siz;
43        tree[p].tag += x;
44    }
45
46    void Apply_Reverse(int p)
47    {
48        if ( !p )
49            return;
50        swap(tree[p].l, tree[p].r);

```

```

51     tree[p].rev = !tree[p].rev;
52 }
53
54 void Push_Down(int p)
55 {
56     if ( tree[p].rev )
57     {
58         Apply_Reverse(tree[p].l);
59         Apply_Reverse(tree[p].r);
60         tree[p].rev = false;
61     }
62     if ( tree[p].tag )
63     {
64         Apply_Add(tree[p].l, tree[p].tag);
65         Apply_Add(tree[p].r, tree[p].tag);
66         tree[p].tag = 0;
67     }
68 }
69
70 // Split into the first k elements and the rest.
71 void Split(int p, int k, int &l, int &r)
72 {
73     if ( !p )
74     {
75         l = 0;
76         r = 0;
77         return;
78     }
79     Push_Down(p);
80     int left = tree[tree[p].l].siz;
81     if ( k > left )
82     {
83         l = p;
84         Split(tree[p].r, k - left - 1, tree[p].r, r);
85     }
86     else
87     {
88         r = p;
89         Split(tree[p].l, k, l, tree[p].l);
90     }
91     Push_Up(p);
92 }
93
94 int Merge(int l, int r)
95 {
96     if ( !l || !r )
97         return l ? l : r;
98     if ( tree[l].pri > tree[r].pri )
99     {
100         Push_Down(l);
101         tree[l].r = Merge(tree[l].r, r);
102         Push_Up(l);
103         return l;
104     }
105     Push_Down(r);
106     tree[r].l = Merge(l, tree[r].l);

```

```

107     Push_Up(r);
108     return r;
109 }
110
111 // Insert after the first k elements, 0 <= k <= size().
112 void Insert(int k, ll value)
113 {
114     assert(0 <= k && k <= Size());
115     assert(tot < N);
116     int p = ++tot, l, r;
117     tree[p] = {0, 0, 1, value, value, 0, rng(), false};
118     Split(root, k, l, r);
119     root = Merge(Merge(l, p), r);
120 }
121
122 // All range operations use 1-based inclusive intervals.
123 void Erase(int l, int r)
124 {
125     assert(1 <= l && l <= r && r <= Size());
126     int x, y, z;
127     Split(root, r, y, z);
128     Split(y, l - 1, x, y);
129     root = Merge(x, z);
130 }
131
132 void Reverse(int l, int r)
133 {
134     assert(1 <= l && l <= r && r <= Size());
135     int x, y, z;
136     Split(root, r, y, z);
137     Split(y, l - 1, x, y);
138     Apply_Reverse(y);
139     root = Merge(Merge(x, y), z);
140 }
141
142 void Add(int l, int r, ll value)
143 {
144     assert(1 <= l && l <= r && r <= Size());
145     int x, y, z;
146     Split(root, r, y, z);
147     Split(y, l - 1, x, y);
148     Apply_Add(y, value);
149     root = Merge(Merge(x, y), z);
150 }
151
152 ll Query(int l, int r)
153 {
154     assert(1 <= l && l <= r && r <= Size());
155     int x, y, z;
156     Split(root, r, y, z);
157     Split(y, l - 1, x, y);
158     ll ans = tree[y].sum;
159     root = Merge(Merge(x, y), z);
160     return ans;
161 }
162

```

```
163 void Collect(int p, vector<ll> &out)
164 {
165     if ( !p )
166         return;
167     Push_Down(p);
168     Collect(tree[p].l, out);
169     out.push_back(tree[p].val);
170     Collect(tree[p].r, out);
171 }
172
173 vector<ll> Values()
174 {
175     vector<ll> answer;
176     Collect(root, answer);
177     return answer;
178 }
179 };
```

第一百一十一章 伸展树

111.1 Splay：可重集合与顺序统计

算法约束见动态版本第 168 页。本节代码独立可抄；采用下列实际接口名。

声明 `Ordered_Splay<N>` 后调用 `Init()`；`N` 为累计新建节点上限。接口采用 `Insert`、`Erase`、`Rank`、`Kth`、`Prev`、`Next`。大对象放在全局或 `static`。

```
1 template <int N> struct Ordered_Splay
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int ch[2]{}, fa = 0, cnt = 0, siz = 0;
8         ll val = 0;
9     };
10
11     Node tree[N + 1];
12     int tot = 0;
13     int root = 0;
14
15     void Init()
16     {
17         root = 0;
18         tot = 0;
19         tree[0] = Node{};
20     }
21
22     int Size() const
23     {
24         return tree[root].siz;
25     }
26
27     void Push_Up(int x)
28     {
29         tree[x].siz = tree[tree[x].ch[0]].siz + tree[x].cnt + tree[tree[x].ch[1]].siz;
30     }
31
32     int New_Node(ll value, int parent)
33     {
34         assert(tot < N);
35         int x = ++tot;
36         tree[x] = {0, 0}, parent, 1, 1, value;
37         return x;
38     }
39
40     void Rotate(int x)
41     {
42         int y = tree[x].fa, z = tree[y].fa;
43         int k = tree[y].ch[1] == x;
44         int w = tree[x].ch[k ^ 1];
```

```

45     if ( z )
46         tree[z].ch[tree[z].ch[1] == y] = x;
47     else
48         root = x;
49     tree[x].fa = z;
50     tree[y].ch[k] = w;
51     if ( w )
52         tree[w].fa = y;
53     tree[x].ch[k ^ 1] = y;
54     tree[y].fa = x;
55     Push_Up(y);
56     Push_Up(x);
57 }
58
59 void Splay(int x)
60 {
61     while ( tree[x].fa )
62     {
63         int y = tree[x].fa, z = tree[y].fa;
64         if ( z )
65         {
66             bool same = (tree[y].ch[1] == x) == (tree[z].ch[1] == y);
67             Rotate(same ? y : x);
68         }
69         Rotate(x);
70     }
71 }
72
73 // Splay the last visited node even when the value does not exist.
74 int Find(ll value)
75 {
76     int x = root, last = 0;
77     while ( x && tree[x].val != value )
78     {
79         last = x;
80         x = tree[x].ch[value > tree[x].val];
81     }
82     if ( x )
83         Splay(x);
84     else if ( last )
85         Splay(last);
86     return x;
87 }
88
89 void Insert(ll value)
90 {
91     int x = root, parent = 0;
92     while ( x && tree[x].val != value )
93     {
94         parent = x;
95         x = tree[x].ch[value > tree[x].val];
96     }
97     if ( x )
98     {
99         ++tree[x].cnt;
100         Push_Up(x);

```



```

101     }
102     else
103     {
104         x = New_Node(value, parent);
105         if ( parent )
106             tree[parent].ch[value > tree[parent].val] = x;
107         else
108             root = x;
109     }
110     Splay(x);
111 }
112
113 bool Erase(ll value)
114 {
115     int x = Find(value);
116     if ( !x )
117         return false;
118     if ( tree[x].cnt > 1 )
119     {
120         --tree[x].cnt;
121         Push_Up(x);
122         return true;
123     }
124     int l = tree[x].ch[0], r = tree[x].ch[1];
125     if ( l )
126         tree[l].fa = 0;
127     if ( r )
128         tree[r].fa = 0;
129     if ( !l )
130     {
131         root = r;
132         return true;
133     }
134     root = l;
135     x = l;
136     while ( tree[x].ch[1] )
137         x = tree[x].ch[1];
138     Splay(x);
139     tree[x].ch[1] = r;
140     if ( r )
141         tree[r].fa = x;
142     Push_Up(x);
143     return true;
144 }
145
146 int Rank(ll value)
147 {
148     int x = root, last = 0, ans = 1;
149     while ( x )
150     {
151         last = x;
152         if ( value <= tree[x].val )
153             x = tree[x].ch[0];
154         else
155         {
156             ans += tree[tree[x].ch[0]].siz + tree[x].cnt;

```

```

157         x = tree[x].ch[1];
158     }
159 }
160 if ( last )
161     Splay(last);
162 return ans;
163 }
164
165 ll Kth(int k)
166 {
167     assert(1 <= k && k <= Size());
168     int x = root;
169     while ( true )
170     {
171         int left = tree[tree[x].ch[0]].siz;
172         if ( k <= left )
173             x = tree[x].ch[0];
174         else if ( k <= left + tree[x].cnt )
175         {
176             Splay(x);
177             return tree[x].val;
178         }
179         else
180         {
181             k -= left + tree[x].cnt;
182             x = tree[x].ch[1];
183         }
184     }
185 }
186
187 optional<ll> Predecessor(ll value)
188 {
189     int x = root, last = 0, answer = 0;
190     while ( x )
191     {
192         last = x;
193         if ( tree[x].val < value )
194         {
195             answer = x;
196             x = tree[x].ch[1];
197         }
198         else
199             x = tree[x].ch[0];
200     }
201     if ( last )
202         Splay(last);
203     if ( !answer )
204         return nullopt;
205     Splay(answer);
206     return tree[answer].val;
207 }
208
209 optional<ll> Successor(ll value)
210 {
211     int x = root, last = 0, answer = 0;
212     while ( x )

```

```
213     {
214         last = x;
215         if ( tree[x].val > value )
216         {
217             answer = x;
218             x = tree[x].ch[0];
219         }
220         else
221             x = tree[x].ch[1];
222     }
223     if ( last )
224         Splay(last);
225     if ( !answer )
226         return nullopt;
227     Splay(answer);
228     return tree[answer].val;
229 }
230 };
```

第一百一十二章 状态序列维护

112.1 状态序列 Treap：分组区间 GCD

算法约束见动态版本第 173 页。本节代码独立可抄；采用下列实际接口名。

声明 `Gcd_Sequence_Treap<N>` 后调用 `Init(seed)`；`N` 为峰值元素数。`Build` 重建序列，`Insert`、`Erase`、`Set`、`Toggle`、`Query` 对应动态版接口。大对象放在全局或 `static`。

```
1 template <int N> struct Gcd_Sequence_Treap
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0, tag = 0;
8         int cnt[2]{};
9         U val = 0, g[2]{}, pri = 0;
10    };
11
12    Node tree[N + 1];
13    int tot = 0;
14    vector<int> recycled;
15    mt19937_64 rng;
16    int root = 0;
17
18    void Init(U seed = 712367821)
19    {
20        root = 0;
21        tot = 0;
22        tree[0] = Node{};
23        recycled.clear();
24        rng.seed(seed);
25    }
26
27    int Size() const
28    {
29        return tree[root].siz;
30    }
31
32    int New_Node(U value, int tag)
33    {
34        assert(tag == 0 || tag == 1);
35        int p;
36        if ( recycled.empty() )
37        {
38            assert(tot < N);
39            p = ++tot;
40            tree[p] = Node{};
41        }
42        else
43        {
44            p = recycled.back();
```

```

45         recycled.pop_back();
46         tree[p] = Node{};
47     }
48     tree[p].siz = 1;
49     tree[p].val = value;
50     tree[p].tag = tag;
51     tree[p].cnt[tag] = 1;
52     tree[p].g[tag] = value;
53     tree[p].pri = rng();
54     return p;
55 }
56
57 void Push_Up(int p)
58 {
59     int l = tree[p].l, r = tree[p].r;
60     tree[p].siz = tree[l].siz + tree[r].siz + 1;
61     for ( int t = 0; t < 2; t++ )
62     {
63         tree[p].cnt[t] = tree[l].cnt[t] + tree[r].cnt[t] + (tree[p].tag == t);
64         tree[p].g[t] = gcd(tree[l].g[t], tree[r].g[t]);
65         if ( tree[p].tag == t )
66             tree[p].g[t] = gcd(tree[p].g[t], tree[p].val);
67     }
68 }
69
70 void Split(int p, int k, int &l, int &r)
71 {
72     if ( !p )
73     {
74         l = 0;
75         r = 0;
76         return;
77     }
78     int left = tree[tree[p].l].siz;
79     if ( k > left )
80     {
81         l = p;
82         Split(tree[p].r, k - left - 1, tree[p].r, r);
83     }
84     else
85     {
86         r = p;
87         Split(tree[p].l, k, l, tree[p].l);
88     }
89     Push_Up(p);
90 }
91
92 int Merge(int l, int r)
93 {
94     if ( !l || !r )
95         return l ? l : r;
96     if ( tree[l].pri > tree[r].pri )
97     {
98         tree[l].r = Merge(tree[l].r, r);
99         Push_Up(l);
100        return l;

```

```

101     }
102     tree[r].l = Merge(l, tree[r].l);
103     Push_Up(r);
104     return r;
105 }
106
107 void Rebuild(int p)
108 {
109     if ( !p )
110         return;
111     Rebuild(tree[p].l);
112     Rebuild(tree[p].r);
113     Push_Up(p);
114 }
115
116 void Release(int p)
117 {
118     if ( !p )
119         return;
120     Release(tree[p].l);
121     Release(tree[p].r);
122     recycled.push_back(p);
123 }
124
125 // O(n) GCD calls; replaces the current sequence.
126 void Build(const vector<pair<U, int>> &v)
127 {
128     assert(v.size() <= N);
129     tot = 0;
130     tree[0] = Node{};
131     recycled.clear();
132     root = 0;
133     vector<int> st;
134     for ( auto [value, tag] : v )
135     {
136         int p = New_Node(value, tag), last = 0;
137         while ( !st.empty() && tree[st.back()].pri < tree[p].pri )
138         {
139             last = st.back();
140             st.pop_back();
141         }
142         tree[p].l = last;
143         if ( !st.empty() )
144             tree[st.back()].r = p;
145         st.push_back(p);
146     }
147     if ( st.empty() )
148         return;
149     root = st.front();
150     Rebuild(root);
151 }
152
153 // Insert after the first k entries. Range operations below are 1-based.
154 void Insert(int k, U value, int tag)
155 {
156     assert(0 <= k && k <= Size());

```

```

157     int p = New_Node(value, tag), l, r;
158     Split(root, k, l, r);
159     root = Merge(Merge(l, p), r);
160 }
161
162 void Erase(int l, int r)
163 {
164     assert(1 <= l && l <= r && r <= Size());
165     int x, y, z;
166     Split(root, r, y, z);
167     Split(y, l - 1, x, y);
168     root = Merge(x, z);
169     Release(y);
170 }
171
172 void Set(int k, U value)
173 {
174     assert(1 <= k && k <= Size());
175     int x, y, z;
176     Split(root, k, y, z);
177     Split(y, k - 1, x, y);
178     tree[y].val = value;
179     Push_Up(y);
180     root = Merge(Merge(x, y), z);
181 }
182
183 void Toggle(int k)
184 {
185     assert(1 <= k && k <= Size());
186     int x, y, z;
187     Split(root, k, y, z);
188     Split(y, k - 1, x, y);
189     tree[y].tag ^= 1;
190     Push_Up(y);
191     root = Merge(Merge(x, y), z);
192 }
193
194 optional<U> Query(int l, int r, int tag)
195 {
196     assert(1 <= l && l <= r && r <= Size());
197     assert(tag == 0 || tag == 1);
198     int x, y, z;
199     Split(root, r, y, z);
200     Split(y, l - 1, x, y);
201     optional<U> answer;
202     if ( tree[y].cnt[tag] )
203         answer = tree[y].g[tag];
204     root = Merge(Merge(x, y), z);
205     return answer;
206 }
207 };

```

第一百一十三章 一般图匹配

113.1 一般图最大基数匹配：带花树

算法约束见动态版本第 177 页。本节代码独立可抄；采用下列实际接口名。

声明 `General_Matching<N>` 后调用 `Init(n)`，`N` 为点数上限。`Insert` 加边，`Solve` 返回匹配边数，`mate` 保存配对点，点号仍为 0-based。大对象放在全局或 `static`。

```
1 template <int N> struct General_Matching
2 {
3     int n;
4     vector<int> g[N];
5     int mate[N], pre[N], base[N], q[N];
6     bool outer[N], flower[N], seen[N];
7
8     void Init(int vertices)
9     {
10         assert(0 <= vertices && vertices <= N);
11         n = vertices;
12         for ( int u = 0; u < n; u++ )
13             g[u].clear();
14         fill(mate, mate + n, -1);
15     }
16
17     void Insert(int u, int v)
18     {
19         assert(0 <= u && u < n && 0 <= v && v < n);
20         if ( u == v )
21             return;
22         g[u].push_back(v);
23         g[v].push_back(u);
24     }
25
26     int Lca(int u, int v)
27     {
28         fill(seen, seen + n, false);
29         while ( true )
30         {
31             u = base[u];
32             seen[u] = true;
33             if ( mate[u] == -1 )
34                 break;
35             u = pre[mate[u]];
36         }
37         while ( !seen[base[v]] )
38             v = pre[mate[base[v]]];
39         return base[v];
40     }
41
42     void Mark(int u, int b, int v)
43     {
44         while ( base[u] != b )
```



```

45     {
46         flower[base[u]] = true;
47         flower[base[mate[u]]] = true;
48         pre[u] = v;
49         v = mate[u];
50         u = pre[v];
51     }
52 }
53
54 bool Augment(int root)
55 {
56     fill(outer, outer + n, false);
57     fill(pre, pre + n, -1);
58     iota(base, base + n, 0);
59     int tail = 0;
60     q[tail++] = root;
61     outer[root] = true;
62     for ( int head = 0; head < tail; head++ )
63     {
64         int u = q[head];
65         for ( int v : g[u] )
66         {
67             if ( base[u] == base[v] || mate[u] == v )
68                 continue;
69             if ( v == root || (mate[v] != -1 && pre[mate[v]] != -1) )
70             {
71                 int b = Lca(u, v);
72                 fill(flower, flower + n, false);
73                 Mark(u, b, v);
74                 Mark(v, b, u);
75                 for ( int x = 0; x < n; x++ )
76                 {
77                     if ( !flower[base[x]] )
78                         continue;
79                     base[x] = b;
80                     if ( !outer[x] )
81                     {
82                         outer[x] = true;
83                         q[tail++] = x;
84                     }
85                 }
86             }
87             else if ( pre[v] == -1 )
88             {
89                 pre[v] = u;
90                 if ( mate[v] == -1 )
91                 {
92                     while ( v != -1 )
93                     {
94                         int p = pre[v];
95                         int next = mate[p];
96                         mate[v] = p;
97                         mate[p] = v;
98                         v = next;
99                     }
100                     return true;

```

```
101         }
102         int x = mate[v];
103         outer[x] = true;
104         q[tail++] = x;
105     }
106 }
107 }
108 return false;
109 }
110
111 // Rebuild the maximum matching; mate[u] is -1 for an unmatched vertex.
112 int Solve()
113 {
114     fill(mate, mate + n, -1);
115     int answer = 0;
116     for ( int u = 0; u < n; u++ )
117     {
118         if ( mate[u] == -1 && Augment(u) )
119             answer++;
120     }
121     return answer;
122 }
123 };
```

第一百一十四章 线性递推

114.1 Berlekamp–Massey：最短线性递推

算法约束见动态版本第 180 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Prime p; normalized input. s[i] = sum(c[j-1] * s[i-j]), i >= c.size().
2 vector<int> Berlekamp_Massey(const vector<int> &s, int p = 998244353)
3 {
4     assert(p >= 2);
5     vector<int> c{1}, b{1};
6     int len = 0, shift = 1, last = 1;
7     for ( int i = 0; i < (int)s.size(); i++ )
8     {
9         assert(0 <= s[i] && s[i] < p);
10        long long d = s[i];
11        for ( int j = 1; j <= len; j++ )
12            d = (d + 1LL * c[j] * s[i - j]) % p;
13        if ( d == 0 )
14        {
15            shift++;
16            continue;
17        }
18        long long inv = 1, x = last;
19        for ( int e = p - 2; e; e >>= 1 )
20        {
21            if ( e & 1 )
22                inv = inv * x % p;
23            x = x * x % p;
24        }
25        long long factor = d * inv % p;
26        auto old = c;
27        if ( c.size() < b.size() + shift )
28            c.resize(b.size() + shift);
29        for ( int j = 0; j < (int)b.size(); j++ )
30            c[j + shift] = (c[j + shift] - factor * b[j] % p + p) % p;
31        if ( 2 * len <= i )
32        {
33            len = i + 1 - len;
34            b = old;
35            last = d;
36            shift = 1;
37        }
38        else
39            shift++;
40    }
41    c.resize(len + 1);
42    c.erase(c.begin());
43    for ( int &x : c )
44        x = x == 0 ? 0 : p - x;
45    return c;
46 }
```

114.2 特征多项式求线性递推第 k 项

算法约束见动态版本第 182 页。本节代码独立可抄；采用下列实际接口名。

```

1 // init has exactly k entries; c[j-1] multiplies a[n-j]. Modulus need not be prime.
2 int Recurrence_Nth(const vector<int> &init,
3                   const vector<int> &c,
4                   unsigned long long n,
5                   int p = 998244353)
6 {
7     assert(p >= 2 && init.size() == c.size());
8     int k = c.size();
9     if ( !k )
10         return 0;
11     for ( int x : init )
12         assert(0 <= x && x < p);
13     for ( int x : c )
14         assert(0 <= x && x < p);
15     if ( n < (unsigned)k )
16         return init[n];
17     auto multiply = [&](const vector<int> &a, const vector<int> &b)
18     {
19         vector<int> t(2 * k - 1);
20         for ( int i = 0; i < k; i++ )
21         {
22             for ( int j = 0; j < k; j++ )
23                 t[i + j] = (t[i + j] + 1LL * a[i] * b[j]) % p;
24         }
25         for ( int i = 2 * k - 2; i >= k; i-- )
26         {
27             for ( int j = 1; j <= k; j++ )
28                 t[i - j] = (t[i - j] + 1LL * t[i] * c[j - 1]) % p;
29         }
30         t.resize(k);
31         return t;
32     };
33     vector<int> answer(k), x(k);
34     answer[0] = 1;
35     if ( k == 1 )
36         x[0] = c[0];
37     else
38         x[1] = 1;
39     while ( n )
40     {
41         if ( n & 1 )
42             answer = multiply(answer, x);
43         x = multiply(x, x);
44         n >>= 1;
45     }
46     long long value = 0;
47     for ( int i = 0; i < k; i++ )
48         value = (value + 1LL * answer[i] * init[i]) % p;
49     return value;
50 }

```

第三部分

数学速查与维护

第一百一十五章 数学速查：条件先于公式

115.1 整除、卷积与筛法

对正整数定义狄利克雷卷积

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d).$$

单位元为 $\varepsilon(n) = [n = 1]$ ，常函数 $1(n) = 1$ ，恒等函数 $\text{id}(n) = n$ 。关键恒等式是 $\mu * 1 = \varepsilon$ 、 $\varphi * 1 = \text{id}$ 。若 $F(n) = \sum_{d|n} f(d)$ ，则 $f(n) = \sum_{d|n} \mu(d)F(n/d)$ 。

$$\sum_{i=1}^a \sum_{j=1}^b [\gcd(i, j) = 1] = \sum_{d=1}^{\min(a, b)} \mu(d) \lfloor a/d \rfloor \lfloor b/d \rfloor.$$

整除分块：当左端点为 l 、商为 $q = \lfloor n/l \rfloor$ ，右端点为 $r = \lfloor n/q \rfloor$ 。同时包含多个商时，取各自右端点的最小值。杜教筛由卷积恒等式获得

$$M(n) = 1 - \sum_{l=2}^n M(\lfloor n/l \rfloor), \quad \Phi(n) = \frac{n(n+1)}{2} - \sum_{l=2}^n \Phi(\lfloor n/l \rfloor),$$

求和按等商区间合并；第二式求和中没有额外乘以 l 。

115.2 模运算的适用条件

模逆元存在当且仅当 $\gcd(a, m) = 1$ 。费马逆元 a^{p-2} 仅适用于素数模数及 $a \not\equiv 0$ 。对正模数 m, n ，同余式 $x \equiv a \pmod{m}$ 、 $x \equiv b \pmod{n}$ 可合并当且仅当 $\gcd(m, n) \mid (b - a)$ ；新模数为 $\text{lcm}(m, n)$ 。

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p} \quad (p \text{ 为素数, } n_i, k_i \text{ 是 } p \text{ 进制位}).$$

Lucas 不能直接套在合数模数。阶乘与逆阶乘预处理要求上限小于模数；超过模数会出现阶乘为零。扩展欧拉降幂中，非互质情况不能直接将指数对 $\varphi(m)$ 取模；需要判断原指数是否达到足够阈值。大整数指数可以在读入时同时维护余数与是否超过阈值。

115.3 任意数组的批量逆元

设 $s_i = \prod_{j < i} a_j \pmod{m}$ ， $s_0 = 1$ 。仅对总乘积 s_n 求一次逆元，随后从右向左处理。若当前 $t = s_{i+1}^{-1}$ ，则

$$a_i^{-1} = t s_i \pmod{m}, \quad s_i^{-1} = t a_i \pmod{m}.$$

先保存原 a_i 再覆盖输出，避免误用已写入的逆元更新 t 。总乘积与 m 互素当且仅当每一项均与 m 互素，因此一次扩展欧几里得也能判断整个批次是否可行；模数无需为素数。实现见第 94 页。

接口返回可选向量：无解是空的可选值，空输入是有效的空向量，两者不同。模数为 1 时按接口约定返回全零。调用者可以移动不再使用的输入数组，避免额外复制。P5431 的驱动程序计算 $\sum_{i=1}^n k^i/a_i$ ，注意权重从 k^1 开始。五百万项规模采用缓冲的字符读取，并移动输入以控制峰值内存。双版本已做乘法证书和非互素拒绝的穷举检查，以及 64 位极值、空数组、模数 1 的检查；完整驱动另用逐项逆元和几何级数进行对拍。P5431 双版本均通过全部 5 个测试点；动态版记录 297561197，传统版记录 297561804。最慢单点分别为 386、384 毫秒，峰值内存均为 76.80 MB。合数模数、不可逆输入和完整 64 位范围以上述本地测试验证。

该方法可处理任意可逆数组。连续整数 $1, 2, \dots, N$ 在素数模数下的专用线性递推仍另列为待补。来源：<https://oi-wiki.org/math/number-theory/inverse/>。

115.4 扩展 CRT：逐式合并

实现见第 87 页的 `crt`，传统版对应 `Crt`。初始取 $r = 0, m = 1$ ，依次合并 $x \equiv b \pmod{n}$ 。先把余数归一化。设 $g = \gcd(m, n)$ ，由扩展欧几里得得到 $mu + nv = g$ 。写 $x = r + mt$ ，原问题化为 $mt \equiv b - r \pmod{n}$ 。若 $g \nmid (b - r)$ 则无解，否则

$$q = n/g, \quad t \equiv \frac{b-r}{g}u \pmod{q}, \quad 0 \leq t < q.$$

新周期为 mq ，新余数为 $(r + mt) \bmod (mq)$ 。所有可行 t 相差 q ，故新余数加新周期的任意整数倍恰好是全部解；逐式合并时这一不变量保持成立。模数不必互素，模数 1 也可直接参与。

代码要求正模数且最终周期不超过有符号 64 位上限；中间乘法与扩展欧几里得系数使用 128 位整数。无解返回假；有解而周期越界抛出异常。失败时旧周期不变，旧余数已归一化。 k 条方程的时间为 $O(k \log M)$ ，其中 M 是输入及中间模数的上界；递归欧几里得使用 $O(\log M)$ 栈空间。P4777 的完整程序按“模数、余数”读入，题目保证有解且周期不超过 10^{18} 。本地验证包含双版本小模数穷举、任意精度整数对拍、有符号极值、无解与越界，以及十万条同余式的完整程序测试。双版本均已通过 P4777 全部 15 个测试点；动态版记录 297521102，传统版记录 297521173。

kuangbin 的“解同余方程组”是模 7 的线性方程组消元应用，见后面的工厂工期一节。来源：<https://oi-wiki.org/math/number-theory/crt/>；WIDA 数论模板的 `excrt` 节。

115.5 Garner：总周期可以超过整数范围

设正模数 m_i 两两互素， $P = \prod_i m_i$ 。经典构造为 $x = \sum_i b_i P_i (P_i^{-1} \bmod m_i) \bmod P$ ，其中 $P_i = P/m_i$ 。模 m_i 时其他项都为零，本项为 b_i ；两解之差被所有 m_i 整除，因而被 P 整除。但直接写此式需要存储整个 P ，并防止乘法溢出。

Garner 改用混合进制

$$x = d_0 + d_1 m_0 + d_2 m_0 m_1 + \cdots, \quad 0 \leq d_i < m_i.$$

在确定第 i 位前，对每个后续模数维护已确定部分的余数 v_j ，以及当前位权 $c_j = \prod_{t < i} m_t \bmod m_j$ 。于是

$$d_i = (b_i - v_i) c_i^{-1} \bmod m_i.$$

把 $c_j d_i$ 加到 v_j ，再将 c_j 乘 m_i 即可进入下一位。互素性保证逆元存在；模数 1 的位固定为 0，可直接保留。位数范围保证重建的 x 恰在 $[0, P)$ ，因此就是最小非负解。实现见第 95 页。

若只需 $x \bmod T$ ，从高位向低位做 Horner 计算：每步将当前值乘对应模数，加上该位，再对 T 取模。 T 无需与 m_i 互素，因为这里不对 T 求逆。这返回的是规范解 x 的余数；一般解 $x + kP$ 对 T 的余数可能不同。例如模 3 余 2、模 5 余 3，规范解是 8；对 7 取模为 1，但另一个解 23 的余数是 2。若用多模 NTT 恢复整数系数，仍须证明真实系数位于所选规范区间，不能在乘积不够大时直接把 CRT 的代表元当成原整数。

双版本本地验证使用任意精度整数的经典 CRT 求和作为独立对拍，检查混合进制的完整重建、负余数、有符号极值、非互素输入拒绝和 1000 个模数。P1495 完整程序按题目保证的 $P \leq 10^{18}$ 输出规范解，双版本均通过全部 11 个测试点；动态版记录 297521594，传统版记录 297521658。超出该题范围的能力以上述本地测试验证。来源：<https://oi-wiki.org/math/number-theory/crt/>。

115.6 素数模平方根

奇素数 p 下，对 $n \not\equiv 0 \pmod{p}$ ，Euler 判别式 $n^{(p-1)/2} \equiv 1 \pmod{p}$ 当且仅当存在平方根，否则为 -1 。当 $p \equiv 3 \pmod{4}$ 且有解时， $r = n^{(p+1)/4}$ 为一根，另一根是 $p - r$ 。

Cipolla 选择 a 使 $w = a^2 - n$ 是非平方，令 $\omega^2 = w$ 。在扩域中取 $\alpha = a + \omega$ ，有 $\alpha^p = a - \omega$ ，故 $\alpha^{p+1} = n$ 。令 $\beta = \alpha^{(p+1)/2}$ ，则 $\beta^2 = n$ 且 $\beta^{p-1} = n^{(p-1)/2} = 1$ ，所以 β 落在原素数域中。实现见第 106 页；零和模数 2 单独处理。模数为合数时不能照搬此判别和扩域构造。

115.7 乘法阶与原根

对 $\gcd(a, n) = 1$, 阶是最小的正整数 d 使 $a^d \equiv 1 \pmod{n}$, 且 $d \mid \varphi(n)$ 。从 $d = \varphi(n)$ 开始, 对每个素因子 q , 只要 $a^{d/q} \equiv 1$ 就继续除去 q , 所得即为实际阶。对于不可逆的 a , 接口返回 0, 表示此处没有定义乘法阶。

原根要求阶达到 $\varphi(n)$ 。只需对其每个不同素因子 q 检查 $a^{\varphi(n)/q} \not\equiv 1$: 若实际阶是真因子, 商中必含某个这样的 q 。必须先检查 $\gcd(a, n) = 1$; 例如模 9 的候选 3 虽通过上述非 1 判定, 却不是原根。 $n \geq 2$ 时, 原根恰存在于 $2, 4, p^k, 2p^k$, 其中 p 是奇素数。原根存在且取一个原根 g 后, g^t 的阶是 $\varphi(n)/\gcd(t, \varphi(n))$, 所以枚举与 $\varphi(n)$ 互素的指数即可得到全部原根, 共 $\varphi(\varphi(n))$ 个; 这些数按大小输出时还需排序。

实现见第 97 页。本地验证穷举小模数的幂循环, 与双版本的阶、最小值和完整集合逐项比较; 另验证大素数、奇素数幂、无原根合数和有符号输入边界。P6091 的完整程序输出排序后第 $d, 2d, \dots$ 个原根, 零个输出时仍保留空行。双版本均通过全部 11 个测试点; 动态版记录 297558740, 传统版记录 297559082。求阶、最小原根的最小性及更大输入范围仍以上述本地证据验证。若 $d = \text{ord}_n(a)$, 则 $\text{ord}_n(a^k) = d/\gcd(d, k)$ 。两个可逆元素乘积的阶整除各自阶的最小公倍数; 两阶互素时, 乘积的阶等于两阶之积。来源: <https://oi-wiki.org/math/number-theory/primitive-root/>; kuangbin 2.17。

115.8 Carmichael: 最小公共指数

$\lambda(n)$ 是模 n 的所有可逆元素之阶的最小公倍数, 约定 $\lambda(1) = 1$ 。它整除 $\varphi(n)$, 但可以严格更小, 例如 $\lambda(8) = 2$ 而 $\varphi(8) = 4$ 。按互素的素数幂分解 $n = \prod_i p_i^{e_i}$ 后, 由 CRT, 对每个分量同时成立的幂等式也对 n 成立, 因此

$$\lambda(n) = \text{lcm}_i \lambda(p_i^{e_i}), \quad \lambda(p^e) = \begin{cases} 2^{e-2}, & p = 2, e \geq 3, \\ p^{e-1}(p-1), & \text{其他情形} \end{cases}$$

奇素数幂的单位群有原根, 公共指数即其大小。模 2^e 的奇数可表示为 $\pm 5^k$, 当 $e \geq 3$ 时 5 的阶为 2^{e-2} , 这给出二幂情形的上界和达到上界的元素。实现见第 100 页, 试除版本适用于有符号 32 位正模数。

对可逆底数可把指数对 $\lambda(n)$ 取模; 非互素底数仍须另行处理。例如模 8 下 $2^2 = 4$, 但 $2^{2 \bmod \lambda(8)} = 1$, 直接降幂会错误。另有 $\lambda(\text{lcm}(a, b)) = \text{lcm}(\lambda(a), \lambda(b))$, 这也提供了独立的组合检查条件。

合数 n 是 Carmichael 数当且仅当 $\lambda(n) \mid (n-1)$ 。等价的 Korselt 判别是: n 无平方因子, 且每个素因子 p 都满足 $(p-1) \mid (n-1)$ 。素数也满足前一整除式, 判定时必须排除素数。这些合数对所有互素底数都通过 Fermat 判定, 不能用单纯的 Fermat 检验代替素性判定。无平方因子再配合 CRT, 也使所有整数 a 满足 $a^n \equiv a \pmod{n}$ 。

双版本以小模数的全部实际阶之最小公倍数对拍, 另检验大二幂、奇素数幂、32 位边界及公共指数证书。

UVa10006 完整程序对全部合法输入进行本地 Korselt 对拍; 在线 AC 待补。

来源: <https://oi-wiki.org/math/number-theory/primitive-root/>;

应用题: <https://onlinejudge.org/external/100/10006.pdf>。

115.9 Lucas: 按素数进制逐位计算

在特征为素数 p 的域中, $(1+x)^p = 1+x^p$ 。若 $n = \sum_i n_i p^i$, 则

$$(1+x)^n = \prod_i (1+x^{p^i})^{n_i}.$$

每个因子选择的次数在 $0, \dots, n_i$ 之间, 严格小于 p , 因此位数表示唯一。比较 x^k 的系数即得 $\binom{n}{k} = \prod_i \binom{n_i}{k_i} \pmod{p}$; 若某位 $k_i > n_i$, 答案立即为零。仅预处理 $0!, \dots, (p-1)!$ 及其逆元, 这些数均非零。 $p!$ 为零, 不能把阶乘表延长后继续用除法公式。实现见第 101 页。

预处理后每位只做常数乘法, 适合固定小素数模数下的多次查询。接口接收 n, k , 不接收 “ n, m 表示选 n 个” 的隐含约定; P3807 的驱动程序显式调用 `choose(n+m, n)`。题目范围内求和安全, 完整 64 位输入自行求和时仍要检查溢出。本地验证包含 Pascal 三角、精确大整数、小模数重建和最大无符号 64 位下标; P3807 双版本均已通过全部 25 个测试点; 动态版记录 297556209, 传统版记录 297556596。完整 64 位参数及额外边界以上述本地对拍验证。来源: <https://oi-wiki.org/math/number-theory/lucas/>。

115.10 合数模数组合数与插值

对素数幂 $q = p^e$, 记 $v_p(n!) = \sum_{j \geq 1} \lfloor n/p^j \rfloor$, $U_p(n) = n!/p^{v_p(n!)}$ 。设 $t = v_p(n!) - v_p(k!) - v_p((n-k)!)$ 。当 $t \geq e$ 时组合数模 q 为零; 否则

$$\binom{n}{k} \equiv U_p(n) U_p(k)^{-1} U_p(n-k)^{-1} p^t \pmod{q}.$$

这里仅对与 p 互素的单位部分求逆。各素数幂上的结果通过 CRT 合并, 对应第 103 页的实现。

在素数域中, n 个模 p 两两不同的横坐标确定唯一的次数小于 n 的多项式。定义 $w_i = y_i / \prod_{j \neq i} (x_i - x_j)$, 则

$$f(k) = \sum_{i=0}^{n-1} w_i \prod_{j < i} (k - x_j) \prod_{j > i} (k - x_j).$$

该式不除以 $k - x_i$, 查询点命中样本点时仍然成立。当 $x_i = i$ 时, 分母为 $i!(n-1-i)!(-1)^{n-1-i}$, 要求 $n \leq p$ 。实现见第 108 页。对整数幂前缀和 $S_d(k) = \sum_{i=1}^k i^d$, 当 $0 \leq d \leq p-2$ 时, 它在模 p 意义下可用次数至多 $d+1$ 的多项式表示; 取 $0, 1, \dots, d+1$ 的样本即可。不能只因某序列前几项能插值, 就断言后续仍由该多项式给出。

115.11 阶乘表: 组合数与排列数

当素数模数为 p 且 $0 \leq k \leq n < p$ 时, 阶乘均非零, 因此

$$C(n, k) = n! (k!)^{-1} ((n-k)!)^{-1}, \quad P(n, k) = n! ((n-k)!)^{-1} \pmod{p}.$$

将表从旧上限扩展到 N , 先正向补齐阶乘, 只对 $N!$ 求一次逆元, 再用 $(i-1)!^{-1} = i(i!)^{-1}$ 反向补齐新逆阶乘项即可。不需要每一项分别快速幂求逆。较小上限的初始化请求保留已有表。查询前显式扩表, 之后查询为常数时间; 内部两表重分配时旧元素引用不能继续使用。实现见第 93 页。

WIDA 的“组合数 (Lucas)”文件实际提供的是普通阶乘、逆阶乘、组合数和排列数, 没有逐位 Lucas 递推。本书以此节覆盖该文件的数学功能, 用显式扩表取代查询时自动扩表, 适用范围仍须限制在模数以内。验证使用独立 Pascal 三角和下降阶乘, 另检查百万项表及扩表前后的原查询结果。

115.12 组合计数

容斥: $|\bigcup A_i| = \sum_{\emptyset \neq S} (-1)^{|S|+1} |\bigcap_{i \in S} A_i|$ 。二项式反演: 若 $b_n = \sum_{k=0}^n \binom{n}{k} a_k$, 则 $a_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k$ 。

$$\sum_k \binom{a}{k} \binom{b}{n-k} = \binom{a+b}{n}, \quad C_n = \binom{2n}{n} - \binom{2n}{n+1}.$$

差分形式的 Catalan 公式避免直接除以 $n+1$, 在模数下尤其有用, 但组合数本身仍须正确求模。第二类 Stirling 数满足

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\} + k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}, \quad S(0, 0) = 1.$$

第一类无符号 Stirling 数把系数 k 换为 $n-1$ 。

球与盒	空盒允许	空盒禁止
不同球, 不同盒	k^n	$k!S(n, k)$
相同球, 不同盒	$\binom{n+k-1}{k-1}$	$\binom{n-1}{k-1}$
不同球, 相同盒	$\sum_{j=0}^k S(n, j)$	$S(n, k)$

上表默认 $k \geq 1$; $n = 0, k = 0$ 等边界应按原问题约定单独处理。

115.13 群作用计数

有限群 G 作用在有限集合 X 上时, 轨道数为

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|.$$

对位置置换的 q 色染色, $|\text{Fix}(g)| = q^{c(g)}$, 其中 $c(g)$ 为循环数。环形项链只商去旋转; 手链还商去反射, 二者群不同, 不能共用分母。模数整除 $|G|$ 时不能使用模逆元, 须先在合适的整数模数下求和再做精确除法。

115.14 生成函数与递推

普通生成函数 $A(x) = \sum a_n x^n$ 的乘法对应普通卷积。指数生成函数 $A(x) = \sum a_n x^n / n!$ 的乘法对应 $c_n = \sum_k \binom{n}{k} a_k b_{n-k}$ 。形式幂级数求逆要求常数项可逆; $\log A$ 取 $A(0) = 1$, $\exp A$ 取 $A(0) = 0$ 。有限域中积分涉及 $1/i$, 必须避免 i 被特征整除。

$$B_{\text{new}} = B(2 - AB) \pmod{x^{2m}}, \quad (\log A)' = A'/A.$$

BM 从样本得到最短线性递推。要对未来项使用, 必须另外证明序列确实具有不超过相应阶数的线性递推; 少量数值吻合不是证明。矩阵转移、有限状态计数常可提供阶数上界。

115.15 线性代数与生成树

域上的线性方程组一致时, 解集为一个特解加零空间, 零空间维数为未知数个数减秩。浮点消元中的秩依赖误差阈值; 模素数消元是精确域运算; 合数模数下不可随意除以非零元。无向图 Laplacian 的对角元为度数, 非对角元为负的边重数。删去同一行列后的行列式为生成树数。自环不参与生成树; 重边需要累加。只有一个顶点时空余子式行列式约定为 1。有向图需要重新明确入度/出度 Laplacian 与根向内/向外树的方向, 不能直接复用无向封装。

115.16 模线性方程组：工厂工期

POJ2947 (问题 UVa1564) 给出每位工人制造的产品类型和起止星期。设第 j 种产品需 x_j 天, 第 i 条记录中出现 a_{ij} 次, 以星期一为 0、星期日为 6, 则

$$\sum_j a_{ij} x_j \equiv \text{finish}_i - \text{start}_i + 1 \pmod{7}.$$

起止两天均计入; 同一种产品重复出现必须累加系数。跨星期无需特判, 模数类型负责归一化负数。每个余数在 $[3, 9]$ 中恰有一个代表, 因此模方程的解数就是原问题的解数。

直接使用第 115 页的模高斯消元, 模数取 7。增广矩阵有 m 行、 $n+1$ 列, 最后一列是右端项。先判断 `consistent`: 假时输出无解; 为真且秩小于 n 时输出多解; 否则读取特解, 余数小于 3 的分量加 7。不能先以秩不足判断多解, 因为零系数行仍可能有非零右端项。本题所有非零系数均可逆; 合数模数不具备这一性质, 不能直接套用此消元。

完整双版本程序位于 `verify/poj/2947.*.cpp`。验证包含所有起止星期组合、产品重复 7 次导致系数归零、矛盾的零行、独立枚举所有 3~9 天赋值, 以及用可逆行变换构造的 300 阶稠密系统。该应用验证覆盖分类及唯一解输出, 不代替零空间基等其他接口的验证。来源: <https://onlinejudge.org/external/15/1564.pdf>; kuangbin 2.10.2。

115.17 概率、期望与博弈

期望线性性不要求独立: $\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$ 。方差可加需要协方差为零; 独立是充分条件。非负整数变量 $\mathbb{E}[X] = \sum_{k \geq 1} \Pr(X \geq k)$ 。吸收过程的期望方程 $E_i = 1 + \sum_j p_{ij} E_j$ 需要检查是否几乎必然吸收及期望是否有限;

不能对闭合非吸收分量直接求唯一有限解。有限无环公平组合游戏在正常玩法下

$$\text{SG}(u) = \text{mex}\{\text{SG}(v) : u \rightarrow v\}.$$

独立子游戏的 SG 值异或为零当且仅当先手必败。反常玩法、含环游戏、双方可走步不同的游戏不能直接套这个结论。

115.18 数值算法与几何

排序必须使用严格弱序；不能以“差小于 eps 则相等”定义通用排序比较器。叉积与行列式容易发生抵消；整数题优先精确整数判定，浮点构造单独处理误差。对于简单格点多边形（无孔），Pick 定理 $A = I + B/2 - 1$ ，边界格点数 $B = \sum \gcd(|\Delta x|, |\Delta y|)$ ；不要重复计入边端点。自适应积分的局部误差估计需要函数足够规则；奇点、间断与强振荡必须先分段处理。三分搜索要求单峰性，整数域最后枚举剩余区间；二分答案要求判定单调性。

来源索引 本章为重新组织的公式速查，扩展范围依据 <https://oi-wiki.org/math/>。小模数 exLucas 与有限域拉格朗日插值已有双风格实现和本地验证；素数模平方根已有 Cipolla 实现；仍需补齐二次剩余的其他方法与定理、Min25、Newton 插值、子集卷积、行列式进阶、数值积分等内容及验证。本章公式不能代替这些待完成代码。

115.19 线性递推：系数方向与外推条件

约定 $a_i = \sum_{j=1}^k c_{j-1} a_{i-j}$ ，初值是 $a_0, \dots, a_{k-1}$ 。对应特征多项式为

$$F(x) = x^k - c_0 x^{k-1} - c_1 x^{k-2} - \dots - c_{k-1}.$$

若 $x^n \bmod F(x) = \sum_{i=0}^{k-1} r_i x^i$ ，则 $a_n = \sum_{i=0}^{k-1} r_i a_i$ 。代码由高次向低次消去多项式项，第 j 个系数写入原次数减 j 的位置，避免倒置系数。例如 Fibonacci 初值为 $[0, 1]$ ，递推系数为 $[1, 1]$ 。

BM 在域上求有限前缀的最短递推。本书实现要求素数模数；一般的模合数环不能保证非零偏差存在逆元。仅凭若干样本无法证明真实序列以后一直遵循同一个规律。若事先知道序列满足阶数至多 d 的齐次线性递推，至少给出前 $2d$ 项，再用恢复出的系数求后续项。系数向量末尾的零不能擅自删除：它可能对应非平稳的初始段，改变阶数会改变递推开始适用的位置。

第一百一十六章 来源与覆盖维护

116.1 参考基线

- kuangbin, ACM 常用算法模板, 2018 年版本: <https://kuangbin.github.io/2018/08/01/ACM-template/>。
- WIDA XCPC: <https://github.com/hh2048/XCPC>, 参考快照 484e9e0。
- OI Wiki 数学: <https://oi-wiki.org/math/>。

本项目重新实现的代码与公式速查不能冒用原作者的验证记录。完整覆盖审计应为每个来源条目记录实现文件、替代关系与验证证据, 保留遗漏项直至完成。

116.2 后续覆盖项

尚需完成的范围以覆盖表为准, 包括动态树、平衡树、一般图匹配、最小树形图、上下界流、高阶几何构造、半平面交、三维凸包、数值积分、复杂数论及多项式、搜索与动态规划等。不能用通用模块的存在推断其覆盖了所有原书例题。

算法索引

AC_Automaton, 259
Affine_Segment_Tree, 207
AffineSegTree, 25
AhoCorasick, 75
Arborescence, 145

Basis_Intersection, 205
basis_intersection, 23
Basis_Sum_Intersection, 206
basis_sum_intersection, 24
Batch_Inverse, 278
batch_inverse, 94
Berlekamp-Massey, 367
Berlekamp_Massey, 360
berlekamp_massey, 180
Biconnected, 4
Biconnected_Graph, 187
Binary_Indexed_Tree, 198
Binomial, 93
Bipartite_Matching, 220
BipartiteMatching, 38
Blossom, 177
Bounded_Circulation, 215
BoundedCirculation, 32
Bridge_Augmentation, 195
bridge_augmentation, 12
Burnside, 367

Carmichael, 283, 365
carmichael, 100
Catalan, 366
Centroid_Pairs, 248
CentroidPairs, 64
Cipolla, 364
Combination, 277
CRT, 363

Dijkstra, 35
Dinic, 28
Directed_Euler, 226
Directed_MST, 326
DirectedEuler, 44
Dirichlet convolution, 363
Discrete_Log, 301

DiscreteLog, [119](#)
Disjoint_Set, [196](#)
DSU, [13](#)
Du_Jiao, [300](#)
DuJiao, [118](#)
Dynamic_Kth, [320](#)
DynamicKth, [139](#)

Edge_Compression, [193](#)
EdgeCompression, [10](#)
Euler theorem, [363](#)
Euler_LCA, [242](#)
EulerLCA, [58](#)
Ex_Lucas, [285](#)
exCRT, [364](#)
ExLucas, [103](#)
exLucas, [366](#)
expectation, [367](#)

Fenwick, [15](#)

Garner, [279](#), [364](#)
garner, [95](#)
Gcd_Sequence_Treap, [353](#)
GcdSequenceTreap, [173](#)
General_Matching, [357](#)
generating functions, [367](#)
Geometry_Extra, [331](#)
GeometryExtra, [151](#)
Global_Min_Cut, [328](#)

Heavy_Light_Decomposition, [238](#)
HLD, [54](#)
Hungarian, [144](#)

inclusion-exclusion, [366](#)
Integer_Geometry, [302](#)
Integer_Geometry_3D, [334](#)
IntegerGeometry, [121](#)
IntegerGeometry3D, [154](#)

Kmp_Match, [257](#)
kmp_match, [73](#)
Korselt criterion, [365](#)
Kuhn_Munkres, [325](#)

Lagrange, [108](#)
Lagrange interpolation, [366](#)
Lagrange_Interpolation, [290](#)
LazySeg, [15](#)

Lex_Two_SAT, [224](#)
LexTwoSAT, [42](#)
Li_Chao_Tree, [309](#)
LiChao, [128](#)
Lifting_LCA, [244](#)
LiftingLCA, [60](#)
Linear_Algebra, [297](#)
Linear_Sieve, [275](#)
LinearAlgebra, [115](#)
LinearSieve, [91](#)
Link_Cut_Tree, [337](#)
LinkCutTree, [157](#)
Low_Link, [222](#)
Lowlink, [40](#)
Lucas, [101](#), [365](#)
Lucas theorem, [363](#)
Lucas_Theorem, [284](#)

Manacher, [258](#)
manacher, [74](#)
Matrix Tree theorem, [367](#)
Max_Plus_Matrix, [309](#)
Maximum_Closure, [216](#)
maximum_closure, [34](#)
MaxPlusMatrix, [128](#)
Min_Cost_Flow, [212](#)
MinCostFlow, [30](#)
Minimum_Rotation, [259](#)
minimum_rotation, [75](#)
Mixed_Euler_Orientation, [233](#)
mixed_euler_orientation, [49](#)
Mixed_Euler_Trail, [235](#)
mixed_euler_trail, [51](#)
Mobius inversion, [363](#)
Mod_Int, [276](#)
Mod_Sqrt, [288](#)
mod_sqrt, [106](#)
ModInt, [92](#)
multiplicative order, [365](#)

Network_Flow, [210](#)
Number_Theory, [271](#)
NumberTheory, [87](#)
numerical stability, [368](#)

Odd_Cycle_Vertices, [190](#)
odd_cycle_vertices, [7](#)
Offline_LCA, [240](#)
OfflineLCA, [56](#)

Ordered_Splay, [348](#)
Ordered_Treap, [341](#)
OrderedSplay, [168](#)
OrderedTreap, [161](#)
Orient_Edges, [194](#)
orient_edges, [11](#)

Palindromic_Tree, [269](#)
PalindromicTree, [85](#)
Path_Intersection, [247](#)
path_intersection, [63](#)
Persistent_Array, [314](#)
Persistent_Distinct, [323](#)
Persistent_Kth, [311](#)
PersistentArray, [133](#)
PersistentDistinct, [142](#)
PersistentKth, [130](#)
Pick theorem, [368](#)
Pollard_Rho, [274](#)
PollardRho, [90](#)
Polya, [367](#)
Polynomial, [110](#), [292](#)
Position_Basis, [203](#)
PositionBasis, [21](#)
Prefix_Function, [257](#)
prefix_function, [73](#)
Prefix_Lcs, [267](#)
prefix_lcs, [83](#)
Primitive_Root, [280](#)
PrimitiveRoot, [97](#)

quadratic residue, [364](#)

rank, [367](#)
Real_Geometry, [305](#)
RealGeometry, [124](#)
Recurrence_Nth, [361](#)
recurrence_nth, [182](#)
Removal_Components, [192](#)
removal_components, [9](#)
Rollback_DSU, [197](#)
RollbackDSU, [14](#)

SCC, [36](#)
Segment_Tree, [199](#)
Sequence_Treap, [344](#)
SequenceTreap, [163](#)
Shortest_Path, [217](#)
Sprague-Grundy, [367](#)

[Square_Counts](#), 268
[square_counts](#), 84
[Stirling numbers](#), 366
[StoerWagner](#), 147
[Strong_Component](#), 218
[Subtree_Colors](#), 251
[SubtreeColors](#), 67
[Suffix_Array](#), 262
[Suffix_Automaton](#), 264
[Suffix_LCP](#), 266
[SuffixArray](#), 78
[SuffixAutomaton](#), 80
[SuffixLCP](#), 82
[Support_Hull](#), 335
[SupportHull](#), 155

[Tarjan_SCC](#), 185
[TarjanSCC](#), 2
[Tree_Path_Kth](#), 317
[TreePathKth](#), 136
[Two_SAT](#), 219
[TwoSAT](#), 37

[Undirected_Euler](#), 230
[UndirectedEuler](#), 47

[Virtual_Tree](#), 254
[VirtualTree](#), 70

[Weighted_Matching](#), 329
[WeightedMatching](#), 149
[Widget Factory](#), 367
[Word_Chain](#), 229
[word_chain](#), 46

[Xor_Basis](#), 201
[Xor_Walk](#), 236
[XorBasis](#), 18
[XorWalk](#), 52

[Z_Function](#), 258
[z_function](#), 74

[原根](#), 365

[扩展中国剩余定理](#), 364
[批量逆元](#), 363
[排列数](#), 366
[模高斯消元](#), 367
[混合进制](#), 364

线性递推, [368](#)

阶乘表, [366](#)